The Hidden Sharpe Ratio

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Abstract

We estimate the maximal Sharpe ratio in large cross-sections of assets directly, without constructing the efficient portfolio. Covariance estimation in this setting requires regularization, which suppresses the contribution of small-eigenvalue directions. Our estimator progressively recovers this contribution, yielding tight upper and lower bounds with asymptotic confidence intervals. We further show that ridge-regularized Markowitz portfolios, the construction underlying many existing high-Sharpe benchmarks, deliver out-of-sample Sharpe ratios that fall short of the true maximum by an explicit, non-vanishing multiplicative factor. Applied to U.S. equities, the method delivers an estimated maximal annualized Sharpe ratio of approximately 7.4, or 5.0 once micro-cap stocks are excluded, substantially exceeding existing portfolio-based benchmarks of 1.0–4.0. Adding corporate bonds raises the maximum, though not by a statistically significant margin. Nonlinear portfolio weights dominate linear ones, and characteristics introduced after 2004 contribute little additional Sharpe ratio.[1]

Keywords: maximal Sharpe ratio, efficient portfolio, high-dimensional inference, random matrix theory, risk-return tradeoff

JEL: C12, C40, C55, C57, G12

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[1]The implementation of the bounds can be found on Github

1 Introduction

How large is the maximal Sharpe ratio attainable from financial markets? It is the Sharpe ratio of the mean-variance efficient portfolio. This portfolio traces the efficient frontier and is the unique tradeable factor that prices the cross-section of assets, both conditionally and unconditionally (Hansen and Richard, 1987). Its Sharpe ratio pins down the minimum volatility of the stochastic discount factor (Hansen and Jagannathan, 1991). Yet despite decades of effort, the answer remains elusive. The maximal Sharpe ratio squared equals $\mathcal{R}_*^2 = \mu'\Sigma^{-1}\mu$, where μ and Σ are the mean vector and covariance matrix of asset returns. In a world of P assets and T observations, the naive plug-in estimator, essentially the Gibbons, Ross and Shanken (1989) (GRS) statistic, is severely biased when P is of the same order as T, and ceases to exist altogether when $P > T$. Modern empirical asset pricing routinely considers thousands of factors built from firm characteristics in pursuit of spanning the efficient portfolio, placing us squarely in the regime where P/T is large and conventional estimators break down.

In this paper, we estimate the maximal Sharpe ratio directly, without constructing the efficient portfolio. Our method builds on the ridge-regularized, debiased GRS statistic of Chernov, Kelly, Malamud and Schwab (2025). We extend it to deliver tight, two-sided bounds on $\mathcal{R}_*^2$ with asymptotic confidence intervals. These results hold when the number of assets and the number of observations grow at the same rate, $P/T \to c > 0$, which we call the proportional regime. We apply the method to a large cross-section of U.S. equities and corporate bonds, using 132 stock characteristics from Jensen, Kelly and Pedersen (2023), 65 bond characteristics from Dickerson, Robotti and Rossetti (2026a), and nonlinear random-feature portfolios à la Didisheim, Ke, Kelly and Malamud (2024). We document three sets of findings.

First, for U.S. equities we estimate a maximal annualized Sharpe ratio near 7.4, well above

the 1.0–4.0 range that portfolio-based methods deliver.[2] In rolling windows, the estimate ranges between 7 and 11. Micro-cap stocks raise liquidity concerns, so we also estimate the maximum on the universe above the 20th NYSE percentile, where it is 5.0. Both figures dominate the portfolio-based literature on the matching universe: the closest comparison, to methods that share our construction and include micro-caps, places the all-stock Sharpe ratio near 3.7, while the micro-excluded estimates elsewhere cluster near one. The true investment opportunity set expands well beyond what portfolio-based methods capture.

Ridge-regularized Markowitz portfolios, the construction underlying many existing high-Sharpe benchmarks, cannot attain this maximum. We show theoretically that the out-of-sample squared Sharpe ratio of any such portfolio falls short of the true maximum by a multiplicative factor that does not shrink with sample size. In the U.S. equity data, this factor is about 1.8 on both the full cross-section and the micro-excluded universe, equivalently $\sqrt{1.8} \approx 1.34$ on the Sharpe ratio itself, so the ridge ceiling sits near 5.5 against our 7.4 all-stock estimate and near 3.7 against our 5.0 ex-micro estimate. Ridge-regularized Markowitz portfolios therefore capture about three-quarters of the maximum in either universe. The realized literature sits below these ceilings on its matching universe: the 4.0 at the top of the micro-inclusive literature approaches the 5.5 ceiling, while the micro-excluded methods top out near 1.7, well short of the 3.7 ceiling. Even the 4.0 figures imply a frictionless Sharpe ratio an order of magnitude above what equilibrium asset pricing models generate. Our estimates widen this existing gap.

Second, adding corporate bonds to equities raises the maximal Sharpe ratio meaningfully but not statistically significantly, consistent with the view that equity factors largely span corporate bond risk premia (Dickerson, Julliard and Mueller, 2026b). Chen, Roussanov, Wang and Zou (2024b) reach a complementary conclusion across stocks, bonds, and options: cross-asset pure-alpha Sharpe ratios do not decay as more factors are added.

[2]This estimate is the Sharpe ratio of the efficient portfolio spanned by our managed-portfolio universe. Our framework bypasses the construction of this portfolio. The figure is reported gross of trading costs, as are all portfolio-based benchmarks we compare against.

Third, nonlinear portfolio weights yield substantially higher maximal Sharpe ratios than their linear counterparts, indicating that the frontier accessible through linear factor models leaves significant gains on the table. Characteristics introduced after 2004, a period that produced several widely studied anomalies, contribute little additional Sharpe ratio.

Three theoretical contributions underpin these empirical findings. To recover the Sharpe ratio that portfolio-based methods miss, we introduce *higher-order debiased GRS statistics*, a new family of statistical objects. These statistics progressively restore the contributions of small eigenvalues that ridge regularization suppresses. Each statistic is constructed by differentiating the debiased GRS statistic with respect to the penalty parameter and applying a recursive bias correction whose terms involve Bell polynomials.

Building on the higher-order debiased GRS statistics, we derive a sharp characterization of $\mathcal{R}^2_*$. The maximal Sharpe ratio squared expands into an infinite series of terms, each estimable from the data, with successive terms exhibiting sub-exponential decay. Truncating the series yields a lower bound; an upper bound follows under a mild regularity condition requiring only that Sharpe ratios associated with small eigenvalues do not dominate those associated with larger ones. Because $\mathcal{R}^2_*$ is the squared Hansen–Jagannathan bound on stochastic discount factor volatility, these are Hansen–Jagannathan bounds in high dimensions: they deliver that classical object in the regime $P > T$, where the plug-in does not exist. We further show that ridge-regularized Markowitz portfolios cannot reach even our lower bound, and the shortfall persists as the sample grows. Lopez-Lira and Roussanov (2023) and Chen, Roussanov and Wang (2025a) reach the corresponding empirical conclusion: portfolios that hedge or orthogonalize against the dominant common factors earn Sharpe ratios well above what dominant-PC methods deliver, the empirical analogue of our structural shortfall result.

To put these bounds to work empirically, we establish a joint Central Limit Theorem for the vector of higher-order debiased GRS statistics under a random-design framework

for expected returns. The asymptotic covariance is pivotal (it can be estimated without knowledge of the true μ or Σ), yielding confidence intervals for both bounds and, hence, a conservative confidence interval for $\mathcal{R}_*^2$.

Our framework imposes only mild regularity conditions on the eigenvalue distribution of the portfolio covariance. Our estimator applies whether factor exposures are linear or nonlinear, constant or time-varying, and whether the covariance of returns is sparse or dense. Other high-dimensional methods typically assume either a finite factor model or sparsity. Managed portfolios built from many characteristics generate infinitely many strong factors (Chernov, Kelly, Malamud and Schwab, 2026), and our framework accommodates them. This flexibility matters for the maximal Sharpe ratio: a finite factor model caps it at the Sharpe ratio of the included factors, whereas in our data the Sharpe ratio spreads across many eigendirections. Sparsity of the precision matrix or the mean return vector (Caner, Medeiros and Vasconcelos, 2023) can hold for individual assets, but forming portfolios generically eliminates it (Chernov et al., 2025). Even within the sparsity tradition, Uematsu and Tanaka (2026) acknowledge that "strong factors can destroy sparsity in real data," and defactor returns before screening assets via Lasso.

Existing estimates of the maximal Sharpe ratio fall into progressively higher tiers. Methods focusing on the dominant principal components, through Bayesian shrinkage on SDF coefficients (Kozak, Nagel and Santosh, 2020), Risk Premium PCA (Lettau and Pelger, 2020), dominant-eigenvalue factor selection (Feng, Giglio and Xiu, 2020), or sparse portfolio optimization (Ao, Li and Zheng, 2019), place the maximal annualized Sharpe ratio in the range 0.7–1.3. Lopez-Lira and Roussanov (2023) document empirically that this tier understates what is attainable: hedging the systematic exposures of characteristic-based strategies roughly doubles their Sharpe ratio.

Machine-learning approaches that construct portfolios from neural networks (Gu, Kelly and Xiu, 2020; Chen, Pelger and Zhu, 2024a), tree-based methods (Bryzgalova, Pelger and

Zhu, 2024; Cong, Feng, He and He, 2025), or random features (Didisheim et al., 2024) report out-of-sample Sharpe ratios between 1.5 and 4.0, with Kelly, Kuznetsov, Malamud and Zhang (2026); Kelly, Kuznetsov, Malamud and Xu (2025) achieving realized values above 4.0.

These tiers also differ by investment universe: the lower tier excludes or down-weights micro-cap stocks, through a market-cap filter (Kozak et al., 2020), a 20th-percentile NYSE cutoff (Lopez-Lira and Roussanov, 2023), or value weighting within externally constructed anomaly portfolios, whereas the higher tier includes micro-caps with equal weight. We report estimates on both, 7.4 on the full cross-section and 5.0 excluding micro-caps, so that each compares against the literature built on the same stocks. These are all portfolio-based estimates: each measures the performance of a constructed portfolio, which can only approach the theoretical maximum from below.

Recent theoretical work places the bulk of this machine-learning literature in the ridge-regularized class directly: Kelly et al. (2026) show that a deep neural network trained to construct the stochastic discount factor converges in population to a ridge-regularized SDF, and tree-based SDFs such as Bryzgalova et al. (2024) and Cong et al. (2025) estimate portfolio weights on their generated basis portfolios using the ridge-penalized estimators. Jagannathan and Ma (2003) show that imposing no-short-sale constraints on the Markowitz optimization acts as implicit covariance shrinkage and lowers out-of-sample risk. Our shortfall result therefore covers them as well.

What explains this shortfall statistically? In the proportional regime, El Karoui (2010) shows that plug-in estimation leads to systematic risk underestimation, and Kan, Wang and Zheng (2024) document the resulting gap between in-sample and out-of-sample Sharpe ratios for multi-factor models when $P < T$. Relatedly, Bodnar, Okhrin and Parolya (2023) develop optimal shrinkage-based portfolio selection in the proportional regime using random matrix theory (RMT) methods similar to those we employ. Kan and Zhou (2007) establish in finite samples that estimation error generates an expected out-of-sample loss for the

plug-in Markowitz portfolio, providing a classical antecedent to the structural shortfall we document in the proportional regime. DeMiguel, Garlappi and Uppal (2009) benchmark fourteen shrinkage-based portfolio constructions against the naive $1/N$ rule and find that none consistently dominates.

Several papers propose corrections for this bias. Da, Nagel and Xiu (2022) derive a correction under the assumption of a diagonal covariance matrix, in which case it takes the simple form P/T; Ao et al. (2019) derive a related correction for debiasing the GRS statistic in a Gaussian setting with $P < T$. Chernov et al. (2025) extend this debiasing approach to general covariance structures using RMT and introduce the ridge-regularized GRS statistic on which our estimator builds. Lu, Yang and Zhang (2024); Meng, Cao and Wang (2025) propose alternative RMT-based corrections for the out-of-sample Sharpe ratios of regularized portfolios under the Gaussianity assumption. Our approach targets $\mathcal{R}_*^2$ directly. Furthermore, we do not assume Gaussianity, and we derive a central limit theorem (CLT). Establishing the CLT requires novel RMT techniques introduced in our paper.

A parallel RMT literature transforms the eigenvalues of $\hat{\Sigma}$ nonlinearly rather than uniformly shifting them (Ledoit and Wolf, 2004, 2017, 2020). Kelly, Malamud, Pourmohammadi and Trojani (2024b) show that approximately optimal spectral shrinkage can be constructed from a combination of ridge penalties, so the ridge shortfall relative to $\mathcal{R}_*^2$ extends approximately to this nonlinear shrinkage class. Our framework targets $\mathcal{R}_*^2$ directly, without constructing an invertible $\hat{\Sigma}^{-1}$. As a byproduct, our higher-order GRS statistics deliver a closed-form out-of-sample Sharpe ratio for any spectrally shrunk Markowitz portfolio, a natural starting point for identifying the GRS-optimal nonlinear shrinkage. Even after debiasing, however, any ridge-regularized estimator suppresses the contribution of small eigenvalues, precisely the terms our higher-order statistics are designed to recover.

Our two-sided bounds on $\mathcal{R}_*^2$ connect to the literature on no-arbitrage upper bounds on Sharpe ratios. Hansen and Jagannathan (1991) establish that the maximal Sharpe ratio

is a lower bound on the volatility of the SDF. Hansen and Jagannathan (1997) introduces the companion Hansen-Jagannathan distance to assess specification errors, and Carrasco and Nokho (2025) extend it to a large cross-section through regularization, under the assumption of a small number of strong factors; a large menu of managed portfolios can instead have infinitely many (Chernov et al., 2025, 2026). We recover the maximal Sharpe ratio, which is the HJ bound, without that assumption. Cochrane and Saá-Requejo (2000), Bernardo and Ledoit (2000), and Černý (2003) use HJ-style bounds for derivative pricing. An earlier upper-bound argument on the maximal Sharpe ratio itself appears in MacKinlay (1995), who contends that squared Sharpe ratios attainable from any risk-based factor model should not substantially exceed a small multiple of the market's. Our upper bound rests on a different economic premise but shares MacKinlay's underlying idea: without economic discipline, the data alone cannot rule out arbitrarily large Sharpe ratios.

Whether any finite upper bound is available at all is a hard question, as recent work on SDF moments illustrates. Using S&P 500 index option prices, Martin and Shi (2026) exploit the convexity of the cumulant-generating function to derive bounds on SDF moments. The bounds diverge before the second moment is reached. If return distributions have sufficiently thick tails, deep out-of-the-money option payoffs have infinite second moments, and the Hansen and Richard (1987) projection of the SDF inherits infinite variance. But options this deep in the tail are not traded. In our framework, strategies with arbitrarily high Sharpe ratios can hide in low-variance principal components. A finite T cannot detect them, and ridge regularization produces finite estimates regardless. The two stories are different (distributional tails versus eigenvalue structure), but both point to the same conclusion: whether the maximal Sharpe ratio is finite cannot be determined from available data. The only reason we can produce an upper bound on $\mathcal{R}_*^2$ is the regularity condition introduced above. It is restrictive but far weaker than the common premise in cross-sectional asset pricing that only the largest eigenvalues carry risk premia.

The paper proceeds as follows. Section 2 introduces the model and defines the risk-return tradeoff function. Section 3 develops the estimator based on higher-order debiased GRS statistics. Section 4 derives the upper and lower bounds on $\mathcal{R}_*^2$ and establishes asymptotic inference. Section 5 contains the empirical analysis, and Section 6 concludes.[3]

2 Model

We work with a large set of portfolios constructed from extensive conditioning information: characteristic-managed portfolios, nonlinear features of characteristics, or any combination that makes the number of test assets P effectively unbounded. Our assumptions and asymptotic results are stated for this regime.

Consider asset excess returns $R_{t+1} \in \mathbb{R}^P$, rescaled by $P^{-1/2}$ throughout. The covariance and mean below refer to this rescaled vector.

Assumption 1 *There exist P assets in the economy. The excess returns $R_{t+1} = (R_{i,t+1})_{i=1}^P$ on P assets satisfy*

$$R_{t+1} \;=\; \mu_P \;+\; \varepsilon_{t+1},\; t = 1, \cdots, T\,, \tag{1}$$

where $\varepsilon_{t+1} = \Sigma_P^{1/2} X_{t+1}$, with $X_{t+1} = (X_{i,t+1})_{i=1}^P$ and $X_{i,t+1}$ i.i.d. across i and t, $E[X_{i,t+1}] = 0$, $E[X_{i,t+1}^2] = 1$, $E[X_{i,t+1}^{4+\varepsilon}] < \infty$ for some $\varepsilon > 0$, and $\mu_P \in \mathbb{R}^P$ and $\Sigma_P \in \mathbb{R}^{P\times P}$. Furthermore, the eigenvalue distributions of Σ_P are asymptotically stable as $P \to \infty$ (see Definition 3 in the Appendix) and converge to a distribution H with compact support, non-degenerate at zero.[4] Furthermore, $\|\mu_P\|$, $P \geq 1$, are uniformly bounded. Finally, we assume that $P/T \to c$ and $T^{1/2}(P/T - c) \to 0$ as $P, T \to \infty$. Everywhere in the sequel, we omit P and use μ and Σ to denote μ_P and Σ_P, respectively.

[3] Appendix M presents extensive simulation evidence.
[4] That is, $H(x = 0) < 1$.

Assumption 1 rules out time variation in the investment opportunity set over the sample. This is stronger than the data likely warrant, but it is the standard premise in cross-sectional asset-pricing tests. In the empirical application (Section 5), we accommodate slowly-moving variation by estimating on rolling windows of $T = 240$ months; within a window, the i.i.d. assumption is a working approximation that matches the convention in the GRS and high-dimensional asset pricing literatures. Appendix M provides an additional large-N justification based on managed portfolios.

For portfolios that exploit extensive conditioning information, Chernov et al. (2026) show that the covariance matrix of a high-dimensional managed-portfolio family has infinitely many eigenvalues growing at rate P, so the $P^{-1/2}$ rescaling above leaves a limiting eigenvalue distribution with positive mass away from zero, as Assumption 1 requires. This growth holds even when the stocks themselves have an approximate factor structure with K factors in the sense of Chamberlain and Rothschild (1982), the structure that underlies the APT. The additional eigenvalues come from weights that vary with the conditioning information, producing payoffs that no fixed combination of the stocks can replicate.

Let

$$\Sigma \;=\; V\Lambda V' \;=\; \sum_{i=1}^{P} \lambda_i v_i v_i' \tag{2}$$

be the eigenvalue (PCA) decomposition of Σ, where $V = [v_1, \ldots, v_P]$ is orthonormal and $\Lambda = \mathrm{diag}(\lambda_i)$ is the diagonal matrix of eigenvalues $\lambda_1 \geq \cdots \geq \lambda_P$.

The same decomposition applies to the population Markowitz portfolio

$$\pi_* \;=\; \Sigma^{-1}\mu \;=\; \sum_{i=1}^{P} \lambda_i^{-1}(v_i'\mu)\, v_i \tag{3}$$

and the maximal Sharpe ratio squared:

$$\mathcal{R}_*^2 \;=\; \frac{(E[\pi_*' R_{T+1}])^2}{\mathrm{Var}[\pi_*' R_{T+1}]} \;=\; \mu'\Sigma^{-1}\mu \;=\; (V'\mu)'\Lambda^{-1}(V'\mu) \;=\; \sum_{i=1}^{P} \mathcal{R}_i^2\,, \tag{4}$$

where

$$\mathcal{R}_i^2 \;=\; \frac{(E[v_i' R_{T+1}])^2}{\mathrm{Var}[v_i' R_{T+1}]} \;=\; \frac{(v_i'\mu)^2}{\lambda_i} \tag{5}$$

are the squared Sharpe ratios of the principal components. The following definition isolates the contribution of the low-risk end of the spectrum to $\mathcal{R}_*^2$.

Definition 1 *The risk-return tradeoff is the function*

$$\mathcal{R}_*^2(a) \;=\; \sum_{\lambda_i \le a} \mathcal{R}_i^2\,. \tag{6}$$

Standard APT-type arguments (Chamberlain and Rothschild, 1982; Kozak et al., 2020) imply that this contribution is small: expected returns compensate systematic risk, so $(v_i'\mu)^2$ decays faster than λ_i as $\lambda_i \to 0$. Discarding it yields the principal-component approximation

$$\mathcal{R}_*^2 \;=\; \sum_{i=1}^{K} \mathcal{R}_i^2 \;+\; \underbrace{\mathcal{R}_*^2(\lambda_{K+1})}_{\approx 0} \;\approx\; \sum_{i=1}^{K} \mathcal{R}_i^2\,, \tag{7}$$

where K is a small number (typically $K \approx 5$).

Both theory and evidence challenge this premise. For high-dimensional managed portfolios, Chernov et al. (2026) show that infinitely many principal components remain priced, so $\mathcal{R}_*^2(a)$ stays positive for every $a > 0$. The weak-factor literature allows small sources of risk to command large risk premiums (Giglio, Xiu and Zhang, 2025); we revisit such configurations as the boundary case of subdiagonality in Section 4. Empirically, Lopez-Lira and Roussanov

(2023) estimate exposures to all common latent factors and document that the top 50 principal components explain roughly half of stock-return variance but carry near-zero risk premium, while a portfolio orthogonal to all systematic risk earns an annualized Sharpe ratio above one, roughly twice that of its unhedged counterpart. Chen et al. (2025a) reach a parallel conclusion through Regressed-PCA, finding that low-dimensional factor models do not span the conditional efficient frontier. The bulk of compensated risk in equities lies outside the dominant eigenvalues.

The estimator we develop in the next section does not require $\mathcal{R}^2_*(a)$ to be small. The function $\mathcal{R}^2_*(a)$ governs how much of the total Sharpe ratio is hidden from any ridge-regularized estimator, and imposing mild structure on its behavior in Section 4 is what ultimately allows us to bound $\mathcal{R}^2_*$ from above.

3 The Estimator

Ridge-regularized Markowitz portfolios are the standard approach in high-dimensional asset pricing. Proposition 4 below shows that the out-of-sample squared Sharpe ratio of any such portfolio is bounded above by $\mathcal{R}^2_*$ divided by a factor strictly greater than one, and that this factor does not approach one as T grows in the proportional regime $P/T \to c > 0$. Existing portfolio-based estimates thus understate $\mathcal{R}^2_*$, and more data does not close the gap. The estimator developed in this section targets $\mathcal{R}^2_*$ directly, without constructing any portfolio.

Ridge regularization, while necessary for stability when P is large relative to T, systematically suppresses the contribution of small eigenvalues to the Sharpe ratio. The suppressed information, however, is recoverable: differentiating the ridge-regularized GRS statistic of Chernov et al. (2025) repeatedly with respect to the penalty parameter recovers precisely what regularization attenuates.

The construction proceeds in three steps. We express $\mathcal{R}^2_*$ as an infinite series in that statistic evaluated at the effective shrinkage (Proposition 1). Each term admits a consistent,

pivotal estimator built from the above derivatives, with Faà di Bruno / Bell-polynomial bias corrections (Theorem 2). Truncating the series and combining the estimators delivers the two-sided bounds on $\mathcal{R}_*^2$ in Section 4. Finally, Proposition 4 shows that ridge-regularized Markowitz portfolios cannot even attain the resulting lower bound.

Let

$$\begin{aligned} \bar{R} &= \frac{1}{T}\sum_{t=1}^{T} R_t \\ \hat{\Sigma} &= \frac{1}{T}\sum_{t=1}^{T} R_t R_t' \; - \; \bar{R}\,\bar{R}' \end{aligned} \tag{8}$$

be the sample means and the sample covariance matrix of asset returns. Throughout, we write $A \approx B$ when $A - B \to 0$ in probability as $T \to \infty$, and $A \leq_p B$ if $\mathbf{1}_{A \geq B} \approx 0$.

A naive way to estimate $\mathcal{R}_*^2$ is to replace μ and Σ^{-1} with their finite-sample counterparts:

$$\mathcal{R}_*^2 \;\approx\; \bar{R}'\hat{\Sigma}^{-1}\bar{R}\,, \tag{9}$$

The quantity on the right of (9) is essentially the classical Gibbons et al. (1989) statistic. It converges to $\mathcal{R}_*^2$ when T is large relative to P, but in the proportional regime $P/T \to c$ two distinct problems arise. First, when $c < 1$, the GRS statistic converges to a biased limit that exceeds $\mathcal{R}_*^2$, a consequence of concentration of measure. Second, the sample covariance $\hat{\Sigma}$ becomes ill-conditioned: its smallest eigenvalue vanishes as $P/T \to 1$ from below, and $\hat{\Sigma}$ is singular when $c \geq 1$, in which case the GRS statistic is not even defined. The $P^{-1/2}$ rescaling introduced before Assumption 1 keeps the population Σ in a compact-support limit, but $\hat{\Sigma}^{-1}$ does not inherit this property. To stabilize the inverse, Chernov et al. (2025) propose the *ridge-penalized GRS statistic*

$$\hat{\Theta}(z) \;=\; \bar{R}'\,(zI + \hat{\Sigma})^{-1}\,\bar{R}\,, \tag{10}$$

where $z > 0$ is the ridge penalty parameter. The population counterpart of this statistic is the ridge-regularized squared Sharpe ratio,

$$\mathcal{B}(z) \;\equiv\; \mu'(zI+\Sigma)^{-1}\mu \;=\; \sum_{i=1}^{P}\frac{(v_i'\mu)^2}{z+\lambda_i} \;=\; \sum_{i=1}^{P}\mathcal{R}_i^2\,\frac{\lambda_i}{z+\lambda_i} \;=\; \sum_{i=1}^{P}\mathcal{R}_i^2\,\frac{1}{1+z/\lambda_i}. \tag{11}$$

Chernov et al. (2025) show that $\hat{\Theta}(z)$ is a biased estimator of $\mathcal{B}(z)$ and that RMT delivers an estimate of the bias. Specifically, they introduce the *debiased GRS statistic*

$$\hat{\Theta}^d(z) \;=\; \frac{\hat{\Theta}(z) \;-\; \hat{\xi}(z;c)}{1+\hat{\xi}(z;c)}\,, \tag{12}$$

where the *bias correction* is given by

$$\hat{\xi}(z;c) \;=\; \frac{1}{1-c+cz\hat{m}(-z)} \;-\; 1\,, \tag{13}$$

$$\hat{m}(-z) \;=\; P^{-1}\,\mathrm{tr}((zI+\hat{\Sigma})^{-1}). \tag{14}$$

The latter object is the Stieltjes transform of the empirical return covariance matrix.

Lemma 1 *In the limit as $P, T \to \infty, P/T \to c$,*

$$\hat{\Theta}^d(z) \;-\mathcal{B}(\hat{Z}_*(z)) \;\to\; 0 \tag{15}$$

in probability, where

$$\hat{Z}_*(z) \;=\; z(1+\hat{\xi}(z)) \tag{16}$$

is the effective shrinkage.

Lemma 1 implies that the over-parametrized regime with $P > T$ imposes significant

barriers to estimating

$$\mathcal{R}_*^2 \;=\; \mathcal{B}(0) \tag{17}$$

because, instead of estimating $\mathcal{B}(z)$, the debiased statistic $\hat{\Theta}^d(z)$ estimates $\mathcal{B}(\hat{Z}_*(z))$. As Chernov et al. (2025) show, $\lim_{z\downarrow 0}\hat{Z}_*(z) = \hat{Z}_*(0) > 0$ when $P > T$. Intuitively, when $P > T$, $\hat{\xi}(z)$ diverges as $z \downarrow 0$ at a rate that exactly cancels the vanishing z, keeping $\hat{Z}_*(z) = z(1+\hat{\xi}(z))$ bounded away from zero; this cancellation is what sustains debiasing at $z = 0$. Thus, instead of estimating $\mathcal{B}(0)$, we end up estimating $\mathcal{B}(\hat{Z}_*(0))$.

The presence of this effective shrinkage, even when the true shrinkage z vanishes, reflects both advantages and challenges of over-parametrized models. Effective shrinkage is an advantage because it underlies the implicit regularization ($\hat{Z}_*(z) > z$) that is responsible for the superior performance of over-parametrized models, whereby the model automatically shrinks the estimated portfolio, efficiently reducing the impact of estimation noise on portfolio performance.[5]

It is also a challenge, because effective shrinkage imposes a fundamental limit on what can be learned in finite samples. Indeed, even for $z \to 0$, we can only estimate[6]

$$\mathcal{B}(\hat{Z}_*(z)) \leq \underbrace{\mathcal{R}_*^2 - \mathcal{R}_*^2(a)}_{large\ \ eigenvalues} + \underbrace{\frac{1}{1+\hat{Z}_*(z)/a}}_{effective\ \ shrinkage} \underbrace{\mathcal{R}_*^2(a)}_{small\ \ eigenvalues}. \tag{18}$$

The bound is tight when $\hat{Z}_*(z)/a$ is small. When $\hat{Z}_*(z) \gg a$, the second term collapses to near zero, and the contribution of $\mathcal{R}_*^2(a)$ is "shadowed" by the effective shrinkage: we

[5] This mechanism underlies the virtue of complexity documented in Kelly, Malamud and Zhou (2024a); Kelly et al. (2026).

[6] The inequality can be established by evaluating (11) at $z = \hat{Z}_*(z)$ and splitting it between large eigenvalues, $\lambda_i > a$, and small eigenvalues, $\lambda_i \leq a$. The split can be further simplified by using $1/(1+\hat{Z}_*(z)/\lambda_i) \leq 1$ for the large eigenvalues, and $\hat{Z}_*(z)/\lambda_i \geq \hat{Z}_*(z)/a$ for the small ones.

cannot recover the contribution of small eigenvalues. In summary: $\mathcal{B}(\hat{Z}_*(z))$ is estimable from the data; $\mathcal{R}_*^2 = \mathcal{B}(0)$ is not.

Our solution to this problem is based on *higher-order GRS statistics*:

$$\frac{z^j(-1)^j}{j!}\frac{d^j}{dz^j}\hat{\Theta}(z) \;=\; \underbrace{z^j\bar{R}'(z\hat{I}+\hat{\Sigma})^{-j-1}\bar{R}}_{j-th\ order\ GRS\ statistic} \tag{19}$$

and their population counterparts,

$$\mathcal{B}_j(z) \;=\; z^j\mu'(zI+\Sigma)^{-j-1}\mu\,. \tag{20}$$

In particular, $\mathcal{B}(z) = \mathcal{B}_0(z)$. For any vector $\boldsymbol{\mathcal{B}} = (\mathcal{B}_0, \cdots, \mathcal{B}_J)$, we let

$$\mathcal{D}_J(\boldsymbol{\mathcal{B}}) \;=\; \sum_{j=0}^{J}\mathcal{B}_j \tag{21}$$

be the partial sum of $\boldsymbol{\mathcal{B}}$. The following identity allows us to express $\mathcal{R}_*^2$ via the ensemble of $\mathcal{B}_j(\hat{Z}_*(z)),\ j \geq 0$.

Proposition 1 *The maximal Sharpe ratio admits the expansion*

$$\mathcal{R}_*^2 \;=\; \lim_{J\to\infty}\mathcal{D}_J(\boldsymbol{\mathcal{B}}) \;=\; \sum_{j=0}^{\infty}\mathcal{B}_j(\hat{Z}_*(z))\,, \tag{22}$$

where $\boldsymbol{\mathcal{B}} \;=\; (\mathcal{B}_0, \cdots, \mathcal{B}_J)(\hat{Z}_*(z))$. *Furthermore,* $\mathcal{B}_j$ *is monotone decreasing in* j.

The intuition behind (22) is as follows: estimating the contributions of small eigenvalues ($\mathcal{R}_*^2(a)$ for small a) requires statistics that offset the shadowing impact of $\hat{Z}_*(z)$ in (18). Such statistics rely on a *multi-scale decomposition*, where each additional term extracts additional variation in Sharpe ratios induced by small eigenvalues.

We now show that, analogously to (15), $\mathcal{B}_j(z)$ admits a pivotal estimator for any $j \geq 1$ in terms of debiased statistics.

Theorem 2 (Higher-Order Debiased GRS Statistics) *The statistics are defined recursively as follows:*

$$\hat{\Theta}_0^d(z) \;=\; \hat{\Theta}^d(z), \tag{23}$$

$$\hat{\Theta}_j^d(z) \;=\; \frac{1}{j!(-\hat{Z}_*'(z))^j}\left(\frac{d^j}{dz^j}\left(\hat{\Theta}^d(z)\right) - \sum_{k=1}^{j-1}(-1)^k(k!)\hat{\Theta}_k^d(z)\beta_{j,k}(z)\right),\; j \geq 1, \tag{24}$$

where

$$\beta_{j,k}(z) \;=\; B_{j,k}\left(\hat{Z}_*'(z), \hat{Z}_*''(z), \cdots, \hat{Z}_*^{(j-k+1)}(z)\right) \tag{25}$$

and $B_{j,k}(\cdot)$ are Bell polynomials.[7] *Then,*

$$\hat{Z}_*(z)^j\,\hat{\Theta}_j^d(z) \;-\; \mathcal{B}_j(\hat{Z}_*(z)) \;\to\; 0 \tag{26}$$

in probability and, hence,

$$\mathcal{D}_J(\boldsymbol{\mathcal{B}}) \;-\; \mathcal{D}_J(\hat{\boldsymbol{\Theta}}) \;\to\; 0 \tag{27}$$

in probability, with $\hat{\boldsymbol{\Theta}} \;=\; (\hat{Z}_*^j\hat{\Theta}_j^d)_{j=0}^J$. We write $\hat{\Theta}_j = \hat{Z}_*^j\hat{\Theta}_j^d$ for the j-th component of $\hat{\boldsymbol{\Theta}}$.

Formula (26) is a direct extension of (12): debiasing requires both additive and multiplicative

[7] The Bell polynomials $B_{j,k}(x_1, \ldots, x_{j-k+1})$ represent the coefficients in the expansion of the k-th power of a power series. They are defined by the identity:

$$\frac{1}{k!}\left(\sum_{i=1}^{\infty} x_i \frac{t^i}{i!}\right)^k = \sum_{j=k}^{\infty} B_{j,k}(x_1, \ldots, x_{j-k+1})\frac{t^j}{j!}.$$

corrections that depend on $\hat{\xi}$ through $\hat{Z}_*$. For example,

$$\hat{\Theta}_1^d(z) \;=\; -\frac{1}{\hat{Z}_*'(z)}\frac{d}{dz}\hat{\Theta}^d(z)\,, \tag{28}$$

and

$$\hat{\Theta}_2^d(z) \;=\; \frac{1}{2(\hat{Z}_*'(z))^2}\left(\frac{d^2}{dz^2}\hat{\Theta}^d(z) + \hat{Z}_*''(z)\hat{\Theta}_1^d(z)\right)\,. \tag{29}$$

We refer to $\hat{\Theta}_j^d(z)$ as *higher-order debiased GRS statistics.*

Proposition 1 decomposes $\mathcal{R}_*^2$ into an infinite series of population quantities $\mathcal{B}_j$, each capturing progressively finer detail in the eigenvalue–Sharpe ratio relationship, and Theorem 2 shows that each of these quantities admits a consistent, pivotal estimator constructed entirely from the data. The practical question is how to turn this into a usable estimator and, ultimately, into bounds on $\mathcal{R}_*^2$.

4 Bounding the Risk-Return Tradeoff

This section converts the expansion (22) into a two-sided, asymptotically consistent interval for $\mathcal{R}_*^2$. The argument proceeds in three steps. Truncating (22) after J terms yields a lower bound $\mathcal{H}_J^L$; sub-exponential decay of the residual $\mathcal{B}_j$ makes the truncation tight at small J. An upper bound requires a restriction on how much Sharpe ratio can hide in small eigenvalues, which we formalize as a subdiagonal condition and combine with the residual control to obtain $\mathcal{H}_J^U$. Central Limit Theorems for the debiased estimators $\hat{\Theta}_j^d(z)$ then convert both bounds into pivotal confidence bands. The three subsections address these steps in order.

4.1 The Lower Bound

The representation (22) and the monotone decrease of $\mathcal{B}_j$ naturally suggest a candidate estimator: we fix a finite $J \geq 0$, truncate (22) after J terms, and construct the estimator

using (27). Because $\mathcal{B}_j(\hat{Z}_*(z))$ is monotone decreasing in j, we know that if the estimated $\mathcal{B}_k$ is sufficiently small, $\mathcal{B}_{k+1}$ will be even smaller, which is useful for implementing the truncation.

The truncated estimator can be improved by taking into account the subtle dependence of $\mathcal{B}_j$ on j.

Corollary 3 (Sub-Exponential Decay) *The ratios of consecutive terms satisfy*

$$\frac{\mathcal{B}_{j-1}(\hat{Z}_*(z))}{\mathcal{B}_j(\hat{Z}_*(z))} \;>\; \frac{\mathcal{B}_j(\hat{Z}_*(z))}{\mathcal{B}_{j+1}(\hat{Z}_*(z))} \;>\; 1\,. \tag{30}$$

Consequently, the decay rate diminishes, yielding

$$\mathcal{B}_{j+k}(\hat{Z}_*(z)) \;\geq\; \mathcal{B}_j(\hat{Z}_*(z)) \left(\frac{\mathcal{B}_j(\hat{Z}_*(z))}{\mathcal{B}_{j-1}(\hat{Z}_*(z))}\right)^k . \tag{31}$$

Summing this over $k \geq 0$ gives the following lower bound:

$$\begin{aligned} \mathcal{R}_*^2 \;&\geq\; \mathcal{H}_J^L(\boldsymbol{\mathcal{B}}) \\ \mathcal{H}_J^L(\boldsymbol{\mathcal{B}}(z)) \;&=\; \mathcal{D}_{J-1}(\boldsymbol{\mathcal{B}}(z)) \;+\; \mathcal{B}_J(\hat{Z}_*(z)) \frac{1}{1 - \frac{\mathcal{B}_J(\hat{Z}_*(z))}{\mathcal{B}_{J-1}(\hat{Z}_*(z))}}\,. \end{aligned} \tag{32}$$

By Theorem 2, replacing $\boldsymbol{\mathcal{B}}$ in (32) with $\hat{\boldsymbol{\Theta}}$ provides an asymptotically consistent, pivotal lower bound for $\mathcal{R}_*^2$.

A natural next question: can existing portfolio construction methods attain out-of-sample Sharpe ratios above the lower bound on $\mathcal{R}_*^2$ from Corollary 3? The leading candidate is the ridge-regularized Markowitz portfolio, which replaces the unknown population covariance with its sample analogue and introduces explicit regularization. This class is broader than it first appears: Kelly et al. (2026) show that deep neural network SDFs converge in population to a ridge-regularized SDF, and Cong et al. (2025) estimate basis-portfolio weights via the ridge estimator of Didisheim et al. (2024). The structural shortfall we derive generalizes the

classical finite-sample loss of the plug-in Markowitz portfolio documented by Kan and Zhou (2007) to the proportional regime.

Proposition 4 *Let*

$$\hat{\pi}_T(z) \;=\; (zI + \hat{\Sigma})^{-1}\bar{R}. \tag{33}$$

Then, its out-of-sample squared Sharpe ratio satisfies

$$\frac{(E_T[\hat{\pi}_T(z)'R_{T+1}])^2}{\mathrm{Var}_T[\hat{\pi}_T(z)'R_{T+1}]} \;\leq_p\; \frac{1}{\hat{Z}'_*(z)}\mathcal{H}_1^L(\boldsymbol{\mathcal{B}}) \;\leq\; \frac{1}{\hat{Z}'_*(z)}\mathcal{R}_*^2 \tag{34}$$

as $P, T \to \infty,\ P/T \to c$, *where*

$$\hat{Z}'_*(z) \;=\; 1 + \hat{\xi}(z;c) + z\hat{\xi}'(z;c) \;\geq\; 1\,. \tag{35}$$

The factor $\hat{Z}'_*(z) \geq 1$ is the derivative in z of the effective shrinkage $\hat{Z}_*(z)$ defined in Lemma 1;[8] the bound $\mathcal{H}_1^L(\boldsymbol{\mathcal{B}}) = \mathcal{B}_0 + \mathcal{B}_1/(1 - \mathcal{B}_1/\mathcal{B}_0)$ from Corollary 3 at $J = 1$. The size of the multiplicative shortfall $1/\hat{Z}'_*(z)$ is characterized by the following.

Lemma 2 *The effective shrinkage* $\hat{Z}_*(z)$ *is monotone increasing in* z, *and its derivative* $\hat{Z}'_*(z)$ *is monotone decreasing. For* $c > 1$,

$$\hat{Z}'_*(z) \;=_p\; \frac{1}{1 - c\int \frac{x^2}{(\hat{Z}_*(z)+x)^2}\,H(dx)} \;\geq_p\; \frac{c}{c - \left(1 - \frac{z}{\hat{Z}_*(z)}\right)^2}, \tag{36}$$

where H *is the limiting eigenvalue distribution of* Σ *from Assumption 1. In particular,*

$$\hat{Z}'_*(0) \;\geq_p\; \frac{c}{c-1} \qquad \textit{and} \qquad \lim_{z\to\infty} \hat{Z}'_*(z) \;=\; 1, \tag{37}$$

[8] Chen, Kelly and Malamud (2025b) refer to the quantity $\hat{Z}'_*(z) - 1$ as the *Limits-to-Learning Gap (LLG)* and show that it controls the gap between out-of-sample Mean Squared Error of ridge regression and the population R^2. The same LLG emerges in our analysis of the Sharpe ratio.

with equality in the $z = 0$ bound if and only if H is supported on a single point.

Proposition 4 and Lemma 2 together deliver a structural conclusion on ridge-regularized Markowitz portfolios. For $c > 1$, the Limits-to-Learning Gap $\hat{Z}'_*(z) - 1$ is bounded below by $1/(c-1)$ at $z = 0$, is strictly positive for any finite z, and vanishes only as $z \to \infty$. Ridge portfolios, therefore, cannot attain even our $J = 1$ lower bound, and the shortfall is structural: for a fixed c, more data does not close it. The total gap between the out-of-sample Sharpe ratio of a ridge-regularized portfolio and $\mathcal{R}_*^2$ decomposes into two sources: a multiplicative shortfall $1/\hat{Z}'_*(z)$ relative to our lower bound, and a residual distance from the bound to $\mathcal{R}_*^2$ which a higher J narrows but does not close at any finite order. Neither is estimation error. Despite strong realized out-of-sample performance of portfolios in this class (Didisheim et al., 2024; Hellum, Jensen, Kelly and Malamud, 2026), Section 5 shows that at empirical values of $\hat{Z}'_*(z)$, the structural factor alone accounts for most of the gap to the existing literature.

4.2 Hansen–Jagannathan Bound in High Dimensions

The maximal Sharpe ratio also bounds every candidate stochastic discount factor. The Hansen and Jagannathan (1991) (HJ) inequality states that any stochastic discount factor M_{t+1}, for instance one implied by an equilibrium macroeconomic model, satisfies

$$\frac{\mathrm{Var}[M_{t+1}]^{1/2}}{E[M_{t+1}]} \;\geq\; \frac{|E[\pi' R_{t+1}]|}{\mathrm{Var}[\pi' R_{t+1}]^{1/2}} \tag{38}$$

for any portfolio π, with all moments taken in population. The population Markowitz portfolio attains the tightest such bound,

$$\frac{\mathrm{Var}[M_{t+1}]^{1/2}}{E[M_{t+1}]} \;\geq\; \mathcal{R}_*\,, \tag{39}$$

so $\mathcal{R}_*$ is the HJ bound itself. The empirical literature estimates it with a plug-in: with a low-dimensional set of portfolios and $P/T \approx 0$, $\bar{R}'\hat{\Sigma}^{-1}\bar{R} \approx \mathcal{R}_*^2$. As the factor zoo enlarged the cross-section, P/T ceased to be negligible: the plug-in overfits in sample and overstates the population $\mathcal{R}_*$. The literature turned to *out-of-sample* Sharpe ratios, which by Proposition 4 are themselves HJ lower bounds. The out-of-sample Sharpe ratio of any feasible, look-ahead-free strategy satisfies

$$\frac{(E[\hat{\pi}_T' R_{T+1}])^2}{\mathrm{Var}[\hat{\pi}_T' R_{T+1}]} \;\leq\; \frac{\mathrm{Var}[M_{T+1}]^{1/2}}{E[M_{T+1}]}, \tag{40}$$

and a good enough model and asset menu would make $\hat{\pi}_T$ approximate the population Markowitz portfolio.

Proposition 4 shows that this route is blocked. Corollary 3 delivers the bound directly,

$$\frac{\mathrm{Var}[M_{t+1}]^{1/2}}{E[M_{t+1}]} \;\geq_p\; \mathcal{H}_J^L(\hat{\mathbf{\Theta}}(z)) \tag{41}$$

for any $J \geq 0$. Corollary 3 is thus a high-dimensional HJ bound, valid for $P > T$.

4.3 The Upper Bound

By itself, the expansion (22) cannot deliver an upper bound on $\mathcal{R}_*^2$. If some eigenvalues of Σ vanish while the corresponding principal components carry non-zero expected returns, $\mathcal{R}_*^2 = \infty$, but every $\mathcal{B}_j(\hat{Z}_*(z))$ is finite, and $\mathcal{D}_J$ is bounded. An upper bound therefore requires structure on the contribution of small eigenvalues, which we impose through the risk-return tradeoff $\mathcal{R}_*^2(a)$ in (6).

We first bound $\mathcal{R}_*^2$ from above in terms of the non-estimable contribution $\mathcal{R}_*^2(a)$:

Corollary 5 *The maximal Sharpe ratio satisfies*

$$\mathcal{R}_*^2 \;\leq \mathcal{D}_{J-1}(\boldsymbol{\mathcal{B}}(z)) \;+\; \mathcal{B}_J(\hat{Z}_*(z))\frac{a + \hat{Z}_*(z)}{a} \;+\; \mathcal{R}_*^2(a) \tag{42}$$

and, hence,

$$\mathcal{R}_*^2(a) \;\geq\; \mathcal{H}_J^L(\boldsymbol{\mathcal{B}}(z_2)) - \mathcal{D}_{J-1}(\boldsymbol{\mathcal{B}}(z)) \;-\; \mathcal{B}_J(\hat{Z}_*(z))\frac{a+\hat{Z}_*(z)}{a} \tag{43}$$

for any $J \geq 1$ *and any* $z, z_2 > 0$

The upper bound (42) has a transparent structure. The first term, $\mathcal{D}_{J-1}(\boldsymbol{\mathcal{B}})$, is the partial sum of the first J estimable terms, the same quantity that anchors the lower bound. The second term, $\mathcal{B}_J(\hat{Z}_*(z))\frac{a+\hat{Z}_*(z)}{a}$, bounds what the first J terms leave out for components with eigenvalues above a. Each such component reveals at least the fraction $a/(a+\hat{Z}_*(z))$ of its unrecovered Sharpe ratio in the next term $\mathcal{B}_J$. Its unrecovered part is therefore at most $(a+\hat{Z}_*(z))/a$ times its contribution to $\mathcal{B}_J$. Below a, this fraction can be arbitrarily small, which is why the third term is needed. The third term, $\mathcal{R}_*^2(a)$, captures the Sharpe ratio carried by eigenvalues below the threshold a; unlike the first two, it cannot be estimated directly and requires further structure on the risk-return tradeoff for a finite bound. The bound (43) follows by combining (42) with the lower bound (32); it underlies the test of subdiagonality in Appendix K.

To turn (42) into a finite, estimable bound, we restrict the non-estimable term through a subdiagonal condition on the risk-return tradeoff: for a threshold a below the average eigenvalue $\bar{\lambda} = P^{-1}\operatorname{tr}(\Sigma)$, the share of $\mathcal{R}_*^2$ carried by eigenvalues below a is small.

Definition 2 *Given an* $a < \bar{\lambda} = P^{-1}\operatorname{tr}(\Sigma)$, *the risk-return tradeoff is a-subdiagonal if*

$$\mathcal{R}_*^2(a) \;\leq\; \frac{a}{\bar{\lambda}}\,\mathcal{R}_*^2\,. \tag{44}$$

The name describes a picture: plotted against $a/\bar{\lambda}$, the share $\mathcal{R}_*^2(a)/\mathcal{R}_*^2$ must lie below the diagonal (Figure 1). For example, if every principal component carries the same Sharpe ratio and the eigenvalues are spread uniformly on $[0, 2\bar{\lambda}]$, the share below a is $a/(2\bar{\lambda})$, half

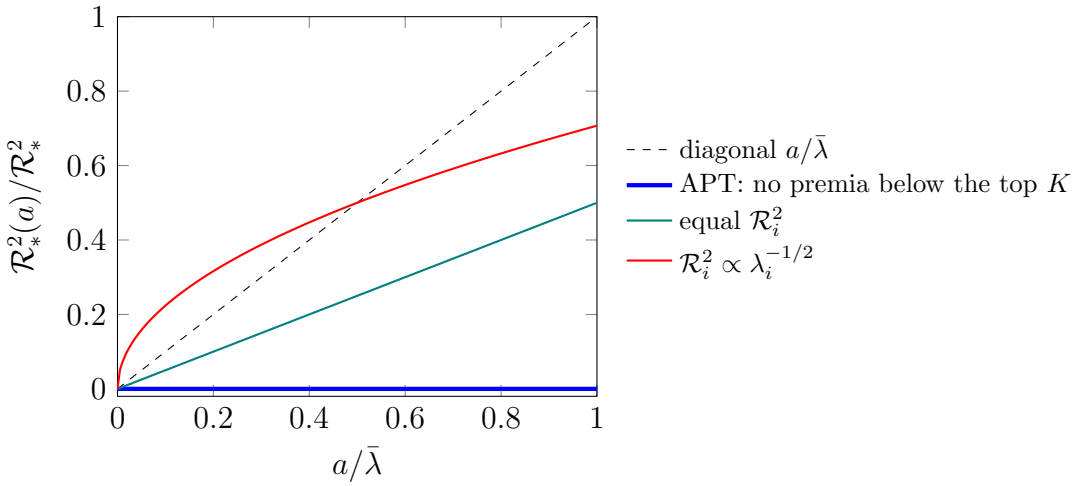


Figure 1: Subdiagonality at a requires the share curve $\mathcal{R}^2_*(a)/\mathcal{R}^2_*$ to lie below the dashed diagonal at $a/\bar{\lambda}$. Except for the APT case, the eigenvalues are spread uniformly on $[0, 2\bar{\lambda}]$. With equal Sharpe ratios across components the condition holds at every a; with $\mathcal{R}^2_i \propto \lambda_i^{-1/2}$ it fails for $a/\bar{\lambda} < 0.5$.

the diagonal, and the condition holds. It fails at small a if Sharpe ratios rise as variance falls, for instance $\mathcal{R}^2_i \propto \lambda_i^{-1/2}$ with the same eigenvalues: the share is then $(a/(2\bar{\lambda}))^{1/2}$, which exceeds $a/\bar{\lambda}$ for $a < \bar{\lambda}/2$.

Subdiagonality is strictly weaker than the standard APT assumption, under which all of the risk premium is carried by a fixed set of top principal components and $\mathcal{R}^2_*(a) = 0$ below an eigenvalue cutoff. The APT assumption is the standard setting in empirical asset pricing; Feng et al. (2020) and Kozak et al. (2020), for example, focus on the top ten principal components. Our condition admits a larger class of DGPs, since the Sharpe ratio can be distributed across the spectrum, as it is for the high-dimensional managed portfolios of Chernov et al. (2026). At the upper endpoint $a = \bar{\lambda}$, the condition holds mechanically; at smaller a, it imposes a substantive restriction and rules out factor configurations in which an intermediate eigenvalue $\lambda_k < \bar{\lambda}$ carries a non-proportional share of $\mathcal{R}^2_*$, a case that standard APT excludes by fiat. If the true DGP is of this kind, the upper bound can understate $\mathcal{R}^2_*$; the lower bound $\mathcal{H}^L_J$ remains valid unconditionally.

Substituting (44) into (42) immediately yields the following result:

Corollary 6 *Suppose the risk-return tradeoff is a-subdiagonal. Then,*

$$\mathcal{R}_*^2 \;\leq\; \frac{\mathcal{D}_{J-1}(\boldsymbol{\mathcal{B}}) \;+\; \mathcal{B}_J(\hat{Z}_*(z))\frac{a+\hat{Z}_*(z)}{a}}{1-\frac{a}{\bar{\lambda}}}. \tag{45}$$

Equation (42) bounds $\mathcal{R}_*^2$ by two estimable quantities plus the non-estimable $\mathcal{R}_*^2(a)$. Subdiagonality replaces $\mathcal{R}_*^2(a)$ by $(a/\bar{\lambda})\mathcal{R}_*^2$, and solving for $\mathcal{R}_*^2$ produces (45). The bound trades off its two a-dependent pieces. As $a \to \bar{\lambda}$, the denominator $1 - a/\bar{\lambda}$ vanishes and $\mathcal{H}_J^U$ diverges; as $a \to 0$, the remainder $\mathcal{B}_J(\hat{Z}_*(z))\frac{a+\hat{Z}_*(z)}{a}$ diverges instead. The bound is therefore finite on $(0, \bar{\lambda})$ and attains an interior minimum. We set a to minimize the bound; the minimizer $a_{\min,J}$ has a closed form (Appendix K).

Empirically, we can also check for subdiagonality at this threshold: if it holds, the upper bound cannot fall below the lower bound, as (43) shows. When the upper bound at $a_{\min,J}$ does fall below the lower bound, we lower the threshold to a_{corner} (see (K.3)), the largest value below $a_{\min,J}$ at which the two bounds meet, so that on this corner branch the upper bound equals the lower bound. The check is necessary but not sufficient: passing it does not establish subdiagonality. Algorithm 2 states the rule.

Together, the lower and upper bounds give a two-sided interval for $\mathcal{R}_*^2$:

$$\begin{aligned}
&\mathcal{H}_J^L(\boldsymbol{\mathcal{B}}) \;\leq\; \mathcal{R}_*^2 \;\leq\; \mathcal{H}_J^U(\boldsymbol{\mathcal{B}}), \quad \text{with} \\
&\mathcal{H}_J^L(\boldsymbol{\mathcal{B}}) \;=\; \mathcal{D}_{J-1}(\boldsymbol{\mathcal{B}}) \;+\; \mathcal{B}_J(\hat{Z}_*(z))\frac{1}{1-\frac{\mathcal{B}_J(\hat{Z}_*(z))}{\mathcal{B}_{J-1}(\hat{Z}_*(z))}} \\
&\mathcal{H}_J^U(\boldsymbol{\mathcal{B}}) \;=\; \frac{\mathcal{D}_{J-1}(\boldsymbol{\mathcal{B}}) \;+\; \mathcal{B}_J(\hat{Z}_*(z))\frac{a_J^*(\boldsymbol{\mathcal{B}})+\hat{Z}_*(z)}{a_J^*(\boldsymbol{\mathcal{B}})}}{1-\frac{a_J^*(\boldsymbol{\mathcal{B}})}{\bar{\lambda}}}
\end{aligned} \tag{46}$$

Equation (46) delivers the paper's main result: a two-sided, asymptotically consistent interval for $\mathcal{R}_*^2$. The lower bound exploits the sub-exponential decay of $\mathcal{B}_j$; the upper bound

requires only that the risk-return tradeoff be subdiagonal. All quantities on both sides are estimable from the data. In practice, we replace $\boldsymbol{\mathcal{B}}$ with $\hat{\boldsymbol{\Theta}}$ and $\bar{\lambda}$ with its sample analogue $P^{-1}\operatorname{tr}(\hat{\Sigma})$[9] to obtain fully data-driven lower and upper bounds on $\mathcal{R}_*^2$.

4.4 Confidence bounds

Confidence bands for the two-sided interval (46) require three ingredients: a probabilistic model for μ, a strengthened moment condition on returns, and a pivotal representation of the asymptotic covariance. We develop each in turn; the formal Central Limit Theorems are in Appendix H.

The probabilistic model is Assumption 2, which takes $\mu = P^{-1/2}\Sigma_{\mu,P}^{1/2}\nu$ with ν_i i.i.d., mean zero, unit variance, and Gaussian-like moments $E[\nu_i^3] = 0$, $E[\nu_i^4] = 3$. This random-design framework is the standard device for pivotal inference in the proportional regime: the moment conditions govern how Sharpe ratio spreads across the eigendirections of Σ, which in turn governs the limiting distribution of the debiased estimators. An investigator applying our bands to a specific dataset is implicitly treating that dataset as one realization of the data-generating process specified by Assumptions 1 and 2; the bands are valid unconditionally, not conditional on μ.

The CLT requires strengthening the moment condition in Assumption 1 to $E[X_{i,t+1}^{8+\varepsilon}] < \infty$ for some $\varepsilon > 0$. The higher moments are needed so that quadratic forms in the sample covariance concentrate around their population counterparts at a rate fast enough to sustain the pivotal limit.

Theorem H.1 establishes that there exists a pivotal covariance estimator Σ_Θ, constructed entirely from observed data, such that

$$\Sigma_\Theta^{-1/2} T^{1/2} \big(\hat{\boldsymbol{\Theta}} - \boldsymbol{\mathcal{B}}\big) \tag{47}$$

[9] Lemma A.5 shows that $P^{-1}\operatorname{tr}(\hat{\Sigma}) - \bar{\lambda}$ converges to zero at the rate $1/T$ instead of the standard $1/T^{1/2}$, simplifying the CLT derivations.

converges to $N(0, I)$. The proof requires new tools in RMT that, to our knowledge, have not appeared in the literature. Subsequently, the standard delta method delivers asymptotic confidence bands in terms of the gradients of the functions (46). The dependence of a_J^* on $\mathcal{B}$ enters the gradient of $\mathcal{H}_J^U$; see Corollary H.2 for details.

A collection of asymptotic algebraic identities transforms the naive, non-pivotal estimator of Σ_Θ in (47) into a pivotal representation. This transformation hinges on a set of subtle cancellations, formalized in Lemma H.6, which eliminate nuisance dependence on high-dimensional latent structure and render the limiting distribution tractable. The full estimation and inference procedure is collected in Algorithm 1 in Appendix J; we use it in both the simulation exercises of Appendix M and the empirical application of Section 5.

5 Empirical Analysis

We now take our estimator to the data. Appendix M reports a Monte Carlo study; we summarize here the findings that bear on our empirical design. Calibrating a nonlinear data-generating process to the JKP characteristics, we find that at the sample sizes and complexity ratios $c = P/T$ relevant here both bounds concentrate near the true Sharpe ratio, with confidence bands that narrow as T grows and contain $\mathcal{R}_*^2$ in more than 97.5% of simulated economies. The data-driven threshold respects subdiagonality throughout and accounts for only 10% to 36% of the upper bound across the grid, so the interval rests primarily on estimable quantities. We use $J = 1$: estimating the second-order refinement precisely requires samples far longer than ours, and even then it tightens the bounds only marginally.

In this section, we implement three exercises. First, we estimate the maximal Sharpe ratio using a broad cross-section of characteristic-managed equity portfolios. Second, we quantify how adding corporate bonds to equities affects the attainable Sharpe ratio. Third, we evaluate the role of nonlinearities in portfolio weights. All estimates in this section

are produced using Algorithm 1 in Appendix J. Figures and empirical statements in this section report the maximal Sharpe ratio (MSR), defined as MSR $\equiv \sqrt{12\,\mathcal{R}_*^2}$, with the $\sqrt{12}$ annualizing the monthly $\mathcal{R}_*^2$. The bound estimates $\mathcal{H}_J^L, \mathcal{H}_J^U$ and their one-sided 97.5% confidence bands are reported after the same $x \mapsto \sqrt{12\,x}$ transformation.

5.1 Data and Test Assets

We construct managed portfolios from NYSE/AMEX/NASDAQ stocks with CRSP share codes 10–12 using the broad set of firm characteristics described below, and use these portfolios as test assets for estimating the maximal Sharpe ratio. The characteristics are the panel of 132 firm characteristics for U.S. equities over 1963–2025 compiled by Jensen et al. (2023), hereafter JKP. As in JKP, we define mega, large, small, and micro stocks using the 80th, 50th, 20th, and 1st percentiles of NYSE stocks' market caps. Nano caps are the remaining stocks.

To ensure a stock-month panel with a consistent set of available characteristics over time, we follow the approach of Didisheim et al. (2024). First, we restrict the raw dataset to a subset of 132 characteristics that have fewer than 30% missing values over the full sample. Next, we drop "nano" stocks whose market capitalization is below the first percentile of the NYSE market capitalization distribution. Then, we exclude stock-month observations with missing values for more than a third of the characteristics. Finally, we cross-sectionally rank-standardize each characteristic into the $[-0.5, 0.5]$ interval and replace remaining missing values with zero (the cross-sectional median under rank-standardization). After this filtering procedure, the resulting $N_t \times 132$ matrix of characteristics each month constitutes the conditioning set X_t used in our empirical analysis. These characteristics are far from independent: the leading quarter of their principal components accounts for roughly 80% of their total variance, a low-dimensional core that the nonlinear random features below

expand into a much broader span. We also use $R_{t+1} = (R_{i,t+1})_{i=1}^{N_t}$ to denote the vector of next-month stock returns in excess of the risk-free rate.

Our simulation results guide our empirical design. A large P is needed in principle: more managed portfolios span a broader portion of the mean-variance frontier, tightening the lower bound on $\mathcal{R}_*^2$. We adopt $P = 5{,}000$ managed portfolios in the empirical exercises. A large T tightens both bounds, but in practice we estimate rolling-window Sharpe ratios, and excessively long windows smooth over time variation in the investment opportunity set. We therefore set $T = 240$ months (20 years), which strikes a balance: long enough for the CLT approximation and confidence intervals to perform well, as our simulations confirm, yet short enough to capture economically meaningful shifts in the risk-return tradeoff. The resulting complexity ratio $c = P/T \approx 21$ falls in a range where the upper bound, though wider than at lower c, remains informative, and the data-driven a^* mitigates the associated regularization penalty.

We generate a large number of stock characteristics $S_t = (S_{i,t})_{i=1}^{N_t} \in \mathbb{R}^{N_t \times P}$ through transformations of $X_{i,t}$, and build the corresponding managed portfolios (hereafter, factors), whose returns R_{t+1}^F are given by

$$R_{t+1}^F = \frac{1}{N_t} S_t' R_{t+1} \in \mathbb{R}^P . \tag{48}$$

We consider two choices for S_t in (48). (i) $S_t = X_t \in \mathbb{R}^{N_t \times 132}$. The corresponding managed portfolio returns are

$$\frac{1}{N_t} X_t' R_{t+1}, \tag{49}$$

which we refer to as *JKP factors*, or linearly managed portfolios. (ii) $S_t = \text{ReLU}\left(\frac{1}{2}\sqrt{\frac{2}{d}} X_t \Omega'\right) \in \mathbb{R}^{N_t \times P}$, where $\Omega \in \mathbb{R}^{P \times d}$, $P = 5{,}000$, with i.i.d. entries $\Omega_{p,j} \sim N(0,1)$, drawn once at the beginning of the sample and kept fixed throughout all rolling-window

estimations. These are nonlinear random features (RF), as studied in Didisheim et al. (2024), hereafter DKKM. The corresponding managed portfolio returns are

$$\frac{1}{N_t}\text{ReLU}\left(\frac{1}{2}\sqrt{\frac{2}{d}}X_t\Omega'\right)' R_{t+1}, \tag{50}$$

which we refer to as non-linearly managed portfolios. The CLT bands of Theorem H.1 rely on Assumption 1, which is natural for our test assets because they are managed portfolios. Individual-stock conditional means move with observable characteristics, whereas portfolios formed from a broad cross-section average that variation away; Didisheim et al. (2024) show that managed-portfolio returns satisfy Assumption 1 as the number of underlying stocks $N \to \infty$.[10]

In addition, we consider corporate bonds using the 65 bond characteristics from Dickerson et al. (2026a) over the period January 1986 to December 2025. These predictors are constructed from a mix of Lehman, BAML/ICE, and TRACE data, supplemented by equity-based characteristics from Chen and Zimmermann (2022) and Jensen et al. (2023). We apply the same procedure as for stocks: cross-sectionally rank-standardize each characteristic into $[-0.5, 0.5]$, replace missing values with zero, and generate 5,000 ReLU random-feature managed portfolios from the resulting characteristic matrix as in (48), with R_{t+1} replaced by the vector of bond excess returns.

Dickerson et al. (2026a) consider two alternative definitions of bond returns. The first is the standard excess return, defined as the total bond return minus the one-month risk-free rate from Kenneth French's website. The second, following van Binsbergen, Nozawa and Schwert (2024), is the duration-hedged return, obtained by subtracting the return on a duration-matched portfolio of U.S. Treasury bonds from the total bond return, thereby

[10] As an illustration, suppose that $S_{i,t}$ are independent across i and such that $E_t[R_{i,t+1}] = S_{i,t}$. Then $N_t^{-1}E_t[R'_{t+1}S_t] = N_t^{-1}\|S_t\|^2$ converges to a non-random limit as $N_t \to \infty$ by the law of large numbers, no matter how persistent S_t is over time.

removing interest-rate risk. We conduct our analysis using factor returns constructed under both definitions.

5.2 Equity portfolios

We estimate maximal Sharpe ratios from equity managed portfolios using a 240-month rolling window. For each window, we report the lower bound with its lower confidence limit and the upper bound with its upper confidence limit, using $J = 1$ and the ridge-penalty grid $z \in [10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}]$. The smallest value, $z = 10^{-4}$, is as small as the numerical stability of $(zI+\hat{\Sigma})^{-1}$ permits; larger penalties widen the bounds without improving coverage, a tradeoff we calibrate in the simulations (Appendix M). The data-driven threshold $a_J^*(\hat{\boldsymbol{\Theta}}(z_k))$ is from Appendix K, and Appendix L combines the bounds across the grid, taking the largest lower bound and the smallest upper bound and correcting the confidence limits for the search over z. Full-sample estimates, which pool the entire 1963–2025 window and so correspond to a lower complexity ratio ($c \approx 6.6$ against $c \approx 21$ for the rolling windows), are shown as dotted horizontal lines. We consider mega, large, small, and micro stocks separately, all together, and all together excluding micro stocks.

Figure 2 shows that the upper and lower bounds of the MSR are close to each other. The bands are narrow at the rolling-window complexity ratio $c = 5{,}000/240 \approx 21$, matching the simulation evidence: at the same c, the simulated bounds concentrate near the true Sharpe ratio (Figure 6).

How much independent information the upper bound carries depends on which branch of the threshold-selection algorithm binds. When the unconstrained minimizer $a_{\min,J}$ delivers an upper bound above the benchmark lower bound, the two bounds are computed from different quantities and their proximity is informative. When it does not, which signals that subdiagonality fails at $a_{\min,J}$, Algorithm 2 retreats to the threshold a_{corner} at which the upper bound equals the benchmark lower bound $\mathcal{H}_1^L(\boldsymbol{\mathcal{B}}(z_2))$ exactly. On that corner branch

the two point estimates coincide by construction, and only the confidence bands separate them. Figure 3 reports how often each branch binds: rarely for mega stocks, but in the large majority of windows for the combined and micro-excluded universes. For those universes, the closeness of the bounds in Figure 2 reflects the selection rule, and the informative content of the interval rests on the lower bound $\mathcal{H}_1^L$, which holds without auxiliary conditions.

The full-sample maximal Sharpe ratio is 7.4 for the combined stock universe. It falls monotonically with size, from 7.6 for micro to 4.9 for small, 3.6 for large, and 2.2 for mega, so the combined figure rests on the smallest names. Micro-cap stocks are thinly traded and costly to transact, and a long line of cross-sectional research excludes them so that estimates reflect an implementable universe (Fama and French, 2008; Kozak et al., 2020; Lopez-Lira and Roussanov, 2023). We follow that practice and add a sixth estimate, on the universe above the 20th NYSE percentile (mega, large, and small), labeled "all ex micro." Its full-sample MSR is 5.0, with confidence limits $[4.3, 5.6]$. Excluding micro stocks lowers the headline from 7.4 to 5.0, a sizable reduction that still leaves the estimate far above what portfolio-based methods report on the same universe. All portfolios except mega exhibit a decline in MSR after approximately 2004, consistent with the weakening of many anomaly-based strategies in recent years.

These estimates compare with existing benchmarks as follows. The U.S. market portfolio delivers an annualized Sharpe ratio of approximately 0.35–0.45, and good-deal bounds (Cochrane and Saá-Requejo, 2000; Bernardo and Ledoit, 2000; Černý, 2003) cap the plausible maximum at roughly twice this level, SR $\leq$ 0.7–1.0. Estimates from the recent high-dimensional literature are larger, and they sort by investment universe. The methods closest to ours build managed portfolios from the full cross-section, micro-caps included: Didisheim et al. (2024), hereafter DKKM, whose equal-weighted random-feature construction we replicate, report an all-stock Sharpe ratio of 3.7; Kelly et al. (2026) and Kelly et al. (2025) reach roughly 4.0 on the same universe; Gu et al. (2020) obtain an equal-weighted 2.4 across

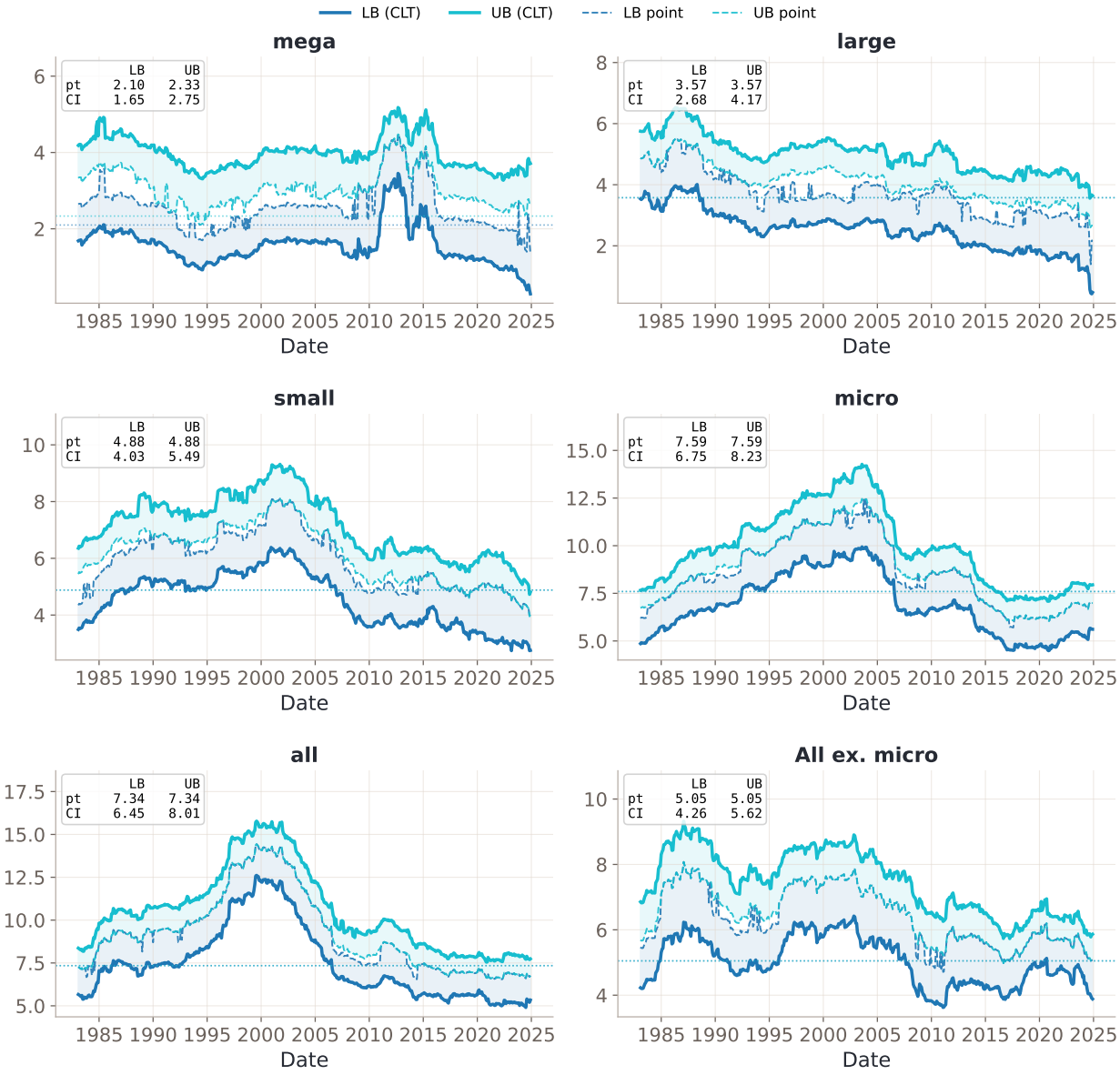


Figure 2: Maximal Sharpe ratio dynamics with $T = 240$ for $P = 5{,}000$ nonlinearly managed stock portfolios, full sample estimates reported in a box in each panel are from January 1963 through December 2025. The full-sample MSR ranges from 2.2 (mega) to 7.6 (micro), equals 7.4 for all stocks and 5.0 once micro stocks are excluded (the "all ex micro" panel, the universe above the 20th NYSE percentile), with a broad decline after 2004.

all of CRSP; and Cong et al. (2025) an out-of-sample 3.9 from value-weighted tree portfolios. Our combined-universe estimate of 7.4 exceeds each. Because our universe and construction match DKKM's, the gap to their 3.7 reflects methodology.

Removing micro-caps brings our estimate onto the universe used by a second group, which reports much smaller numbers. Kozak et al. (2020) exclude micro-caps at construction and estimate 0.7–1.3, and Lopez-Lira and Roussanov (2023), who drop everything below the 20th NYSE percentile, reach roughly one. The micro-inclusive methods land in the same range once they repeat the exercise without micro-caps: DKKM's Sharpe ratio falls to 1.6 on mega and large stocks alone, and Gu et al. (2020)'s equal-weighted spread to 1.7 once the bottom 20% of NYSE stocks are dropped. Our 5.0 on the micro-excluded universe stands well above all of these, so the gap to the portfolio-based literature survives the exclusion of micro-caps. All of these estimates are portfolio-based: they report the Sharpe ratio of a particular constructed portfolio. Proposition 4 shows that ridge-regularized Markowitz portfolios, the construction underlying many of these benchmarks, fall short of the true maximum by a multiplicative factor $1/\hat{Z}'_*(z) < 1$ that does not shrink with T. Our bounds, by contrast, estimate the maximum directly. The step from the 0.35–0.45 market Sharpe ratio to the 4.0 figures in Kelly et al. (2026, 2025) has already pushed the implied Hansen-Jagannathan bound well beyond the range that equilibrium models generate. Our estimates are the population quantity that portfolio-based efforts approach from below.

Proposition 4 and Lemma 2 together account for where the gap ends up. Lemma 2 places a floor on the structural factor: $\hat{Z}'_*(0) \geq c/(c-1)$, which at the full-sample ratio $c = P/T \approx 6.6$ equals 1.18. We estimate $\hat{Z}'_*(z) \approx 1.8$ at $z = 10^{-4}$ for both the combined universe and the micro-excluded universe, with a range of 1.8 (micro) to 1.9 (mega) across the individual size groups. Because $\hat{Z}'_*(z)$ is decreasing in z (Lemma 2), $\hat{Z}'_*(0) \geq 1.8$, well above the $c/(c-1)$ floor. The gap between 1.18 and 1.8 reflects the empirical eigenvalue

distribution H : the aspect ratio determines the floor, and the shape of H determines the magnitude.

The ridge-regularized Markowitz portfolio therefore achieves at most $7.4/\sqrt{1.8} \approx 5.5$ on the full cross-section and $5.0/\sqrt{1.8} \approx 3.7$ on the micro-excluded universe. The 4.0 figures of Kelly et al. (2026, 2025) sit modestly below the all-stock ceiling, whereas the micro-excluded methods top out near 1.7, well short of the 3.7 ceiling. Our 7.4 estimate is the Sharpe ratio attainable by trading the efficient portfolio, whose identity our framework leaves implicit. Both the 7.4 and the 1.0–4.0 benchmarks are gross of trading costs. To the extent that the efficient portfolio puts weight on illiquid securities, particularly micro and small stocks, the realized net-of-cost Sharpe ratio would be lower; this caveat applies symmetrically to the portfolio-based literature.

The 7.4 headline admits a decomposition that separates the part resting on data from the part resting on the untestable subdiagonality condition. The lower bound $\mathcal{H}_1^L$ draws only on the estimable quantities $\hat{\Theta}_0$ and $\hat{\Theta}_1$ and lies close to 7 for the combined universe in the full sample, so the bulk of the Sharpe-ratio news is carried by the lower bound and does not depend on subdiagonality. The upper bound $\mathcal{H}_1^U$ adds the untestable component, and the ratio $a^*/\bar{\lambda}$ from Appendix K quantifies exactly the share of that bound attributable to it. In the simulations this share lies between 0.1 and 0.36 at the (P, T) combinations we study, including the empirical $(P, T) = (5{,}000,\ 240)$ (Figure 7). In the data, however, the threshold rests on the corner branch in most windows for this universe. The upper bound is then pinned to the lower bound, and the decomposition is not informative. For the headline, what matters is that the step from the 1.0–4.0 portfolio-based figures to our 7.4 estimate is carried by $\mathcal{H}_1^L$, which does not depend on subdiagonality.

Figure 3 counts the covariance eigenvalues above the threshold a^*, the directions the upper bound retains rather than assigns to the subdiagonal tail. Out of the at most $T = 240$ nonzero sample eigenvalues, $\#\{\lambda_i > a^*\}$ runs from roughly 50 to 180 and trends upward

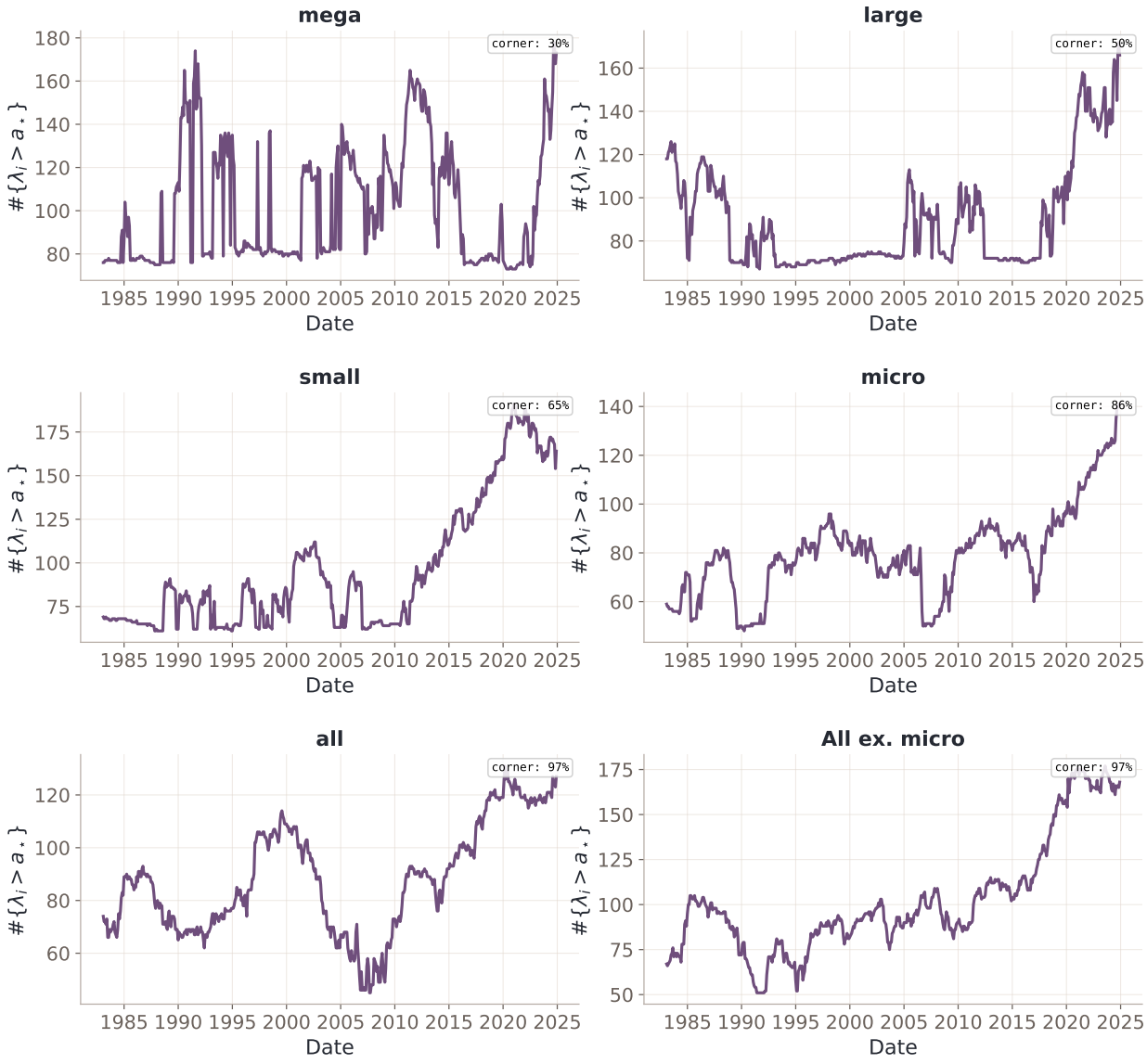


Figure 3: Number of managed-portfolio covariance eigenvalues above the data-driven threshold, $\#\{\lambda_i > a_J^*\}$, in real time for $P = 5{,}000$ nonlinearly managed stock portfolios with $T = 240$, across the six size groups of Figure 2. The threshold a_J^* is selected by the algorithm of Appendix K. The "corner" percentage in each panel is the share of windows in which that algorithm uses the subdiagonality-validated value a_{corner} rather than the unconstrained minimizer $a_{\min}$.

over the sample, steeply so after 2010 for the small, micro, all, and all-ex-micro universes. The threshold refers to the population spectrum, while the count uses sample eigenvalues, which are heavily dispersed at $P/T \approx 21$; the count is therefore indicative of how many population components lie above a^*. The priced subspace is high-dimensional and widening: Sharpe ratio spreads across many eigendirections, the empirical counterpart to the infinite factor structure that high-dimensional managed portfolios generate (Chernov et al., 2026).

The count also shows that subdiagonality is a mild restriction, in the sense that it binds only deep in the spectrum. With 50 to 180 eigenvalues above a^*, the threshold falls well past the leading ten to fifty principal components that the dominant-eigenvalue literature retains (Kozak et al., 2020; Feng et al., 2020; Lettau and Pelger, 2020). That literature assigns zero risk premium below its cutoff; subdiagonality asks only that the eigendirections beyond a^* not carry disproportionately large Sharpe ratio. Granting these very-low-rank directions a modest premium rather than none is the weaker assumption, and it governs precisely the part of the spectrum the literature discards.

The corner percentages report how often the threshold rests on the subdiagonality-validity boundary a_{corner} rather than the unconstrained minimizer, rising from 30% of windows for mega stocks to 97% for the all and all-ex-micro universes as the cross-section broadens. A corner window is one in which subdiagonality fails at the unconstrained minimizer $a_{\min,J}$, so that the threshold must be lowered until the bound is again consistent with the data. These frequent failures are consistent with a mild restriction, because subdiagonality is a family of conditions indexed by the threshold: failing at $a_{\min,J}$ does not imply failing at the lower a_{corner}. The failures are also informative, since they show that small-eigenvalue directions carry more Sharpe ratio than the allowance at $a_{\min,J}$ permits. Their cost falls on the upper bound. In corner windows it is pinned to the lower bound and adds no information beyond it, so in the broad universes the estimate rests on the lower bound alone, and only its confidence limit holds without the subdiagonality assumption.

5.3 Equities and bonds

Do corporate bonds expand the efficient frontier beyond what equities alone can achieve? Equivalently, does the maximal Sharpe ratio rise when bond portfolios are added to stock portfolios? Figure 4 presents the results for duration-hedged (left column) and unhedged bonds (right column).

Overall, MSRs increase when bonds are combined with stocks. For the combined universe, adding bonds raises the MSR from 7.7 to 8.7.[11] The lower 97.5% confidence band on the stocks-plus-bonds lower bound overlaps with the upper 97.5% confidence band on the equity-only upper bound in Figure 4, so the improvement, while economically meaningful, is not statistically significant at conventional levels. When we use unhedged bonds, implicitly adding information from Treasury bonds, the situation is similar.

The comparison holds the equity side fixed. Adding $P = 5{,}000$ bond portfolios to $P = 5{,}000$ equity portfolios doubles the complexity ratio from $c \approx 21$ to $c \approx 42$. We keep the equity feature count at $P_{\text{eq}} = 5{,}000$, matching the stocks-only analysis rather than halving it to preserve the combined c, because the question of interest is the incremental contribution of bonds on top of the equity frontier already characterized above. A matched-c design with $P_{\text{eq}} = P_{\text{bd}} = 2{,}500$ would shrink the equity nonlinear span simultaneously with the bond addition, conflating the two channels and shifting the baseline against which the increment is measured; at $P_{\text{eq}} = 2{,}500$ the equity-only frontier itself differs from the one we study in the stocks-only analysis, so the bond contribution evaluated against it would be a different economic object. Higher c can inflate the upper bound, but that inflation is not a concern here: the comparison rests on the lower bound of the combined (high-c) MSR against the upper bound of the equity-only (low-c) MSR.

The fact that the MSR for all securities combined does not improve significantly echoes the findings of Dickerson et al. (2026b). These authors consider linear factor models

[11] The equity-only number is slightly different from Figure 2 because the sample starts later, in 1986, due to bond data availability.

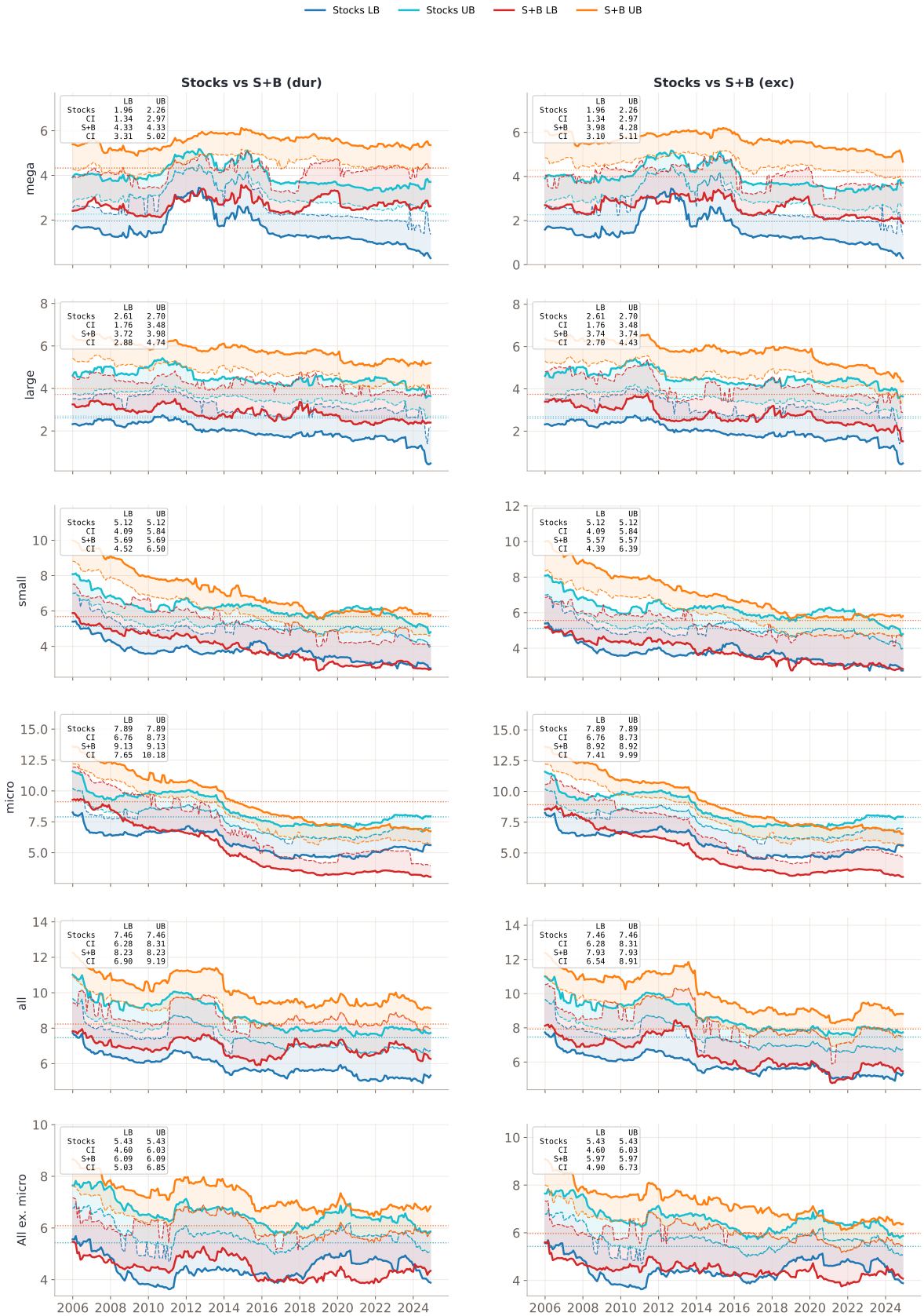


Figure 4: Maximal Sharpe ratio dynamics with $T = 240$ for $P = 5{,}000$ nonlinearly managed stock and $P = 5{,}000$ bond portfolios, full sample estimates reported in a box in each panel are from January 1986 through December 2025. Duration-hedged (unhedged) bond returns are displayed in the left (right) column. Adding bonds raises the MSR, an economically meaningful but statistically insignificant improvement.

and conclude that equity and nontradable factors can explain duration-adjusted corporate bond risk premia. This conclusion suggests, consistent with our findings, that stocks and bonds span the same frontier. Chen et al. (2024b) document the corresponding factor-structure result through their joint stocks-bonds-options analysis, finding strong common factor structure across asset classes. Our framework clarifies why: if bond and equity managed portfolios load on similar eigenvalue directions, combining them increases P but does not broaden the spectrum in a way that generates new Sharpe ratio for the higher-order terms to capture. Duration hedging removes interest-rate exposure but does not materially alter the cross-sectional structure of bond characteristics, explaining the similarity of the two panels.

5.4 Linear vs non-linear managed portfolio weights

We examine the role of nonlinearity by tracking the MSR of managed portfolios $S_t' R_{t+1}$ under two choices of weights, $S_t \propto X_t$ and $S_t \propto \text{ReLU}(X_t \Omega')$. Both use real-time characteristics: at each date t, only characteristics that were known by t enter X_t. We estimate the MSR at each point in time on an expanding window, whose first observation, in January 1980, corresponds to an initial sample of $T = 205$ months. Figure 5 reports the findings.

First, with the exception of mega stocks, nonlinear weights in managed portfolios result in substantially higher MSRs. Virtually all research on factors assumes a linear mapping from characteristics to portfolio weights. Our results document a large margin of improvement from abandoning this classic framework. Our framework identifies the source of this improvement. With 132 linear portfolios, managed portfolio returns span at most 132 eigenvalue directions, capping the attainable Sharpe ratio. Nonlinear transformations expand this span to $P = 5{,}000$ directions, broadening the eigenvalue spectrum. The additional Sharpe ratio sits in small-eigenvalue directions that linear portfolios do not span, and the higher-order terms of our expansion recover it. Scaling up P in the linear case would not

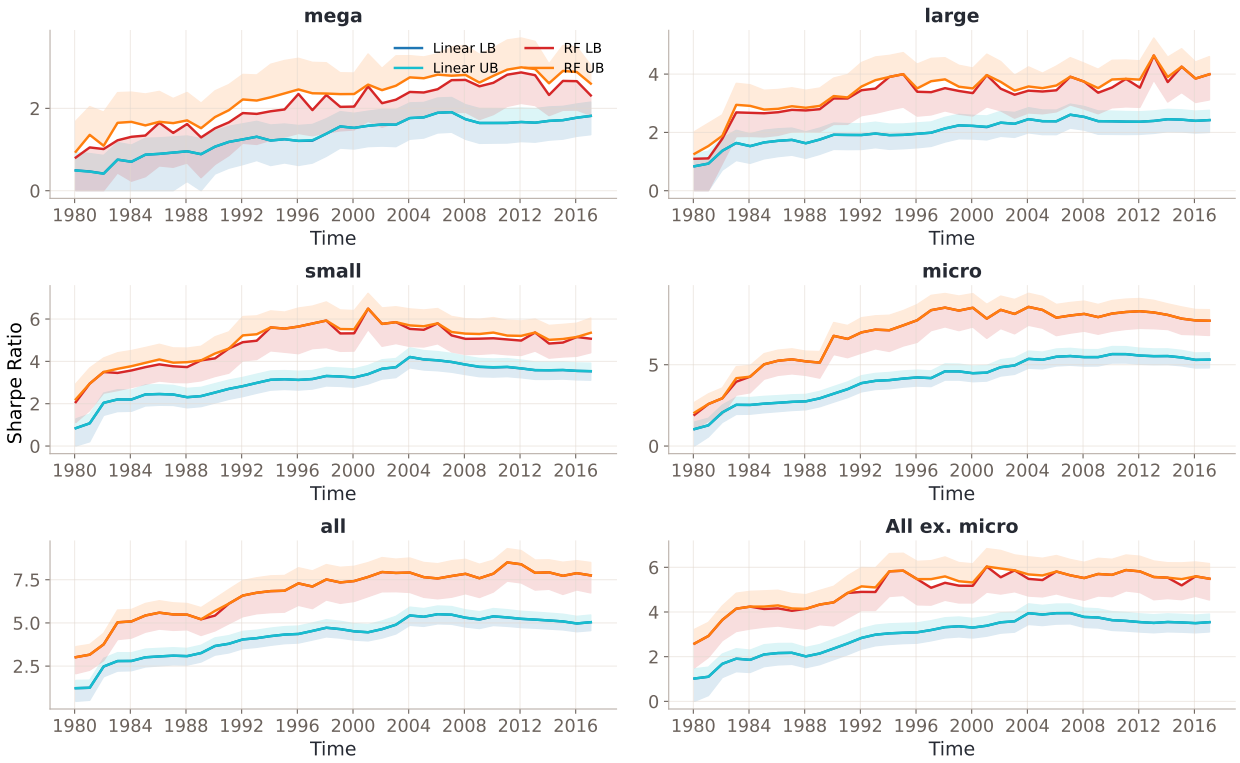


Figure 5: Maximal Sharpe ratio dynamics for 132 linearly and $P = 5{,}000$ nonlinearly managed stock portfolios in real time. Nonlinear weights produce substantially higher MSRs in every size group except mega, and characteristics introduced after 2004 contribute little additional Sharpe ratio.

change this: any linear combination of the 132 characteristics stays in a 132-dimensional span. The $P = 132$ linear versus $P = 5{,}000$ nonlinear contrast therefore isolates the marginal value of nonlinearity. Raising P in the nonlinear case keeps opening new directions, and the simulations confirm the bounds remain well behaved at $P = 5{,}000$ (Appendix M).

Second, regardless of whether weights are linear or nonlinear, characteristics discovered after 2004 add little to the MSR. The real-time design makes this directly testable: at each date t, the MSR is constructed from only those characteristics known by t. We operationalize "known by t" using the original-publication year that Jensen et al. (2023) record for each characteristic in the JKP metadata, admitting a characteristic into the date-t sample only if its publication year is no later than t. As Figure 5 shows, the upper 97.5% band for the 2004

MSR overlaps the lower 97.5% band for the MSR at every subsequent date, across all size groups and for both linear and nonlinear weights. The null that post-2004 characteristics raise the MSR therefore cannot be rejected at conventional levels.

This finding holds despite the post-2004 vintage containing several influential anomalies: gross profitability (Novy-Marx, 2013), asset growth (Cooper, Gulen and Schill, 2008), idiosyncratic volatility (Ang, Hodrick, Xing and Zhang, 2006), betting against beta (Frazzini and Pedersen, 2014), and operating profitability (Fama and French, 2015). The proliferation of published anomalies over the past two decades has not materially widened the investment frontier already spanned by earlier characteristics: novel as these anomalies appear, their priced components can be expressed as functions of characteristics already in the cross-section.

6 Conclusion

This paper develops a method for estimating the maximal Sharpe ratio in high dimensions directly, without constructing the efficient portfolio. The method rests on a new family of higher-order debiased GRS statistics that progressively recover the contributions of small eigenvalues suppressed by ridge regularization. We show that the maximal Sharpe ratio squared expands into an infinite series of such terms with sub-exponential decay, and that imposing subdiagonality on the risk-return tradeoff yields a two-sided confidence interval for $\mathcal{R}_*^2$.

Applied to U.S. equities, the estimated maximal Sharpe ratios are substantially larger than existing benchmarks. The gap is structural: Proposition 4 shows that the Sharpe ratios of ridge-regularized Markowitz portfolios fall short of our lower bound on $\mathcal{R}_*^2$ by a factor that cannot be closed within this class. The lower bound, which requires no auxiliary condition, accounts for nearly all of the estimated maximum. The upper bound also requires subdiagonality of the risk-return tradeoff, and for the broadest universes our threshold-

selection rule pins it to the lower bound. The proximity of the two bounds should therefore not be read as independent evidence of sharp identification.

The empirics yield two further findings. First, nonlinear portfolio weights yield substantially higher maximal Sharpe ratios than their linear counterparts, indicating that the conventional linear mapping from characteristics to portfolio weights leaves considerable gains unexploited. Second, stock characteristics introduced after 2004, a period that produced several widely studied anomalies, add little to the investment frontier.

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A Auxiliary Moment Bounds

Many proofs in this paper are based on the classic "concentration of quadratic forms" lemma. See, e.g., Didisheim et al. (2024) for a proof.

Lemma A.1 *Let A be a uniformly bounded matrix and let $Y_t = \varepsilon_t' A \varepsilon_t$. Then,*

$$\mathrm{Var}\left[\frac{1}{P} Y_t\right] \leq C \frac{1}{P^2} \|\Sigma^{1/2} A \Sigma^{1/2}\|_F^2 \;\leq\; C \frac{1}{P} \|\Sigma\|^2 \|A\|^2, \tag{A.1}$$

where $\|\cdot\|_F$ is the Frobenius norm and $C = \max\{2,\ \mathbb{E}[X_{ti}^4] - 1\}$.[12] *Hence,*

$$\frac{1}{P} Y_t \approx \mathbb{E}\left[\frac{1}{P} Y_t\right] = \frac{1}{P} \operatorname{tr}(A\Sigma)\,. \tag{A.2}$$

The following lemma is a direct consequence of Lemma A.1.

Lemma A.2 *Suppose that $u = (u_i)_{i=1}^P$ where u_i are i.i.d., with $E[u_i] = 0$, $E[u_i^2] = \sigma^2$, $E[u_i^4] < \infty$. Suppose also A_P is a sequence of symmetric random matrices that is independent of u and is such that $E[(P^{-1}\operatorname{tr}(A_P))^2] < K$ and $\lim P^{-2} E[\operatorname{tr}(A_P^2)] = 0$. Then,*

$$\frac{1}{P} u' A_P u \;-\; P^{-1}\sigma^2 \operatorname{tr}(A_P) \;\to\; 0 \tag{A.3}$$

in L_2 and, hence, in probability.

Lemma A.3 (Fourth moment of a linear form) *Let $Y = (Y_1, \ldots, Y_P)'$ have independent coordinates with*

$$E[Y_i] = 0, \qquad E[Y_i^2] = 1, \qquad \sup_i E|Y_i|^4 < \infty.$$

[12] The exact variance is $2\operatorname{tr}(\tilde{A}^2) + (E[X_{ti}^4] - 3)\sum_i \tilde{a}_{ii}^2$ with $\tilde{A} = \Sigma^{1/2} A \Sigma^{1/2}$, and $\sum_i \tilde{a}_{ii}^2 \leq \operatorname{tr}(\tilde{A}^2) = \|\tilde{A}\|_F^2$. The constant $E[X_{ti}^4] - 1$ is the right one when $E[X_{ti}^4] \geq 3$; otherwise the second term is negative and the bound $2\|\tilde{A}\|_F^2$ applies.

Then there exists $C < \infty$, depending only on $\sup_i E|Y_i|^4$, such that for every deterministic vector $a \in \mathbb{R}^P$,

$$E[(a'Y)^4] \le C\|a\|^4.$$

Consequently, if a is random but independent of Y, then

$$E[(a'Y)^4] \le C\, E\|a\|^4.$$

Proof of Lemma A.3. Write

$$a'Y = \sum_{i=1}^{P} a_i Y_i.$$

Expanding the fourth power and using independence and centering, only index configurations in which each index appears at least twice contribute. Hence

$$E[(a'Y)^4] = \sum_{i=1}^{P} a_i^4 E[Y_i^4] + 6 \sum_{1 \le i < j \le P} a_i^2 a_j^2 E[Y_i^2] E[Y_j^2].$$

Therefore

$$E[(a'Y)^4] \le \Big(\sup_i E[Y_i^4] \Big) \sum_i a_i^4 + 6 \sum_{i<j} a_i^2 a_j^2.$$

Since

$$\sum_i a_i^4 \le \Big(\sum_i a_i^2 \Big)^2 = \|a\|^4, \qquad \sum_{i<j} a_i^2 a_j^2 \le \frac{1}{2} \Big(\sum_i a_i^2 \Big)^2 = \frac{1}{2}\|a\|^4,$$

the claim follows. □

Lemma A.4 (Fourth moment of a centered quadratic form) *Let $Y = (Y_1, \dots, Y_P)'$*

have independent coordinates with

$$E[Y_i] = 0, \qquad E[Y_i^2] = 1, \qquad \sup_i E|Y_i|^8 < \infty.$$

Then there exists $C < \infty$*, depending only on* $\sup_i E|Y_i|^8$*, such that for every deterministic symmetric matrix* $B \in \mathbb{R}^{P\times P}$*,*

$$E\Big[\big(Y'BY - \operatorname{tr}(B)\big)^4\Big] \le C\Big((\operatorname{tr}(B^2))^2 + \operatorname{tr}(B^4)\Big).$$

Proof of Lemma A.4. Write

$$Y'BY - \operatorname{tr}(B) = \sum_{i=1}^{P} b_{ii}(Y_i^2 - 1) + 2\sum_{1\le i<j\le P} b_{ij}Y_iY_j =: D + O.$$

By $(x+y)^4 \le 8x^4 + 8y^4$,

$$E[(Y'BY - \operatorname{tr}(B))^4] \le 8E[D^4] + 8E[O^4].$$

Step 1: bound for D**.** Set $U_i := b_{ii}(Y_i^2 - 1)$. The variables U_i are independent and centered. Expanding $(\sum_i U_i)^4$ and using again that only index patterns with all indices paired contribute,

$$E[D^4] = \sum_i E[U_i^4] + 6\sum_{i<j} E[U_i^2]E[U_j^2].$$

Since $E|Y_i|^8$ is uniformly bounded, both $E[(Y_i^2-1)^2]$ and $E[(Y_i^2-1)^4]$ are uniformly bounded. Hence

$$E[D^4] \le C\sum_i b_{ii}^4 + C\Big(\sum_i b_{ii}^2\Big)^2 \le C\Big(\sum_i b_{ii}^2\Big)^2 \le C(\operatorname{tr}(B^2))^2.$$

Step 2: bound for O. Write

$$O = 2 \sum_{i<j} b_{ij} Y_i Y_j.$$

Expand O^4. A generic monomial has the form

$$b_{ij} b_{kl} b_{mn} b_{rs}\, Y_i Y_j Y_k Y_l Y_m Y_n Y_r Y_s.$$

By independence and centering, the expectation of such a monomial vanishes unless every index appearing in $\{i, j, k, l, m, n, r, s\}$ appears at least twice. Thus only finitely many pairing patterns contribute. Each contributing pattern can be bounded by a constant multiple of either

$$\sum_{i<j} b_{ij}^4 \qquad \text{or} \qquad \Big(\sum_{i<j} b_{ij}^2 \Big)^2,$$

because all mixed moments of total degree 8 are uniformly bounded by the eighth-moment assumption.

Therefore

$$E[O^4] \le C \sum_{i<j} b_{ij}^4 + C \Big(\sum_{i<j} b_{ij}^2 \Big)^2.$$

Now

$$\sum_{i<j} b_{ij}^4 \le \sum_{i,j} b_{ij}^4 \le \operatorname{tr}(B^4),$$

since B is symmetric, and

$$\sum_{i<j} b_{ij}^2 \le \frac{1}{2}\sum_{i\neq j} b_{ij}^2 \le \frac{1}{2}\operatorname{tr}(B^2).$$

Hence

$$E[O^4] \le C\operatorname{tr}(B^4) + C(\operatorname{tr}(B^2))^2.$$

Combining the bounds for D and O proves the result. □

Lemma A.5 (Normalized trace of the sample covariance) *Let, for each P, T,*

$$\hat{\Sigma}_{P,T} := \frac{1}{T}\varepsilon'\varepsilon \in \mathbb{R}^{P\times P}, \qquad \varepsilon = \begin{pmatrix} \varepsilon_1' \\ \vdots \\ \varepsilon_T' \end{pmatrix} \in \mathbb{R}^{T\times P},$$

where

$$\varepsilon_t = \Sigma_P^{1/2} X_t, \qquad t = 1, \dots, T.$$

Assume the following.

(i) For each P, $\Sigma_P \in \mathbb{R}^{P\times P}$ is deterministic, symmetric, positive semidefinite, and uniformly bounded in operator norm:

$$\sup_{P\ge 1} \|\Sigma_P\| < \infty.$$

(ii) $X_1, \dots, X_T \in \mathbb{R}^P$ are independent across t, and for each t,

$$X_t = (X_{1t}, \dots, X_{Pt})',$$

with i.i.d. coordinates satisfying

$$E[X_{1t}] = 0, \qquad E[X_{1t}^2] = 1, \qquad E[X_{1t}^4] < \infty.$$

(iii) As $P, T \to \infty$,

$$\frac{P}{T} \to c \in (0, \infty).$$

Then

$$\frac{1}{P}\operatorname{tr}(\hat{\Sigma}_{P,T}) - \frac{1}{P}\operatorname{tr}(\Sigma_P) \to 0 \qquad \text{in } L^2 \text{ and hence in probability.}$$

More precisely,

$$E\left[\left(\frac{1}{P}\operatorname{tr}(\hat{\Sigma}_{P,T}) - \frac{1}{P}\operatorname{tr}(\Sigma_P)\right)^2\right] = O\left(\frac{1}{PT}\right) = O\left(\frac{1}{P^2}\right),$$

so that

$$\frac{1}{P}\operatorname{tr}(\hat{\Sigma}_{P,T}) - \frac{1}{P}\operatorname{tr}(\Sigma_P) = O_p(P^{-1}).$$

The same rate holds for the centered sample covariance $\hat{\Sigma} = \hat{\Sigma}_{P,T} - \bar{\varepsilon}\bar{\varepsilon}'$ *used in the main text: by Lemma D.1,*

$$P^{-1}\operatorname{tr}(\bar{\varepsilon}\bar{\varepsilon}') = P^{-1}\|\bar{\varepsilon}\|^2, \qquad E\big[P^{-1}\|\bar{\varepsilon}\|^2\big] = \frac{\operatorname{tr}(\Sigma_P)}{PT} = \frac{\bar{\lambda}}{T} = O(P^{-1}),$$

so that $P^{-1}\operatorname{tr}(\hat{\Sigma}) - P^{-1}\operatorname{tr}(\Sigma_P) = O_p(P^{-1})$ *as well. Note that this second contribution is of the same order as the first and is deterministic to leading order rather than mean zero.*

Proof of Lemma A.5. We divide the proof into several steps.

Step 1: Algebraic reduction. Since

$$\hat{\Sigma}_{P,T} = \frac{1}{T}\varepsilon'\varepsilon,$$

we have, by cyclicity of the trace,

$$\operatorname{tr}(\hat{\Sigma}_{P,T}) = \frac{1}{T}\operatorname{tr}(\varepsilon'\varepsilon) = \frac{1}{T}\operatorname{tr}(\varepsilon\varepsilon') = \frac{1}{T}\sum_{t=1}^{T}\varepsilon_t'\varepsilon_t.$$

Therefore

$$\frac{1}{P}\operatorname{tr}(\hat{\Sigma}_{P,T}) - \frac{1}{P}\operatorname{tr}(\Sigma_P) = \frac{1}{PT}\sum_{t=1}^{T}\Big(\varepsilon_t'\varepsilon_t - \operatorname{tr}(\Sigma_P)\Big). \tag{A.4}$$

Now, for each t,

$$\varepsilon_t'\varepsilon_t = X_t'\Sigma_P X_t.$$

Hence if we define

$$Y_t := X_t'\Sigma_P X_t - \operatorname{tr}(\Sigma_P),$$

then (A.4) becomes

$$\frac{1}{P}\operatorname{tr}(\hat{\Sigma}_{P,T}) - \frac{1}{P}\operatorname{tr}(\Sigma_P) = \frac{1}{PT}\sum_{t=1}^{T}Y_t. \tag{A.5}$$

We next verify that $(Y_t)_{t=1}^T$ are centered and independent. Independence follows immediately from the independence of $X_1, \ldots, X_T$. For centering, write

$$X_t'\Sigma_P X_t = \sum_{i,j=1}^{P}(\Sigma_P)_{ij}X_{it}X_{jt}.$$

Since the coordinates are i.i.d. with mean zero and variance one,

$$E[X_{it}X_{jt}] = \delta_{ij},$$

and therefore

$$E[X_t'\Sigma_P X_t] = \sum_{i,j=1}^{P} (\Sigma_P)_{ij} E[X_{it}X_{jt}] = \sum_{i=1}^{P} (\Sigma_P)_{ii} = \operatorname{tr}(\Sigma_P).$$

Thus $E[Y_t] = 0$.

Step 2: Second-moment reduction. Since the Y_t are independent and centered, (A.5) yields

$$\begin{aligned} E\left[\left(\frac{1}{P}\operatorname{tr}(\hat{\Sigma}_{P,T}) - \frac{1}{P}\operatorname{tr}(\Sigma_P)\right)^2\right] &= E\left[\left(\frac{1}{PT}\sum_{t=1}^{T} Y_t\right)^2\right] \\ &= \frac{1}{P^2T^2}\sum_{t=1}^{T} E[Y_t^2] \\ &= \frac{1}{P^2T} E[Y_1^2], \end{aligned} \tag{A.6}$$

where the cross terms vanish because $E[Y_tY_s] = E[Y_t]E[Y_s] = 0$ for $t \neq s$.

Thus, it suffices to show that

$$E[Y_1^2] = O(P).$$

Step 3: Variance of a quadratic form. Fix P, and write $\Sigma = \Sigma_P$, $X = X_1$, $Y = Y_1 = X'\Sigma X - \operatorname{tr}(\Sigma)$. Expanding,

$$X'\Sigma X = \sum_{i,j=1}^{P} \sigma_{ij} X_i X_j.$$

Hence

$$E[(X'\Sigma X)^2] = \sum_{i,j,k,\ell=1}^{P} \sigma_{ij}\sigma_{k\ell}\, E[X_i X_j X_k X_\ell].$$

Because the coordinates (X_i) are independent, centered, and have unit variance, the expectation $E[X_i X_j X_k X_\ell]$ vanishes unless each index appears at least twice. Thus only the following configurations contribute:

1. $i = j \neq k = \ell$,
2. $i = k \neq j = \ell$,
3. $i = \ell \neq j = k$,
4. $i = j = k = \ell$.

Collecting these cases carefully, one obtains the standard identity

$$\mathrm{Var}(X'\Sigma X) = 2\,\mathrm{tr}(\Sigma^2) + (E[X_1^4] - 3)\sum_{i=1}^{P} \sigma_{ii}^2. \tag{A.7}$$

For completeness, we verify this formula. First,

$$E[X'\Sigma X] = \mathrm{tr}(\Sigma),$$

so

$$\mathrm{Var}(X'\Sigma X) = E[(X'\Sigma X)^2] - \mathrm{tr}(\Sigma)^2.$$

Now

$$\begin{aligned} E[(X'\Sigma X)^2] &= \sum_{i,j,k,\ell} \sigma_{ij}\sigma_{k\ell} E[X_i X_j X_k X_\ell] \\ &= \sum_{i\neq k} \sigma_{ii}\sigma_{kk} E[X_i^2]E[X_k^2] + \sum_{i\neq j} \sigma_{ij}\sigma_{ij} E[X_i^2]E[X_j^2] + \sum_{i\neq j} \sigma_{ij}\sigma_{ji} E[X_i^2]E[X_j^2] \\ &\quad + \sum_i \sigma_{ii}^2 E[X_i^4]. \end{aligned}$$

Since Σ is symmetric, $\sigma_{ij} = \sigma_{ji}$, and $E[X_i^2] = 1$, this becomes

$$E[(X'\Sigma X)^2] = \sum_{i\neq k} \sigma_{ii}\sigma_{kk} + 2\sum_{i\neq j} \sigma_{ij}^2 + E[X_1^4] \sum_i \sigma_{ii}^2.$$

Using

$$\sum_{i\neq k} \sigma_{ii}\sigma_{kk} = \Big(\sum_i \sigma_{ii}\Big)^2 - \sum_i \sigma_{ii}^2 = \operatorname{tr}(\Sigma)^2 - \sum_i \sigma_{ii}^2,$$

we obtain

$$E[(X'\Sigma X)^2] = \operatorname{tr}(\Sigma)^2 + 2\sum_{i\neq j} \sigma_{ij}^2 + (E[X_1^4] - 1) \sum_i \sigma_{ii}^2.$$

Subtracting $\operatorname{tr}(\Sigma)^2$, and observing that

$$\operatorname{tr}(\Sigma^2) = \sum_{i,j} \sigma_{ij}^2 = \sum_{i\neq j} \sigma_{ij}^2 + \sum_i \sigma_{ii}^2,$$

gives

$$\operatorname{Var}(X'\Sigma X) = 2\operatorname{tr}(\Sigma^2) + (E[X_1^4] - 3) \sum_{i=1}^{P} \sigma_{ii}^2,$$

which is (A.7).

Consequently,

$$E[Y_1^2] = \mathrm{Var}(X'\Sigma_P X) \le 2\,\mathrm{tr}(\Sigma_P^2) + |E[X_{11}^4] - 3| \sum_{i=1}^{P} (\Sigma_P)_{ii}^2. \tag{A.8}$$

Step 4: Uniform $O(P)$ bound. Let $M := \sup_{P\ge 1} \|\Sigma_P\| < \infty$. Since Σ_P is positive semidefinite, all of its eigenvalues lie in $[0, M]$. Therefore

$$\mathrm{tr}(\Sigma_P^2) \le \|\Sigma_P\|\,\mathrm{tr}(\Sigma_P) \le M\,\mathrm{tr}(\Sigma_P).$$

Moreover,

$$\mathrm{tr}(\Sigma_P) = \sum_{j=1}^{P} \lambda_j(\Sigma_P) \le MP,$$

hence

$$\mathrm{tr}(\Sigma_P^2) \le M^2 P. \tag{A.9}$$

Next, for each i,

$$0 \le (\Sigma_P)_{ii} = e_i' \Sigma_P e_i \le \|\Sigma_P\| \le M,$$

so

$$\sum_{i=1}^{P} (\Sigma_P)_{ii}^2 \le M^2 P. \tag{A.10}$$

Combining (A.8), (A.9), and (A.10), we obtain

$$E[Y_1^2] \le CP$$

for some constant $C < \infty$ independent of P, T.

Substituting into (A.6) yields

$$E\left[\left(\frac{1}{P}\operatorname{tr}(\hat{\Sigma}_{P,T}) - \frac{1}{P}\operatorname{tr}(\Sigma_P)\right)^2\right] \leq \frac{C}{PT}.$$

This proves the L^2 bound

$$\frac{1}{P}\operatorname{tr}(\hat{\Sigma}_{P,T}) - \frac{1}{P}\operatorname{tr}(\Sigma_P) = O_{L^2}\big((PT)^{-1/2}\big).$$

Step 5: Use the aspect-ratio condition. Since $P/T \to c \in (0, \infty)$, there exist constants $0 < a < b < \infty$ such that for all sufficiently large P, T,

$$a \leq \frac{P}{T} \leq b.$$

Equivalently,

$$\frac{1}{T} \leq \frac{1}{aP}.$$

Hence

$$\frac{1}{PT} \leq \frac{1}{aP^2},$$

and therefore

$$E\left[\left(\frac{1}{P}\operatorname{tr}(\hat{\Sigma}_{P,T}) - \frac{1}{P}\operatorname{tr}(\Sigma_P)\right)^2\right] = O(P^{-2}).$$

It follows that

$$\frac{1}{P}\operatorname{tr}(\hat{\Sigma}_{P,T}) - \frac{1}{P}\operatorname{tr}(\Sigma_P) = O_{L^2}(P^{-1}),$$

and thus also

$$\frac{1}{P}\operatorname{tr}(\hat{\Sigma}_{P,T}) - \frac{1}{P}\operatorname{tr}(\Sigma_P) = O_p(P^{-1}).$$

In particular,

$$\frac{1}{P}\operatorname{tr}(\hat{\Sigma}_{P,T}) - \frac{1}{P}\operatorname{tr}(\Sigma_P) \to 0 \qquad \text{in probability.}$$

This completes the proof. □

B Random Matrix Theory Master Theorems

A key object of RMT is the empirical spectral distribution $H_A(x)$ of a symmetric matrix $A \in \mathbb{R}^{P\times P}$:

$$H_A(x) \;=\; \frac{1}{P}\sum_{i=1}^{P} \mathbf{1}_{\lambda_i(A)\le x}, \tag{B.1}$$

where $\lambda_i(A)$ are the eigenvalues of the matrix A.

Definition 3 *Given a sequence of symmetric matrices $\Sigma_P \in \mathbb{R}^{P\times P}$, $P \to \infty$, we say that their eigenvalue distributions H_{Σ_P} are asymptotically stable if there exists a distribution H with compact support, non-degenerate at zero,*[13] *such that*

$$\sqrt{P} D_W(H_{\Sigma_P}, H) \to 0, \quad \text{as } P \to \infty,$$

[13] That is, $H(x = 0) < 1$.

where $D_W(F_1, F_2)$ denotes the Wasserstein distance between distributions F_1 and F_2, defined as

$$D_W(F_1, F_2) = \sup_f \left\{ \left| \int f \, dF_1 - \int f \, dF_2 \right| : f \text{ is 1-Lipschitz} \right\}.$$

One trivial example of asymptotic stability is constant: if $\Sigma_P = I_{P\times P}$, then H_{Σ_P} is the distribution function of the point mass δ_1 at one. The true Stieltjes transform admits an integral form representation given by

$$m_{\Sigma_P}(-z) \;=\; \int_0^\infty \frac{1}{x+z} dH_{\Sigma_P}(x), \;\; z > 0, \tag{B.2}$$

and, hence, under asymptotic stability,

$$\lim_{P\to\infty} m_{\Sigma_P}(-z) \tag{B.3}$$

exists and is given by

$$m(-z) \;=\; \int_0^\infty \frac{1}{x+z} dH(x)\,. \tag{B.4}$$

Note that by construction, for $z > 0$, we have that $m(-z) > 0$.

In the classic case of zero complexity, $\hat{\Sigma}_P$ converges to the true covariance matrix of residuals and, hence, so does the Stieltjes transform:

$$\lim_{T,P\to\infty,\; P/T\to 0} \hat{m}(-z) \;=\; m(-z)\,. \tag{B.5}$$

A cornerstone result of RMT is that when $P/T \to c > 0$, (B.5) needs to be adjusted: $\hat{m}(-z)$ also converges to a deterministic limit that can be expressed in terms of $m_{\Sigma_P}(-z)$.

Theorem B.1 (Bai and Zhou (2008)) *For each $z > 0$,*

$$\lim_{T,P\to\infty, P/T\to c} \hat{m}(-z) \;=\; m(-z;c) \tag{B.6}$$

exists in probability and $m(-z;c)$ is the unique positive solution to the nonlinear master equation

$$m(-z;c) = \frac{1}{1 \;-\; c \;+\; c\,z\,m(-z;c)}\, m_{\Sigma_P}\left(\frac{-z}{1 \;-\; c \;+\; c\,z\,m(-z;c)}\right). \tag{B.7}$$

As a result, the random quantities $\hat{\xi}, \hat{\xi}', \hat{\Delta}$ converge to their theoretical (constant) counterparts in probability:

$$\begin{aligned}
\hat{\xi}(z;c) \;&\to\; \xi(z;c) \;=\; (1 - c + czm(-z;c))^{-1} \;-\; 1 \\
\hat{\xi}'(z;c) \;&\to\; \xi'(z;c) \;=\; \left(\frac{1}{1 - c(1 - zm(-z;c))}\right)^2 c(zm'(-z;c) - m(-z;c)) \\
\hat{\Delta}(z;c) \;&\to\; \Delta(z;c) \;=\; 2(\xi(z;c) + z\xi'(z;c))(1 + \xi(z;c))^2\,,
\end{aligned} \tag{B.8}$$

where

$$\hat{\Delta}(z;c) \;=\; 2(\hat{\xi}(z;c) + z\hat{\xi}'(z;c))(1 + \hat{\xi}(z;c))^2\,. \tag{B.9}$$

The Master equation (B.7) is a non-linear fixed-point equation. This equation admits a unique positive solution for $z > 0$, that can be found using Newton's method.

Theorem B.2 (Li, Aue and Paul (2020)) *Let*

$$\hat{W}(z) \;=\; \bar{\varepsilon}'(zI + T^{-1}\varepsilon'\varepsilon - \bar{\varepsilon}\bar{\varepsilon}')^{-1}\bar{\varepsilon},$$

where

$$\bar{\varepsilon} \;=\; \frac{1}{T}\sum_t \varepsilon_t = \Sigma^{1/2}\bar{X}, \quad \bar{X} = T^{-1}\sum_t X_t, \quad T^{-1}\varepsilon'\varepsilon \;=\; \frac{1}{T}\sum_t \varepsilon_t\varepsilon_t'\,.$$

Note also that $\bar{R} = \mu + \bar{\varepsilon}$. *Then,* $T^{1/2}(\hat{W}(z) - \hat{\xi}(z;c)),\ z \in \mathbb{C}\setminus\mathbb{R}_-$ *converges to a Gaussian process with mean zero and the covariance kernel*

$$2\frac{z_1\xi(z_1;c) - z_2\xi(z_2;c)}{z_1 - z_2}(1+\xi(z_1;c))(1+\xi(z_2;c))\,. \tag{B.10}$$

In particular,

$$\frac{T^{1/2}(\hat{W}(z) - \hat{\xi}(z;c))}{\hat{\Delta}(z;c)^{1/2}} \;\to\; N(0,1) \tag{B.11}$$

in distribution, where

$$\hat{\Delta}(z;c) \;=\; 2(\hat{\xi}(z;c) + z\hat{\xi}'(z;c))(1+\hat{\xi}(z;c))^2\,. \tag{B.12}$$

C Vanishing Cross Terms

The goal of this section is to prove the following theorem.

Theorem C.1 *Let* $\hat{\Sigma} := T^{-1}\varepsilon'\varepsilon = T^{-1}\sum_{t=1}^T \varepsilon_t\varepsilon_t'$ *denote the uncentered second-moment matrix of the residuals (the object written* $\hat{\Psi}_\varepsilon$ *elsewhere in this appendix), and strengthen the moment condition of Assumption 1 to* $E[X_{i,t}^{8+\varepsilon}] < \infty$ *for some* $\varepsilon > 0$. *Then*

$$\mu_P'(zI_P + \hat{\Sigma})^{-1}\bar{\varepsilon} \to 0 \qquad \text{in } L^2.$$

Equivalently,

$$E\left[\left(\mu_P'(zI_P + \hat{\Sigma})^{-1}\bar{\varepsilon}\right)^2\right] \to 0.$$

The corresponding statement for the centered $\hat{\Sigma} = \hat{\Sigma} - \bar{\varepsilon}\bar{\varepsilon}'$ *follows by one application of the Sherman–Morrison formula.*

Proof of Theorem C.1. Fix $z > 0$, and write

$$A := zI_P + \hat{\Sigma}.$$

We must show

$$M_{P,T} := \mu_P' A^{-1}\bar{\varepsilon} \to 0 \qquad \text{in } L^2.$$

Step 1: leave-one-out representation. For each t, define

$$\hat{\Sigma}_{T,t} := \hat{\Sigma} - \frac{1}{T}\varepsilon_t\varepsilon_t', \qquad Q_t := (zI_P + \hat{\Sigma}_{T,t})^{-1}.$$

Then

$$A = (zI_P + \hat{\Sigma}_{T,t}) + \frac{1}{T}\varepsilon_t\varepsilon_t' = Q_t^{-1} + \frac{1}{T}\varepsilon_t\varepsilon_t'.$$

By the Sherman–Morrison formula,

$$A^{-1}\varepsilon_t = \frac{Q_t\varepsilon_t}{1 + q_t}, \qquad q_t := \frac{1}{T}\varepsilon_t' Q_t \varepsilon_t.$$

Therefore

$$M_{P,T} = \frac{1}{T}\sum_{t=1}^{T} \frac{\alpha_t}{1+q_t}, \qquad \alpha_t := \mu_P' Q_t \varepsilon_t. \tag{C.1}$$

Also define

$$\xi_t := \frac{1}{T}\operatorname{tr}(\Sigma_P Q_t).$$

We decompose

$$\frac{\alpha_t}{1+q_t} = \frac{\alpha_t}{1+\xi_t} + \alpha_t\left(\frac{1}{1+q_t} - \frac{1}{1+\xi_t}\right) =: \beta_t + r_t.$$

Thus

$$M_{P,T} = \frac{1}{T}\sum_{t=1}^{T}\beta_t + \frac{1}{T}\sum_{t=1}^{T} r_t. \tag{C.2}$$

We show that both averages vanish in L^2.

Step 2: preliminary bounds. Since $\hat{\Sigma}_{T,t} \succeq 0$,

$$\|Q_t\| \le z^{-1}. \tag{C.3}$$

Because $\sup_P \|\Sigma_P\| < \infty$ and $\sup_P \|\mu_P\| < \infty$, there exists $C < \infty$ such that

$$\|\Sigma_P^{1/2} Q_t \mu_P\| \le C \qquad \text{a.s. for all } t, P, T.$$

Conditional on Q_t, the vector $\Sigma_P^{1/2} Q_t \mu_P$ is deterministic and independent of X_t. Since

$$\alpha_t = \mu_P' Q_t \Sigma_P^{1/2} X_t = (\Sigma_P^{1/2} Q_t \mu_P)' X_t,$$

Lemma A.3 yields

$$\sup_{P,T,t} E[\alpha_t^4] \leq C. \tag{C.4}$$

In particular,

$$\sup_{P,T,t} E[\alpha_t^2] \leq C. \tag{C.5}$$

Next, conditional on Q_t,

$$T(q_t - \xi_t) = \varepsilon_t' Q_t \varepsilon_t - \operatorname{tr}(\Sigma_P Q_t) = X_t' B_t X_t - \operatorname{tr}(B_t), \qquad B_t := \Sigma_P^{1/2} Q_t \Sigma_P^{1/2}.$$

The matrix B_t is symmetric, positive semidefinite, and

$$\|B_t\| \leq \|\Sigma_P\| \|Q_t\| \leq C.$$

Moreover,

$$\operatorname{tr}(B_t^2) \leq \|B_t\| \operatorname{tr}(B_t) \leq C \operatorname{tr}(B_t) \leq CP,$$

and

$$\operatorname{tr}(B_t^4) \leq \|B_t\|^2 \operatorname{tr}(B_t^2) \leq CP.$$

Hence, by Lemma A.4,

$$E\left[(q_t - \xi_t)^4 \,\middle|\, Q_t\right] \leq \frac{C}{T^4}\Big((\operatorname{tr}(B_t^2))^2 + \operatorname{tr}(B_t^4)\Big) \leq \frac{C(P^2 + P)}{T^4}.$$

Taking expectations and using $P/T \to c$,

$$\sup_t \|q_t - \xi_t\|_4 \le CT^{-1/2}. \tag{C.6}$$

Step 3: the remainder average $\frac{1}{T}\sum r_t$. Since $(1+q_t)(1+\xi_t) \ge 1$,

$$|r_t| = |\alpha_t| \frac{|q_t - \xi_t|}{(1+q_t)(1+\xi_t)} \le |\alpha_t||q_t - \xi_t|.$$

Therefore, by Hölder, (C.4), and (C.6),

$$\|r_t\|_2 \le \|\alpha_t\|_4 \|q_t - \xi_t\|_4 \le CT^{-1/2}.$$

Hence, by Minkowski,

$$\left\| \frac{1}{T}\sum_{t=1}^{T} r_t \right\|_2 \le \frac{1}{T}\sum_{t=1}^{T} \|r_t\|_2 \le CT^{-1/2} \to 0.$$

Thus

$$\frac{1}{T}\sum_{t=1}^{T} r_t \to 0 \qquad \text{in } L^2. \tag{C.7}$$

Step 4: second moments of the principal terms β_t. Since $(1+\xi_t) \ge 1$,

$$|\beta_t| \le |\alpha_t|.$$

Hence, by (C.5),

$$\sup_t E[\beta_t^2] \le C. \tag{C.8}$$

We now estimate the covariances $E[\beta_t \beta_s]$ for $t \neq s$.

Step 5: leave-two-out approximation for β_t. Fix $t \neq s$, and define

$$\hat{\Sigma}_{T,t,s} := \hat{\Sigma} - \frac{1}{T}\varepsilon_t \varepsilon_t' - \frac{1}{T}\varepsilon_s \varepsilon_s', \qquad Q_{ts} := (zI_P + \hat{\Sigma}_{T,t,s})^{-1},$$

$$\xi_t^{(s)} := \frac{1}{T}\operatorname{tr}(\Sigma_P Q_{ts}), \qquad \widetilde{\beta}_t^{(s)} := \frac{\mu_P' Q_{ts} \varepsilon_t}{1 + \xi_t^{(s)}}.$$

Since Q_{ts} is independent of ε_t, and $E[\varepsilon_t] = 0$,

$$E[\widetilde{\beta}_t^{(s)} \mid Q_{ts}] = 0. \tag{C.9}$$

Likewise,

$$E[\widetilde{\beta}_s^{(t)} \mid Q_{ts}] = 0.$$

Moreover, conditional on Q_{ts}, the variables $\widetilde{\beta}_t^{(s)}$ and $\widetilde{\beta}_s^{(t)}$ only depend on ε_t and ε_s, respectively, and hence are independent. Therefore

$$E[\widetilde{\beta}_t^{(s)} \widetilde{\beta}_s^{(t)}] = 0. \tag{C.10}$$

We next compare β_t and $\widetilde{\beta}_t^{(s)}$. Since

$$Q_t^{-1} = Q_{ts}^{-1} + \frac{1}{T}\varepsilon_s \varepsilon_s',$$

Sherman–Morrison gives

$$Q_t = Q_{ts} - \frac{T^{-1} Q_{ts} \varepsilon_s \varepsilon_s' Q_{ts}}{1 + b_s}, \qquad b_s := \frac{1}{T}\varepsilon_s' Q_{ts} \varepsilon_s.$$

Thus

$$Q_t - Q_{ts} = -\frac{T^{-1} Q_{ts} \varepsilon_s \varepsilon_s' Q_{ts}}{1 + b_s}. \tag{C.11}$$

Hence

$$\begin{aligned} \beta_t - \widetilde{\beta}_t^{(s)} &= \frac{\mu_P'(Q_t - Q_{ts})\varepsilon_t}{1+\xi_t} + \mu_P' Q_{ts} \varepsilon_t \left(\frac{1}{1+\xi_t} - \frac{1}{1+\xi_t^{(s)}} \right) \\ &=: u_{ts} + v_{ts}. \end{aligned} \tag{C.12}$$

Step 5.1: bound for u_{ts}. By (C.11),

$$u_{ts} = -\frac{1}{T} \frac{\mu_P' Q_{ts} \varepsilon_s}{1+b_s} \frac{\varepsilon_s' Q_{ts} \varepsilon_t}{1+\xi_t}.$$

Since $1 + b_s \geq 1$ and $1 + \xi_t \geq 1$,

$$|u_{ts}| \leq \frac{1}{T} |\mu_P' Q_{ts} \varepsilon_s| \, |\varepsilon_s' Q_{ts} \varepsilon_t|.$$

Conditional on Q_{ts}, the vector $\Sigma_P^{1/2} Q_{ts} \mu_P$ is deterministic and independent of X_s, so by Lemma A.3,

$$\|\mu_P' Q_{ts} \varepsilon_s\|_4 \leq C.$$

Also, conditional on (Q_{ts}, ε_s), the quantity $\varepsilon_s' Q_{ts} \varepsilon_t$ is a linear form in ε_t, so Lemma A.3 gives

$$E\left[(\varepsilon_s' Q_{ts} \varepsilon_t)^4 \,\middle|\, Q_{ts}, \varepsilon_s \right] \leq C \|\Sigma_P^{1/2} Q_{ts} \varepsilon_s\|^4 \leq C \|\varepsilon_s\|^4.$$

Taking expectations,

$$\|\varepsilon_s' Q_{ts} \varepsilon_t\|_4 \le C\|\varepsilon_s\|_4 \le CP^{1/2}.$$

Therefore

$$\|u_{ts}\|_2 \le \frac{1}{T}\|\mu_P' Q_{ts}\varepsilon_s\|_4 \|\varepsilon_s' Q_{ts}\varepsilon_t\|_4 \le C\frac{P^{1/2}}{T} = O(T^{-1/2}). \tag{C.13}$$

Step 5.2: bound for v_{ts}. Using

$$\left|\frac{1}{1+\xi_t} - \frac{1}{1+\xi_t^{(s)}}\right| \le |\xi_t - \xi_t^{(s)}|,$$

we get

$$|v_{ts}| \le |\mu_P' Q_{ts}\varepsilon_t|\,|\xi_t - \xi_t^{(s)}|.$$

Now

$$\xi_t - \xi_t^{(s)} = \frac{1}{T}\operatorname{tr}\big(\Sigma_P(Q_t - Q_{ts})\big).$$

Using (C.11),

$$\xi_t - \xi_t^{(s)} = -\frac{1}{T^2}\frac{\varepsilon_s' Q_{ts}\Sigma_P Q_{ts}\varepsilon_s}{1+b_s}.$$

Hence

$$|\xi_t - \xi_t^{(s)}| \le \frac{1}{T^2}\varepsilon_s' Q_{ts}\Sigma_P Q_{ts}\varepsilon_s \le \frac{C}{T^2}\|\varepsilon_s\|^2,$$

because $\|Q_{ts}\| \le z^{-1}$ and $\|\Sigma_P\| \le C$. Therefore

$$\|\xi_t - \xi_t^{(s)}\|_4 \le \frac{C}{T^2}\|\|\varepsilon_s\|^2\|_4 = \frac{C}{T^2}\|\varepsilon_s\|_8^2 \le \frac{CP}{T^2} = O(T^{-1}).$$

Also,

$$\|\mu_P' Q_{ts}\varepsilon_t\|_4 \le C$$

as above. Hence

$$\|v_{ts}\|_2 \le \|\mu_P' Q_{ts}\varepsilon_t\|_4 \, \|\xi_t - \xi_t^{(s)}\|_4 \le CT^{-1}. \tag{C.14}$$

Combining (C.12), (C.13), and (C.14),

$$\|\beta_t - \widetilde{\beta}_t^{(s)}\|_2 \le CT^{-1/2}. \tag{C.15}$$

Step 6: covariance bound for β_t, β_s. Using (C.10),

$$E[\beta_t\beta_s] = E\big[(\beta_t - \widetilde{\beta}_t^{(s)})\beta_s\big] + E\big[\widetilde{\beta}_t^{(s)}(\beta_s - \widetilde{\beta}_s^{(t)})\big].$$

Therefore, by Cauchy–Schwarz, (C.8), and (C.15),

$$|E[\beta_t\beta_s]| \le \|\beta_t - \widetilde{\beta}_t^{(s)}\|_2\|\beta_s\|_2 + \|\widetilde{\beta}_t^{(s)}\|_2\|\beta_s - \widetilde{\beta}_s^{(t)}\|_2 \le CT^{-1/2}.$$

Thus, for $t \ne s$,

$$|\mathrm{Cov}(\beta_t, \beta_s)| \le CT^{-1/2}. \tag{C.16}$$

Step 7: the principal average $\frac{1}{T}\sum \beta_t$. We have

$$E\left[\left(\frac{1}{T}\sum_{t=1}^{T}\beta_t\right)^2\right] = \frac{1}{T^2}\sum_{t=1}^{T}E[\beta_t^2] + \frac{1}{T^2}\sum_{t\neq s}E[\beta_t\beta_s].$$

By (C.8),

$$\frac{1}{T^2}\sum_{t=1}^{T}E[\beta_t^2] \leq \frac{C}{T}.$$

By (C.16),

$$\frac{1}{T^2}\sum_{t\neq s}|E[\beta_t\beta_s]| \leq \frac{T(T-1)}{T^2}CT^{-1/2} \leq CT^{-1/2}.$$

Hence

$$\frac{1}{T}\sum_{t=1}^{T}\beta_t \to 0 \qquad \text{in } L^2. \tag{C.17}$$

Step 8: conclusion. Finally, (C.2), (C.7), and (C.17) imply

$$M_{P,T} = \frac{1}{T}\sum_{t=1}^{T}\beta_t + \frac{1}{T}\sum_{t=1}^{T}r_t \to 0 \qquad \text{in } L^2.$$

This is exactly the claim of the theorem. □

D Main RMT Asymptotic Results

We will now need the following result from Kelly et al. (2024a).

Proposition D.1 *We have*

$$\lim_{T\to\infty} \frac{1}{T} \operatorname{tr}((zI + \hat{\Sigma})^{-1}\Sigma) \;\to\; \xi(z;c) \tag{D.1}$$

almost surely and

$$\lim_{T\to\infty} \frac{1}{T} \varepsilon'_{T+1}(zI + \hat{\Sigma})^{-1}\varepsilon_{T+1} \;\to\; \xi(z;c) \tag{D.2}$$

in probability, where

$$\xi(z;c) = \frac{1 - zm(-z;c)}{c^{-1} - 1 + zm(-z;c)}. \tag{D.3}$$

Similarly,

$$\lim_{T\to\infty} \frac{1}{T} \operatorname{tr}((zI + \hat{\Sigma})^{-2}\Sigma) \;\to\; -\xi'(z;c) \tag{D.4}$$

almost surely, where

$$\begin{aligned} \xi'(z;c) &= \frac{d}{dz}\left(\frac{1 - zm(-z;c)}{c^{-1} - 1 + zm(-z;c)}\right) \\ &= \frac{d}{dz}\left(-1 + \frac{1}{1 - c + czm(-z;c)}\right) \\ &= -\frac{c\Big(m(-z;c) - zm'(-z;c)\Big)}{\Big(1 - c + czm(-z;c)\Big)^2} \end{aligned} \tag{D.5}$$

We will also define

$$\hat{Z}_*(z;c) \;=\; \frac{z}{1 - c + czm(-z;c)} \;=\; z(1 + \xi(z;c)),\; \hat{Z}'_* = 1 + \xi + z\xi'\,. \tag{D.6}$$

to be the implicit shrinkage. We follow the main text in writing $\hat{Z}_*$ for this object even

though, as defined here through the limiting Stieltjes transform $m(-z;c)$, it is deterministic; its sample counterpart $z(1+\hat{\xi}(z;c))$ of Lemma 1 differs from it by $o_p(1)$ and, by Lemma H.1, by $O_p(P^{-1})$, so the two may be used interchangeably at the $T^{-1/2}$ scale relevant for the CLT.

Lemma D.1 *Define*

$$\bar{\varepsilon} = \frac{1}{T}\sum_{t=1}^{T}\varepsilon_t.$$

Then

$$E[\bar{\varepsilon}'\bar{\varepsilon}] \;=\; \frac{\mathrm{tr}(\Sigma)}{T}.$$

In particular, under Assumption 1 the right-hand side stays bounded, being of order $P/T \to c$, and $P^{-1}E[\bar{\varepsilon}'\bar{\varepsilon}] = O(T^{-1}) \to 0$.[14]

Proof of Lemma D.1. Since $\varepsilon_t = \Sigma^{1/2}X_t$ and $E[X_tX_t'] = I$, we have

$$E[\varepsilon_t] = \Sigma^{1/2}E[X_t] = 0$$

and

$$E[\varepsilon_t\varepsilon_t'] = \Sigma^{1/2}E[X_tX_t']\Sigma^{1/2} = \Sigma.$$

Hence,

$$E[\bar{\varepsilon}'\bar{\varepsilon}] = E\left[\left(\frac{1}{T}\sum_{t=1}^{T}\varepsilon_t\right)'\left(\frac{1}{T}\sum_{s=1}^{T}\varepsilon_s\right)\right] = \frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}E[\varepsilon_t'\varepsilon_s].$$

[14] The stronger conclusion $E[\bar{\varepsilon}'\bar{\varepsilon}] \to 0$ would require $\mathrm{tr}(\Sigma)/T \to 0$, which never holds here: the $P^{-1/2}$ rescaling of Section 2 keeps $\mathrm{tr}(\Sigma) = P\bar{\lambda}$ of order P, so $\mathrm{tr}(\Sigma)/T \to c\bar{\lambda} > 0$. Every use of this lemma below carries a compensating factor of P^{-1}.

Because the vectors ε_t are independent across t and have mean zero, for $t \neq s$,

$$E[\varepsilon_t'\varepsilon_s] = E[\varepsilon_t]'E[\varepsilon_s] = 0.$$

Therefore, only the diagonal terms remain:

$$E[\bar{\varepsilon}'\bar{\varepsilon}] = \frac{1}{T^2}\sum_{t=1}^{T} E[\varepsilon_t'\varepsilon_t] = \frac{1}{T}E[\varepsilon_1'\varepsilon_1].$$

Now

$$E[\varepsilon_1'\varepsilon_1] = E[\mathrm{tr}(\varepsilon_1\varepsilon_1')] = \mathrm{tr}\big(E[\varepsilon_1\varepsilon_1']\big) = \mathrm{tr}(\Sigma).$$

Combining the above identities yields

$$E[\bar{\varepsilon}'\bar{\varepsilon}] = \frac{\mathrm{tr}(\Sigma)}{T}.$$

□

Theorem D.2 (Rubio and Mestre (2011)) *Suppose that* $R_t = \mu + \Sigma^{1/2}X_t$, *where* $X_t = (X_{i,t})$ *where* $E[X_{i,t}] = 0$, $E[X_{i,t}^2] = 1$, $E[X_{i,t}^{8+\varepsilon}] < \infty$ *are independent and identically distributed, and* Σ *is a positive semi-definite, uniformly bounded signal covariance matrix. Let* $R = (R_t)_{t=1}^T \in \mathbb{R}^{T\times P}$ *and define*

$$\begin{aligned} \bar{R} &= R'\mathbf{1}/T \\ \hat{\Sigma} &= T^{-1}R'R - \bar{R}\bar{R}' \\ \hat{\Psi} &= T^{-1}R'R\,. \end{aligned} \tag{D.7}$$

Then, in the limit as $P, T \to \infty, P/T \to c$, *for any uniformly bounded sequence of non-*

random matrices A_P, we have

$$P^{-1}z\,\mathrm{tr}(A_P(zI+\underbrace{\hat{\Sigma}}_{random})^{-1})\;-P^{-1}\hat{Z}_*\,\mathrm{tr}(A_P(\hat{Z}_*I+\underbrace{\Sigma}_{deterministic})^{-1})\;\to\;0 \tag{D.8}$$

and

$$P^{-1}z\,\mathrm{tr}(A_P(zI+\underbrace{\hat{\Psi}}_{random})^{-1})\;-P^{-1}\hat{Z}_*\,\mathrm{tr}(A_P(\hat{Z}_*I+\underbrace{\Sigma}_{deterministic})^{-1})\;\to\;0 \tag{D.9}$$

in probability.

Similarly, for any sequence of uniformly bounded vectors ν, we have

$$z\nu'(zI+\underbrace{\hat{\Sigma}}_{random})^{-1}\mu\;-\hat{Z}_*\nu'(\hat{Z}_*I+\underbrace{\Sigma}_{deterministic})^{-1}\mu\;\to\;0 \tag{D.10}$$

in probability.[15]

Proof of Theorem D.2. Let

$$\hat{\Psi}_\varepsilon\;=\;T^{-1}\varepsilon'\varepsilon\,. \tag{D.11}$$

Then, Theorem 1 in Rubio and Mestre (2011) implies that

$$\begin{aligned}&P^{-1}z\,\mathrm{tr}(A_P(zI+\hat{\Psi}_\varepsilon)^{-1})\;-P^{-1}\hat{Z}_*\,\mathrm{tr}(A_P(\hat{Z}_*I+\Sigma)^{-1})\;\to\;0\\&z\nu'(zI+\hat{\Psi}_\varepsilon)^{-1}\mu\;-\hat{Z}_*\nu'(\hat{Z}_*I+\Sigma)^{-1}\mu\;\to\;0\end{aligned} \tag{D.12}$$

Our results for R require a special treatment because R has non-zero mean μ.

[15] This result is not in general true for $\hat{\Psi}$ because the non-zero mean of R has a non-trivial impact on this quantity.

We first note that

$$\begin{aligned}\hat{\Sigma} \;&=\; T^{-1}R'R \;-\; \bar{R}\bar{R}' \\ &=\; T^{-1}(R-\bar{R})'(R-\bar{R}) \;=\; T^{-1}(\varepsilon-\bar{\varepsilon})'(\varepsilon-\bar{\varepsilon}) \;=\; \hat{\Psi}_\varepsilon \;-\; \bar{\varepsilon}\bar{\varepsilon}'\end{aligned} \tag{D.13}$$

and, by the Sherman-Morrison formula,

$$(zI+\hat{\Sigma})^{-1} \;=\; (zI+\hat{\Psi}_\varepsilon)^{-1} + \frac{(zI+\hat{\Psi}_\varepsilon)^{-1}\bar{\varepsilon}\,\bar{\varepsilon}'(zI+\hat{\Psi}_\varepsilon)^{-1}}{1-\bar{\varepsilon}'(zI+\hat{\Psi}_\varepsilon)^{-1}\bar{\varepsilon}}\,. \tag{D.14}$$

Hence,

$$\begin{aligned}P^{-1}z\,\mathrm{tr}(A_P(zI+\hat{\Sigma})^{-1}) \;&=\; P^{-1}z\,\mathrm{tr}(A_P\left((zI+\hat{\Psi}_\varepsilon)^{-1} + \frac{(zI+\hat{\Psi}_\varepsilon)^{-1}\bar{\varepsilon}\,\bar{\varepsilon}'(zI+\hat{\Psi}_\varepsilon)^{-1}}{1-\bar{\varepsilon}'(zI+\hat{\Psi}_\varepsilon)^{-1}\bar{\varepsilon}}\right)) \\ &=\; P^{-1}z\,\mathrm{tr}(A_P(zI+\hat{\Psi}_\varepsilon)^{-1}) \\ &+\; P^{-1}z\,\mathrm{tr}(A_P\frac{(zI+\hat{\Psi}_\varepsilon)^{-1}\bar{\varepsilon}\,\bar{\varepsilon}'(zI+\hat{\Psi}_\varepsilon)^{-1}}{1-\bar{\varepsilon}'(zI+\hat{\Psi}_\varepsilon)^{-1}\bar{\varepsilon}}) \\ &=\; P^{-1}z\,\mathrm{tr}(A_P(zI+\hat{\Psi}_\varepsilon)^{-1}) \\ &+\; P^{-1}z(\frac{\bar{\varepsilon}'(zI+\hat{\Psi}_\varepsilon)^{-1}A_P(zI+\hat{\Psi}_\varepsilon)^{-1}\bar{\varepsilon}}{1-\bar{\varepsilon}'(zI+\hat{\Psi}_\varepsilon)^{-1}\bar{\varepsilon}})\end{aligned} \tag{D.15}$$

By Theorem B.2 and the Sherman-Morrison formula,

$$\bar{\varepsilon}'(zI+\hat{\Psi}_\varepsilon)^{-1}\bar{\varepsilon} \;=\; \frac{\bar{\varepsilon}'(zI+\hat{\Psi}_\varepsilon-\bar{\varepsilon}\bar{\varepsilon}')^{-1}\bar{\varepsilon}}{1+\bar{\varepsilon}'(zI+\hat{\Psi}_\varepsilon-\bar{\varepsilon}\bar{\varepsilon}')^{-1}\bar{\varepsilon}} \;\to\; \frac{\xi(z;c)}{1+\xi(z;c)} \;<\; 1 \tag{D.16}$$

in probability. At the same time,

$$P^{-1}E[|\bar{\varepsilon}'(zI+\hat{\Psi}_\varepsilon)^{-1}A_P(zI+\hat{\Psi}_\varepsilon)^{-1}\bar{\varepsilon}|] \;\le\; P^{-1}\|A_P\|z^{-2}E[\bar{\varepsilon}'\bar{\varepsilon}] \;=\; P^{-1}T^{-1}\|A_P\|z^{-2}\,\mathrm{tr}(\Sigma) \tag{D.17}$$

by Lemma D.1, and the right-hand side is $O(T^{-1})$ because $\operatorname{tr}(\Sigma) = O(P)$. For $\nu'(zI+\hat{\Sigma})^{-1}\mu$, the argument is similar and reduces to bounding

$$E[|\bar{\varepsilon}'(zI+\hat{\Psi}_\varepsilon)^{-1}\nu\mu'(zI+\hat{\Psi}_\varepsilon)^{-1}\bar{\varepsilon}|] \;\leq\; (E[(\bar{\varepsilon}'(zI+\hat{\Psi}_\varepsilon)^{-1}\nu)^2]E[(\mu'(zI+\hat{\Psi}_\varepsilon)^{-1}\bar{\varepsilon})^2])^{1/2} \quad \text{(D.18)}$$

and it converges to zero in probability by Theorem C.1 (with an application of the Sherman-Morrison formula).

Thus, it remains to deal with

$$\begin{aligned} & P^{-1}z\operatorname{tr}(A_P(zI+\hat{\Psi})^{-1}) \;=\; P^{-1}z\operatorname{tr}(A_P(zI+\hat{\Sigma}+\bar{R}\bar{R}')^{-1}) \\ &= \; P^{-1}z\operatorname{tr}(A_P(zI+\hat{\Sigma})^{-1}) - P^{-1}z\operatorname{tr}(A_P\frac{(zI+\hat{\Sigma})^{-1}\bar{R}\bar{R}'(zI+\hat{\Sigma})^{-1}}{1+\bar{R}'(zI+\hat{\Sigma})^{-1}\bar{R}}) \,. \end{aligned} \quad \text{(D.19)}$$

Our goal is to show that the Sherman-Morrison correction is negligible.

We have

$$\begin{aligned} & |P^{-1}z\operatorname{tr}(A_P(zI+\hat{\Sigma})^{-1}\bar{R}\bar{R}'(zI+\hat{\Sigma})^{-1})| \\ &= \; P^{-1}z|\bar{R}'(zI+\hat{\Sigma})^{-1}A_P(zI+\hat{\Sigma})^{-1}\bar{R}| \\ &\leq \; P^{-1}z\|A_P\|z^{-2}\|\bar{R}\|^2 \,. \end{aligned} \quad \text{(D.20)}$$

Now,

$$E[\|\bar{R}\|^2] \;=\; \|\mu\|^2 \;+\; E[\bar{\varepsilon}'\bar{\varepsilon}] \;=\; \|\mu\|^2 \;+\; T^{-1}\operatorname{tr}(\Sigma) \quad \text{(D.21)}$$

is bounded, and the claim follows. □

Informally, we can rewrite this theorem as

$$\begin{aligned} z(zI+\hat{\Sigma})^{-1} \;&\approx\; \hat{Z}_*(\hat{Z}_*I+\Sigma)^{-1} \\ z(zI+\hat{\Psi})^{-1} \;&\approx\; \hat{Z}_*(\hat{Z}_*I+\Sigma)^{-1} \,, \end{aligned} \quad \text{(D.22)}$$

with an important caveat that (D.10) holds for $\hat{\Sigma}$ but not for $\hat{\Psi}$.

E A Key Second-Order Expression

The results of this section come in two forms. The diagonal case $z_1 = z_2$ holds under Assumption 1 alone. The off-diagonal case $z_1 \neq z_2$ additionally requires a random-design structure on expected returns, which we isolate here.

Condition (R). *$\mu = P^{-1/2}\Sigma_\mu^{1/2}u$, where $u = (u_i)_{i=1}^P$ has i.i.d. coordinates with $E[u_i] = 0$, $E[u_i^2] = 1$, $E[u_i^4] < \infty$, and $\Sigma_\mu = \Sigma_{\mu,P}$ is a uniformly bounded sequence of nonnegative definite matrices.*

Condition (R) is implied by Assumption 2, which is in force throughout Appendix H; every off-diagonal result below is therefore available wherever the Central Limit Theorem is applied. It is *not* imposed anywhere in the main text, and in particular the bounds of Section 4 and Proposition 4 do not rely on it: those use only the diagonal case.

Lemma E.1 (i) (Diagonal case.) *Under Assumption 1, for every $z > 0$,*

$$\mu'(zI + \hat{\Sigma})^{-1}\Sigma(zI + \hat{\Sigma})^{-1}\mu \;\approx\; (1 + \xi(z))^2 \frac{d}{dz}(zq(z))\,. \tag{E.1}$$

(ii) (Off-diagonal case.) *If in addition Condition (R) holds, then for all $z_1, z_2 > 0$ with $z_1 \neq z_2$,*

$$\mu'(z_1I + \hat{\Sigma})^{-1}\Sigma(z_2I + \hat{\Sigma})^{-1}\mu \;\approx\; (1 + \xi(z_1))(1 + \xi(z_2))\frac{z_2q(z_2) - z_1q(z_1)}{z_2 - z_1}\,. \tag{E.2}$$

In both statements

$$q(z) \;=\; \mu'(zI + \hat{\Sigma})^{-1}\mu \;\approx\; (1 + \xi)\mu'(\hat{Z}_*I + \Sigma)^{-1}\mu \tag{E.3}$$

and

$$q'(z) \;\approx\; \frac{d}{dz}((1+\xi)\mu'(\hat{Z}_* I + \Sigma)^{-1}\mu)\,. \tag{E.4}$$

The right-hand side of (E.2) *converges to that of* (E.1) *as* $z_2 \to z_1$*, so the two statements agree where both apply; the content of* (i) *is that the diagonal case needs no random-design structure.*

Proof of Lemma E.1. We have by Sherman-Morrison and Theorem C.1

$$\mu'(z_1 I + \hat{\Sigma})^{-1}\mu \;\approx\; \mu'(z_1 I + \hat{\Psi}_\varepsilon)^{-1}\mu \tag{E.5}$$

where $\hat{\Psi}_\varepsilon = T^{-1}\varepsilon'\varepsilon$, $\hat{\Sigma}_\varepsilon = \hat{\Psi}_\varepsilon - \bar{\varepsilon}\bar{\varepsilon}' = \hat{\Sigma}$. Thus

$$\begin{aligned}
q(z_1) \;&\approx\; \mu'(z_1 I + \hat{\Psi}_\varepsilon)^{-1}\mu \\
&=\; \mu'(z_1 I + \hat{\Psi}_\varepsilon)^{-1}(\hat{\Psi}_\varepsilon + z_2 I)(z_2 I + \hat{\Psi}_\varepsilon)^{-1}\mu \\
&=\; \mu'(z_1 I + \hat{\Psi}_\varepsilon)^{-1}\hat{\Psi}_\varepsilon(z_2 I + \hat{\Psi}_\varepsilon)^{-1}\mu \\
&+\; \mu'(z_1 I + \hat{\Psi}_\varepsilon)^{-1} z_2 (z_2 I + \hat{\Psi}_\varepsilon)^{-1}\mu \\
&=\; \mu'(z_1 I + \hat{\Psi}_\varepsilon)^{-1}\hat{\Psi}_\varepsilon(z_2 I + \hat{\Psi}_\varepsilon)^{-1}\mu \\
&+\; z_2(z_2 - z_1)^{-1}(\mu'(z_1 I + \hat{\Psi}_\varepsilon)^{-1}\mu - \mu'(z_2 I + \hat{\Psi}_\varepsilon)^{-1}\mu)
\end{aligned} \tag{E.6}$$

so that

$$\begin{aligned}
&\mu'(z_1 I + \hat{\Psi}_\varepsilon)^{-1}\hat{\Psi}_\varepsilon(z_2 I + \hat{\Psi}_\varepsilon)^{-1}\mu \\
&\quad=\; q(z_1) - z_2(z_2 - z_1)^{-1}(q(z_1) - q(z_2)) \;=\; \frac{z_2 q(z_2) - z_1 q(z_1)}{z_2 - z_1}
\end{aligned} \tag{E.7}$$

In particular, Theorem D.2 implies that the quantity $\mu'(z_1 I + \hat{\Psi}_\varepsilon)^{-1}\hat{\Psi}_\varepsilon(z_2 I + \hat{\Psi}_\varepsilon)^{-1}\mu$

concentrates around its expectation. Thus,

$$\mu'(z_1 I + \hat{\Psi}_\varepsilon)^{-1}\hat{\Psi}_\varepsilon(z_2 I + \hat{\Psi}_\varepsilon)^{-1}\mu - E[\mu'(z_1 I + \hat{\Psi}_\varepsilon)^{-1}\hat{\Psi}_\varepsilon(z_2 I + \hat{\Psi}_\varepsilon)^{-1}\mu] \to 0 \tag{E.8}$$

almost surely.

Thus, we need to compute

$$\begin{aligned}
&E[\mu'(z_1 I + \hat{\Psi}_\varepsilon)^{-1}\hat{\Psi}_\varepsilon(z_2 I + \hat{\Psi}_\varepsilon)^{-1}\mu] \\
&\underbrace{=}_{symmetry} E[\mu'(z_1 I + \hat{\Psi})^{-1}\varepsilon_t\varepsilon_t'(z_2 I + \hat{\Psi})^{-1}\mu] \\
&\underbrace{\approx}_{Lemma\ E.12} (1+\xi(z_1))^{-1}(1+\xi(z_2))^{-1}E[\mu'(z_1 I + \hat{\Psi}_{\varepsilon,T,t})^{-1}\varepsilon_t\varepsilon_t'(z_2 I + \hat{\Psi}_{\varepsilon,T,t})^{-1}\mu] \\
&\approx (1+\xi(z_1))^{-1}(1+\xi(z_2))^{-1}E[\mu'(z_1 I + \hat{\Psi}_\varepsilon)^{-1}\Sigma(z_2 I + \hat{\Psi}_\varepsilon)^{-1}\mu] \\
&\approx (1+\xi(z_1))^{-1}(1+\xi(z_2))^{-1}E[\mu'(z_1 I + \hat{\Sigma})^{-1}\Sigma(z_2 I + \hat{\Sigma})^{-1}\mu],
\end{aligned} \tag{E.9}$$

where the last step follows Sherman-Morrison and Theorem C.1. It remains to prove that

$$E[\mu'(z_1 I + \hat{\Psi}_\varepsilon)^{-1}\Sigma(z_2 I + \hat{\Psi}_\varepsilon)^{-1}\mu] \approx \mu'(z_1 I + \hat{\Psi}_\varepsilon)^{-1}\Sigma(z_2 I + \hat{\Psi}_\varepsilon)^{-1}\mu\,. \tag{E.10}$$

When $\mu = P^{-1/2}\Sigma_\mu^{1/2}u$, this follows directly from concentration of quadratic forms and Lemma E.9.

However, without imposing additional structure on μ, we only prove this result for $z_1 = z_2$. In this case, the proof requires an additional trick.

Lemma E.2 *We have*

$$\mu'(zI + \hat{\Psi}_\varepsilon)^{-1}\Sigma(zI + \hat{\Psi}_\varepsilon)^{-1}\mu = -\frac{d}{d\alpha}\mu'(zI + \alpha\Sigma + \hat{\Psi}_\varepsilon)^{-1}\mu|_{\alpha=0} \tag{E.11}$$

Proof of Lemma E.2. Write $A(\alpha) = zI + \alpha\Sigma + \hat{\Psi}_\varepsilon$, which is invertible for every $\alpha \geq 0$

because $A(\alpha) \succeq zI$. Differentiating the identity $A(\alpha)A(\alpha)^{-1} = I$ gives $\frac{d}{d\alpha}A(\alpha)^{-1} = -A(\alpha)^{-1}A'(\alpha)A(\alpha)^{-1} = -A(\alpha)^{-1}\Sigma A(\alpha)^{-1}$, and evaluating at $\alpha = 0$, where $A(0) = zI + \hat{\Psi}_\varepsilon$, yields the claim. □

Our goal is to show that

$$\mu'(zI + \alpha\Sigma + \hat{\Psi}_\varepsilon)^{-1}\mu - F(\alpha) \;\to\; 0 \tag{E.12}$$

for some deterministic $F(\alpha)$ and then differentiate it to imply

$$\frac{d}{d\alpha}\mu'(zI + \alpha\Sigma + \hat{\Psi}_\varepsilon)^{-1}\mu - \frac{d}{d\alpha}F(\alpha) \;\to\; 0 \tag{E.13}$$

To do this, we will need a sequence of technical lemmas.

Lemma E.3 (One-sided differentiation of a deterministic equivalent) *Fix $\bar{\alpha} > 0$. Let $F_T, F : [0, \bar{\alpha}] \to \mathbb{R}$ be such that:*

1. *for every T, the function F_T is twice continuously differentiable on $[0, \bar{\alpha}]$;*
2. *the function F is twice continuously differentiable on $[0, \bar{\alpha}]$;*
3. *for every $\alpha \in [0, \bar{\alpha}]$,*

$$F_T(\alpha) - F(\alpha) \to 0 \qquad \text{as } T \to \infty;$$

4. *there exists a constant $C < \infty$ such that*

$$\sup_T \sup_{\alpha \in [0,\bar{\alpha}]} \left| F_T''(\alpha) - F''(\alpha) \right| \leq C.$$

Then

$$F_T'(0) - F'(0) \to 0.$$

More generally, for every $\alpha_0 \in [0, \bar{\alpha})$*,*

$$F_T'(\alpha_0) - F'(\alpha_0) \to 0.$$

Proof. Define the error

$$R_T(\alpha) := F_T(\alpha) - F(\alpha).$$

Then $R_T \in C^2([0, \bar{\alpha}])$, and by assumption,

$$R_T(\alpha) \to 0 \qquad \text{for every } \alpha \in [0, \bar{\alpha}].$$

Moreover,

$$\sup_T \sup_{\alpha \in [0,\bar{\alpha}]} |R_T''(\alpha)| \le C.$$

We first prove that $R_T'(0) \to 0$. Fix $h \in (0, \bar{\alpha}]$. By Taylor's theorem, for each T there exists $\xi_{T,h} \in (0, h)$ such that

$$R_T(h) = R_T(0) + hR_T'(0) + \frac{h^2}{2} R_T''(\xi_{T,h}).$$

Hence

$$R_T'(0) = \frac{R_T(h) - R_T(0)}{h} - \frac{h}{2} R_T''(\xi_{T,h}).$$

Using the uniform bound on R_T'', we obtain

$$|R_T'(0)| \le \frac{|R_T(h)| + |R_T(0)|}{h} + \frac{Ch}{2}.$$

Now let $T \to \infty$. Since $R_T(h) \to 0$ and $R_T(0) \to 0$ for each fixed h,

$$\limsup_{T\to\infty} |R_T'(0)| \le \frac{Ch}{2}.$$

Because this holds for every $h \in (0, \bar{\alpha}]$, letting $h \downarrow 0$ yields

$$\lim_{T\to\infty} R_T'(0) = 0.$$

That is,

$$F_T'(0) - F'(0) \to 0.$$

The same argument applies at any $\alpha_0 \in [0, \bar{\alpha})$: for $h > 0$ such that $\alpha_0 + h \le \bar{\alpha}$, Taylor's theorem gives

$$R_T(\alpha_0 + h) = R_T(\alpha_0) + hR_T'(\alpha_0) + \frac{h^2}{2} R_T''(\xi_{T,h}),$$

for some $\xi_{T,h} \in (\alpha_0, \alpha_0 + h)$, and therefore

$$|R_T'(\alpha_0)| \le \frac{|R_T(\alpha_0 + h)| + |R_T(\alpha_0)|}{h} + \frac{Ch}{2}.$$

Letting $T \to \infty$ and then $h \downarrow 0$ yields

$$R_T'(\alpha_0) \to 0.$$

This proves the claim. □

The deterministic equivalent appearing in Lemma E.5 below is available in closed form, which is what allows us to bound its second derivative directly rather than through an abstract smoothness hypothesis. Both lemmas concern

$$F_T(\alpha) \;:=\; \mu'\big(zI + \alpha\Sigma + \Sigma^{1/2}XX'\Sigma^{1/2}/T\big)^{-1}\mu, \qquad \alpha \in [0, \bar{\alpha}], \tag{E.14}$$

and its *transformed* form: writing $Q(\alpha) := (zI + \alpha\Sigma)^{1/2}$,

$$\tilde{\mu}(\alpha) := Q(\alpha)^{-1}\mu = (zI+\alpha\Sigma)^{-1/2}\mu, \qquad \tilde{\Sigma}(\alpha) := Q(\alpha)^{-1}\Sigma Q(\alpha)^{-1} = \Sigma(zI+\alpha\Sigma)^{-1}, \tag{E.15}$$

one has $F_T(\alpha) = \tilde{\mu}(\alpha)'\big(I + \tilde{\Sigma}(\alpha)^{1/2}XX'\tilde{\Sigma}(\alpha)^{1/2}/T\big)^{-1}\tilde{\mu}(\alpha)$, which is the standard model at ridge parameter one. We write $F(\alpha)$ for its deterministic equivalent. Throughout, let

$$\varphi(s) \;:=\; \frac{1}{T}\operatorname{tr}\big[\Sigma(zI + s\Sigma)^{-1}\big] \;=\; \frac{1}{T}\sum_{i=1}^{P}\frac{\lambda_i}{z + s\lambda_i}\,, \qquad s \geq 0\,, \tag{E.16}$$

where $\lambda_1, \ldots, \lambda_P$ are the eigenvalues of Σ, and set

$$M := \sup_P \|\Sigma_P\| < \infty, \qquad L := \sup_P \|\mu_P\| < \infty, \qquad \bar{c} := \sup_{P,T} P/T < \infty, \qquad K_0 := \frac{\bar{c}M}{z}\,. \tag{E.17}$$

The quantity φ depends on (P, T); all bounds below are uniform in (P, T).

Lemma E.4 (Closed form for the deterministic equivalent) *Fix $z > 0$ and $\bar{\alpha} > 0$. For every $\alpha \in [0, \bar{\alpha}]$ the scalar equation*

$$s \;=\; \alpha \;+\; \frac{1}{1 + \varphi(s)} \tag{E.18}$$

has a unique solution $s = g(\alpha) \in (0, \bar{\alpha}+1]$, *and the deterministic equivalent of Lemma E.5 is*

$$F(\alpha) \;=\; \mu'\big(zI + g(\alpha)\Sigma\big)^{-1}\mu\,. \tag{E.19}$$

Moreover g *is real-analytic on* $[0, \bar{\alpha}]$, *and*

$$\frac{1}{1+K_0} \;\le\; g(\alpha) \;\le\; \bar{\alpha}+1, \qquad 1 \;\le\; g'(\alpha) \;\le\; 1+K_0, \qquad |g''(\alpha)| \;\le\; C_g\,, \tag{E.20}$$

with $C_g < \infty$ *depending only on* $z, M, \bar{c}$.

Proof of Lemma E.4. Step 1: the closed form. Apply Theorem D.2 to the transformed model of Lemma E.5, that is, with population covariance $\tilde{\Sigma}(\alpha) = \Sigma(zI+\alpha\Sigma)^{-1}$, mean vector $\tilde{\mu}(\alpha) = (zI+\alpha\Sigma)^{-1/2}\mu$, and ridge parameter equal to one. Writing $\kappa := 1+\xi_\alpha$ for the corresponding value of $1+\xi$ (so that the effective shrinkage is $\hat{Z}_* = \kappa$), the theorem gives

$$F(\alpha) \;=\; \kappa\,\tilde{\mu}(\alpha)'\big(\kappa I + \tilde{\Sigma}(\alpha)\big)^{-1}\tilde{\mu}(\alpha)\,. \tag{E.21}$$

Since Σ and $(zI+\alpha\Sigma)$ commute,

$$\big(\kappa I + \tilde{\Sigma}(\alpha)\big)^{-1} \;=\; (zI+\alpha\Sigma)\big(\kappa(zI+\alpha\Sigma)+\Sigma\big)^{-1}, \tag{E.22}$$

so the two factors $(zI+\alpha\Sigma)^{-1/2}$ cancel and

$$F(\alpha) \;=\; \kappa\,\mu'\big(\kappa z I + (\kappa\alpha+1)\Sigma\big)^{-1}\mu \;=\; \mu'\Big(zI + \Big(\alpha+\tfrac{1}{\kappa}\Big)\Sigma\Big)^{-1}\mu\,, \tag{E.23}$$

which is (E.19) with $g(\alpha) = \alpha + \kappa^{-1}$. It remains to identify κ. By Proposition D.1 applied to the transformed model, $\xi_\alpha = \kappa\,T^{-1}\,\mathrm{tr}[\tilde{\Sigma}(\kappa I+\tilde{\Sigma})^{-1}]$, and the same commutation gives $\tilde{\Sigma}(\kappa I+\tilde{\Sigma})^{-1} = \Sigma(\kappa z I + (\kappa\alpha+1)\Sigma)^{-1} = \kappa^{-1}\Sigma(zI+g(\alpha)\Sigma)^{-1}$. Hence $\xi_\alpha = \varphi(g(\alpha))$, and

$g(\alpha) = \alpha + (1+\varphi(g(\alpha)))^{-1}$ is exactly (E.18). Specializing to $\alpha = 0$ recovers $g(0) = (1+\xi)^{-1}$ and $z/g(0) = \hat{Z}_*(z)$, consistent with (D.6).

Step 2: existence and uniqueness. Each summand of φ is positive and strictly decreasing in s, so $\varphi > 0$ is strictly decreasing, and

$$0 < \varphi(s) \;\le\; \frac{1}{T}\sum_{i=1}^{P}\frac{\lambda_i}{z} \;\le\; \frac{P}{T}\cdot\frac{M}{z} \;\le\; K_0 \qquad \text{for all } s \ge 0. \tag{E.24}$$

Put $\psi(\delta) := \varphi\big(\alpha + (1+\delta)^{-1}\big)$; a solution of (E.18) corresponds to a fixed point $\delta = \psi(\delta)$, via $s = \alpha + (1+\delta)^{-1}$. As a composition of two decreasing maps, ψ is increasing, and by (E.24) it maps $[0, K_0]$ into itself; since it is continuous, $h(\delta) := \delta - \psi(\delta)$ satisfies $h(0) < 0 \le h(K_0)$, so a fixed point exists in $(0, K_0]$.

For uniqueness, note first the elementary bound

$$|\varphi'(s)| \;=\; \frac{1}{T}\sum_i \Big(\frac{\lambda_i}{z+s\lambda_i}\Big)^2 \;\le\; \frac{1}{s}\cdot\frac{1}{T}\sum_i \frac{\lambda_i}{z+s\lambda_i} \;=\; \frac{\varphi(s)}{s}, \qquad s > 0, \tag{E.25}$$

where we used $s\lambda_i/(z+s\lambda_i) < 1$. At any fixed point we have $s = \alpha + (1+\varphi(s))^{-1} \ge (1+\varphi(s))^{-1}$ because $\alpha \ge 0$, so

$$\psi'(\delta) \;=\; \frac{|\varphi'(s)|}{(1+\delta)^2} \;=\; \frac{|\varphi'(s)|}{(1+\varphi(s))^2} \;\le\; \frac{\varphi(s)}{s\,(1+\varphi(s))^2} \;\le\; \frac{\varphi(s)}{1+\varphi(s)} \;<\; 1\,. \tag{E.26}$$

Thus $h' > 0$ at every fixed point. If $\delta_1 < \delta_2$ were two fixed points with δ_1 the smallest and δ_2 the next, then $h > 0$ immediately to the right of δ_1 and $h(\delta_2) = 0$ would force $h'(\delta_2) \le 0$, a contradiction. Hence the fixed point is unique, and $g(\alpha) = \alpha + (1+\delta)^{-1} \in (0, \bar{\alpha}+1]$. Combining (E.24) with $g(\alpha) \ge (1+\varphi)^{-1}$ gives the lower bound in (E.20).

Step 3: smoothness and derivative bounds. Let $G(s) := (1+\varphi(s))^{-1}$ and $\Phi(s, \alpha) := s - \alpha - G(s)$, so that (E.18) reads $\Phi(g(\alpha), \alpha) = 0$. Since $z > 0$ and $\lambda_i \ge 0$, φ is real-analytic on a neighbourhood of $[0, \infty)$, hence so is G; and $G'(s) = |\varphi'(s)|/(1+\varphi(s))^2$, which by (E.26)

satisfies $0 \le G'(g(\alpha)) \le \varphi/(1+\varphi)$ at the solution. Therefore

$$\partial_s \Phi\big(g(\alpha), \alpha\big) \;=\; 1 - G'(g(\alpha)) \;\in\; \Big[\tfrac{1}{1+K_0},\, 1\Big]\,, \tag{E.27}$$

using (E.24). The implicit function theorem then makes g real-analytic with

$$g'(\alpha) \;=\; \frac{1}{1 - G'(g(\alpha))} \;\in\; [1,\, 1+K_0]\,, \qquad g''(\alpha) \;=\; G''(g(\alpha))\,\big(g'(\alpha)\big)^3\,. \tag{E.28}$$

Finally, for $s \ge 0$,

$$|\varphi'(s)| \le \frac{1}{T}\sum_i \frac{\lambda_i^2}{z^2} \le \frac{\bar{c}M^2}{z^2}, \qquad |\varphi''(s)| = \frac{2}{T}\sum_i \frac{\lambda_i^3}{(z+s\lambda_i)^3} \le \frac{2\bar{c}M^3}{z^3}, \tag{E.29}$$

and $1 + \varphi \ge 1$, so

$$|G''| \;=\; \Big| -\frac{\varphi''}{(1+\varphi)^2} + \frac{2(\varphi')^2}{(1+\varphi)^3} \Big| \;\le\; \frac{2\bar{c}M^3}{z^3} + \frac{2\bar{c}^2M^4}{z^4}\,, \tag{E.30}$$

whence $|g''| \le C_g := \big(2\bar{c}M^3 z^{-3} + 2\bar{c}^2M^4 z^{-4}\big)(1+K_0)^3$. All bounds depend only on $z, M, \bar{c}$. □

Lemma E.5 *Fix $z > 0$ and $\bar{\alpha} > 0$, and define*

$$F_T(\alpha) := \mu'\big(zI + \alpha\Sigma + \Sigma^{1/2}XX'\Sigma^{1/2}/T\big)^{-1}\mu, \qquad \alpha \in [0, \bar{\alpha}].$$

Let

$$Q(\alpha) := (zI + \alpha\Sigma)^{1/2}, \qquad \tilde{\mu}(\alpha) := Q(\alpha)^{-1}\mu, \qquad \tilde{\Sigma}(\alpha) := Q(\alpha)^{-1}\Sigma Q(\alpha)^{-1} = \Sigma(zI+\alpha\Sigma)^{-1}.$$

Then

$$F_T(\alpha) = \tilde{\mu}(\alpha)'\Big(I + \tilde{\Sigma}(\alpha)^{1/2} X X' \tilde{\Sigma}(\alpha)^{1/2}/T\Big)^{-1} \tilde{\mu}(\alpha).$$

Let

$$F(\alpha) = m\big(\tilde{\mu}(\alpha), \tilde{\Sigma}(\alpha)\big),$$

where $m(\tilde{\mu}, \tilde{\Sigma})$ is the deterministic-equivalent functional for the standard model

$$\tilde{\mu}'\Big(I + \tilde{\Sigma}^{1/2} X X' \tilde{\Sigma}^{1/2}/T\Big)^{-1} \tilde{\mu}.$$

(Theorem D.2). Assume

$$\sup_P \|\Sigma\| < \infty, \qquad \sup_P \|\mu\| < \infty.$$

Then there exists a constant $C < \infty$ such that

$$\sup_T \sup_{\alpha \in [0, \bar{\alpha}]} |R_T''(\alpha)| \leq C, \qquad R_T(\alpha) := F_T(\alpha) - F(\alpha).$$

Proof. We bound F_T'' and F'' separately.

Step 1: bound on F_T''. Let

$$A_T(\alpha) := zI + \alpha\Sigma + \Sigma^{1/2} X X' \Sigma^{1/2}/T.$$

Since $A_T(\alpha) \succeq zI$,

$$\|A_T(\alpha)^{-1}\| \leq z^{-1}.$$

Differentiating twice gives

$$F_T'(\alpha) = -\mu' A_T(\alpha)^{-1}\Sigma A_T(\alpha)^{-1}\mu,$$

$$F_T''(\alpha) = 2\mu' A_T(\alpha)^{-1}\Sigma A_T(\alpha)^{-1}\Sigma A_T(\alpha)^{-1}\mu.$$

Hence

$$|F_T''(\alpha)| \le 2\|\mu\|^2\|A_T(\alpha)^{-1}\|^3\|\Sigma\|^2 \le 2z^{-3}\|\mu\|^2\|\Sigma\|^2.$$

Therefore

$$\sup_T \sup_{\alpha\in[0,\bar{\alpha}]} |F_T''(\alpha)| \le C_1$$

for some finite constant C_1.

Step 2: bounds on $\tilde{\mu}(\alpha)$ and $\tilde{\Sigma}(\alpha)$. Since $Q(\alpha)^2 = zI + \alpha\Sigma \succeq zI$,

$$\|Q(\alpha)^{-1}\| \le z^{-1/2}.$$

Therefore

$$\|\tilde{\mu}(\alpha)\| \le z^{-1/2}\|\mu\|, \qquad \|\tilde{\Sigma}(\alpha)\| \le z^{-1}\|\Sigma\|.$$

Because $Q(\alpha)$ is a function of Σ, it commutes with Σ, so

$$\tilde{\mu}(\alpha) = (zI + \alpha\Sigma)^{-1/2}\mu, \qquad \tilde{\Sigma}(\alpha) = \Sigma(zI + \alpha\Sigma)^{-1}.$$

Thus

$$\tilde{\mu}'(\alpha) = -\frac{1}{2}(zI + \alpha\Sigma)^{-3/2}\Sigma\mu,$$

$$\tilde{\mu}''(\alpha) = \frac{3}{4}(zI + \alpha\Sigma)^{-5/2}\Sigma^2\mu,$$

and

$$\tilde{\Sigma}'(\alpha) = -\Sigma^2(zI + \alpha\Sigma)^{-2}, \qquad \tilde{\Sigma}''(\alpha) = 2\Sigma^3(zI + \alpha\Sigma)^{-3}.$$

Hence

$$\sup_{\alpha\in[0,\bar{\alpha}]} \|\tilde{\mu}(\alpha)\| + \sup_{\alpha\in[0,\bar{\alpha}]} \|\tilde{\mu}'(\alpha)\| + \sup_{\alpha\in[0,\bar{\alpha}]} \|\tilde{\mu}''(\alpha)\| \le C_\mu,$$

and

$$\sup_{\alpha\in[0,\bar{\alpha}]} \|\tilde{\Sigma}(\alpha)\| + \sup_{\alpha\in[0,\bar{\alpha}]} \|\tilde{\Sigma}'(\alpha)\| + \sup_{\alpha\in[0,\bar{\alpha}]} \|\tilde{\Sigma}''(\alpha)\| \le C_\Sigma,$$

for finite constants C_μ, C_Σ depending only on $z, \bar{\alpha}, \sup\|\Sigma\|, \sup\|\mu\|$.

Step 3: bound on F''. By Lemma E.4, $F = f \circ g$ with

$$f(s) := \mu'(zI + s\Sigma)^{-1}\mu,$$

and g real-analytic on $[0, \bar{\alpha}]$ with $1 \le g' \le 1 + K_0$ and $|g''| \le C_g$. Differentiating f exactly as in Step 1, for every $s \ge 0$

$$f'(s) = -\mu'(zI+s\Sigma)^{-1}\Sigma(zI+s\Sigma)^{-1}\mu, \qquad f''(s) = 2\mu'(zI+s\Sigma)^{-1}\Sigma(zI+s\Sigma)^{-1}\Sigma(zI+s\Sigma)^{-1}\mu,$$

so that, using $\|(zI + s\Sigma)^{-1}\| \le z^{-1}$ for $s \ge 0$,

$$|f'(s)| \le \frac{L^2 M}{z^2}, \qquad |f''(s)| \le \frac{2L^2 M^2}{z^3}.$$

By the chain rule, $F''(\alpha) = f''(g(\alpha))\, g'(\alpha)^2 + f'(g(\alpha))\, g''(\alpha)$, whence

$$\sup_{\alpha \in [0,\bar{\alpha}]} |F''(\alpha)| \;\le\; \frac{2L^2 M^2}{z^3}(1 + K_0)^2 + \frac{L^2 M}{z^2}\, C_g \;=:\; C_2\,,$$

a finite constant depending only on $z, \bar{\alpha}, M, L, \bar{c}$. In particular F is twice continuously differentiable on $[0, \bar{\alpha}]$, as required by hypothesis (2) of Lemma E.3. No smoothness assumption on the deterministic-equivalent functional is needed: the closed form (E.19) reduces the question to the scalar fixed point (E.18).

Step 4: conclusion. Since

$$R_T''(\alpha) = F_T''(\alpha) - F''(\alpha),$$

we get

$$|R_T''(\alpha)| \le |F_T''(\alpha)| + |F''(\alpha)| \le C_1 + C_2.$$

Therefore

$$\sup_T \sup_{\alpha \in [0,\bar{\alpha}]} |R_T''(\alpha)| \le C_1 + C_2 < \infty.$$

□

Conclusion of the proof of Lemma E.1. For each fixed $\alpha \in [0, \bar{\alpha}]$, Theorem D.2 applied to the pair $(\tilde{\mu}(\alpha), \tilde{\Sigma}(\alpha))$ of Lemma E.5 gives $F_T(\alpha) - F(\alpha) \to 0$, which is hypothesis (3) of Lemma E.3; Lemma E.5 supplies hypothesis (4). Lemma E.3 therefore yields $F_T'(0) - F'(0) \to$

0, and Lemma E.2 identifies $-F_T'(0)$ with $\mu'(zI+\hat{\Psi}_\varepsilon)^{-1}\Sigma(zI+\hat{\Psi}_\varepsilon)^{-1}\mu$. Hence that quadratic form converges to the deterministic limit $-F'(0)$, which establishes (E.10) for $z_1 = z_2 = z$ and, with the display preceding it, completes the proof of part (i) of Lemma E.1. Part (ii), the off-diagonal case $z_1 \neq z_2$, was established earlier in this proof under Condition (R). □

Lemma F.1 below implies

Lemma E.6 *We have*

$$\mu'(zI+\hat{\Sigma})^{-1}\mu \;\approx\; \hat{\Theta}(z) \;-\; \hat{\xi}(z) \tag{E.31}$$

Lemma E.7 (i) (Diagonal case.) *Under Assumption 1, for every $z > 0$,*

$$\mu'(zI+\hat{\Sigma})^{-1}\Sigma(zI+\hat{\Sigma})^{-1}\mu \;\approx\; (1+\hat{\xi}(z))^2(\hat{\Theta}(z)-\hat{\xi}(z)+z(\hat{\Theta}'(z)-\hat{\xi}'(z))) \tag{E.32}$$

(ii) (Off-diagonal case.) *If in addition Condition* (R) *holds, then*

$$\mu'(z_1I+\hat{\Sigma})^{-1}\Sigma(z_2I+\hat{\Sigma})^{-1}\mu \;\approx\; (1+\hat{\xi}(z_1))(1+\hat{\xi}(z_2))\frac{z_2(\hat{\Theta}(z_2)-\hat{\xi}(z_2))-z_1(\hat{\Theta}(z_1)-\hat{\xi}(z_1))}{z_2-z_1} \tag{E.33}$$

for any $z_1, z_2 \in \mathbb{C}\setminus\mathbb{R}_-$. Both statements are pivotal: the right-hand sides involve only $\hat{\Theta}, \hat{\xi}$ and their derivatives, all computable from the data.

Proof of Lemma E.7. Part (i) follows from Lemma E.1(i) and Lemma E.6, and part (ii) from Lemma E.1(ii) and Lemma E.6, in each case by substituting $q(z) \approx \hat{\Theta}(z) - \hat{\xi}(z)$. □

E.1 Technical Inequalities

We will also need

Lemma E.8 *Let $q \geq 1$ and let $f_N(z)$ be a sequence of random functions such that f_N is analytic in the right half-plane $\mathbb{C}_+ = \{z : \Re z > 0\}$ and*

$$E[|f_N(z)|^q] \;<\; K(q, \Re z) N^{-q/2}, \tag{E.34}$$

where $\Re z$ is the real part of z, for all z with positive real part, and where $K(q, \cdot)$ is bounded on compact subsets of $(0, \infty)$. Then,

$$E[|f_N^{(n)}(z)|^q] \;\leq\; \tilde{K}(q, \Re z) N^{-q/2} \tag{E.35}$$

for some other constant $\tilde{K}$.

Proof of Lemma E.8. Fix $z_0 \in \mathbb{C}_+$ and let Γ be the circle of radius $R = 0.5\,\Re z_0$ centered at z_0, so that $\Re z \geq 0.5\,\Re z_0$ for every $z \in \Gamma$. By the Cauchy integral formula,

$$f_N^{(n)}(z_0) = \frac{n!}{2\pi i} \int_\Gamma \frac{f_N(z)}{(z - z_0)^{n+1}}\, dz\,. \tag{E.36}$$

Parametrizing Γ by $z = z_0 + Re^{i\theta},\ \theta \in [0, 2\pi]$, so that $dz = iRe^{i\theta}\, d\theta$, we obtain

$$|f_N^{(n)}(z_0)| \;\leq\; \frac{n!}{2\pi} \int_0^{2\pi} \frac{|f_N(z_0 + Re^{i\theta})|}{R^{n+1}}\, R\, d\theta \;=\; \frac{n!}{R^n}\, \frac{1}{2\pi} \int_0^{2\pi} |f_N(z_0 + Re^{i\theta})|\, d\theta\,. \tag{E.37}$$

Raising to the power $q \geq 1$ and applying Jensen's inequality to the probability measure $d\theta/(2\pi)$,

$$|f_N^{(n)}(z_0)|^q \;\leq\; \left(\frac{n!}{R^n}\right)^q \frac{1}{2\pi} \int_0^{2\pi} |f_N(z_0 + Re^{i\theta})|^q\, d\theta\,, \tag{E.38}$$

Hence,

$$E[|f_N^{(n)}(z_0)|^q] \;\leq\; \left(\frac{n!}{R^n}\right)^q \frac{1}{2\pi} \int_0^{2\pi} E[|f_N(z_0 + Re^{i\theta})|^q]\, d\theta \;\leq\; \tilde{K}(q, \Re z_0) N^{-q/2} \tag{E.39}$$

with $\tilde{K}(q, \Re z_0) = (n!/R^n)^q \sup_{z\in\Gamma} K(q, \Re z)$, which is finite because $K(q,\cdot)$ is bounded on the compact set $\{\Re z : z \in \Gamma\} \subset (0,\infty)$. The proof of Lemma E.8 is complete. □

The following results follow by the standard Martingale arguments from the results in Bai and Silverstein (2010).

Lemma E.9 (Law of Large Numbers For Resolvents) *Let $Q_1(P), \cdots, Q_K(P)$, $K \in \mathbb{N}$, be uniformly bounded sequences of* **non-random** *matrices. Suppose also that $R_t \in \mathbb{R}^P$ are independent over time and let $z_1, \cdots, z_K \in \mathbb{C} \setminus \mathbb{R}_-$. Define*

$$q_T(z;P) \;=\; P^{-1}\,\mathrm{tr}\left(Q_1(P)(z_1 I+\hat{\Sigma})^{-1}Q_2(P)(z_2 I+\hat{\Sigma})^{-1}\cdots Q_K(P)(z_K I+\hat{\Sigma})^{-1}\right). \quad \text{(E.40)}$$

Then,

$$q_T(z;P) - E[q_T(z;P)] \;\to\; 0 \quad \text{(E.41)}$$

almost surely as $T, P \to \infty$, with $P/T > \varepsilon$ for some fixed $\varepsilon > 0$.

Furthermore, by Lemma E.8,

$$D_z^k q_T(z;P) - E[D_z^k q_T(z;P)] \;\to\; 0 \quad \text{(E.42)}$$

where D_z^k is any mixed derivative of any order.

We will also need

Lemma E.10 (Sherman-Morrison formula) *We have*

$$(M+uu')^{-1} = M^{-1} - \frac{M^{-1}uu'M^{-1}}{1+u'M^{-1}u} \quad \text{(E.43)}$$

We thus get

Lemma E.11 *Let $\hat{\Psi} = T^{-1}R'R$. We have*

$$(zI + \hat{\Psi})^{-1} = (zI + \hat{\Psi}_{T,t})^{-1} - \frac{(zI + \hat{\Psi}_{T,t})^{-1}R_t R_t'(zI + \hat{\Psi}_{T,t})^{-1}}{T(1 + \widehat{\xi_t})} \tag{E.44}$$

and, hence,

$$(zI + \hat{\Psi})^{-1}R_t \;=\; (zI + \hat{\Psi}_{T,t})^{-1}R_t \frac{1}{1 + \hat{\xi}_t} \tag{E.45}$$

where we have defined

$$\begin{aligned} \hat{\Psi}_{T,t} \;&=\; \sum_{\tau \neq t} T^{-1}R_\tau R_\tau' \\ \hat{\xi}_t(z) \;&=\; T^{-1}R_t'(zI + \hat{\Psi}_{T,t})^{-1}R_t\,, \end{aligned} \tag{E.46}$$

E.2 Convergence of Expectations

Roadmap: In many of the proofs below, we would like to use the following intuitive argument: If $Y_P - E[Y_P] \to 0$, then

$$E[Z_P Y_P] \;\approx E[Z_P E[Y_P]] \;=\; E[Z_P]E[Y_P]. \tag{E.47}$$

The next two lemmas make this intuition rigorous.

Lemma E.12 *Suppose that $Y = Y_P$ is such that $Y_P - E[Y_P] \to 0$ in L_2, and $Y_P \geq 0$. Then,*

$$E[X/(1 + Y)] \;-\; E[X]/(1 + E[Y]) \;\to\; 0 \tag{E.48}$$

if $E[|X|^2]$ is uniformly bounded.

Proof of Lemma E.12. We have

$$\begin{aligned} |E[X/(1+Y)] \; - E[X]/(1+E[Y])| \; &= \; |E[\frac{X(E[Y]-Y)}{(1+Y)(1+E[Y])}]| \\ \leq \; E[\frac{|X||(Y-E[Y])|}{(1+Y)(1+E[Y])}] \; &\leq \; E[|X||Y-E[Y]|] \\ \leq \; E[X^2]^{1/2} \operatorname{Var}[Y]^{1/2} \, . \end{aligned} \tag{E.49}$$

□

Lemma E.13 *Suppose that* $Y(j) \geq 0, j = 1, \cdots, J$, *are such that* $\operatorname{Var}[Y(j)] \; \leq \; KP^{-1}$ *and* $E[Y_j]$ *are bounded. Then,*

$$|E[\frac{X}{(1+Y_1)\cdots(1+Y_J)}] \; - \; \frac{E[X]}{(1+E[Y_1])\cdots(1+E[Y_J])}| \; \leq \; K(J)P^{-1/2}E[X^2]^{1/2} \tag{E.50}$$

for some $K(J) > 0$.

Proof of Lemma E.13. Let Y be such that

$$1 + Y_J \; = \; (1+Y_1)\cdots(1+Y_J) \, . \tag{E.51}$$

Let also

$$1 + Z_J \; = \; (1+E[Y_1])\cdots(1+E[Y_J]) \, . \tag{E.52}$$

Then,

$$|E[\frac{X}{(1+Y_1)\cdots(1+Y_J)}] \; - \; \frac{E[X]}{1+Z}| \; = \; |E[\frac{Y-Z}{(1+Y)(1+Z)}X]| \; \leq \; E[(Y-Z)^2/(1+Y)^2]^{1/2}E[X^2]^{1/2} \, . \tag{E.53}$$

We now prove by induction in J that $E[(Y - Z_J)^2/(1+Y_J)^2] \leq \kappa(J)P^{-1}$. We have

$$(1+Y_1)\cdots(1+Y_J) \;=\; (1+E[Y_1])(1+Y_2)\cdots(1+Y_J) + (Y_1-E[Y_1])(1+Y_2)\cdots(1+Y_J)\,. \quad \text{(E.54)}$$

Thus, defining $Z_J = (1+E[Y_1])\cdots(1+E[Y_J]) - 1,\ Z_{J-1} = (1+E[Y_2])\cdots(1+E[Y_J]) - 1$, we get

$$Y_J - Z_J \;=\; (1+E[Y_1])(Y_{J-1} - Z_{J-1}) \;+\; (Y_1 - E[Y_1])Y_{J-1}\,. \quad \text{(E.55)}$$

Hence, since $1 \leq Y_{J-1} \leq Y_J$, we get

$$\begin{aligned} E[(Y_J - Z_J)^2/(1+Y)^2] \;&\leq\; 2\,E[(1+E[Y_1])^2(Y_{J-1} - Z_{J-1})^2/(1+Y_J)^2] \\ &+\; E[(Y_1 - E[Y_1])^2 Y_{J-1}^2/(1+Y_J)^2] \\ &\leq\; 2(1+E[Y_1])^2\kappa(J-1)P^{-1} \;+\; E[(Y_1 - E[Y_1])^2] \;\leq\; (2(1+E[Y_1])^2\kappa(J-1) + K)P^{-1} \end{aligned} \quad \text{(E.56)}$$

and thus we have proved the induction step.

□

F Proof of Theorem 2 and other results from Section 3

Proof of Proposition 1. We have

$$x^{-1} \;=\; \sum_{j=0}^{\infty} Z_*^j (Z_* + x)^{-j-1} \quad \text{(F.1)}$$

and, hence, the same is true for matrices:

$$\Sigma^{-1} \;=\; \sum_{j=0}^{\infty} Z_*^j (Z_* I + \Sigma)^{-j-1} \quad \text{(F.2)}$$

and we get the main claim. The fact that $\mathcal{B}_j$ is monotone decreasing in j follows because $Z_*^j(Z_*I+\Sigma)^{-j-1}$ is monotone decreasing in the sense of positive semi-definite order. □

Lemma 1 follows directly from Lemma F.1 below.

Lemma F.1 *We have*

$$\hat{\Theta}(z) \;=\; \hat{\mathcal{B}}(z) \;+\; 2\hat{\mathcal{I}}(z) \;+\; \hat{\mathcal{V}}(z) \tag{F.3}$$

where

$$\begin{aligned}
\hat{\mathcal{B}}(z) &= \mu'(zI+\hat{\Sigma})^{-1}\mu \\
\hat{\mathcal{I}}(z) &= \mu'(zI+\hat{\Sigma})^{-1}\bar{\varepsilon} \\
\hat{\mathcal{V}}(z) &= \bar{\varepsilon}'(zI+\hat{\Sigma})^{-1}\bar{\varepsilon}\,.
\end{aligned} \tag{F.4}$$

We also have $\hat{\mathcal{V}}(z) \approx \hat{\xi}(z;c)$ *and*

$$\hat{\Theta}(z) - \hat{\xi}(z;c) \;\approx\; (1+\hat{\xi}(z;c))\,\mu'(\hat{Z}_*(z)I+\Sigma)^{-1}\mu\,. \tag{F.5}$$

Proof of Lemma F.1. The fact that $\hat{\mathcal{V}}(z) \approx \hat{\xi}(z)$ follows Theorem B.2. The main result that $\hat{\mathcal{I}}(z) \approx 0$ follows from Theorem C.1. The fact that $\hat{\mathcal{B}}(z) \approx (\hat{Z}_*/z)\mu'(\hat{Z}_*I+\Sigma)^{-1}\mu$ follows from Theorem D.2. □

F.1 Higher-Order GRS Statistics

Lemma F.2 (Faà di Bruno's formula) *We have*

$$\frac{d^j}{dz^j} f(g(z)) = \sum_{k=1}^{j} f^{(k)}(g(z)) B_{j,k}(g'(z), g''(z), \cdots, g^{(j-k+1)}(z)) \tag{F.6}$$

where $B_{j,k}$ are Bell polynomials given by

$$
\begin{aligned}
&B_{j,k}(x_1, x_2, \ldots, x_{j-k+1}) = \\
&\sum_{\substack{j_1+j_2+\cdots+j_{j-k+1}=k \\ 1j_1+2j_2+\cdots+(j-k+1)j_{j-k+1}=j}} \frac{j!}{j_1!j_2!\cdots j_{j-k+1}!} \left(\frac{x_1}{1!}\right)^{j_1} \left(\frac{x_2}{2!}\right)^{j_2} \cdots \left(\frac{x_{j-k+1}}{(j-k+1)!}\right)^{j_{j-k+1}} \\
&= j! \sum_{\substack{j_1+j_2+\cdots+j_{j-k+1}=k \\ 1j_1+2j_2+\cdots+(j-k+1)j_{j-k+1}=j}} \prod_{i=1}^{j-k+1} \frac{x_i^{j_i}}{(i!)^{j_i}\, j_i!}.
\end{aligned}
\tag{F.7}
$$

Lemma F.3 *Let*

$$
\tilde{\mathcal{B}}_0(z) \;=\; \mu'(\hat{Z}_*(z)I + \Sigma)^{-1}\mu \tag{F.8}
$$

and define

$$
\tilde{\mathcal{B}}_j(z) \;=\; \mu'(\hat{Z}_*(z)I + \Sigma)^{-j-1}\mu \tag{F.9}
$$

Then,

$$
\tilde{\mathcal{B}}_j(z) = \frac{1}{j!(-\hat{Z}'_*(z))^j} \left(\frac{d^j}{dz^j}\left(\tilde{\mathcal{B}}_0(z)\right) - \sum_{k=1}^{j-1} (-1)^k (k!) \tilde{\mathcal{B}}_k(z) B_{j,k}\left(\hat{Z}'_*(z), \hat{Z}''_*(z), \cdots, \hat{Z}^{(j-k+1)}_*(z)\right) \right) \tag{F.10}
$$

Proof of Lemma F.3. By Faà di Bruno's formula of Lemma F.2, we have

$$
\frac{d^j}{dz^j}\tilde{\mathcal{B}}_0(z) = \sum_{k=1}^{j} \tilde{\mathcal{B}}_0^{(k)}(z) B_{j,k}\left(\hat{Z}'_*(z), \hat{Z}''_*(z), \cdots, \hat{Z}^{(j-k+1)}_*(z)\right), \tag{F.11}
$$

where we have defined

$$\tilde{\mathcal{B}}_0^{(k)}(z) = \frac{d^k}{d\hat{Z}_*^k}\mu'(\hat{Z}_*(z)I + \Sigma)^{-1}\mu = (-1)^k(k!)\tilde{\mathcal{B}}_k(z)\,. \tag{F.12}$$

We get,

$$\begin{aligned}\frac{d^j}{dz^j}\tilde{\mathcal{B}}_0(z) &= \sum_{k=1}^{j}(-1)^k(k!)\tilde{\mathcal{B}}_k(z)B_{j,k}\left(\hat{Z}_*'(z),\hat{Z}_*''(z),\cdots,\hat{Z}_*^{(j-k+1)}(z)\right)\\ &\underbrace{=}_{B_{j,j}=(\hat{Z}_*'(z))^j}\sum_{k=1}^{j-1}(-1)^k(k!)\tilde{\mathcal{B}}_k(z)B_{j,k}\left(\hat{Z}_*'(z),\hat{Z}_*''(z),\cdots,\hat{Z}_*^{(j-k+1)}(z)\right)+(-1)^j(j!)\tilde{\mathcal{B}}_j(z)(\hat{Z}_*'(z))^j\end{aligned} \tag{F.13}$$

Thus,

$$\tilde{\mathcal{B}}_j(z) = \frac{1}{j!(-\hat{Z}_*'(z))^j}\left(\frac{d^j}{dz^j}\tilde{\mathcal{B}}_0(z) - \sum_{k=1}^{j-1}(-1)^k(k!)\tilde{\mathcal{B}}_k(z)B_{j,k}\left(\hat{Z}_*'(z),\hat{Z}_*''(z),\cdots,\hat{Z}_*^{(j-k+1)}(z)\right)\right) \tag{F.14}$$

The proof of Lemma F.3 is complete. □

Proof of Theorem 2. We have

$$\hat{\Theta}^d(z) \;\approx\; \tilde{\mathcal{B}}(z) \tag{F.15}$$

by Lemma F.1 and the claim follows from Lemmas F.3 and E.8. □

G Proofs: Properties of $\mathcal{B}_j$

Proof of Corollary 3. We have

$$\mathcal{B}_j(z) \;=\; z^j \mu'(zI+\Sigma)^{-j-1}\mu \;=\; z^j \sum_i (z+\lambda_i)^{-j-1}\tilde{\mu}_i^2 \tag{G.1}$$

where $\tilde{\mu} = U'\mu$ with $\Sigma = U\,\mathrm{diag}(\lambda)U'$. Our goal is to prove that

$$\frac{\mathcal{B}_{j-1}}{\mathcal{B}_j} \;>\; \frac{\mathcal{B}_j}{\mathcal{B}_{j+1}} \tag{G.2}$$

which is equivalent to

$$(\sum_i (z+\lambda_i)^{-j-1}\tilde{\mu}_i^2)^2 < \sum_i (z+\lambda_i)^{-j}\tilde{\mu}_i^2 \sum_i (z+\lambda_i)^{-j-2}\tilde{\mu}_i^2 \tag{G.3}$$

which follows from the Cauchy-Schwarz inequality. Strictness requires at least two distinct λ_i with $\tilde{\mu}_i \neq 0$, which is generic under Assumption 1. □

Proof of Proposition 4. We have

$$\begin{aligned} E_T[\hat{\pi}_T(z)'R_{T+1}] \;&=\; \hat{\pi}_T(z)'\mu \\ &=\; (\bar{\varepsilon}+\mu)'(zI+\hat{\Sigma})^{-1}\mu \\ &\underbrace{\approx}_{Theorem\ C.1}\; \mu'(zI+\hat{\Sigma})^{-1}\mu \\ &\underbrace{\approx}_{Theorem\ D.2}\; (1+\xi)\mu'(\hat{Z}_*(z)I+\Sigma)^{-1}\mu\,. \end{aligned} \tag{G.4}$$

At the same time

$$\begin{aligned} \mathrm{Var}_T[\hat{\pi}_T(z)'R_{T+1}] \;&=\; \hat{\pi}_T(z)'\Sigma\hat{\pi}_T(z) \\ &=\; (\bar{\varepsilon}+\mu)'(zI+\hat{\Sigma})^{-1}\Sigma(zI+\hat{\Sigma})^{-1}(\bar{\varepsilon}+\mu)\,. \end{aligned} \tag{G.5}$$

By Theorem C.1 the cross-term is negligible and hence

$$\mathrm{Var}_T[\hat{\pi}_T(z)'R_{T+1}] \approx \mu'(zI+\hat{\Sigma})^{-1}\Sigma(zI+\hat{\Sigma})^{-1}\mu \;+\; \bar{\varepsilon}'(zI+\hat{\Sigma})^{-1}\Sigma(zI+\hat{\Sigma})^{-1}\bar{\varepsilon} \tag{G.6}$$

By Lemma E.1(i), which requires no random-design structure on μ, we have

$$\mu'(zI+\hat{\Sigma})^{-1}\Sigma(zI+\hat{\Sigma})^{-1}\mu \;\approx\; (1+\xi(z))^2\frac{d}{dz}(zq(z)) \tag{G.7}$$

where

$$q(z) \;=\; \mu'(zI+\hat{\Sigma})^{-1}\mu \;\approx\; (1+\xi)\mu'(\hat{Z}_*I+\Sigma)^{-1}\mu \tag{G.8}$$

and

$$q'(z) \;\approx\; \frac{d}{dz}((1+\xi)\mu'(\hat{Z}_*I+\Sigma)^{-1}\mu) \;=\; \xi'\mathcal{B}_0-(1+\xi)(\hat{Z}_*'/\hat{Z}_*)\mathcal{B}_1 \tag{G.9}$$

Thus,

$$\begin{aligned}&\mu'(zI+\hat{\Sigma})^{-1}\Sigma(zI+\hat{\Sigma})^{-1}\mu \;\approx\; (1+\xi(z))^2((1+\xi)\mathcal{B}_0+z(\xi'\mathcal{B}_0-(1+\xi)(\hat{Z}_*'/\hat{Z}_*)\mathcal{B}_1))\\ &=\; (1+\xi(z))^2\hat{Z}_*'(\mathcal{B}_0-\mathcal{B}_1)\,.\end{aligned} \tag{G.10}$$

At the same time, by the Sherman-Morrison formula,

$$\bar{\varepsilon}'(zI+\hat{\Sigma})^{-1}\Sigma(zI+\hat{\Sigma})^{-1}\bar{\varepsilon} \;=\; -\frac{d}{d\alpha}(\bar{\varepsilon}'(zI+\alpha\Sigma+\hat{\Sigma})^{-1}\bar{\varepsilon})|_{\alpha=0}\,. \tag{G.11}$$

Furthermore,

$$\begin{aligned}F_T(\alpha) \;&=\; \bar{\varepsilon}'(zI+\alpha\Sigma+\hat{\Sigma})^{-1}\bar{\varepsilon} \;=\; \bar{X}'\Sigma^{1/2}(zI+\alpha\Sigma+\Sigma^{1/2}XX'\Sigma^{1/2}/T)^{-1}\Sigma^{1/2}\bar{X}\\ &=\; \bar{X}'\tilde{\Sigma}^{1/2}(I+\tilde{\Sigma}^{1/2}XX'\tilde{\Sigma}^{1/2}/T)^{-1}\tilde{\Sigma}^{1/2}\bar{X}\end{aligned} \tag{G.12}$$

Thus, $F_T(\alpha)$ concentrates at $\xi(\alpha)$ by Theorem B.2. The same argument applied to the derivative implies that $F_T'(\alpha)$ concentrates at $(d/d\alpha)\xi(\alpha)$, so that $-F_T'(0)$ concentrates at $-(d/d\alpha)\xi(\alpha)|_{\alpha=0}$. Since this limit is independent of the distribution of X, we can compute it assuming X is Gaussian. Since it concentrates, we can just directly compute its expectation. In the Gaussian setting, $\bar{\varepsilon}$ is independent of $\hat{\Sigma}$ and, hence,

$$\begin{aligned} & E[\bar{\varepsilon}'(zI+\hat{\Sigma})^{-1}\Sigma(zI+\hat{\Sigma})^{-1}\bar{\varepsilon}] \\ &= \; T^{-1}\operatorname{tr} E[(zI+\hat{\Sigma})^{-1}\Sigma(zI+\hat{\Sigma})^{-1}\Sigma] \end{aligned} \tag{G.13}$$

By Lemma 25 in Didisheim et al. (2024),

$$T^{-1}\operatorname{tr} E[(zI+\hat{\Sigma})^{-1}\Sigma(zI+\hat{\Sigma})^{-1}\Sigma] \approx \; (\xi + z\xi')(1+\xi)^2$$

Since $\hat{Z}_*' = 1 + \xi + z\xi'$, the two variance terms combine to

$$\operatorname{Var}_T[\hat{\pi}_T(z)'R_{T+1}] \;\approx\; (1+\xi)^2\Big(\hat{Z}_*'(\mathcal{B}_0 - \mathcal{B}_1) \;+\; (\hat{Z}_*' - 1)\Big)\,, \tag{G.14}$$

while the squared mean is $(1+\xi)^2\mathcal{B}_0^2$. The factors $(1+\xi)^2$ cancel, so

$$\frac{(E_T[\hat{\pi}_T(z)'R_{T+1}])^2}{\operatorname{Var}_T[\hat{\pi}_T(z)'R_{T+1}]} \;\approx\; \frac{\mathcal{B}_0^2}{\hat{Z}_*'(\mathcal{B}_0 - \mathcal{B}_1) \;+\; (\hat{Z}_*' - 1)} \;\leq\; \frac{\mathcal{B}_0^2}{\hat{Z}_*'(\mathcal{B}_0 - \mathcal{B}_1)}\,, \tag{G.15}$$

the inequality holding because $\hat{Z}_*' \geq 1$ by Lemma 2. Finally, at $J = 1$ Corollary 3 gives

$$\mathcal{H}_1^L(\boldsymbol{\mathcal{B}}) \;=\; \mathcal{B}_0 + \frac{\mathcal{B}_1}{1 - \mathcal{B}_1/\mathcal{B}_0} \;=\; \frac{\mathcal{B}_0^2}{\mathcal{B}_0 - \mathcal{B}_1}\,, \tag{G.16}$$

which is well defined because $\mathcal{B}_1 < \mathcal{B}_0$ by Proposition 1. Substituting into (G.15) yields

$$\frac{(E_T[\hat{\pi}_T(z)'R_{T+1}])^2}{\operatorname{Var}_T[\hat{\pi}_T(z)'R_{T+1}]} \;\leq_p\; \frac{1}{\hat{Z}_*'(z)}\mathcal{H}_1^L(\boldsymbol{\mathcal{B}}) \;\leq\; \frac{1}{\hat{Z}_*'(z)}\mathcal{R}_*^2\,, \tag{G.17}$$

the last step by Corollary 3. The proof of Proposition 4 is complete. Note that the inequality in (G.15) is strict whenever $\hat{Z}'_* > 1$: the term $\hat{Z}'_* - 1$ is the contribution of the estimation noise $\bar{\varepsilon}$ to the portfolio's out-of-sample variance, so the shortfall documented here is if anything understated. □

Lemma G.1 (Lower bound for $Z'_*(z)$ via Cauchy–Schwarz) *We have*

$$Z'_*(z) = \frac{1}{1 - c\int \frac{x^2}{(Z_*(z)+x)^2}\, H(dx)} \geq \frac{1}{1 - c\left(\int \frac{x}{Z_*(z)+x}\, H(dx)\right)^2}. \tag{G.18}$$

Using the fixed point (G.20), this can be rewritten as

$$Z'_*(z) \geq \frac{1}{1 - \frac{1}{c}\left(1 - \frac{z}{Z_*(z)}\right)^2} = \frac{c}{c - \left(1 - \frac{z}{Z_*(z)}\right)^2}. \tag{G.19}$$

In particular, this bound is sharper than the crude estimate $Z'_*(z) \geq 1$.

Proof of Lemmas G.1 and 2. Substitute

$$m(-z; c) = \frac{1}{c}\tilde{m}(z; c) - \left(\frac{1}{c} - 1\right)\frac{1}{z}$$

into the master equation (B.7). Setting $Z_*(z) := 1/\tilde{m}(z; c)$, a direct calculation shows that $Z_*(z)$ satisfies

$$\frac{z}{Z_*(z)} - 1 + c\int \frac{x}{Z_*(z) + x}\, H(dx) = 0. \tag{G.20}$$

Define

$$Z := Z_*(z), \qquad A(Z) := \int \frac{x}{Z + x}\, H(dx), \qquad B(Z) := \int \frac{x^2}{(Z + x)^2}\, H(dx).$$

Then the fixed point equation is

$$\frac{z}{Z} - 1 + cA(Z) = 0. \tag{G.21}$$

We first differentiate (G.20) implicitly with respect to z. Since

$$\frac{d}{dZ}\frac{x}{Z+x} = -\frac{x}{(Z+x)^2},$$

we obtain

$$\frac{1}{Z} - \frac{z}{Z^2}Z'_*(z) - c\left(\int \frac{x}{(Z+x)^2}\,H(dx)\right)Z'_*(z) = 0.$$

Hence

$$Z'_*(z) = \frac{1/Z}{z/Z^2 + c\int \frac{x}{(Z+x)^2}\,H(dx)}.$$

Multiplying numerator and denominator by Z yields

$$Z'_*(z) = \frac{1}{z/Z + cZ\int \frac{x}{(Z+x)^2}\,H(dx)}. \tag{G.22}$$

Now use the algebraic identity

$$\frac{Zx}{(Z+x)^2} = \frac{x}{Z+x} - \frac{x^2}{(Z+x)^2}.$$

Substituting this into (G.22) gives

$$Z'_*(z) = \frac{1}{z/Z + cA(Z) - cB(Z)}.$$

By the fixed point (G.21), the first two terms in the denominator satisfy

$$\frac{z}{Z} + cA(Z) = 1.$$

Therefore

$$Z'_*(z) = \frac{1}{1 - cB(Z)} = \frac{1}{1 - c\int \frac{x^2}{(Z+x)^2}\, H(dx)}. \tag{G.23}$$

Next, apply Cauchy–Schwarz to the nonnegative function

$$u(x) := \frac{x}{Z + x}$$

under the probability measure H. Since H has total mass one,

$$\left(\int u\, dH\right)^2 \le \int u^2\, dH.$$

That is,

$$A(Z)^2 \le B(Z).$$

Multiplying by $c > 0$ and subtracting from 1,

$$1 - cB(Z) \le 1 - cA(Z)^2.$$

Since the reciprocal is decreasing on $(0, \infty)$, (G.23) implies

$$Z'_*(z) \ge \frac{1}{1 - cA(Z)^2}.$$

This proves (G.18).

Finally, from (G.21),

$$A(Z) = \frac{1 - z/Z}{c},$$

so

$$1 - cA(Z)^2 = 1 - \frac{1}{c}\left(1 - \frac{z}{Z}\right)^2.$$

Substituting this into the previous bound yields (G.19).

To see that this improves on the crude lower bound $Z'_*(z) \geq 1$, note simply that

$$1 - cA(Z)^2 \leq 1,$$

hence

$$\frac{1}{1 - cA(Z)^2} \geq 1.$$

□

Corollary G.1 *For any threshold $a > 0$ and any $j \geq 1$,*

$$\mathcal{B}_j(\hat{Z}_*(z)) \;\leq\; \mathcal{R}^2_*(a) \;+\; \frac{\hat{Z}_*(z)}{a + \hat{Z}_*(z)} \mathcal{B}_{j-1}(\hat{Z}_*(z))\,. \tag{G.24}$$

Proof of Corollary G.1. For any threshold α, we have

$$\begin{aligned}
\mathcal{B}_j(\hat{Z}_*(z)) \;&=\; \sum_i \mathcal{R}_i^2 \frac{\lambda_i \hat{Z}_*(z)^j}{(\lambda_i + \hat{Z}_*(z))^{j+1}} \;=\; \sum_{\lambda_i \le a} \mathcal{R}_i^2 \frac{\lambda_i \hat{Z}_*(z)^j}{(\lambda_i + \hat{Z}_*(z))^{j+1}} \;+\; \sum_{\lambda_i > a} \mathcal{R}_i^2 \frac{\lambda_i \hat{Z}_*(z)^j}{(\lambda_i + \hat{Z}_*(z))^{j+1}} \\
&\le \sum_{\lambda_i \le a} \mathcal{R}_i^2 \;+\; \sum_{\lambda_i > a} \mathcal{R}_i^2 \frac{\lambda_i \hat{Z}_*(z)^j}{(\lambda_i + \hat{Z}_*(z))^{j+1}} \\
&\le \sum_{\lambda_i \le a} \mathcal{R}_i^2 \;+\; \frac{\hat{Z}_*(z)}{a + \hat{Z}_*(z)} \sum_{\lambda_i > a} \mathcal{R}_i^2 \frac{\lambda_i \hat{Z}_*(z)^{j-1}}{(\lambda_i + \hat{Z}_*(z))^j} \\
&\le \sum_{\lambda_i \le a} \mathcal{R}_i^2 \;+\; \frac{\hat{Z}_*(z)}{a + \hat{Z}_*(z)} \sum_{i} \mathcal{R}_i^2 \frac{\lambda_i \hat{Z}_*(z)^{j-1}}{(\lambda_i + \hat{Z}_*(z))^j} \\
&= \mathcal{R}_*^2(a) \;+\; \frac{\hat{Z}_*(z)}{a + \hat{Z}_*(z)} \mathcal{B}_{j-1}(\hat{Z}_*(z))\,,
\end{aligned} \tag{G.25}$$

and the claim follows. □

Proof of Corollary 5. Consider the decomposition

$$\Psi \;=\; \Psi_a + \Psi_{>a} \tag{G.26}$$

corresponding to eigenvalues below a and above a. Then,

$$\mathcal{R}_*^2 \;=\; \mathcal{R}_*^2(a) \;+\; \mathcal{R}_*^2(>a) \tag{G.27}$$

and all $\mathcal{B}_j$ admit the same decomposition:

$$\mathcal{B}_j = \mathcal{B}_{j,a} + \mathcal{B}_{j,>a}\,. \tag{G.28}$$

Then, applying Corollary G.1 to $\Psi_{>a}$, we get (since it has no eigenvalues below a) that

$$0 \geq \mathcal{B}_{j+k,>a}(\hat{Z}_*(z)) - \left(\frac{\hat{Z}_*(z)}{a+\hat{Z}_*(z)}\right)^k \mathcal{B}_{j,>a}(\hat{Z}_*(z))\,. \tag{G.29}$$

and, hence,

$$\sum_{k=0}^{\infty} \mathcal{B}_{j+k,>a}(\hat{Z}_*(z)) \leq \mathcal{B}_{j,>a}(\hat{Z}_*(z))\frac{a+\hat{Z}_*(z)}{a} \tag{G.30}$$

because

$$\sum_{k=0}^{\infty} \left(\frac{\hat{Z}_*(z)}{a+\hat{Z}_*(z)}\right)^k = \frac{a+\hat{Z}_*(z)}{a} \tag{G.31}$$

Thus,

$$\begin{aligned}
\mathcal{R}_*^2 &= \mathcal{R}_*^2(a) + \mathcal{R}_*^2(>a) \\
&= \mathcal{R}_*^2(a) + \sum_{j=0}^{\infty} \mathcal{B}_{j,>a}(\hat{Z}_*(z)) \\
&\leq \mathcal{R}_*^2(a) + \sum_{j=0}^{J-1} \mathcal{B}_{j,>a}(\hat{Z}_*(z)) \\
&+ \mathcal{B}_{J,>a}(\hat{Z}_*(z))\frac{a+\hat{Z}_*(z)}{a} \\
&\leq \mathcal{R}_*^2(a) + \sum_{j=0}^{J-1} \mathcal{B}_j(\hat{Z}_*(z)) \\
&+ \mathcal{B}_J(\hat{Z}_*(z))\frac{a+\hat{Z}_*(z)}{a}
\end{aligned} \tag{G.32}$$

The proof of Corollary 5 is complete. □

G.1 Tight upper bound

Corollary G.2 *The tightest upper bound on $\mathcal{R}_*^2$ is*

$$\mathcal{R}_*^2 \;\leq\; \frac{\mathcal{D}_{J-1}(\boldsymbol{\mathcal{B}}) \;+\; \mathcal{B}_J(\hat{Z}_*(z))\frac{a_J^*(\boldsymbol{\mathcal{B}})+\hat{Z}_*(z)}{a_J^*(\boldsymbol{\mathcal{B}})}}{1-\frac{a_J^*(\boldsymbol{\mathcal{B}})}{\bar{\lambda}}} \tag{G.33}$$

where

$$a_{\min,J}(\boldsymbol{\mathcal{B}}) = \frac{-\mathcal{B}_J(\hat{Z}_*(z))\hat{Z}_*(z) \;+\; \sqrt{\left(\mathcal{B}_J(\hat{Z}_*(z))\hat{Z}_*(z)\right)^2 \;+\; \mathcal{B}_J(\hat{Z}_*(z))\hat{Z}_*(z)\,\mathcal{D}_J(\boldsymbol{\mathcal{B}})\,\bar{\lambda}}}{\mathcal{D}_J(\boldsymbol{\mathcal{B}})} \tag{G.34}$$

Proof of Corollary G.2. We are looking for a minimizing the upper bound

$$\arg\min_{a\in(0,\bar{\lambda})} \frac{\mathcal{D}_{J-1}(\boldsymbol{\mathcal{B}}) \;+\; \mathcal{B}_J(\hat{Z}_*(z))\frac{a+\hat{Z}_*(z)}{a}}{1-\frac{a}{\bar{\lambda}}} \tag{G.35}$$

The FOC reduces to a quadratic equation

$$\begin{aligned}
&\frac{\mathcal{D}_{J-1}(\boldsymbol{\mathcal{B}}) \;+\; \mathcal{B}_J(\hat{Z}_*(z))\frac{a+\hat{Z}_*(z)}{a}}{\left(1-\frac{a}{\bar{\lambda}}\right)^2}\frac{1}{\bar{\lambda}} - \frac{\mathcal{B}_J(\hat{Z}_*(z))\frac{\hat{Z}_*(z)}{a^2}}{1-\frac{a}{\bar{\lambda}}} = 0 \\
\Longrightarrow\; &\left(\mathcal{D}_{J-1}(\boldsymbol{\mathcal{B}}) \;+\; \mathcal{B}_J(\hat{Z}_*(z))\frac{a+\hat{Z}_*(z)}{a}\right)\frac{1}{\bar{\lambda}} - \mathcal{B}_J(\hat{Z}_*(z))\frac{\hat{Z}_*(z)}{a^2}\left(1-\frac{a}{\bar{\lambda}}\right) = 0 \\
\Longrightarrow\; &\mathcal{D}_{J-1}(\boldsymbol{\mathcal{B}}) \;+\; \mathcal{B}_J(\hat{Z}_*(z))\frac{a+\hat{Z}_*(z)}{a} - \mathcal{B}_J(\hat{Z}_*(z))\frac{\hat{Z}_*(z)}{a^2}\bar{\lambda} + \mathcal{B}_J(\hat{Z}_*(z))\frac{\hat{Z}_*(z)}{a} = 0 \\
\Longrightarrow\; &g(a) = a^2\underbrace{\left(\mathcal{D}_{J-1}(\boldsymbol{\mathcal{B}}) \;+\mathcal{B}_J(\hat{Z}_*(z))\right)}_{\mathcal{D}_J(\boldsymbol{\mathcal{B}})} + 2a\mathcal{B}_J(\hat{Z}_*(z))\hat{Z}_*(z) - \mathcal{B}_J(\hat{Z}_*(z))\hat{Z}_*(z)\bar{\lambda} = 0
\end{aligned} \tag{G.36}$$

The solutions are

$$a = \frac{-\mathcal{B}_J(\hat{Z}_*(z))\hat{Z}_*(z) \pm \sqrt{\mathcal{B}_J(\hat{Z}_*(z))^2\hat{Z}_*(z)^2 + \mathcal{D}_J(\boldsymbol{\mathcal{B}})\mathcal{B}_J(\hat{Z}_*(z))\hat{Z}_*(z)\bar{\lambda}}}{\mathcal{D}_J(\boldsymbol{\mathcal{B}})} \tag{G.37}$$

We select the positive root since a is a threshold on eigenvalues ($a \geq 0$), and only the positive solution is admissible

$$a_{\min,J}(\boldsymbol{\mathcal{B}}) = \frac{-\mathcal{B}_J(\hat{Z}_*(z))\hat{Z}_*(z) + \sqrt{\mathcal{B}_J(\hat{Z}_*(z))^2\hat{Z}_*(z)^2 + \mathcal{D}_J(\boldsymbol{\mathcal{B}})\mathcal{B}_J(\hat{Z}_*(z))\hat{Z}_*(z)\bar{\lambda}}}{\mathcal{D}_J(\boldsymbol{\mathcal{B}})} \quad \text{(G.38)}$$

To verify that the solution lies in $(0, \bar{\lambda})$, we study the sign of the quadratic $g(a)$, the LHS of the FOC. At $a = 0$

$$g(0) = -\mathcal{B}_J(\hat{Z}_*(z))\hat{Z}_*(z)\bar{\lambda} < 0 \quad \text{(G.39)}$$

and at $a = \bar{\lambda}$

$$g(\bar{\lambda}) = \bar{\lambda}^2\mathcal{D}_J(\boldsymbol{\mathcal{B}}) + \bar{\lambda}\mathcal{B}_J(\hat{Z}_*(z))\hat{Z}_*(z) \geq 0 \quad \text{(G.40)}$$

Moreover, $g(a)$ is strictly increasing on $a \in (0, \infty)$ because

$$g'(a) = 2a \underbrace{\left(\mathcal{D}_{J-1}(\boldsymbol{\mathcal{B}}) \ + \mathcal{B}_J(\hat{Z}_*(z))\right)}_{\mathcal{D}_J(\boldsymbol{\mathcal{B}})} + 2\mathcal{B}_J(\hat{Z}_*(z))\hat{Z}_*(z) > 0 \quad \text{(G.41)}$$

Thus, its unique positive root $a_J^*(\boldsymbol{\mathcal{B}})$ lies in $(0, \bar{\lambda})$. Lastly, since $\mathcal{B}_J(\hat{Z}_*(z))\hat{Z}_*(z) \geq 0$, we have

$$\lim_{a \downarrow 0} \frac{\mathcal{D}_{J-1}(\boldsymbol{\mathcal{B}}) \ + \ \mathcal{B}_J(\hat{Z}_*(z))\frac{a+\hat{Z}_*(z)}{a}}{1 - \frac{a}{\bar{\lambda}}} = +\infty \quad \text{(G.42)}$$

Moreover,

$$\lim_{a \uparrow \bar{\lambda}} \frac{\mathcal{D}_{J-1}(\boldsymbol{\mathcal{B}}) \ + \ \mathcal{B}_J(\hat{Z}_*(z))\frac{a+\hat{Z}_*(z)}{a}}{1 - \frac{a}{\bar{\lambda}}} = +\infty \quad \text{(G.43)}$$

Since the FOC admits a unique positive solution, and the objective diverges at the boundaries, the computed solution is global minimum on $(0, \bar{\lambda})$. □

H Central Limit Theorem

Everywhere in this section, we make the following assumption.

Assumption 2 *Let $\mu = P^{-1/2}\Sigma_{\mu,P}^{1/2}\nu$, where $\nu = (\nu_i)_{i=1}^P$ has i.i.d. coordinates, with $E[\nu_i] = 0$, $E[\nu_i^2] = 1$, $E[\nu_i^3] = 0$, $E[\nu_i^4] = 3$, and $\Sigma_{\mu,P}$, $P \geq 1$, is a sequence of uniformly bounded, nonnegative definite matrices. In the sequel, we abuse the notation and use Σ_μ to denote $\Sigma_{\mu,P}$.*

The main result of this section is the following accompanying CLT for Theorem 2.

Theorem H.1 *Let $\hat{\boldsymbol{\Theta}} = (\hat{Z}_*^j \hat{\Theta}_j^d)_{j=0}^J$. Then, under Assumption 2, there exists a pivotal estimator Σ_Θ given in (H.85), such that*

$$\Sigma_\Theta^{-1/2} T^{1/2} \big(\hat{\boldsymbol{\Theta}} - \boldsymbol{\mathcal{B}}\big) \tag{H.1}$$

converges to $N(0, I)$, where $\boldsymbol{\mathcal{B}} = (\mathcal{B}_0, \cdots, \mathcal{B}_J)(\hat{Z}_*(z))$.

Since $\mathcal{H}_J^L$ and $\mathcal{H}_J^U$ in (46) are smooth functions of $\boldsymbol{\mathcal{B}}$, we can use the standard delta method to derive asymptotic confidence intervals for those bounds. Define

$$\begin{aligned}
\mathcal{H}_J^L(\boldsymbol{\mathcal{B}}) &= \mathcal{D}_{J-1}(\boldsymbol{\mathcal{B}}) + \mathcal{B}_J(\hat{Z}_*(z)) \frac{1}{1 - \frac{\mathcal{B}_J(\hat{Z}_*(z))}{\mathcal{B}_{J-1}(\hat{Z}_*(z))}} \\
\mathcal{H}_J^U(\boldsymbol{\mathcal{B}}) &= \frac{\mathcal{D}_{J-1}(\boldsymbol{\mathcal{B}}) + \mathcal{B}_J(\hat{Z}_*(z)) \frac{a_J^*(\boldsymbol{\mathcal{B}}) + \hat{Z}_*(z)}{a_J^*(\boldsymbol{\mathcal{B}})}}{1 - \frac{a_J^*(\boldsymbol{\mathcal{B}})}{\bar{\lambda}}} \\
a_J^*(\boldsymbol{\mathcal{B}}) &= \frac{-\mathcal{B}_J(\hat{Z}_*(z))\hat{Z}_*(z) + \sqrt{\mathcal{B}_J(\hat{Z}_*(z))^2 \hat{Z}_*(z)^2 + \mathcal{D}_J(\boldsymbol{\mathcal{B}})\mathcal{B}_J(\hat{Z}_*(z))\hat{Z}_*(z)\bar{\lambda}}}{\mathcal{D}_J(\boldsymbol{\mathcal{B}})}
\end{aligned} \tag{H.2}$$

Below, we sometimes abuse the notation and write $\mathcal{H}_J^U(\boldsymbol{\mathcal{B}}) = \mathcal{H}_J^U(\boldsymbol{\mathcal{B}}; a)$. Theorem H.1 combined with the delta method allows us to establish an asymptotically consistent estimator for this lower bound:

Corollary H.2 (CLT for upper and lower bounds) *Define*

$$\mathcal{L}_J^I(z) = \mathcal{H}_J^I(\boldsymbol{\mathcal{B}}), \tag{H.3}$$

and their asymptotically consistent estimators

$$\hat{\mathcal{L}}_J^I(z) = \mathcal{H}_J^I(\hat{\boldsymbol{\Theta}}(z)),\ I = L, U\,. \tag{H.4}$$

Then,

$$\hat{\sigma}_{I,J}^{-1} T^{1/2}(\hat{\mathcal{L}}_J^I(z) - \mathcal{L}_J^I(z)) \to \mathcal{N}(0,1)\,, \tag{H.5}$$

where

$$\hat{\sigma}_{I,J}^2 = \nabla \mathcal{H}_J^I(\hat{\boldsymbol{\Theta}})' \Sigma_\Theta \nabla \mathcal{H}_J^I(\hat{\boldsymbol{\Theta}}) \tag{H.6}$$

for $I = L, U$.

H.1 Super-Consistency of Traces

Everywhere in this paper, we compute asymptotics of traces and quadratic forms. Quadratic forms of the form $P^{-1}\nu' A \nu$ converge to $P^{-1}\operatorname{tr}(A)$ at the standard rate of $P^{-1/2}$ by Lemma A.1. By contrast, traces (expressions as in Lemma E.9) converge much faster, at the rate of P^{-1}, as follows from the remarkable results of Bai and Silverstein (2004):

Lemma H.1 (Bai and Silverstein (2004) super-consistency) *Let*

$$q_T(z;P) \;=\; P^{-1}\operatorname{tr}(Q(P)(zI+\hat{\Sigma})^{-1})\,, \tag{H.7}$$

where $Q(P)$ is a uniformly bounded sequence of matrices. We have

$$P(q_T(z;P) - \lim_{T,P\to\infty,P/T\to c} E[q_T(z;P)]) \tag{H.8}$$

is uniformly bounded in L_2 as $P\to\infty$

This *super-consistency* of traces implies that in our calculations for the CLT that involve such traces, we can simply replace $\hat{\Sigma}$ with Σ because the overall convergence rate is $T^{-1/2}$ (due to the quadratic forms) and, hence, the errors from the trace estimation are negligible.

H.2 Concentration of Quadratic Forms

Lemma H.2 (Corollary 2.4 in Srivastava (2009)) *Suppose that $E[\varepsilon_t^4]=3\sigma_\varepsilon^4$. For any sequence of bounded matrices A_P independent of ε,*

$$\frac{T^{-1/2}(\varepsilon' A_P\varepsilon \;-\; \sigma_\varepsilon^2\operatorname{tr}(A))}{\tau(A)^{1/2}} \;\to\; \mathcal{N}(0,1) \tag{H.9}$$

in distribution, where

$$\tau(A) \;=\; 2T^{-1}\operatorname{tr}(A^2)\sigma_\varepsilon^4\,. \tag{H.10}$$

Lemma H.3 *Suppose that $E[\varepsilon_t]=0$, $E[\varepsilon_t^2]=1$, $E[\varepsilon_t^3]=0$ and $E[\varepsilon_t^4]=3$. Let a_T be a sequence of bounded vectors and A_T a sequence of bounded random matrices, all of them being independent of ε_t. Then, $(a_T'\varepsilon\|a_T\|^{-1},\; T^{-1/2}(\varepsilon' A_T\varepsilon-\sigma_\varepsilon^2\operatorname{tr}(A_T))/\sqrt{2T^{-1}\sigma_\varepsilon^4\operatorname{tr}(A_T^2)})$ converges to a $\mathcal{N}(0,I)$ distribution. That is, these two random variables are asymptotically independent standard Normals.*

Proof of Lemma H.3. The proof of Lemma H.3 is completely analogous to that of the main result in Srivastava (2009) and follows closely the classical argument from De Jong (1987), plus the Cramer-Wold device (Billingsley (1995)). Without loss of generality, we normalize $\sigma_\varepsilon^2 = 1$. Let $\tilde{a}_T = a_T/\|a_T\|$, $\tilde{A}_T = A_T/\sqrt{2T^{-1}\operatorname{tr}(A_T^2)}$.

With Cramer-Wold, we build for any vector (q_1, q_2)

$$S(T) = q_1\tilde{a}_T'\varepsilon + q_2T^{-1/2}(\varepsilon'\tilde{A}_T\varepsilon - \operatorname{tr}(\tilde{A}_T)) \tag{H.11}$$

As in De Jong (1987), we define

$$S(t) = \sum_{\tau\leq t}\left(q_1\tilde{a}(\tau)\varepsilon_\tau + q_2T^{-1/2}(\varepsilon_\tau^2 - 1)\tilde{A}_{\tau,\tau} + q_2T^{-1/2}2\sum_{\tau_1<\tau}\varepsilon_{\tau_1}\varepsilon_\tau\tilde{A}_{\tau_1,\tau}\right) \tag{H.12}$$

By direct calculation, $S(t)$ is a Martingale and then, standard arguments based on the Lindeberg CLT imply asymptotic normality. □

We will also use the following multivariate extension.

Lemma H.4 *Suppose that* $E[\varepsilon_t] = 0$, $E[\varepsilon_t^2] = 1$, $E[\varepsilon_t^3] = 0$ *and* $E[\varepsilon_t^4] = 3$. *Let* $a_{k,T}$ *be a sequence of bounded vectors and* $A_{k,T}$ *a sequence of bounded, symmetric random matrices, all of them being independent of* ε_t; $k = 1, \cdots, K$. *Then,* $((a_{k,T}'\varepsilon\|a_{k,T}\|^{-1}, T^{-1/2}(\varepsilon'A_{k,T}\varepsilon - \sigma_\varepsilon^2\operatorname{tr}(A_{k,T}))/\sqrt{2T^{-1}\sigma_\varepsilon^4\operatorname{tr}(A_{k,T}^2)}))_{k=1}^K$ *converges to a* $\mathcal{N}(0, \Sigma(\sigma_\varepsilon^2))$ *distribution. The structure of the covariance matrix* $\Sigma(\sigma_\varepsilon)$ *is as follows: any of the linear components,* $a_{k,T}'\varepsilon\|a_{k,T}\|^{-1}$, *has zero covariance with any quadratic component; by contrast, quadratic terms are correlated with*

$$\operatorname{Cov}(\varepsilon'A\varepsilon,\ \varepsilon'B\varepsilon) = 2\operatorname{tr}(AB) \tag{H.13}$$

while linear terms give

$$\mathrm{Cov}(\varepsilon' a,\ \varepsilon' b) \ = \ a' b\,. \tag{H.14}$$

Proof of Lemma H.4. The joint normality follows by the same Cramer-Wold argument.

The only claim to prove is the covariance formula. We have

$$\mathrm{Var}[\varepsilon'(A+B)\varepsilon] \ = \ 2\,\mathrm{tr}((A+B)^2) \ = \ 2\,\mathrm{tr}(A^2+B^2+2AB) \tag{H.15}$$

and, by the variance decomposition,

$$\mathrm{Var}[\varepsilon'(A+B)\varepsilon] \ = \ \mathrm{Var}[\varepsilon' A\varepsilon] + \mathrm{Var}[\varepsilon' B\varepsilon] + 2\mathrm{Cov}(\varepsilon' A\varepsilon, \varepsilon' B\varepsilon) \tag{H.16}$$

Comparing, we get the required identity. □

H.3 Computing the Error Covariances

Lemma H.5 *Suppose Condition* (R) *holds. We have*

$$\begin{aligned} &P^{-1}\,\mathrm{tr}\,E[\Sigma_\mu(z_1 I+\hat{\Sigma})^{-1}\Sigma(z_2 I+\hat{\Sigma})^{-1}] \\ &\approx \ (1+\xi(z_1))(1+\xi(z_2))\frac{z_2 q(z_2)-z_1 q(z_1)}{z_2-z_1} \end{aligned} \tag{H.17}$$

where

$$q(z) \ = \ \mu'(zI+\hat{\Sigma})^{-1}\mu \tag{H.18}$$

Proof of Lemma H.5. The claim follows directly from Lemma E.1(ii) and the concentration of quadratic forms, which under Condition (R) gives $\mu' A\mu = P^{-1}u'\Sigma_\mu^{1/2}A\Sigma_\mu^{1/2}u \approx P^{-1}\,\mathrm{tr}(A\Sigma_\mu)$ for any uniformly bounded A independent of u. □

H.4 Computing Σ_Θ in Theorem H.1

Theorem H.3 *We have that*

$$T^{1/2}\Delta(z) \;=\; T^{1/2}(\hat{\Theta}^d - \mu'(\hat{Z}_*(z;c)I + \Sigma)^{-1}\mu) \tag{H.19}$$

converges to a Gaussian process for $z > 0$, with a covariance kernel $\sigma(z_1, z_2) \approx \hat{\sigma}(z_1, z_2)$ with

$$\begin{aligned}
\hat{\sigma}(z_1, z_2) \;&=\; 2\left(\frac{z_2(\hat{\Theta}(z_2) \;-\; \hat{\xi}(z_2)) - z_1(\hat{\Theta}(z_1) \;-\; \hat{\xi}(z_1))}{z_2 - z_1}\right)^2 \\
&\times\; \frac{z_2 - z_1}{\hat{Z}_*(z_2) - \hat{Z}_*(z_1)} \\
&+\; 4\frac{z_2(\hat{\Theta}(z_2) \;-\; \hat{\xi}(z_2)) - z_1(\hat{\Theta}(z_1) \;-\; \hat{\xi}(z_1))}{z_2 - z_1} \\
&+\; 2\frac{z_2\hat{\xi}(z_2) - z_1\hat{\xi}(z_1)}{z_2 - z_1}\,.
\end{aligned} \tag{H.20}$$

The rest of the section is devoted to the proof of this Theorem.

H.5 Variance and Covariance Decompositions

We have

$$\hat{\Theta}(z) \;=\; \mu'(zI + \hat{\Sigma})^{-1}\mu \;+\; 2\mu'(zI + \hat{\Sigma})^{-1}\bar{\varepsilon} \;+\; \bar{\varepsilon}'(zI + \hat{\Sigma})^{-1}\bar{\varepsilon} \tag{H.21}$$

Now,

$$\begin{aligned}
&\hat{\Theta} - \hat{\xi} - (1 + \hat{\xi})\mu'(\hat{Z}_*I + \Sigma)^{-1}\mu \\
&=\; (\mu'(zI + \hat{\Sigma})^{-1}\mu - (1 + \hat{\xi})\mu'(\hat{Z}_*I + \Sigma)^{-1}\mu) \\
&+\; 2\mu'(zI + \hat{\Sigma})^{-1}\bar{\varepsilon} \;+\; (\bar{\varepsilon}'(zI + \hat{\Sigma})^{-1}\bar{\varepsilon} - \hat{\xi})\,.
\end{aligned} \tag{H.22}$$

Thus,

$$\begin{aligned}
&\mathrm{Var}[\hat{\Theta} - \hat{\xi} - (1+\hat{\xi})\mu'(\hat{Z}_* I + \Sigma)^{-1}\mu] \\
&\approx \ \mathrm{Var}[(\mu'(zI+\hat{\Sigma})^{-1}\mu - (1+\hat{\xi})\mu'(\hat{Z}_* I + \Sigma)^{-1}\mu)] \\
&+ \ \mathrm{Var}[2\mu'(zI+\hat{\Sigma})^{-1}\bar{\varepsilon}] \ + \ \mathrm{Var}[(\bar{\varepsilon}'(zI+\hat{\Sigma})^{-1}\bar{\varepsilon} - \hat{\xi})]\,,
\end{aligned} \tag{H.23}$$

where the cross-covariances are asymptotically negligible by Lemma H.4: the first term is a centered quadratic form in the vector ν of Assumption 2, the third a centered quadratic form in $\bar{X}$, and the middle term is bilinear in the two, so that under $E[\nu_i^3] = 0$ every pairwise covariance is of smaller order than the diagonal terms. Let

$$\begin{aligned}
\hat{\mathcal{C}}(z) &= \mu'((zI+\hat{\Sigma})^{-1} - (\hat{Z}_*/z)(\hat{Z}_* I + \Sigma)^{-1})\mu \\
\hat{\mathcal{I}}(z) &= \mu'(zI+\hat{\Sigma})^{-1}\bar{\varepsilon} \\
\hat{W}(z) &= \bar{\varepsilon}'(zI+\hat{\Sigma})^{-1}\bar{\varepsilon}\,.
\end{aligned} \tag{H.24}$$

Then, define

$$\xi_i \ = \ \hat{\xi}(z_i), \ \hat{Z}_*(i) \ = \ \hat{Z}_*(z_i). \tag{H.25}$$

Then,

$$\begin{aligned}
P\mathrm{Var}[\hat{\mathcal{C}}(z)] &= P^{-1} 2\,\mathrm{tr}\, E\Big[\Big(((zI+\hat{\Sigma})^{-1} - (\hat{Z}_*/z)(\hat{Z}_* I + \Sigma)^{-1})\Sigma_\mu\Big)^2\Big] \\
P\mathrm{Cov}[\hat{\mathcal{C}}(z_1), \hat{\mathcal{C}}(z_2)] &= P^{-1} 2\,\mathrm{tr}\, E\Big[\Big(((z_1 I+\hat{\Sigma})^{-1} - (1+\xi(z_1))(\hat{Z}_*(z_1) I + \Sigma)^{-1})\Sigma_\mu\Big) \\
&\quad \times \Big(((z_2 I+\hat{\Sigma})^{-1} - (1+\xi(z_2))(\hat{Z}_*(z_2) I + \Sigma)^{-1})\Sigma_\mu\Big)\Big]\,.
\end{aligned} \tag{H.26}$$

Then,

$$\begin{aligned} TE[\widehat{\mathcal{I}}(z)^2] &= E[\mu'(zI+\hat{\Sigma})^{-1}(T\bar{\varepsilon}\bar{\varepsilon}')(zI+\hat{\Sigma})^{-1}\mu] \\ &\approx P^{-1}\operatorname{tr} E[(zI+\hat{\Sigma})^{-1}(T\bar{\varepsilon}\bar{\varepsilon}')(zI+\hat{\Sigma})^{-1}\Sigma_\mu] \end{aligned} \tag{H.27}$$

and the same argument as in the proof of Theorem C.1 implies that

$$\begin{aligned} TE[\widehat{\mathcal{I}}(z)^2] &\approx E[\mu'(zI+\hat{\Sigma})^{-1}\Sigma(zI+\hat{\Sigma})^{-1}\mu] \\ TE[\widehat{\mathcal{I}}(z_1)\widehat{\mathcal{I}}(z_2)] &\approx E[\mu'(z_1I+\hat{\Sigma})^{-1}\Sigma(z_2I+\hat{\Sigma})^{-1}\mu] \end{aligned} \tag{H.28}$$

Note that all the quantities involving traces of products of $(zI+\hat{\Sigma})^{-1}$ converge to their asymptotic limits as the rate of T^{-1} by Lemma H.1 and, hence, from the point of view of the $T^{-1/2}$-CLT, can be viewed as deterministic so that covariances of all the terms produced by the randomness originating from such traces is asymptotically negligible.

Finally, by Theorem B.2,

$$T\mathrm{Var}[\hat{W}(z)-\hat{\xi}(z)] = 2(\xi+z\xi')(1+\xi)^2 \tag{H.29}$$

and

$$TE[(\hat{W}(z_1)-\hat{\xi}(z_1))(\hat{W}(z_2)-\hat{\xi}(z_2)] = 2\frac{z_1\xi(z_1)-z_2\xi(z_2)}{z_1-z_2}(1+\xi(z_1))(1+\xi(z_2)) \tag{H.30}$$

Lemma H.6 (Pivotal Estimator for $\mathrm{Cov}(\hat{C}(z_1),\hat{C}(z_2))$) *We have*

$$\begin{aligned} &P\mathrm{Cov}(\hat{C}(z_1),\hat{C}(z_2)) \\ &\approx 2c\cdot(1+\hat{\xi}(z_1))(1+\hat{\xi}(z_2))\frac{\Big(z_2(\hat{\Theta}(z_2)-\hat{\xi}(z_2))-z_1(\hat{\Theta}(z_1)-\hat{\xi}(z_1))\Big)^2}{z_2-z_1} \\ &\times \frac{1}{\hat{Z}_*(z_2)-\hat{Z}_*(z_1)} \end{aligned} \tag{H.31}$$

Proof of Lemma H.6. Everywhere in this proof, we use the following observation that follows directly from the Sherman-Morrison formula

Lemma H.7 *We have*

$$P^{-1}\operatorname{tr}(A_1(z_1I+\hat{\Sigma})^{-1}\cdots A_K(z_KI+\hat{\Sigma})^{-1}) \;=\; P^{-1}\operatorname{tr}(A_1(z_1I+\hat{\Psi})^{-1}\cdots A_K(z_KI+\hat{\Psi})^{-1}) + O(P^{-1}) \tag{H.32}$$

where $\hat{\Psi} = T^{-1}R'R$.

This lemma allows us to replace $\hat{\Sigma}$ with $\hat{\Psi}$, and the latter can be used with the Sherman-Morrison formula.

We have

$$\begin{aligned} P\mathrm{Cov}[\hat{\mathcal{C}}(z_1),\hat{\mathcal{C}}(z_2)] = P^{-1}2\operatorname{tr} E\Big[&\Big(((z_1I+\hat{\Sigma})^{-1} - (1+\xi(z_1))(\hat{Z}_*(z_1)I+\Sigma)^{-1})\Sigma_\mu\Big) \\ &\times\Big(((z_2I+\hat{\Sigma})^{-1} - (1+\xi(z_2))(\hat{Z}_*(z_2)I+\Sigma)^{-1})\Sigma_\mu\Big)\Big]. \end{aligned} \tag{H.33}$$

By the Master Theorem D.2,

$$P^{-1}\operatorname{tr}(A(zI+\hat{\Sigma})^{-1}) \approx\; (1+\xi(z))\,P^{-1}\operatorname{tr}(A(\hat{Z}_*(z)I+\Sigma)^{-1})\,. \tag{H.34}$$

Applying this theorem, we get

$$\begin{aligned} &0.5P\mathrm{Cov}[\hat{\mathcal{C}}(z_1),\hat{\mathcal{C}}(z_2)] \\ &\approx\; P^{-1}\operatorname{tr}(\Sigma_\mu(z_1I+\hat{\Sigma})^{-1}\Sigma_\mu(z_2I+\hat{\Sigma})^{-1}) \\ &-(1+\xi(z_1))(1+\xi(z_2))P^{-1}\operatorname{tr}(\Sigma_\mu(\hat{Z}_*(z_1)I+\Sigma)^{-1}\Sigma_\mu(\hat{Z}_*(z_2)I+\Sigma)^{-1}) \end{aligned} \tag{H.35}$$

Let

$$\begin{aligned} q(A;z) &= P^{-1}\operatorname{tr} E[(zI+\hat{\Sigma})^{-1}A] \\ Q(A;z_1;z_2) &= P^{-1}\operatorname{tr} E[(z_1I+\hat{\Sigma})^{-1}\Sigma_\mu(z_2I+\hat{\Sigma})^{-1}A] \end{aligned} \tag{H.36}$$

Lemma H.8 (Normality of the resolvent trace functionals) *Let* $\Omega := \mathbb{C}\setminus\mathbb{R}_-$, *where* $\mathbb{R}_- := (-\infty, 0]$. *The families*

$$\{q_P(A;\cdot)\}_{P\geq 1} \subset \mathcal{O}(\Omega)$$

and

$$\{Q_P(A;\cdot,\cdot)\}_{P\geq 1} \subset \mathcal{O}(\Omega^2)$$

are locally uniformly bounded. Consequently, by Montel's theorem, they are relatively compact in the topology of uniform convergence on compact subsets of Ω *and* Ω^2, *respectively.*

Proof of Lemma H.8. Fix a compact set $K \subset \Omega$. Since $K \cap \mathbb{R}_- = \varnothing$ and K is compact,

$$\delta_K := \inf_{z\in K,\ x\geq 0} |z+x| > 0.$$

Because $\hat{\Sigma}_P$ is positive semidefinite, its spectrum is contained in $[0,\infty)$. Hence, by the spectral theorem, for every $z \in K$,

$$\|(zI+\hat{\Sigma}_P)^{-1}\| \leq \delta_K^{-1}.$$

Therefore,

$$\begin{aligned}|q_P(A;z)| &\le P^{-1}\epsilon\left|\mathrm{tr}\{(zI+\hat{\Sigma}_P)^{-1}A_P\}\right| \\ &\le P^{-1}\epsilon\,\|(zI+\hat{\Sigma}_P)^{-1}\|\,\mathrm{tr}\,|A_P| \\ &\le \delta_K^{-1}P^{-1}\,\mathrm{tr}\,|A_P| \le C_A\delta_K^{-1}.\end{aligned}$$

Thus $\{q_P(A;\cdot)\}_{P\ge1}$ is uniformly bounded on K.

Now fix compact sets $K_1, K_2 \subset \Omega$. Define

$$\delta_{K_i} := \inf_{z\in K_i,\ x\ge0}|z+x| > 0, \qquad i=1,2.$$

Then for $z_i \in K_i$,

$$\|(z_iI+\hat{\Sigma}_P)^{-1}\| \le \delta_{K_i}^{-1}.$$

Using again the trace inequality $|\,\mathrm{tr}(BA)| \le \|B\|\,\mathrm{tr}\,|A|$, we obtain

$$\begin{aligned}|Q_P(A;z_1,z_2)| &\le P^{-1}\epsilon\left|\mathrm{tr}\{(z_1I+\hat{\Sigma}_P)^{-1}\Sigma_{\mu,P}(z_2I+\hat{\Sigma}_P)^{-1}A_P\}\right| \\ &\le P^{-1}\epsilon\,\|(z_1I+\hat{\Sigma}_P)^{-1}\Sigma_{\mu,P}(z_2I+\hat{\Sigma}_P)^{-1}\|\,\mathrm{tr}\,|A_P| \\ &\le \delta_{K_1}^{-1}\delta_{K_2}^{-1}\|\Sigma_{\mu,P}\|P^{-1}\,\mathrm{tr}\,|A_P| \\ &\le C_\mu C_A\,\delta_{K_1}^{-1}\delta_{K_2}^{-1}.\end{aligned}$$

Hence $\{Q_P(A;\cdot,\cdot)\}_{P\ge1}$ is uniformly bounded on $K_1\times K_2$. Since every compact subset of Ω^2 is contained in $K_1\times K_2$ for suitable compact $K_1, K_2 \subset \Omega$, the family is locally uniformly bounded on Ω^2.

For each P, analyticity of q_P and Q_P follows from the analyticity of the resolvent $z \mapsto (zI+\hat{\Sigma}_P)^{-1}$ on Ω, together with finite-dimensional trace and expectation operations. Local uniform boundedness therefore implies normality by Montel's theorem. Consequently every

sequence admits a subsequence converging uniformly on compact subsets of the relevant domain. This is precisely relative compactness in the compact-open topology. □

We have

$$
\begin{aligned}
q(A\Sigma_\mu; z_2) &= P^{-1} \operatorname{tr} E[(z_1 I + \hat{\Sigma})(z_1 I + \hat{\Sigma})^{-1} \Sigma_\mu (z_2 I + \hat{\Sigma})^{-1} A] \\
&= z_1 P^{-1} \operatorname{tr} E[(z_1 I + \hat{\Sigma})^{-1} \Sigma_\mu (z_2 I + \hat{\Sigma})^{-1} A] \\
&+ P^{-1} \operatorname{tr} E[\hat{\Sigma}(z_1 I + \hat{\Sigma})^{-1} \Sigma_\mu (z_2 I + \hat{\Sigma})^{-1} A] \,.
\end{aligned} \tag{H.37}
$$

Now, by Lemma H.7,

$$
\begin{aligned}
&P^{-1} \operatorname{tr} E[\hat{\Sigma}(z_1 I + \hat{\Sigma})^{-1} \Sigma_\mu (z_2 I + \hat{\Sigma})^{-1} A] \\
&= P^{-1} \operatorname{tr} E[R_t R_t' (z_1 I + \hat{\Sigma})^{-1} \Sigma_\mu (z_2 I + \hat{\Sigma})^{-1} A] \\
&= P^{-1} \operatorname{tr} E[(1 + \hat{\xi}_t(z_1))^{-1} R_t R_t' (z_1 I + \hat{\Psi}_{T,t})^{-1} \Sigma_\mu \Big((z_2 I + \hat{\Psi}_{T,t})^{-1} \\
&- (1 + \hat{\xi}_t(z_2))^{-1} T^{-1} (z_2 I + \hat{\Psi}_{T,t})^{-1} R_t R_t' (z_2 I + \hat{\Psi}_{T,t})^{-1} \Big) A] \\
&= P^{-1} \operatorname{tr} E[(1 + \hat{\xi}_t(z_1))^{-1} R_t R_t' (z_1 I + \hat{\Psi}_{T,t})^{-1} \Sigma_\mu \Big((z_2 I + \hat{\Psi}_{T,t})^{-1} \Big) A] \\
&- P^{-1} \operatorname{tr} E[(1 + \hat{\xi}_t(z_1))^{-1} R_t R_t' (z_1 I + \hat{\Psi}_{T,t})^{-1} \Sigma_\mu \Big((1 + \hat{\xi}_t(z_2))^{-1} T^{-1} (z_2 I + \hat{\Psi}_{T,t})^{-1} R_t R_t' (z_2 I + \hat{\Psi}_{T,t})^{-1} \Big) A] \\
&\approx (1 + \xi(z_1))^{-1} P^{-1} \operatorname{tr} E[\Sigma (z_1 I + \hat{\Sigma})^{-1} \Sigma_\mu \Big((z_2 I + \hat{\Sigma})^{-1} \Big) A] \\
&- (1 + \xi(z_1))^{-1} (1 + \xi(z_2))^{-1} P^{-1} \operatorname{tr} E[R_t \underbrace{R_t' (z_1 I + \hat{\Psi}_{T,t})^{-1} \Sigma_\mu T^{-1} (z_2 I + \hat{\Psi}_{T,t})^{-1} R_t}_{\approx T^{-1} \operatorname{tr}(\Sigma (z_1 I + \hat{\Sigma})^{-1} \Sigma_\mu (z_2 I + \hat{\Sigma})^{-1})} R_t' (z_2 I + \hat{\Psi}_{T,t})^{-1} A] \\
&\approx (1 + \xi(z_1))^{-1} Q(A\Sigma; z_1; z_2) \\
&- (1 + \xi(z_1))^{-1} (1 + \xi(z_2))^{-1} c Q(\Sigma; z_1; z_2) P^{-1} \operatorname{tr} E[R_t R_t' (z_2 I + \hat{\Psi}_{T,t})^{-1} A] \\
&\approx (1 + \xi(z_1))^{-1} Q(A\Sigma; z_1; z_2) \\
&- (1 + \xi(z_1))^{-1} (1 + \xi(z_2))^{-1} c Q(\Sigma; z_1; z_2) P^{-1} \operatorname{tr} E[(z_2 I + \hat{\Psi}_{T,t})^{-1} A\Sigma] \\
&\approx (1 + \xi(z_1))^{-1} Q(A\Sigma; z_1; z_2) \\
&- (1 + \xi(z_1))^{-1} (1 + \xi(z_2))^{-1} c Q(\Sigma; z_1; z_2) q(A\Sigma; z_2) \,.
\end{aligned}
$$

(H.38)

Here and below, $q(\cdot;\cdot)$ always denotes the two-argument trace functional just defined; the one-argument $q(z) = \mu'(zI + \hat{\Sigma})^{-1}\mu$ of Lemma E.1 is a different object, related to it by $q(\Sigma_\mu; z) \approx q(z)$ under Assumption 2. Substituting into (H.37), we get

$$\begin{aligned} q(A\Sigma_\mu; z_2) \;&\approx\; z_1 Q(A; z_1, z_2) \\ &+\; (1+\xi(z_1))^{-1} Q(A\Sigma; z_1; z_2) \\ &-\; (1+\xi(z_1))^{-1}(1+\xi(z_2))^{-1} c Q(\Sigma; z_1; z_2) q(A\Sigma; z_2)\,. \end{aligned} \tag{H.39}$$

Then, (H.39) takes the form

$$Q(A; z_1, z_2) \;\approx\; z_1^{-1} q(A\Sigma_\mu; z_2) - \hat{Z}_*(z_1)^{-1} Q(A\Sigma; z_1; z_2) + \hat{Z}_*(z_1)^{-1}(1+\xi(z_2))^{-1} c Q(\Sigma; z_1; z_2) q(A\Sigma; z_2)\,. \tag{H.40}$$

Iterating forward we get

$$\begin{aligned} Q(A; z_1, z_2) \;&\approx\; z_1^{-1} q(A\Sigma_\mu; z_2) - \hat{Z}_*(z_1)^{-1} Q(A\Sigma; z_1; z_2) + \hat{Z}_*(z_1)^{-1}(1+\xi(z_2))^{-1} c Q(\Sigma; z_1; z_2) q(A\Sigma; z_2) \\ Q(A\Sigma; z_1, z_2) \;&\approx\; z_1^{-1} q(A\Sigma\Sigma_\mu; z_2) - \hat{Z}_*(z_1)^{-1} Q(A\Sigma^2; z_1; z_2) + \hat{Z}_*(z_1)^{-1}(1+\xi(z_2))^{-1} c Q(\Sigma; z_1; z_2) q(A\Sigma^2; z_2) \end{aligned} \tag{H.41}$$

so that

$$\begin{aligned} Q(A; z_1, z_2) \;&\approx\; z_1^{-1} q(A\Sigma_\mu; z_2) - \hat{Z}_*(z_1)^{-1} Q(A\Sigma; z_1; z_2) + \hat{Z}_*(z_1)^{-1}(1+\xi(z_2))^{-1} c Q(\Sigma; z_1; z_2) q(A\Sigma; z_2) \\ &\approx\; z_1^{-1} q(A\Sigma_\mu; z_2) - \hat{Z}_*(z_1)^{-1} \\ &\times \left(z_1^{-1} q(A\Sigma\Sigma_\mu; z_2) - \hat{Z}_*(z_1)^{-1} Q(A\Sigma^2; z_1; z_2) + \hat{Z}_*(z_1)^{-1}(1+\xi(z_2))^{-1} c Q(\Sigma; z_1; z_2) q(A\Sigma^2; z_2) \right) \\ &+ \hat{Z}_*(z_1)^{-1}(1+\xi(z_2))^{-1} c Q(\Sigma; z_1; z_2) q(A\Sigma; z_2)\,. \end{aligned}$$

(H.42)

Note now that $q(\cdot)$, $Q(\cdot)$ are linear in A. When $z_1 > (1+\varepsilon)\sup_P \|\Sigma_P\|$, $\hat{Z}_*(z_1) > z_1$ and iterating this approximate identity forward, we get

$$\begin{aligned} Q(A; z_1, z_2) &\approx \sum_{i=0}^{\infty} (-\hat{Z}_*(z_1)^{-1})^i \left(z_1^{-1} q(A\Sigma^i\Sigma_\mu; z_2) + \hat{Z}_*(z_1)^{-1}(1+\xi(z_2))^{-1} c Q(\Sigma; z_1; z_2) q(A\Sigma^{i+1}; z_2) \right) \\ &= z_1^{-1} q(A(I + \hat{Z}_*(z_1)^{-1}\Sigma)^{-1}\Sigma_\mu; z_2) + \hat{Z}_*(z_1)^{-1}(1+\xi(z_2))^{-1} c Q(\Sigma; z_1; z_2) q(A(I + \hat{Z}_*(z_1)^{-1}\Sigma)^{-1}\Sigma; z_2) \\ &= z_1^{-1}\hat{Z}_*(z_1) q(A(\hat{Z}_*(z_1) I + \Sigma)^{-1}\Sigma_\mu; z_2) + (1+\xi(z_2))^{-1} c Q(\Sigma; z_1; z_2) q(A(\hat{Z}_*(z_1) I + \Sigma)^{-1}\Sigma; z_2) \end{aligned} \tag{H.43}$$

for all $z_1 > \sup_P \|\Sigma_P\|$. We now show that (H.43) holds for all $z \in \Omega$. Indeed, $\hat{Z}_*(z_1)$ maps Ω to Ω and is analytic on Ω. Hence, Lemma H.8 implies that both the left and the right-hand sides of (H.43) are compact. Our goal is to show that this approximate identity holds for all $z \in \Omega$. Suppose the contrary. Let A_P be a bounded sequence such that the difference between the left and the right-hand side stays uniformly bounded away from zero for some $z_1 \in \Omega$. By compactness, passing to a subsequence, we get two limiting functions analytic on Ω and coinciding for $z_1 > \sup_P \|\Sigma_P\|$. But then, they coincide everywhere by analyticity, leading to a contradiction.

Now, by the Master Theorem D.2,

$$q(A; z) \approx (1+\xi(z))\, P^{-1} \operatorname{tr}(A(\hat{Z}_*(z) I + \Sigma)^{-1})\,. \tag{H.44}$$

Thus,

$$
\begin{aligned}
&Q(A; z_1, z_2) \\
&\approx\ z_1^{-1}\hat{Z}_*(z_1)(1+\xi(z_2))\, P^{-1}\operatorname{tr}(A(\hat{Z}_*(z_1)I+\Sigma)^{-1}\Sigma_\mu(\hat{Z}_*(z_2)I+\Sigma)^{-1}) \\
&+ (1+\xi(z_2))^{-1} cQ(\Sigma; z_1; z_2)(1+\xi(z_2))\, P^{-1}\operatorname{tr}(A(\hat{Z}_*(z_1)I+\Sigma)^{-1}\Sigma(\hat{Z}_*(z_2)I+\Sigma)^{-1}) \\
&=\ (1+\xi(z_1))(1+\xi(z_2))\, P^{-1}\operatorname{tr}(A(\hat{Z}_*(z_1)I+\Sigma)^{-1}\Sigma_\mu(\hat{Z}_*(z_2)I+\Sigma)^{-1}) \\
&+\ cQ(\Sigma; z_1; z_2) P^{-1}\operatorname{tr}(A(\hat{Z}_*(z_1)I+\Sigma)^{-1}\Sigma(\hat{Z}_*(z_2)I+\Sigma)^{-1})
\end{aligned}
\tag{H.45}
$$

Taking $A = \Sigma_\mu$ and substituting this into (H.35), we get

$$
\begin{aligned}
&0.5P\mathrm{Cov}[\hat{\mathcal{C}}(z_1), \hat{\mathcal{C}}(z_2)] \\
&\approx\ cQ(\Sigma; z_1; z_2) P^{-1}\operatorname{tr}(\Sigma_\mu(\hat{Z}_*(z_1)I+\Sigma)^{-1}\Sigma(\hat{Z}_*(z_2)I+\Sigma)^{-1})
\end{aligned}
\tag{H.46}
$$

Here, by Lemma E.7(ii),

$$
Q(\Sigma; z_1; z_2)\ \approx\ (1+\hat{\xi}(z_1))(1+\hat{\xi}(z_2))\frac{z_2(\hat{\Theta}(z_2)\ -\ \hat{\xi}(z_2)) - z_1(\hat{\Theta}(z_1)\ -\ \hat{\xi}(z_1))}{z_2 - z_1}, \tag{H.47}
$$

and it remains to compute

$$
\begin{aligned}
&P^{-1}\operatorname{tr}(\Sigma_\mu(\hat{Z}_*(z_1)I+\Sigma)^{-1}\Sigma(\hat{Z}_*(z_2)I+\Sigma)^{-1}) \\
&=\ P^{-1}\operatorname{tr}(\Sigma_\mu\Sigma(\hat{Z}_*(z_2)I+\Sigma)^{-1}(\hat{Z}_*(z_1)I+\Sigma)^{-1}) \\
&=\ \frac{1}{\hat{Z}_*(z_2)-\hat{Z}_*(z_1)}\left(P^{-1}\operatorname{tr}(\Sigma_\mu\Sigma(\hat{Z}_*(z_1)I+\Sigma)^{-1}) - P^{-1}\operatorname{tr}(\Sigma_\mu\Sigma(\hat{Z}_*(z_2)I+\Sigma)^{-1})\right).
\end{aligned}
\tag{H.48}
$$

Here,

$$\begin{aligned}
&P^{-1}\operatorname{tr}(\Sigma_\mu\Sigma(\hat{Z}_*(z_1)I+\Sigma)^{-1})\\
&= P^{-1}\operatorname{tr}(\Sigma_\mu(\hat{Z}_*(z_1)I+\Sigma-\hat{Z}_*(z_1)I)(\hat{Z}_*(z_1)I+\Sigma)^{-1})\\
&= P^{-1}\operatorname{tr}(\Sigma_\mu)-\hat{Z}_*(z_1)P^{-1}\operatorname{tr}(\Sigma_\mu(\hat{Z}_*(z_1)I+\Sigma)^{-1}) \qquad \text{(H.49)}\\
&\approx \|\mu\|^2-z_1\mu'(z_1I+\hat{\Sigma})^{-1}\mu\\
&\approx \|\mu\|^2-z_1(\hat{\Theta}(z_1)\ -\ \hat{\xi}(z_1))
\end{aligned}$$

Thus,

$$\begin{aligned}
&P^{-1}\operatorname{tr}(\Sigma_\mu(\hat{Z}_*(z_1)I+\Sigma)^{-1}\Sigma(\hat{Z}_*(z_2)I+\Sigma)^{-1})\\
&= \frac{1}{\hat{Z}_*(z_2)-\hat{Z}_*(z_1)}\left(P^{-1}\operatorname{tr}(\Sigma_\mu\Sigma(\hat{Z}_*(z_1)I+\Sigma)^{-1})-P^{-1}\operatorname{tr}(\Sigma_\mu\Sigma(\hat{Z}_*(z_2)I+\Sigma)^{-1})\right) \qquad \text{(H.50)}\\
&\approx \frac{1}{\hat{Z}_*(z_2)-\hat{Z}_*(z_1)}\left(z_2(\hat{\Theta}(z_2)\ -\ \hat{\xi}(z_2))-z_1(\hat{\Theta}(z_1)\ -\ \hat{\xi}(z_1))\right)
\end{aligned}$$

Furthermore, by Lemma E.1(ii), available here because Assumption 2 implies Condition (R),

$$\begin{aligned}
Q(\Sigma;z_1;z_2) &\approx (1+\hat{\xi}(z_1))(1+\hat{\xi}(z_2))\frac{z_2q(\Sigma_\mu;z_2)-z_1q(\Sigma_\mu;z_1)}{z_2-z_1}\\
&\underbrace{\approx}_{\textit{Lemma E.6}} (1+\hat{\xi}(z_1))(1+\hat{\xi}(z_2))\frac{z_2(\hat{\Theta}(z_2)\ -\ \hat{\xi}(z_2))-z_1(\hat{\Theta}(z_1)\ -\ \hat{\xi}(z_1))}{z_2-z_1} \qquad \text{(H.51)}
\end{aligned}$$

Thus, by (H.46),

$$
\begin{aligned}
&0.5P\mathrm{Cov}[\hat{\mathcal{C}}(z_1),\hat{\mathcal{C}}(z_2)]\\
&\approx\ cQ(\Sigma;z_1;z_2)P^{-1}\,\mathrm{tr}(\Sigma_\mu(\hat{Z}_*(z_1)I+\Sigma)^{-1}\Sigma(\hat{Z}_*(z_2)I+\Sigma)^{-1})\\
&\approx\ c(1+\hat{\xi}(z_1))(1+\hat{\xi}(z_2))\frac{z_2(\hat{\Theta}(z_2)\ -\ \hat{\xi}(z_2))-z_1(\hat{\Theta}(z_1)\ -\ \hat{\xi}(z_1))}{z_2-z_1}\\
&\times\frac{1}{\hat{Z}_*(z_2)-\hat{Z}_*(z_1)}\left(z_2(\hat{\Theta}(z_2)\ -\ \hat{\xi}(z_2))-z_1(\hat{\Theta}(z_1)\ -\ \hat{\xi}(z_1))\right).
\end{aligned}
\tag{H.52}
$$

The proof of Lemma H.6 is complete. □

Lemma H.9 *We have*

$$
\begin{aligned}
&\lim_{z_2\to z_1}(1+\hat{\xi}(z_1))(1+\hat{\xi}(z_2))\frac{z_2(\hat{\Theta}(z_2)-\hat{\xi}(z_2))-z_1(\hat{\Theta}(z_1)-\hat{\xi}(z_1))}{z_2-z_1}\\
&=(1+\hat{\xi}(z_1))^2\left(z_1(\hat{\Theta}'(z_1)-\hat{\xi}'(z_1))+(\hat{\Theta}(z_1)-\hat{\xi}(z_1))\right)\\
&\lim_{z_2\to z_1}(1+\hat{\xi}(z_1))(1+\hat{\xi}(z_2))\frac{z_2\hat{\xi}(z_2)-z_1\hat{\xi}(z_1)}{z_2-z_1}\\
&=(1+\hat{\xi}(z_1))^2\left(z_1\hat{\xi}'(z_1)+\hat{\xi}(z_1)\right)\\
&\lim_{z_2\to z_1}\left((1+\hat{\xi}(z_1))(1+\hat{\xi}(z_2))\frac{\left(z_2(\hat{\Theta}(z_2)\ -\ \hat{\xi}(z_2))-z_1(\hat{\Theta}(z_1)\ -\ \hat{\xi}(z_1))\right)^2}{z_2-z_1}\right.\\
&\left.\times\ \frac{1}{\hat{Z}_*(z_2)-\hat{Z}_*(z_1)}\right)\\
&=(1+\hat{\xi}(z_1))^2\left(z_1(\hat{\Theta}'(z_1)-\hat{\xi}'(z_1))+(\hat{\Theta}(z_1)-\hat{\xi}(z_1))\right)^2\frac{1}{\hat{Z}_*'(z_1)}
\end{aligned}
\tag{H.53}
$$

Proof of Lemma H.9. $\hat{\Theta}$ and $\hat{\xi}(z)$ are differentiable at z_1. Moreover, we know that

$\lim_{z_2\to z_1}\hat{\xi}(z_2), \lim_{z_2\to z_1}\hat{\Theta}(z_2)$ exist and are finite. Thus,

$$\begin{aligned}
&\lim_{z_2\to z_1}(1+\hat{\xi}(z_1))(1+\hat{\xi}(z_2))\frac{z_2(\hat{\Theta}(z_2)-\hat{\xi}(z_2))-z_1(\hat{\Theta}(z_1)-\hat{\xi}(z_1))}{z_2-z_1}\\
&=\lim_{z_2\to z_1}\Big((1+\hat{\xi}(z_1))(1+\hat{\xi}(z_2))\Big)\lim_{z_2\to z_1}\frac{z_2(\hat{\Theta}(z_2)-\hat{\xi}(z_2))-z_1(\hat{\Theta}(z_1)-\hat{\xi}(z_1))}{z_2-z_1}\\
&=(1+\hat{\xi}(z_1))^2\lim_{z_2\to z_1}\frac{z_2(\hat{\Theta}(z_2)-\hat{\xi}(z_2))-z_1(\hat{\Theta}(z_1)-\hat{\xi}(z_1))}{z_2-z_1}\\
&=(1+\hat{\xi}(z_1))^2\Big(z_1(\hat{\Theta}'(z_1)-\hat{\xi}'(z_1))+(\hat{\Theta}(z_1)-\hat{\xi}(z_1))\Big)
\end{aligned}\tag{H.54}$$

Similarly, we have

$$\begin{aligned}
&\lim_{z_2\to z_1}(1+\hat{\xi}(z_1))(1+\hat{\xi}(z_2))\frac{z_2\hat{\xi}(z_2)-z_1\hat{\xi}(z_1)}{z_2-z_1}\\
&=\lim_{z_2\to z_1}(1+\hat{\xi}(z_1))(1+\hat{\xi}(z_2))\lim_{z_2\to z_1}\frac{z_2\hat{\xi}(z_2)-z_1\hat{\xi}(z_1)}{z_2-z_1}\\
&=(1+\hat{\xi}(z_1))^2\lim_{z_2\to z_1}\frac{z_2\hat{\xi}(z_2)-z_1\hat{\xi}(z_1)}{z_2-z_1}\\
&=(1+\hat{\xi}(z_1))^2\Big(z_1\hat{\xi}'(z_1)+\hat{\xi}(z_1)\Big)
\end{aligned}\tag{H.55}$$

Lastly,

$$\begin{aligned}
&\lim_{z_2\to z_1}\left((1+\hat{\xi}(z_1))(1+\hat{\xi}(z_2))\frac{\Big(z_2(\hat{\Theta}(z_2)\ -\ \hat{\xi}(z_2))-z_1(\hat{\Theta}(z_1)\ -\ \hat{\xi}(z_1))\Big)^2}{z_2-z_1}\right.\\
&\left.\times\ \frac{1}{\hat{Z}_*(z_2)-\hat{Z}_*(z_1)}\right)\\
&=\ (1+\hat{\xi}(z_1))^2\lim_{z_2\to z_1}\left(\Big(\frac{\Big(z_2(\hat{\Theta}(z_2)\ -\ \hat{\xi}(z_2))-z_1(\hat{\Theta}(z_1)\ -\ \hat{\xi}(z_1))\Big)}{z_2-z_1}\Big)^2\right.\\
&\left.\times\ \frac{z_2-z_1}{\hat{Z}_*(z_2)-\hat{Z}_*(z_1)}\right)\\
&=\ (1+\hat{\xi}(z_1))^2\Big(z_1(\hat{\Theta}'(z_1)-\hat{\xi}'(z_1))+(\hat{\Theta}(z_1)-\hat{\xi}(z_1))\Big)^2\frac{1}{\hat{Z}_*'(z_1)}\,.
\end{aligned}\tag{H.56}$$

□

Proof of Theorem H.3. Recall that $\hat{\Theta}^d = (\hat{\Theta} - \hat{\xi})/(1+\hat{\xi})$, so that (H.22) gives the exact identity

$$\Delta(z) \;=\; \hat{\Theta}^d(z) - \mu'(\hat{Z}_*(z)I + \Sigma)^{-1}\mu \;=\; \frac{\hat{\mathcal{C}}(z) \;+\; 2\hat{\mathcal{I}}(z) \;+\; (\hat{W}(z) - \hat{\xi}(z))}{1+\hat{\xi}(z)}. \tag{H.57}$$

Because $\hat{\xi}$ is a normalized trace functional, Lemma H.1 lets us treat the denominators as deterministic at the $T^{-1/2}$ scale. The three numerator terms have asymptotically vanishing cross-covariances by Lemma H.4: $\hat{\mathcal{C}}$ is a centered quadratic form in ν, $\hat{W} - \hat{\xi}$ is a centered quadratic form in $\bar{X}$, and $\hat{\mathcal{I}}$ is bilinear in the two, so that under Assumptions 1 and 2 (in particular $E[\nu_i^3] = 0$) each pairwise covariance is $o(T^{-1})$. Hence

$$\begin{aligned} &T\mathrm{Cov}(\Delta(z_1), \Delta(z_2)) \\ &\approx \frac{T\mathrm{Cov}(\hat{\mathcal{C}}(z_1), \hat{\mathcal{C}}(z_2)) \;+\; 4TE[\hat{\mathcal{I}}(z_1)\hat{\mathcal{I}}(z_2)] \;+\; TE[(\hat{W}(z_1) - \hat{\xi}(z_1))(\hat{W}(z_2) - \hat{\xi}(z_2))]}{(1+\hat{\xi}(z_1))(1+\hat{\xi}(z_2))}. \end{aligned} \tag{H.58}$$

Each of the three numerator terms carries a factor $(1+\hat{\xi}(z_1))(1+\hat{\xi}(z_2))$, which the denominator cancels. Indeed, $T\mathrm{Cov}(\hat{\mathcal{C}}(z_1), \hat{\mathcal{C}}(z_2)) = c^{-1}P\mathrm{Cov}(\hat{\mathcal{C}}(z_1), \hat{\mathcal{C}}(z_2))$ since $P/T \to c$, so (H.31) gives

$$\frac{T\mathrm{Cov}(\hat{\mathcal{C}}(z_1), \hat{\mathcal{C}}(z_2))}{(1+\hat{\xi}(z_1))(1+\hat{\xi}(z_2))} \;\approx\; 2\left(\frac{z_2(\hat{\Theta}(z_2) \;-\; \hat{\xi}(z_2)) - z_1(\hat{\Theta}(z_1) \;-\; \hat{\xi}(z_1))}{z_2 - z_1}\right)^2 \frac{z_2 - z_1}{\hat{Z}_*(z_2) - \hat{Z}_*(z_1)}, \tag{H.59}$$

and (H.30) gives

$$\frac{TE[(\hat{W}(z_1) - \hat{\xi}(z_1))(\hat{W}(z_2) - \hat{\xi}(z_2))]}{(1+\hat{\xi}(z_1))(1+\hat{\xi}(z_2))} \;\approx\; 2\,\frac{z_2\hat{\xi}(z_2) - z_1\hat{\xi}(z_1)}{z_2 - z_1}. \tag{H.60}$$

Finally, by (H.28) and Lemma E.7(ii),

$$\begin{aligned} TE[\hat{\mathcal{I}}(z_1)\,\hat{\mathcal{I}}(z_2)] \;&\approx\; \mu'(z_1 I+\hat{\Sigma})^{-1}\Sigma(z_2 I+\hat{\Sigma})^{-1}\mu \\ &\approx\; (1+\hat{\xi}(z_1))(1+\hat{\xi}(z_2))\frac{z_2(\hat{\Theta}(z_2)\;-\;\hat{\xi}(z_2))-z_1(\hat{\Theta}(z_1)\;-\;\hat{\xi}(z_1))}{z_2-z_1}\,, \end{aligned} \tag{H.61}$$

so that the middle term contributes

$$\frac{4TE[\hat{\mathcal{I}}(z_1)\,\hat{\mathcal{I}}(z_2)]}{(1+\hat{\xi}(z_1))(1+\hat{\xi}(z_2))} \;\approx\; 4\,\frac{z_2(\hat{\Theta}(z_2)\;-\;\hat{\xi}(z_2))-z_1(\hat{\Theta}(z_1)\;-\;\hat{\xi}(z_1))}{z_2-z_1}\,. \tag{H.62}$$

Adding the three displays gives exactly $\hat{\sigma}(z_1,z_2)$. The proof of Theorem H.3 is complete. □

H.6 Tightness

The CLT for the higher-order debiased statistics requires upgrading convergence in $C(I)$ for $A(z)=T^{1/2}(\hat{\Theta}(z)-\hat{\xi}(z))$ to convergence of all derivatives. We deliver this upgrade in two forms below.

Theorem H.4 (Convergence in C^1 from Sobolev-type moment bounds) *Let $I=[a,b]\subset\mathbb{R}$ be compact. For each T, let $\Theta_T : I\to\mathbb{R}$ have almost surely C^2 sample paths. Suppose that*

$$\Theta_T \Rightarrow G \qquad \text{in } C(I),$$

where G is a mean-zero Gaussian process with covariance kernel

$$\Sigma(z_1,z_2)=\mathbb{E}[G(z_1)G(z_2)].$$

Assume also that $\Sigma\in C^2(I\times I)$, so that G admits a version with almost surely C^1 sample

paths. Finally, assume that for some $p > 1$,

$$\sup_T \sup_{z \in I} \mathbb{E}|\Theta_T'(z)|^p < \infty, \qquad \sup_T \sup_{z \in I} \mathbb{E}|\Theta_T''(z)|^p < \infty.$$

Then $\{\Theta_T\}$ *is tight in* $C^1(I)$*, and*

$$\Theta_T \Rightarrow G \qquad \text{in } C^1(I).$$

In particular, for every $z \in I$,

$$\Theta_T'(z) \Rightarrow G'(z),$$

and G' *is a mean-zero Gaussian process with covariance kernel*

$$\mathrm{Cov}\big(G'(z_1), G'(z_2)\big) = \frac{\partial^2}{\partial z_1 \partial z_2} \Sigma(z_1, z_2).$$

Moreover, the joint process (Θ_T, Θ_T') *converges weakly in* $C(I) \times C(I)$ *to* (G, G').

Proof of Theorem H.4. We first prove tightness of the derivative process $\{\Theta_T'\}$ in $C(I)$. Fix $z_0 \in I$. By assumption,

$$\sup_T \mathbb{E}|\Theta_T'(z_0)|^p < \infty,$$

so $\{\Theta_T'(z_0)\}$ is tight.

Next, for any $z, w \in I$, the fundamental theorem of calculus and Hölder's inequality give

$$|\Theta_T'(z) - \Theta_T'(w)| \le \int_{z \wedge w}^{z \vee w} |\Theta_T''(u)|\, du \le |z - w|^{1-1/p} \left(\int_I |\Theta_T''(u)|^p \, du \right)^{1/p}.$$

Therefore,

$$\sup_{|z-w|\le\delta} |\Theta_T'(z) - \Theta_T'(w)|^p \le \delta^{p-1} \int_I |\Theta_T''(u)|^p \, du.$$

Taking expectations and using Fubini,

$$\sup_T \mathbb{E}\left[\sup_{|z-w|\le\delta} |\Theta_T'(z) - \Theta_T'(w)|^p \right] \le \delta^{p-1} \int_I \sup_T \mathbb{E}|\Theta_T''(u)|^p \, du \le |I| \, C \, \delta^{p-1},$$

where

$$C := \sup_T \sup_{u\in I} \mathbb{E}|\Theta_T''(u)|^p < \infty.$$

Hence, by Markov's inequality, for every $\varepsilon > 0$,

$$\sup_T \mathbb{P}\left(\sup_{|z-w|\le\delta} |\Theta_T'(z) - \Theta_T'(w)| > \varepsilon \right) \le \varepsilon^{-p} |I| C \delta^{p-1} \longrightarrow 0$$

as $\delta \downarrow 0$. Thus $\{\Theta_T'\}$ is equicontinuous in probability. Together with tightness at z_0, the Arzelà–Ascoli tightness criterion implies that $\{\Theta_T'\}$ is tight in $C(I)$.

Since $\Theta_T \Rightarrow G$ in $C(I)$ by assumption, the sequence (Θ_T, Θ_T') is tight in $C(I) \times C(I)$. Let (F, H) be any subsequential weak limit. Along this subsequence,

$$\Theta_T \Rightarrow F, \qquad \Theta_T' \Rightarrow H,$$

with $F \overset{d}{=} G$.

We now identify H as the derivative of F. For every $z, w \in I$,

$$\Theta_T(z) - \Theta_T(w) = \int_w^z \Theta_T'(u) \, du.$$

The map

$$(f,h) \mapsto f(z) - f(w) - \int_w^z h(u)\,du$$

is continuous on $C(I) \times C(I)$. Therefore, passing to the subsequential limit,

$$F(z) - F(w) = \int_w^z H(u)\,du \qquad \text{almost surely.}$$

Hence F is almost surely continuously differentiable and $F' = H$.

Since $F \overset{d}{=} G$ and G has a C^1 version, every subsequential limit of (Θ_T, Θ_T') is equal in distribution to (G, G'). Therefore the whole sequence converges:

$$(\Theta_T, \Theta_T') \Rightarrow (G, G') \qquad \text{in } C(I) \times C(I),$$

which is equivalent to

$$\Theta_T \Rightarrow G \qquad \text{in } C^1(I).$$

Finally, since G is Gaussian and $\Sigma \in C^2(I \times I)$, the derivative process G' is again Gaussian, and differentiation under the covariance gives

$$\operatorname{Cov}\big(G'(z_1), G'(z_2)\big) = \frac{\partial^2}{\partial z_1 \partial z_2}\Sigma(z_1, z_2).$$

This proves the claim. □

Theorem H.5 (Analytic tightness and convergence of derivatives) *Let $D \subset \mathbb{C}$ be open, and let $\Theta_T : D \to \mathbb{C}$ be random analytic functions. Suppose that, for every compact*

set $K \Subset D$, there exist $p > 1$ and a compact set $K^+ \Subset D$ with $K \subset \mathrm{int}(K^+)$ such that

$$\sup_T \sup_{z \in K^+} \mathbb{E}|\Theta_T(z)|^p < \infty.$$

Assume moreover that

$$\Theta_T \Rightarrow G \qquad \text{in } C(K^+)$$

for every compact $K^+ \Subset D$, where G has almost surely analytic sample paths on D.

Then, for every integer $m \geq 0$ and every compact $K \Subset D$,

$$\Theta_T^{(m)} \Rightarrow G^{(m)} \qquad \text{in } C(K).$$

Equivalently,

$$\Theta_T \Rightarrow G \qquad \text{in } \mathcal{H}(D),$$

where $\mathcal{H}(D)$ is the space of analytic functions endowed with the topology of uniform convergence on compact subsets. In particular, the vector process

$$\left(\Theta_T, \Theta_T', \ldots, \Theta_T^{(m)}\right)$$

converges jointly to

$$\left(G, G', \ldots, G^{(m)}\right)$$

in $C(K)^{m+1}$.

Proof of Theorem H.5. Fix $K \Subset D$. Choose $r > 0$ such that

$$K_r := \{w \in \mathbb{C} : \operatorname{dist}(w, K) \le r\} \Subset D.$$

For $z \in K$, Cauchy's integral formula gives

$$\Theta_T^{(m)}(z) = \frac{m!}{2\pi i} \int_{|\zeta - z| = r} \frac{\Theta_T(\zeta)}{(\zeta - z)^{m+1}} \, d\zeta.$$

Hence

$$|\Theta_T^{(m)}(z)| \le \frac{m!}{r^m} \sup_{\zeta \in K_r} |\Theta_T(\zeta)|.$$

In particular, if we can control the supremum of Θ_T on K_r, then we control all derivatives on K.

To obtain a moment bound for this supremum from pointwise moment bounds, cover K_r by finitely many disks $B(z_j, \rho)$ with $\overline{B(z_j, 2\rho)} \Subset D$. For $w \in B(z_j, \rho)$, the subharmonicity of $|\Theta_T|^p$ gives

$$|\Theta_T(w)|^p \le \frac{1}{\pi\rho^2} \int_{B(w,\rho)} |\Theta_T(u)|^p \, du \le \frac{1}{\pi\rho^2} \int_{B(z_j,2\rho)} |\Theta_T(u)|^p \, du.$$

Therefore

$$\sup_{w \in K_r} |\Theta_T(w)|^p \le \sum_{j=1}^{N} \frac{1}{\pi\rho^2} \int_{B(z_j,2\rho)} |\Theta_T(u)|^p \, du.$$

Taking expectations and using Fubini,

$$\sup_T \mathbb{E} \sup_{w \in K_r} |\Theta_T(w)|^p \le C_{K,r,\rho} \sup_T \sup_{u \in K^+} \mathbb{E}|\Theta_T(u)|^p < \infty,$$

where K^+ is a compact set containing all the disks $B(z_j, 2\rho)$.

Thus, for every $m \geq 0$,

$$\sup_T \mathbb{E}\|\Theta_T^{(m)}\|_{C(K)}^p < \infty.$$

Moreover, applying Cauchy's estimate to one further derivative,

$$|\Theta_T^{(m)}(z) - \Theta_T^{(m)}(w)| \leq |z - w| \sup_{\xi \in K} |\Theta_T^{(m+1)}(\xi)|,$$

up to replacing K by a slightly larger compact set. Hence

$$\sup_T \mathbb{E}\left[\sup_{z \neq w \in K} \frac{|\Theta_T^{(m)}(z) - \Theta_T^{(m)}(w)|^p}{|z - w|^p}\right] < \infty.$$

By Markov's inequality, $\{\Theta_T^{(m)}\}_T$ is equicontinuous in probability on K. Together with the uniform sup-norm moment bound, this implies tightness in $C(K)$.

It remains to identify the limit. Let a subsequence satisfy

$$\left(\Theta_T, \Theta_T', \ldots, \Theta_T^{(m)}\right) \Rightarrow (F_0, F_1, \ldots, F_m) \qquad \text{in } C(K)^{m+1}.$$

Since $\Theta_T \Rightarrow G$, we have $F_0 \stackrel{d}{=} G$. For every closed contour $\gamma \subset K^\circ$,

$$\int_\gamma \Theta_T(z)\, dz = 0,$$

and by continuity of the contour integral,

$$\int_\gamma F_0(z)\, dz = 0.$$

Thus F_0 is analytic. Also, for every $z \in K^\circ$ and every small circle $|\zeta - z| = \rho$ contained in

K°,

$$\Theta_T^{(m)}(z) = \frac{m!}{2\pi i} \int_{|\zeta - z| = \rho} \frac{\Theta_T(\zeta)}{(\zeta - z)^{m+1}} \, d\zeta.$$

Passing to the limit gives

$$F_m(z) = \frac{m!}{2\pi i} \int_{|\zeta - z| = \rho} \frac{F_0(\zeta)}{(\zeta - z)^{m+1}} \, d\zeta = F_0^{(m)}(z).$$

Since $F_0 \stackrel{d}{=} G$, we obtain $F_m \stackrel{d}{=} G^{(m)}$. Therefore, every subsequent limit is

$$(G, G', \ldots, G^{(m)}),$$

and hence the full sequence converges. □

Summarizing, we arrive at the following results.

Corollary H.6 *Let $\Omega = \mathbb{C} \backslash \mathbb{R}_-$. Let $A(z) = T^{1/2}(\hat{\Theta}(z) - \hat{\xi}(z))$. The sequence $\{A(z)\}$ satisfies the technical conditions of Theorem H.5.*

Proof of Corollary H.6. We have

$$\sup_{z \in K} \sup_T E[|A(z)|^2] < \infty \tag{H.63}$$

for any compact subset K of Ω, directly from Theorem H.3. □

Lemma H.10 (Covariance of higher derivatives of a Gaussian process) *Let $I \subset \mathbb{R}$ be an interval and let $G = \{G(z) : z \in I\}$ be a centered Gaussian process with covariance kernel*

$$\Sigma(z_1, z_2) = \mathbb{E}[G(z_1) G(z_2)]. \tag{H.64}$$

Fix integers $j, k \geq 0$ and assume that the mixed partial derivatives $\partial_{z_1}^j \partial_{z_2}^k \Sigma(z_1, z_2)$ exist and are continuous on $I \times I$.

Then the j-th and k-th mean-square derivatives $G^{(j)}(z)$ and $G^{(k)}(z)$ exist for all $z \in I$, define (jointly) Gaussian processes, and their covariance is given by

$$\operatorname{Cov}\left(G^{(j)}(z_1), G^{(k)}(z_2)\right) = \frac{\partial^{j+k}}{\partial z_1^j \, \partial z_2^k} \Sigma(z_1, z_2), \qquad z_1, z_2 \in I. \tag{H.65}$$

Proof of Lemma H.10. We work with mean-square derivatives. For clarity, first treat the case $j = k = 1$ and then iterate.

Define the first mean-square derivative of G at z by

$$G'(z) := \lim_{h \to 0} \frac{G(z+h) - G(z)}{h}, \tag{H.66}$$

where the limit is in L^2, i.e. $\mathbb{E}\big[\big(G(z+h) - G(z)\big)^2\big]/h^2$ converges.

Since G is Gaussian, any finite collection of difference quotients is jointly Gaussian. Thus, it suffices to identify their second moments and show convergence.

For $z_1, z_2 \in I$, consider

$$X_h(z_1) := \frac{G(z_1+h) - G(z_1)}{h}, \qquad Y_{h'}(z_2) := \frac{G(z_2+h') - G(z_2)}{h'}. \tag{H.67}$$

Then

$$\begin{aligned}
\mathbb{E}\big[X_h(z_1) Y_{h'}(z_2)\big] &= \frac{1}{hh'} \mathbb{E}\big[\big(G(z_1+h) - G(z_1)\big)\big(G(z_2+h') - G(z_2)\big)\big] \\
&= \frac{1}{hh'}\Big(\Sigma(z_1+h, z_2+h') - \Sigma(z_1+h, z_2) - \Sigma(z_1, z_2+h') + \Sigma(z_1, z_2)\Big).
\end{aligned}$$

By the assumed C^2-regularity of Σ (in this simple case), the two-variable difference quotient

converges to the mixed partial derivative:

$$\lim_{h,h'\to 0} \frac{\Sigma(z_1+h, z_2+h') - \Sigma(z_1+h, z_2) - \Sigma(z_1, z_2+h') + \Sigma(z_1, z_2)}{hh'} = \frac{\partial^2}{\partial z_1 \partial z_2} \Sigma(z_1, z_2). \tag{H.68}$$

Hence

$$\mathbb{E}\big[X_h(z_1) Y_{h'}(z_2)\big] \xrightarrow[h,h'\to 0]{} \partial_{z_1}\partial_{z_2}\Sigma(z_1, z_2). \tag{H.69}$$

Taking $h' = h \to 0$, this shows that the covariance of the difference quotients converges to $\partial_{z_1}\partial_{z_2}\Sigma(z_1, z_2)$. In particular, the family $\{X_h(z)\}_h$ is Cauchy in L^2 for each fixed z, so the mean-square limit $G'(z)$ exists, and

$$\mathrm{Cov}\big(G'(z_1), G'(z_2)\big) = \partial_{z_1}\partial_{z_2}\Sigma(z_1, z_2). \tag{H.70}$$

Since linear limits of Gaussian vectors are Gaussian, G' is again a Gaussian process.

For higher-order derivatives, one iterates the argument. For example, the j-th derivative at z can be defined via j-th order difference quotients (or by iterating the first derivative), and the covariance between $G^{(j)}(z_1)$ and $G^{(k)}(z_2)$ is obtained by applying the same limit argument j times in z_1 and k times in z_2. The assumed existence and continuity of $\partial_{z_1}^j \partial_{z_2}^k \Sigma$ justify passage to the limit, yielding

$$\mathrm{Cov}\big(G^{(j)}(z_1), G^{(k)}(z_2)\big) = \frac{\partial^{j+k}}{\partial z_1^j \partial z_2^k} \Sigma(z_1, z_2). \tag{H.71}$$

This completes the proof of Lemma H.10. □

We are now ready to prove Theorem H.1.

Proof of Theorem H.1. Recall that

$$\begin{aligned}\Delta(z) &= \frac{\hat{\Theta}(z)-\hat{\xi}(z)}{(1+\hat{\xi}(z;c))}-\mu'(\hat{Z}_*(z;c)I+\Sigma)^{-1}\mu \\ &= \hat{\Theta}_0^d(z)-\mathcal{B}(\hat{Z}_*(z))\,.\end{aligned} \tag{H.72}$$

By Theorem H.3,

$$T^{1/2}\Delta(z) = T^{1/2}(\hat{\Theta}^d-\mu'(\hat{Z}_*(z;c)I+\Sigma)^{-1}\mu) \tag{H.73}$$

converges to a Gaussian process for $z>0$, with a covariance kernel $\sigma(z_1,z_2)\approx\hat{\sigma}(z_1,z_2)$. Applying Lemma H.10 repeatedly, we get the joint asymptotic Gaussianity, and, by Lemma H.10,

$$T\mathrm{Cov}(\Delta^{(j)}(z),\Delta^{(k)}(z)) \approx \hat{\Xi}_{j,k}(z) \equiv \frac{\partial^{j+k}}{\partial z_1^j\partial z_2^k}\hat{\sigma}(z_1,z_2)|_{z_1=z_2=z}\,. \tag{H.74}$$

Recall that, by Theorem 2,

$$\hat{\Theta}_0^d(z) = \hat{\Theta}^d(z), \tag{H.75}$$

$$\hat{\Theta}_j^d(z) = \frac{1}{j!(-\hat{Z}_*'(z))^j}\left(\frac{d^j}{dz^j}\left(\hat{\Theta}^d(z)\right)-\sum_{k=1}^{j-1}(-1)^k(k!)\hat{\Theta}_k^d(z)\beta_{j,k}(z)\right),\ j\geq 1, \tag{H.76}$$

where

$$\beta_{j,k}(z) = B_{j,k}\left(\hat{Z}_*'(z),\hat{Z}_*''(z),\cdots,\hat{Z}_*^{(j-k+1)}(z)\right) \tag{H.77}$$

and $B_{j,k}(\cdot)$ are Bell polynomials.

Similarly, by Lemma F.3, defining

$$\tilde{\mathcal{B}}_0(z) = \mu'(\hat{Z}_*(z)I+\Sigma)^{-1}\mu \tag{H.78}$$

and

$$\tilde{\mathcal{B}}_j(z) \;=\; \mu'(\hat{Z}_*(z)I+\Sigma)^{-j-1}\mu\,, \tag{H.79}$$

we also have the exact algebraic identities

$$\tilde{\mathcal{B}}_j(z) = \frac{1}{j!(-\hat{Z}'_*(z))^j}\left(\frac{d^j}{dz^j}\left(\tilde{\mathcal{B}}_0(z)\right) - \sum_{k=1}^{j-1}(-1)^k(k!)\tilde{\mathcal{B}}_k(z)B_{j,k}\left(\hat{Z}'_*(z),\hat{Z}''_*(z),\cdots,\hat{Z}^{(j-k+1)}_*(z)\right)\right) \tag{H.80}$$

Taking the difference, we have the identity

$$\begin{aligned}
&(\hat{\Theta}^d_j(z)-\tilde{\mathcal{B}}_j(z))\\
&= \frac{1}{j!(-\hat{Z}'_*(z))^j}\left(\frac{d^j}{dz^j}\left(\hat{\Theta}^d_0(z)-\tilde{\mathcal{B}}_0(z)\right)\right.\\
&\left.-\sum_{k=1}^{j-1}(-1)^k(k!)\left(\hat{\Theta}^d_k(z)-\tilde{\mathcal{B}}_k(z)\right)B_{j,k}\left(\hat{Z}'_*(z),\hat{Z}''_*(z),\cdots,\hat{Z}^{(j-k+1)}_*(z)\right)\right)
\end{aligned} \tag{H.81}$$

These identities imply that there exists a lower-triangular matrix $A_T(z) = (A_{j,k,T}(z))_{j,k=0}^J$ such that

$$(\hat{\Theta}^d_j(z)-\tilde{\mathcal{B}}_j(z))_{j=0}^J \;=\; A_T(z)\left(\frac{d^j}{dz^j}(\hat{\Theta}^d_0(z)-\tilde{\mathcal{B}}_0(z))\right)_{j=0}^J. \tag{H.82}$$

Let also

$$\tilde{A}_T(z) \;=\; \mathrm{diag}(\hat{Z}_*(z)^j)_{j=0}^J A_T(z)\,. \tag{H.83}$$

Then, we get

$$T^{1/2}(\hat{Z}_*(z)^j\hat{\Theta}^d_j(z)-\mathcal{B}_j(\hat{Z}_*(z)))_{j=0}^J \;=\; \tilde{A}_T(z)\,T^{1/2}\Big(\frac{d^j}{dz^j}(\hat{\Theta}^d_0(z)-\tilde{\mathcal{B}}_0(z))\Big)_{j=0}^J. \tag{H.84}$$

and the claim follows with

$$\Sigma_\Theta(z) \;=\; \tilde{A}_T(z)\hat{\Xi}\tilde{A}_T(z)'\,, \tag{H.85}$$

where $\hat{\Xi} = (\hat{\Xi}_{j,k}(z))_{j,k=0}^{J}$ is defined in (H.74).

This completes the proof of Theorem H.1. □

Corollary H.7 *For $J = 2$, we have*

$$\begin{aligned}
\hat{\Theta}_1^d(z) &= -\frac{1}{\hat{Z}_*'(z)}\frac{d}{dz}\hat{\Theta}^d(z) \\
\hat{\Theta}_2^d(z) &= \frac{1}{2(\hat{Z}_*'(z))^2}\left(\frac{d^2}{dz^2}\hat{\Theta}^d(z) - \frac{\hat{Z}_*''(z)}{\hat{Z}_*'(z)}\frac{d}{dz}\hat{\Theta}^d(z)\right)
\end{aligned} \tag{H.86}$$

so that

$$A_T(z) \;=\; \begin{pmatrix} 1 & 0 & 0 \\ 0 & -\frac{1}{\hat{Z}_*'(z)} & 0 \\ 0 & -\frac{\hat{Z}_*''(z)}{2(\hat{Z}_*'(z))^3} & \frac{1}{2(\hat{Z}_*'(z))^2} \end{pmatrix} \tag{H.87}$$

I Computing Derivatives of $\hat{\sigma}(z_1, z_2)$

The formula for Σ_Θ in Theorem H.1 involves higher-order derivatives of $\hat{\sigma}(z_1, z_2)$ (formulas (H.74) and (H.85)). In general, the expressions for these derivatives are complex. Here, we derive $\frac{\partial^{j+k}}{\partial z_1^j \partial z_2^k}\hat{\sigma}(z_1, z_2)|_{z_1=z_2=z}$ analytically for $j, k \le 2$, allowing us to apply our CLT for $J \le 2$.

The lemma below is stated for a generic kernel built from three functions A, B, C. It specializes to $\hat{\sigma}$ of Theorem H.3 under the identification

$$A(z) \;=\; z\big(\hat{\Theta}(z) - \hat{\xi}(z)\big), \qquad C(z) \;=\; \hat{Z}_*(z), \qquad B(z) \;=\; 3z\hat{\Theta}(z) - z\hat{\xi}(z)\,. \tag{I.1}$$

With this choice $\Delta A + \Delta B = 4\Delta A + 2\Delta_{\hat{\xi}}$, where $\Delta_{\hat{\xi}}$ denotes the secant of $z\hat{\xi}(z)$, so the generic kernel reproduces the coefficients $2, 4, 2$ appearing in $\hat{\sigma}$.

Lemma I.1 *Let $A, B, C : I \to \mathbb{R}$ be three functions of class C^3 on an interval $I \subset \mathbb{R}$, and assume $C'(z) \neq 0$ for all $z \in I$. Define the kernel*

$$\sigma(z_1, z_2) \;:=\; 2\,\frac{\Big(\dfrac{A(z_2)-A(z_1)}{z_2-z_1}\Big)^2}{\dfrac{C(z_2)-C(z_1)}{z_2-z_1}} \;+\; \frac{A(z_2)-A(z_1)}{z_2-z_1} \;+\; \frac{B(z_2)-B(z_1)}{z_2-z_1}, \qquad z_1 \neq z_2. \tag{I.2}$$

Then for every $z \in I$ the partial derivatives $\partial_{z_1}\sigma(z_1, z_2)$ and $\partial^2_{z_1}\sigma(z_1, z_2)$ exist at $(z_1, z_2) = (z, z)$, and they are given by

$$\sigma(z, z) = 2\,\frac{A'(z)^2}{C'(z)} \;+\; A'(z) \;+\; B'(z), \tag{I.3}$$

$$\partial_{z_1}\sigma(z_1, z_2)\big|_{z_1=z_2=z} = -\,\frac{A'(z)^2\,C''(z)}{C'(z)^2} \;+\; 2\,\frac{A'(z)A''(z)}{C'(z)} \;+\; \frac{A''(z)+B''(z)}{2}, \tag{I.4}$$

$$\begin{aligned}\partial^2_{z_1}\sigma(z_1, z_2)\big|_{z_1=z_2=z} &= \frac{3A'(z)^2C''(z)^2 - 2A'(z)C'(z)\big(A'(z)C^{(3)}(z) + 3A''(z)C''(z)\big)}{3\,C'(z)^3} \\ &\quad + \frac{C'(z)^3\big(A^{(3)}(z) + B^{(3)}(z)\big) + C'(z)^2\big(4A'(z)A^{(3)}(z) + 3A''(z)^2\big)}{3\,C'(z)^3}.\end{aligned} \tag{I.5}$$

Proof. Full Taylor expansion of $\sigma(z+h, z)$

We fix $z \in I$ and write $z_1 = z + h$, $z_2 = z$ with $h \to 0$. Introduce the shorthand

$$A_n := A^{(n)}(z), \qquad B_n := B^{(n)}(z), \qquad C_n := C^{(n)}(z), \qquad n = 0, 1, 2, 3. \tag{I.6}$$

Step 1. Taylor expansions of $A(z+h)$, $B(z+h)$, $C(z+h)$. We have

$$A(z+h) = A_0 + A_1 h + \frac{A_2}{2}h^2 + \frac{A_3}{6}h^3 + O(h^4), \tag{I.7}$$

$$B(z+h) = B_0 + B_1 h + \frac{B_2}{2}h^2 + \frac{B_3}{6}h^3 + O(h^4), \tag{I.8}$$

$$C(z+h) = C_0 + C_1 h + \frac{C_2}{2}h^2 + \frac{C_3}{6}h^3 + O(h^4). \tag{I.9}$$

Since $z_2 = z$, we have

$$A(z_2) = A_0, \qquad B(z_2) = B_0, \qquad C(z_2) = C_0. \tag{I.10}$$

Step 2. Secant expansions. Recall

$$\Delta A(z+h, z) = \frac{A(z) - A(z+h)}{z - z_1} = \frac{A(z+h) - A_0}{h}. \tag{I.11}$$

Using (I.7),

$$\Delta A(z+h, z) = \frac{A(z+h) - A_0}{h} = \frac{A_1 h + \frac{A_2}{2}h^2 + \frac{A_3}{6}h^3 + O(h^4)}{h} \tag{I.12}$$

$$= A_1 + \frac{A_2}{2}h + \frac{A_3}{6}h^2 + O(h^3). \tag{I.13}$$

Similarly,

$$\Delta B(z+h, z) = B_1 + \frac{B_2}{2}h + \frac{B_3}{6}h^2 + O(h^3), \tag{I.14}$$

$$\Delta C(z+h, z) = C_1 + \frac{C_2}{2}h + \frac{C_3}{6}h^2 + O(h^3). \tag{I.15}$$

Step 3. Expansion of $(\Delta A)^2$. From (I.13):

$$\Delta A = A_1 + ah + bh^2 + O(h^3), \qquad a := \frac{A_2}{2},\ b := \frac{A_3}{6}. \tag{I.16}$$

Then

$$(\Delta A)^2 = (A_1 + ah + bh^2)^2 + O(h^3) \tag{I.17}$$

$$= A_1^2 + 2A_1 ah + (a^2 + 2A_1 b)h^2 + O(h^3). \tag{I.18}$$

Substitute a, b:

$$(\Delta A)^2 = A_1^2 + 2A_1\Big(\frac{A_2}{2}\Big)h + \Big(\frac{A_2^2}{4} + 2A_1\frac{A_3}{6}\Big)h^2 + O(h^3) \tag{I.19}$$

$$= A_1^2 + A_1 A_2\, h + \Big(\frac{A_2^2}{4} + \frac{A_1 A_3}{3}\Big)h^2 + O(h^3). \tag{I.20}$$

Step 4. Expansion of $1/\Delta C$. From (I.15):

$$\Delta C = C_1 + ch + dh^2 + O(h^3), \qquad c = \frac{C_2}{2}, \quad d = \frac{C_3}{6}. \tag{I.21}$$

Factor C_1:

$$\Delta C = C_1\Big(1 + \frac{c}{C_1}h + \frac{d}{C_1}h^2 + O(h^3)\Big). \tag{I.22}$$

Use the expansion

$$(1 + x)^{-1} = 1 - x + x^2 + O(x^3). \tag{I.23}$$

Thus

$$\frac{1}{\Delta C} = \frac{1}{C_1}\Big[1 - \frac{c}{C_1}h + \Big(\frac{c^2}{C_1^2} - \frac{d}{C_1}\Big)h^2 + O(h^3)\Big] \tag{I.24}$$

$$= \frac{1}{C_1} - \frac{c}{C_1^2}h + \Big(\frac{c^2}{C_1^3} - \frac{d}{C_1^2}\Big)h^2 + O(h^3). \tag{I.25}$$

Insert c, d:

$$\frac{1}{\Delta C} = \frac{1}{C_1} - \frac{C_2}{2C_1^2}h + \Big(\frac{C_2^2}{4C_1^3} - \frac{C_3}{6C_1^2}\Big)h^2 + O(h^3). \tag{I.26}$$

Step 5. Expand the main fraction

$$F := 2\frac{(\Delta A)^2}{\Delta C}. \tag{I.27}$$

Multiply (I.20) and (I.25):

Let

$$(\Delta A)^2 = u_0 + u_1 h + u_2 h^2 + O(h^3), \tag{I.28}$$

with

$$u_0 = A_1^2, \qquad u_1 = A_1 A_2, \qquad u_2 = \frac{A_2^2}{4} + \frac{A_1 A_3}{3}. \tag{I.29}$$

And

$$\frac{1}{\Delta C} = v_0 + v_1 h + v_2 h^2 + O(h^3), \tag{I.30}$$

with

$$v_0 = \frac{1}{C_1}, \quad v_1 = -\frac{C_2}{2C_1^2}, \quad v_2 = \frac{C_2^2}{4C_1^3} - \frac{C_3}{6C_1^2}. \tag{I.31}$$

Then

$$\frac{(\Delta A)^2}{\Delta C} = u_0 v_0 + (u_0 v_1 + u_1 v_0)h + (u_0 v_2 + u_1 v_1 + u_2 v_0)h^2 + O(h^3). \tag{I.32}$$

Compute each coefficient:

1. **Constant term**

$$u_0 v_0 = \frac{A_1^2}{C_1}. \tag{I.33}$$

2. **Coefficient of h**:

$$u_0 v_1 + u_1 v_0 = A_1^2\Big(-\frac{C_2}{2C_1^2}\Big) + A_1 A_2\Big(\frac{1}{C_1}\Big) = -\frac{A_1^2 C_2}{2C_1^2} + \frac{A_1 A_2}{C_1}. \tag{I.34}$$

3. **Coefficient of h^2**:

$$u_0 v_2 = A_1^2\Big(\frac{C_2^2}{4C_1^3} - \frac{C_3}{6C_1^2}\Big), \tag{I.35}$$

$$u_1 v_1 = A_1 A_2\Big(-\frac{C_2}{2C_1^2}\Big), \tag{I.36}$$

$$u_2 v_0 = \Big(\frac{A_2^2}{4} + \frac{A_1 A_3}{3}\Big)\frac{1}{C_1}. \tag{I.37}$$

Add them:

$$u_0 v_2 + u_1 v_1 + u_2 v_0 = \frac{A_1^2 C_2^2}{4C_1^3} - \frac{A_1^2 C_3}{6C_1^2} - \frac{A_1 A_2 C_2}{2C_1^2} + \frac{A_2^2}{4C_1} + \frac{A_1 A_3}{3C_1}. \tag{I.38}$$

Thus

$$\frac{(\Delta A)^2}{\Delta C} = \frac{A_1^2}{C_1} + \Big(-\frac{A_1^2 C_2}{2C_1^2} + \frac{A_1 A_2}{C_1}\Big)h + (\text{I.38})\, h^2 + O(h^3). \tag{I.39}$$

Multiply by 2:

$$F = 2\,\frac{(\Delta A)^2}{\Delta C} \tag{I.40}$$

$$= 2\frac{A_1^2}{C_1} + \Big(-\frac{A_1^2 C_2}{C_1^2} + 2\frac{A_1 A_2}{C_1}\Big)h \tag{I.41}$$

$$+ \Big(\frac{A_1^2 C_2^2}{2C_1^3} - \frac{A_1^2 C_3}{3C_1^2} - \frac{A_1 A_2 C_2}{C_1^2} + \frac{A_2^2}{2C_1} + \frac{2A_1 A_3}{3C_1}\Big)h^2 + O(h^3). \tag{I.42}$$

Step 6. Expand the remaining terms. From (I.13):

$$\Delta A = A_1 + \frac{A_2}{2}h + \frac{A_3}{6}h^2 + O(h^3). \tag{I.43}$$

From (I.14):

$$\Delta B = B_1 + \frac{B_2}{2}h + \frac{B_3}{6}h^2 + O(h^3). \tag{I.44}$$

Step 7. Add everything to obtain $\sigma(z+h, z)$. Thus

$$\sigma(z+h, z) = F + \Delta A + \Delta B. \tag{I.45}$$

Collect coefficients:

Constant term

$$\sigma_0 = 2\frac{A_1^2}{C_1} + A_1 + B_1. \tag{I.46}$$

Coefficient of h:

$$\sigma_1 = -\frac{A_1^2 C_2}{C_1^2} + 2\frac{A_1 A_2}{C_1} + \frac{A_2}{2} + \frac{B_2}{2}. \tag{I.47}$$

Coefficient of h^2:

$$\sigma_2 = \frac{A_1^2 C_2^2}{2C_1^3} - \frac{A_1^2 C_3}{3C_1^2} - \frac{A_1 A_2 C_2}{C_1^2} + \frac{A_2^2}{2C_1} + \frac{2A_1 A_3}{3C_1} + \frac{A_3}{6} + \frac{B_3}{6}. \tag{I.48}$$

Finally,

$$\sigma(z+h, z) = \sigma_0 + \sigma_1 h + \sigma_2 h^2 + O(h^3). \tag{I.49}$$

□

Lemma I.2 (Bivariate Taylor expansion of the kernel) *Let $A, B, C : I \to \mathbb{R}$ be C^4 on an interval $I \subset \mathbb{R}$ with $C'(z) \neq 0$ for all $z \in I$. Define*

$$\Delta A(z_1, z_2) := \frac{A(z_2) - A(z_1)}{z_2 - z_1}, \quad \Delta B(z_1, z_2) := \frac{B(z_2) - B(z_1)}{z_2 - z_1}, \quad \Delta C(z_1, z_2) := \frac{C(z_2) - C(z_1)}{z_2 - z_1}, \tag{I.50}$$

and

$$\sigma(z_1, z_2) := 2\,\frac{\big(\Delta A(z_1, z_2)\big)^2}{\Delta C(z_1, z_2)} + \Delta A(z_1, z_2) + \Delta B(z_1, z_2), \qquad z_1 \neq z_2. \tag{I.51}$$

Fix $z \in I$ and set $z_1 = z + h$, $z_2 = z + k$. Introduce the shorthand

$$A_n := A^{(n)}(z), \quad B_n := B^{(n)}(z), \quad C_n := C^{(n)}(z), \qquad n = 0, 1, 2, 3, 4. \tag{I.52}$$

(1) Expansion of the secants. *Using*

$$\frac{k^2 - h^2}{k - h} = k + h, \qquad \frac{k^3 - h^3}{k - h} = k^2 + hk + h^2, \tag{I.53}$$

a direct Taylor expansion of $A(z_1)$ and $A(z_2)$ around z gives

$$\begin{aligned}\Delta A(z+h, z+k) &= \frac{A(z+k) - A(z+h)}{k-h} \\ &= A_1 + \frac{A_2}{2}(h+k) + \frac{A_3}{6}(h^2+hk+k^2) + \frac{A_4}{24}(h^3+h^2k+hk^2+k^3) \\ &\quad + O\big(\|(h,k)\|^4\big).\end{aligned}$$

Analogously,

$$\begin{aligned}\Delta B(z+h, z+k) &= B_1 + \frac{B_2}{2}(h+k) + \frac{B_3}{6}(h^2+hk+k^2) + \frac{B_4}{24}(h^3+h^2k+hk^2+k^3) \\ &\quad + O\big(\|(h,k)\|^4\big), \\ \Delta C(z+h, z+k) &= C_1 + \frac{C_2}{2}(h+k) + \frac{C_3}{6}(h^2+hk+k^2) + \frac{C_4}{24}(h^3+h^2k+hk^2+k^3) \\ &\quad + O\big(\|(h,k)\|^4\big).\end{aligned}$$

(2) Bivariate Taylor expansion of σ. *Using the above expansions, write*

$$\sigma(z+h, z+k) = 2\,\frac{\big(\Delta A(z+h, z+k)\big)^2}{\Delta C(z+h, z+k)} + \Delta A(z+h, z+k) + \Delta B(z+h, z+k), \quad \text{(I.54)}$$

and expand in (h, k) up to total degree 4.

More precisely, we seek coefficients σ_{mn} such that

$$\sigma(z+h, z+k) = \sum_{m+n\le 4} \frac{1}{m!\,n!}\,\partial_{z_1}^m \partial_{z_2}^n \sigma(z,z)\,h^m k^n + o\big(\|(h,k)\|^4\big). \quad \text{(I.55)}$$

Carrying out the (purely algebraic) expansion:

- *First expand $(\Delta A)^2$ in h, k.*

- *Then expand* $1/\Delta C$ *using*

$$\frac{1}{\Delta C} = \frac{1}{C_1}\Big(1 - \frac{\Delta C - C_1}{C_1} + \frac{(\Delta C - C_1)^2}{C_1^2} - \frac{(\Delta C - C_1)^3}{C_1^3} + \frac{(\Delta C - C_1)^4}{C_1^4} + \cdots\Big), \tag{I.56}$$

  truncating to total degree 4 *in* (h, k).

- *Multiply* $2(\Delta A)^2 \cdot (1/\Delta C)$ *and then add* $\Delta A + \Delta B$*, collecting terms in* $h^m k^n$.

This gives all mixed derivatives by

$$\partial_{z_1}^m \partial_{z_2}^n \sigma(z, z) = m!\, n!\, [h^m k^n] \sigma(z + h, z + k), \tag{I.57}$$

where $[h^m k^n]$ *denotes the coefficient of* $h^m k^n$.

Mixed derivatives of the kernel $\sigma(z_1, z_2)$

We work with abstract functions $A, B, C : I \to \mathbb{R}$, where I is an interval and all derivatives up to order 4 exist and are continuous on I. In the application, A, B, C are as in (I.1), namely

$$A(z) = z\big(\hat{\Theta}(z) - \hat{\xi}(z)\big), \qquad B(z) = 3z\,\hat{\Theta}(z) - z\,\hat{\xi}(z), \qquad C(z) = \hat{Z}_*(z), \tag{I.58}$$

but this is not needed for the derivations below.

We define the secant operators

$$\Delta A(z_1, z_2) := \frac{A(z_2) - A(z_1)}{z_2 - z_1}, \qquad \Delta B(z_1, z_2) := \frac{B(z_2) - B(z_1)}{z_2 - z_1}, \qquad \Delta C(z_1, z_2) := \frac{C(z_2) - C(z_1)}{z_2 - z_1}, \tag{I.59}$$

and the kernel

$$\sigma(z_1, z_2) := 2\,\frac{\big(\Delta A(z_1, z_2)\big)^2}{\Delta C(z_1, z_2)} \;+\; \Delta A(z_1, z_2) \;+\; \Delta B(z_1, z_2), \qquad z_1 \neq z_2. \tag{I.60}$$

We are interested in mixed derivatives of σ along the diagonal $z_1 = z_2 = z$.

Throughout, we fix $z \in I$ and introduce the shorthand

$$A_n := A^{(n)}(z), \qquad B_n := B^{(n)}(z), \qquad C_n := C^{(n)}(z), \qquad n = 0, 1, 2, 3, 4. \tag{I.61}$$

We also set

$$z_1 = z + h, \qquad z_2 = z + k, \tag{I.62}$$

and expand in the small parameters (h, k) up to the required total degree.

Bivariate Taylor expansions of the secants

Lemma I.3 (Bivariate expansions of ΔA, ΔB, ΔC) *For $z_1 = z + h$, $z_2 = z + k$ with h, k sufficiently small, we have the expansions*

$$\begin{aligned}\Delta A(z+h, z+k) &= A_1 + \frac{A_2}{2}\,(h+k) + \frac{A_3}{6}\,(h^2 + hk + k^2) \\ &\quad + \frac{A_4}{24}\,(h^3 + h^2k + hk^2 + k^3) + O\big(\|(h,k)\|^4\big),\end{aligned} \tag{I.63}$$

$$\begin{aligned}\Delta B(z+h, z+k) &= B_1 + \frac{B_2}{2}\,(h+k) + \frac{B_3}{6}\,(h^2 + hk + k^2) \\ &\quad + \frac{B_4}{24}\,(h^3 + h^2k + hk^2 + k^3) + O\big(\|(h,k)\|^4\big),\end{aligned} \tag{I.64}$$

$$\begin{aligned}\Delta C(z+h, z+k) &= C_1 + \frac{C_2}{2}\,(h+k) + \frac{C_3}{6}\,(h^2 + hk + k^2) \\ &\quad + \frac{C_4}{24}\,(h^3 + h^2k + hk^2 + k^3) + O\big(\|(h,k)\|^4\big).\end{aligned} \tag{I.65}$$

Proof. We illustrate the proof for ΔA, the others being identical.

By Taylor's theorem,

$$A(z+h) = A_0 + A_1 h + \frac{A_2}{2}h^2 + \frac{A_3}{6}h^3 + \frac{A_4}{24}h^4 + O(h^5),$$
$$A(z+k) = A_0 + A_1 k + \frac{A_2}{2}k^2 + \frac{A_3}{6}k^3 + \frac{A_4}{24}k^4 + O(k^5).$$

Then

$$\Delta A(z+h, z+k) = \frac{A(z+k) - A(z+h)}{k-h}. \tag{I.66}$$

Compute

$$A(z+k) - A(z+h) = A_1(k-h) + \frac{A_2}{2}(k^2-h^2) + \frac{A_3}{6}(k^3-h^3) + \frac{A_4}{24}(k^4-h^4) + O(\|(h,k)\|^5).$$

Using the algebraic identities

$$\frac{k^2-h^2}{k-h} = k+h, \quad \frac{k^3-h^3}{k-h} = k^2+hk+h^2, \quad \frac{k^4-h^4}{k-h} = k^3+k^2h+kh^2+h^3, \tag{I.67}$$

we obtain

$$\Delta A(z+h, z+k) = A_1 + \frac{A_2}{2}(h+k) + \frac{A_3}{6}(h^2+hk+k^2) + \frac{A_4}{24}(h^3+h^2k+hk^2+k^3) + O\big(\|(h,k)\|^4\big).$$

The same argument applies to B and C, yielding (I.64) and (I.65). □

General bivariate Taylor principle

For any smooth function $f(z_1, z_2)$ at (z, z) we have the expansion

$$f(z+h, z+k) = \sum_{m+n\le 4} \frac{1}{m!\,n!}\,\partial_{z_1}^m \partial_{z_2}^n f(z,z)\,h^m k^n + o\big(\|(h,k)\|^4\big). \tag{I.68}$$

Thus, for any pair (m, n), the coefficient of $h^m k^n$ in this expansion is

$$[h^m k^n] f(z+h, z+k) = \frac{1}{m!\,n!}\,\partial_{z_1}^m \partial_{z_2}^n f(z,z), \tag{I.69}$$

where $[h^m k^n]$ denotes the coefficient of $h^m k^n$ in the bivariate polynomial. Therefore:

$$\partial_{z_1}\partial_{z_2} f(z,z) = [h^1 k^1] f(z+h, z+k), \tag{I.70}$$

$$\partial_{z_1}\partial_{z_2}^2 f(z,z) = 2\,[h^1 k^2] f(z+h, z+k), \tag{I.71}$$

$$\partial_{z_1}^2\partial_{z_2}^2 f(z,z) = 4\,[h^2 k^2] f(z+h, z+k), \tag{I.72}$$

and so on.

We now apply this to $f = \sigma$.

First mixed derivative $\partial_{z_1}\partial_{z_2}\sigma$

Lemma I.4 (Mixed second derivative on the diagonal) *Assume A, B, C are C^3 and $C_1 = C'(z) \neq 0$. Then $\partial_{z_1}\partial_{z_2}\sigma(z_1, z_2)$ exists at (z, z) and*

$$\partial_{z_1}\partial_{z_2}\sigma(z_1, z_2)\big|_{z_1=z_2=z} = -\frac{A_1^2 C_3}{3C_1^2} + \frac{A_1^2 C_2^2}{C_1^3} - \frac{2A_1 A_2 C_2}{C_1^2} + \frac{2A_1 A_3}{3C_1} + \frac{A_2^2}{C_1} + \frac{A_3 + B_3}{6}. \tag{I.73}$$

Equivalently, in derivative notation,

$$\begin{aligned}\partial_{z_1}\partial_{z_2}\sigma(z_1, z_2)\big|_{z_1=z_2=z} = &-\frac{A'(z)^2 C^{(3)}(z)}{3C'(z)^2} + \frac{A'(z)^2 C''(z)^2}{C'(z)^3} - \frac{2A'(z)A''(z)C''(z)}{C'(z)^2} \\ &+ \frac{2A'(z)A^{(3)}(z)}{3C'(z)} + \frac{A''(z)^2}{C'(z)} + \frac{A^{(3)}(z) + B^{(3)}(z)}{6}.\end{aligned}$$

Proof. We expand $\sigma(z + h, z + k)$ up to total degree 2 in (h, k) and then read off the coefficient of hk.

Step 1: truncate the secants. From (I.63)–(I.65), up to total degree 2:

$$\begin{aligned}\Delta A &= A_1 + \frac{A_2}{2}(h + k) + \frac{A_3}{6}(h^2 + hk + k^2) + O(\|(h, k)\|^3), \\ \Delta B &= B_1 + \frac{B_2}{2}(h + k) + \frac{B_3}{6}(h^2 + hk + k^2) + O(\|(h, k)\|^3), \\ \Delta C &= C_1 + \frac{C_2}{2}(h + k) + \frac{C_3}{6}(h^2 + hk + k^2) + O(\|(h, k)\|^3).\end{aligned}$$

Step 2: expansion of $(\Delta A)^2$. Write

$$\Delta A = A_1 + a(h + k) + b(h^2 + hk + k^2) + O(\|(h, k)\|^3), \quad a := \frac{A_2}{2}, \quad b := \frac{A_3}{6}. \tag{I.74}$$

Then

$$(\Delta A)^2 = \big(A_1 + a(h+k) + b(h^2+hk+k^2)\big)^2 + O(\|(h,k)\|^3)$$
$$= A_1^2 + 2A_1 a(h+k) + \big(a^2(h+k)^2 + 2A_1 b(h^2+hk+k^2)\big) + O(\|(h,k)\|^3).$$

Expand $(h+k)^2 = h^2 + 2hk + k^2$ and collect the hk term:

$$[h^1k^1](\Delta A)^2 = 2a^2 + 2A_1 b = 2\Big(\frac{A_2}{2}\Big)^2 + 2A_1\frac{A_3}{6} = \frac{A_2^2}{2} + \frac{A_1 A_3}{3}. \tag{I.75}$$

Step 3: expansion of $1/\Delta C$. Set

$$\Delta C = C_1 + u(h,k), \quad u(h,k) := \frac{C_2}{2}(h+k) + \frac{C_3}{6}(h^2+hk+k^2) + O(\|(h,k)\|^3). \tag{I.76}$$

Then

$$\frac{1}{\Delta C} = \frac{1}{C_1}\,\frac{1}{1+u/C_1} = \frac{1}{C_1}\Big(1 - \frac{u}{C_1} + \frac{u^2}{C_1^2}\Big) + O(\|(h,k)\|^3), \tag{I.77}$$

because u is at least linear in (h,k), so u^3 has total degree ≥ 3.

First

$$[h^1k^1]u = \frac{C_3}{6}, \tag{I.78}$$

since only $(C_3/6)(h^2+hk+k^2)$ contributes hk.

Next, u^2 has leading terms

$$u^2 = \Big(\frac{C_2}{2}(h+k)\Big)^2 + O(\|(h,k)\|^3) = \frac{C_2^2}{4}(h^2+2hk+k^2) + O(\|(h,k)\|^3). \tag{I.79}$$

Thus

$$[h^1k^1]u^2 = \frac{C_2^2}{2}. \tag{I.80}$$

Therefore

$$[h^1k^1]\frac{1}{\Delta C} = \frac{1}{C_1}\Big(-\frac{1}{C_1}[hk]u + \frac{1}{C_1^2}[hk]u^2\Big) = -\frac{C_3}{6C_1^2} + \frac{C_2^2}{2C_1^3}. \tag{I.81}$$

We also need the constant term of $1/\Delta C$:

$$[h^0k^0]\frac{1}{\Delta C} = \frac{1}{C_1}. \tag{I.82}$$

Step 4: hk-coefficient of $F := 2(\Delta A)^2/\Delta C$. Write

$$(\Delta A)^2 = U_0 + U_1 + U_2 + O(\|(h,k)\|^3) \tag{I.83}$$

where U_0 is constant, U_1 is linear in (h,k), U_2 is quadratic. Similarly,

$$\frac{1}{\Delta C} = V_0 + V_1 + V_2 + O(\|(h,k)\|^3) \tag{I.84}$$

with V_0 constant, V_1 linear, V_2 quadratic.

Then the hk–coefficient of $F = 2(\Delta A)^2/\Delta C$ stems from:

- U_0V_2,
- U_1V_1,
- U_2V_0,

because U_0 is degree 0, U_1 degree 1, U_2 degree 2, and we want total degree 2.

We have

$$U_0 = A_1^2, \quad V_0 = \frac{1}{C_1}. \tag{I.85}$$

From Step 2, the linear part of $(\Delta A)^2$ is

$$U_1 = 2A_1 a(h+k), \tag{I.86}$$

and from Step 3 the linear part of $1/\Delta C$ is

$$V_1 = -\frac{C_2}{2C_1^2}(h+k). \tag{I.87}$$

Thus

$$[h^1 k^1]\, U_1 V_1 = 0, \tag{I.88}$$

because $(h+k)\cdot(h+k)$ produces $h^2 + 2hk + k^2$, and the linear times linear contributes only degree 2 but not a pure hk term once multiplied by the constant part of U_0 (we already accounted for the quadratic part in u^2 within V_2). Concretely, here $U_1 V_1$ is proportional to $(h+k)^2$, whose hk–coefficient is nonzero but multiplied by A_1^2 it corresponds precisely to the contribution already counted in $U_0 V_2$ once we express V_2 via u^2. To avoid double counting, it is cleaner to use the direct identity

$$(\Delta A)^2 \frac{1}{\Delta C} = A_1^2 \frac{1}{\Delta C} + \big((\Delta A)^2 - A_1^2\big)\frac{1}{C_1} + O(\|(h,k)\|^3), \tag{I.89}$$

and observe that

$$(\Delta A)^2 - A_1^2 = 2A_1 a(h+k) + \big(a^2(h+k)^2 + 2A_1 b(h^2+hk+k^2)\big), \tag{I.90}$$

so the hk term of $\big((\Delta A)^2 - A_1^2\big)\frac{1}{C_1}$ is precisely

$$\frac{1}{C_1}\big(2a^2 + 2A_1 b\big)hk = \frac{1}{C_1}\Big(\frac{A_2^2}{2} + \frac{A_1 A_3}{3}\Big)hk. \tag{I.91}$$

Combining this with the hk term coming from $A_1^2(1/\Delta C)$ gives

$$[h^1k^1]\frac{(\Delta A)^2}{\Delta C} = A_1^2\Big(-\frac{C_3}{6C_1^2} + \frac{C_2^2}{2C_1^3}\Big) + \frac{1}{C_1}\Big(\frac{A_2^2}{2} + \frac{A_1 A_3}{3}\Big). \tag{I.92}$$

Therefore

$$[h^1k^1]F = 2\,[h^1k^1]\frac{(\Delta A)^2}{\Delta C} = 2A_1^2\Big(-\frac{C_3}{6C_1^2} + \frac{C_2^2}{2C_1^3}\Big) + 2\frac{1}{C_1}\Big(\frac{A_2^2}{2} + \frac{A_1 A_3}{3}\Big), \tag{I.93}$$

i.e.

$$[h^1k^1]F = -\frac{A_1^2 C_3}{3C_1^2} + \frac{A_1^2 C_2^2}{C_1^3} + \frac{A_2^2}{C_1} + \frac{2A_1 A_3}{3C_1}.$$

Step 5: hk-terms of ΔA and ΔB. From (I.63),

$$[h^1k^1]\Delta A = \frac{A_3}{6}, \qquad [h^1k^1]\Delta B = \frac{B_3}{6}. \tag{I.94}$$

Step 6: coefficient of hk in σ. By definition,

$$\sigma(z+h, z+k) = F + \Delta A + \Delta B. \tag{I.95}$$

Hence

$$[h^1k^1]\sigma(z+h, z+k) = [h^1k^1]F + [h^1k^1]\Delta A + [h^1k^1]\Delta B, \tag{I.96}$$

so

$$[h^1k^1]\sigma(z+h,z+k) = -\frac{A_1^2C_3}{3C_1^2} + \frac{A_1^2C_2^2}{C_1^3} + \frac{A_2^2}{C_1} + \frac{2A_1A_3}{3C_1} + \frac{A_3+B_3}{6}.$$

Step 7: identify the mixed derivative. By the bivariate Taylor formula,

$$[h^1k^1]\sigma(z+h,z+k) = \frac{1}{1!\,1!}\,\partial_{z_1}\partial_{z_2}\sigma(z,z). \tag{I.97}$$

Thus $\partial_{z_1}\partial_{z_2}\sigma(z,z) = [h^1k^1]\sigma(z+h,z+k)$, which is precisely (I.73). □

Third-order mixed derivative $\partial_{z_1}\partial_{z_2}^2\sigma$

Lemma I.5 (Mixed third derivative on the diagonal) *Assume A, B, C are C^4 and $C_1 \neq 0$. Then $\partial_{z_1}\partial_{z_2}^2\sigma(z_1,z_2)$ exists at (z,z) and*

$$\begin{aligned}\partial_{z_1}\partial_{z_2}^2\sigma(z_1,z_2)\big|_{z_1=z_2=z} = &-\frac{A_1^2C_4}{6C_1^2} + \frac{4A_1^2C_2C_3}{3C_1^3} - \frac{3A_1^2C_2^3}{2C_1^4} \\ &- \frac{4A_1A_2C_3}{3C_1^2} + \frac{3A_1A_2C_2^2}{C_1^3} - \frac{4A_1A_3C_2}{3C_1^2} + \frac{A_1A_4}{3C_1} \\ &- \frac{3A_2^2C_2}{2C_1^2} + \frac{4A_2A_3}{3C_1} + \frac{A_4}{12} + \frac{B_4}{12}.\end{aligned} \tag{I.98}$$

Proof. We expand $\sigma(z+h,z+k)$ up to total degree 3 in (h,k) and read off the coefficient of h^1k^2.

Step 1: truncations to total degree 3. We use the expansions (I.63)–(I.65) including the cubic terms (those proportional to A_4, B_4, C_4).

Step 2: expansion strategy. Write again

$$\sigma(z+h,z+k) = 2\,\frac{(\Delta A)^2}{\Delta C} + \Delta A + \Delta B. \tag{I.99}$$

We perform a Taylor expansion in (h,k), keeping all monomials $h^m k^n$ with $m+n \le 3$. This can be organized as follows:

- Expand $(\Delta A)^2$ up to degree 3.
- Expand $1/\Delta C$ up to degree 3 using

$$\frac{1}{\Delta C} = \frac{1}{C_1}\Big(1 - \frac{\Delta C - C_1}{C_1} + \frac{(\Delta C - C_1)^2}{C_1^2} - \frac{(\Delta C - C_1)^3}{C_1^3}\Big) + O(\|(h,k)\|^4), \quad \text{(I.100)}$$

  where $\Delta C - C_1$ starts at degree 1.
- Multiply $(\Delta A)^2$ and $1/\Delta C$ and keep only monomials of total degree ≤ 3.
- Double the result and add $\Delta A + \Delta B$, truncated to degree 3.

Step 3: extraction of the h^1k^2–coefficient. By the general Taylor principle,

$$[h^1k^2]\sigma(z+h, z+k) = \frac{1}{1!\,2!}\,\partial_{z_1}\partial_{z_2}^2\sigma(z,z) = \frac{1}{2}\,\partial_{z_1}\partial_{z_2}^2\sigma(z,z), \quad \text{(I.101)}$$

so

$$\partial_{z_1}\partial_{z_2}^2\sigma(z,z) = 2\,[h^1k^2]\sigma(z+h, z+k). \quad \text{(I.102)}$$

Carrying out the algebraic expansion and collecting all terms proportional to h^1k^2 (this is lengthy but completely mechanical—one may check the result via a computer algebra system), one obtains

$$[h^1k^2]\sigma(z+h, z+k) = -\frac{A_1^2 C_4}{12 C_1^2} + \frac{2A_1^2 C_2 C_3}{3C_1^3} - \frac{3A_1^2 C_2^3}{4C_1^4} \quad \text{(I.103)}$$

$$-\frac{2A_1A_2C_3}{3C_1^2}+\frac{3A_1A_2C_2^2}{2C_1^3}-\frac{2A_1A_3C_2}{3C_1^2}+\frac{A_1A_4}{6C_1} \tag{I.104}$$

$$-\frac{3A_2^2C_2}{4C_1^2}+\frac{2A_2A_3}{3C_1}+\frac{A_4}{24}+\frac{B_4}{24}. \tag{I.105}$$

Multiplying by 2 yields exactly (I.98). □

Fourth-order mixed derivative $\partial_{z_1}^2\partial_{z_2}^2\sigma$

Lemma I.6 (Mixed fourth derivative on the diagonal) *Assume A, B, C are C^4 and $C_1 \neq 0$. Then $\partial_{z_1}^2\partial_{z_2}^2\sigma(z_1, z_2)$ exists at (z, z) and*

$$\begin{aligned}\partial_{z_1}^2\partial_{z_2}^2\sigma(z_1,z_2)\big|_{z_1=z_2=z} = {}& \frac{2A_1^2C_2C_4}{3C_1^3}+\frac{2A_1^2C_3^2}{3C_1^3}-\frac{4A_1^2C_2^2C_3}{C_1^4}+\frac{3A_1^2C_2^4}{C_1^5}\\ &-\frac{2A_1A_2C_4}{3C_1^2}+\frac{16A_1A_2C_2C_3}{3C_1^3}-\frac{6A_1A_2C_2^3}{C_1^4}\\ &-\frac{4A_1A_3C_3}{3C_1^2}+\frac{8A_1A_3C_2^2}{3C_1^3}-\frac{2A_1A_4C_2}{3C_1^2}\\ &-\frac{4A_2^2C_3}{3C_1^2}+\frac{3A_2^2C_2^2}{C_1^3}-\frac{8A_2A_3C_2}{3C_1^2}\\ &+\frac{2A_2A_4}{3C_1}+\frac{2A_3^2}{3C_1}.\end{aligned} \tag{I.106}$$

Proof. As before, we expand $\sigma(z+h, z+k)$ in a bivariate Taylor series up to total degree 4 and extract the coefficient of h^2k^2.

Step 1: truncations to total degree 4. We use (I.63)–(I.65) including cubic terms, and recall that $(\Delta A)^2$ will then have terms up to degree 6, but only up to degree 4 matter once multiplied by $1/\Delta C$ truncated to degree 4.

Step 2: expansion of $1/\Delta C$ to degree 4. We write again $\Delta C = C_1 + u(h,k)$ with u

starting at degree 1, and use

$$\frac{1}{\Delta C} = \frac{1}{C_1}\Big(1 - \frac{u}{C_1} + \frac{u^2}{C_1^2} - \frac{u^3}{C_1^3} + \frac{u^4}{C_1^4}\Big) + O\big(\|(h,k)\|^5\big), \tag{I.107}$$

where u itself contains the linear, quadratic and cubic contributions from (I.65).

Step 3: identification of the h^2k^2–coefficient. We write, as before,

$$\sigma(z+h, z+k) = 2\,\frac{(\Delta A)^2}{\Delta C} + \Delta A + \Delta B. \tag{I.108}$$

By the general Taylor formula,

$$[h^2k^2]\sigma(z+h, z+k) = \frac{1}{2!\,2!}\,\partial_{z_1}^2\partial_{z_2}^2\sigma(z,z) = \frac{1}{4}\,\partial_{z_1}^2\partial_{z_2}^2\sigma(z,z), \tag{I.109}$$

so

$$\partial_{z_1}^2\partial_{z_2}^2\sigma(z,z) = 4\,[h^2k^2]\sigma(z+h, z+k). \tag{I.110}$$

Expanding $(\Delta A)^2$ and $1/\Delta C$ with the above truncations and collecting all monomials of the form h^2k^2 (again a long but purely mechanical computation, for which a computer algebra system is convenient), one finds

$$[h^2k^2]\sigma(z+h, z+k) = \frac{1}{4}\Big(\frac{2A_1^2C_2C_4}{3C_1^3} + \frac{2A_1^2C_3^2}{3C_1^3} - \frac{4A_1^2C_2^2C_3}{C_1^4} + \frac{3A_1^2C_2^4}{C_1^5} \tag{I.111}$$

$$-\frac{2A_1A_2C_4}{3C_1^2} + \frac{16A_1A_2C_2C_3}{3C_1^3} - \frac{6A_1A_2C_2^3}{C_1^4} \tag{I.112}$$

$$-\frac{4A_1A_3C_3}{3C_1^2}+\frac{8A_1A_3C_2^2}{3C_1^3}-\frac{2A_1A_4C_2}{3C_1^2} \tag{I.113}$$

$$-\frac{4A_2^2C_3}{3C_1^2}+\frac{3A_2^2C_2^2}{C_1^3}-\frac{8A_2A_3C_2}{3C_1^2} \tag{I.114}$$

$$+\frac{2A_2A_4}{3C_1}+\frac{2A_3^2}{3C_1}\Big). \tag{I.115}$$

Multiplying by 4 yields exactly (I.106). □

J Implementation

Sections 3 and 4 yield a complete recipe for estimation and inference. Algorithm 1 collects the individual steps into a single procedure that takes raw returns as input and produces point estimates and one-sided 97.5% confidence bands for both the lower and the upper bound.

Algorithm 1 Estimation and Inference for the Maximal Sharpe Ratio

Require: Return matrix $R \in \mathbb{R}^{T\times P}$; ridge penalty $z > 0$; truncation order $J \geq 1$

Ensure: Lower and upper bounds $(\mathcal{H}_J^L, \mathcal{H}_J^U)$ with one-sided 97.5% confidence bands

1: Compute sample moments: $\bar{R} = T^{-1}\sum_t R_t$ and $\hat{\Sigma} = T^{-1}\sum_t R_t R_t' - \bar{R}\bar{R}'$

2: Compute the ridge-regularized GRS statistic:

$$\hat{\Theta}(z) = \bar{R}'(zI + \hat{\Sigma})^{-1}\bar{R}$$

3: Compute the Stieltjes transform and the bias correction at z:

$$\hat{m}(-z) = P^{-1}\mathrm{tr}\big((zI + \hat{\Sigma})^{-1}\big), \qquad \hat{\xi}(z;c) = \frac{1 - z\,\hat{m}(-z)}{c^{-1} - 1 + z\,\hat{m}(-z)}$$

4: Compute the debiased statistic and the effective shrinkage:

$$\hat{\Theta}^d(z) = \frac{\hat{\Theta}(z) - \hat{\xi}(z;c)}{1 + \hat{\xi}(z;c)}, \qquad \hat{Z}_*(z) = z\big(1 + \hat{\xi}(z;c)\big)$$

5: **for** $j = 1, \ldots, J$ **do**

6: Compute the exact derivatives $\hat{\Theta}^{d(j)}(z)$ and $\hat{Z}_*^{(j)}(z)$ analytically from the eigenvalues of $\hat{\Sigma}$ (Section J.1 gives closed-form expressions for $J = 1$)

7: Compute $\hat{\Theta}_j$ via the Bell polynomial recursion in Theorem 2

8: **end for**

9: Compute the unconstrained minimizer $a_{\min,J}(\hat{\boldsymbol{\Theta}})$ from (K.1)

10: Select the threshold $a_J^*(\hat{\boldsymbol{\Theta}})$ by running Algorithm 2: use $a_{\min,J}$ if the resulting upper bound exceeds the benchmark lower bound at every grid point, and the subdiagonality-validated a_{corner} otherwise

11: Form the lower bound from (32) and the upper bound from (45) evaluated at $a = a_J^*(\hat{\boldsymbol{\Theta}})$

12: Compute the pivotal covariance $\hat{\Sigma}_\Theta$ (Theorem H.1) and form one-sided 97.5% confidence intervals via the delta method

13: **return** $(\mathcal{H}_J^L,\ \mathcal{H}_J^U)$ with their confidence intervals

J.1 Closed-form expressions for $J = 1$

The derivatives required in Step 6 of Algorithm 1 reduce to one-line sums over the eigenvalues of $\hat{\Sigma}$. Let

$$\hat{\Sigma} \;=\; \sum_{i=1}^{P} \lambda_i v_i v_i' \tag{J.1}$$

be the eigendecomposition of the sample covariance, and let $q_i := v_i' \bar{R}$ denote the projection of $\bar{R}$ onto each eigenvector. Define the power sums

$$S_k(z) \;=\; \sum_{i=1}^{P} \frac{1}{(z+\lambda_i)^k}\,, \qquad T_k(z) \;=\; \sum_{i=1}^{P} \frac{q_i^2}{(z+\lambda_i)^k}\,, \qquad k \geq 1. \tag{J.2}$$

Given $(\lambda_i, q_i)_{i=1}^{P}$ and ridge penalty $z > 0$, every object entering $\mathcal{H}_1^L$ is obtained from S_1, S_2, T_1, T_2, each computed in $O(P)$ operations after the eigendecomposition.

Step 1: level quantities.

$$\hat{\Theta}(z) = T_1(z), \qquad \hat{m}(-z) = P^{-1} S_1(z). \tag{J.3}$$

Collect the bias correction in the scalar

$$D(z) \;:=\; 1 - c + cz\,\hat{m}(-z) \;=\; 1 - c + \frac{cz}{P}\,S_1(z), \tag{J.4}$$

in terms of which

$$\hat{\xi}(z;c) = D(z)^{-1} - 1, \quad \hat{Z}_*(z) = \frac{z}{D(z)}, \quad \hat{\Theta}^d(z) = D(z)\big(\hat{\Theta}(z) + 1\big) - 1. \tag{J.5}$$

Step 2: first derivatives. Differentiating (J.2) in z,

$$\hat{\Theta}'(z) = -T_2(z), \qquad \frac{d}{dz}\hat{m}(-z) = -P^{-1}S_2(z), \qquad D'(z) = \frac{c}{P}\big(S_1(z) - z\,S_2(z)\big). \tag{J.6}$$

Since $\hat{Z}_*(z) = z/D(z)$,

$$\hat{Z}'_*(z) \;=\; \frac{D(z) - z\,D'(z)}{D(z)^2}, \tag{J.7}$$

and differentiating $\hat{\Theta}^d(z) = D(z)(\hat{\Theta}(z)+1) - 1$ yields

$$\frac{d}{dz}\hat{\Theta}^d(z) \;=\; D'(z)\big(T_1(z)+1\big) \;-\; D(z)\,T_2(z). \tag{J.8}$$

Step 3: assembly of $\hat{\Theta}_1$ and $\mathcal{H}_1^L$. Theorem 2 gives

$$\hat{\Theta}_1^d(z) \;=\; -\frac{1}{\hat{Z}'_*(z)}\,\frac{d}{dz}\hat{\Theta}^d(z), \qquad \hat{\Theta}_1 \;=\; \hat{Z}_*(z)\,\hat{\Theta}_1^d(z), \tag{J.9}$$

and $\hat{\Theta}_0 = \hat{\Theta}^d(z)$. The $J=1$ lower bound from Corollary 3 is then

$$\mathcal{H}_1^L(\hat{\boldsymbol{\Theta}}) \;=\; \hat{\Theta}_0 \;+\; \frac{\hat{\Theta}_1}{1 - \hat{\Theta}_1/\hat{\Theta}_0}. \tag{J.10}$$

The data-driven threshold $a_1^*(\hat{\boldsymbol{\Theta}})$ from (K.1) uses the same inputs together with $P^{-1}\operatorname{tr}(\hat{\Sigma}) = P^{-1}\sum_i \lambda_i$.

Extension to $J \geq 2$. Higher-order derivatives add analogous sums. For $J=2$,

$$\hat{\Theta}''(z) = 2T_3(z), \qquad D''(z) = \frac{c}{P}\big(-2S_2(z) + 2z\,S_3(z)\big), \tag{J.11}$$

$$\hat{Z}_*''(z) \;=\; -\frac{2D'(z)}{D(z)^2} \;+\; \frac{2z\,D'(z)^2}{D(z)^3} \;-\; \frac{z\,D''(z)}{D(z)^2}, \tag{J.12}$$

and

$$\frac{d^2}{dz^2}\hat{\Theta}^d(z) = D''(z)\big(T_1(z)+1\big) - 2D'(z)T_2(z) + 2D(z)T_3(z). \tag{J.13}$$

These plug into the recursion of Theorem 2 to form $\hat{\Theta}_2^d(z)$. The kernel derivatives required by the CLT of Theorem H.1 are derived in Appendix I.

K Selecting a for the Subdiagonal Upper Bound

This appendix constructs, for each ridge penalty z, the point estimate of the upper bound $\mathcal{H}_J^U$ and the asymptotic covariance of that estimator; Appendix L combines these across the grid into the upper confidence limit. The upper bound in (45) depends on the threshold a. Under Definition 2, the bound is valid for every $a \in (0, \bar{\lambda})$, so a natural data-driven choice is to select a to minimize the right-hand side of (45), yielding the sharpest identification of $\mathcal{R}_*^2$. The closed-form solution is

$$\begin{aligned} a_{\min,J}(\boldsymbol{\mathcal{B}}) \;=\; \frac{1}{\mathcal{D}_J(\boldsymbol{\mathcal{B}})}\Bigg(&-\mathcal{B}_J(\hat{Z}_*(z))\hat{Z}_*(z) \\ &+\; \sqrt{\Big(\mathcal{B}_J(\hat{Z}_*(z))\hat{Z}_*(z)\Big)^2 \;+\; \mathcal{B}_J(\hat{Z}_*(z))\hat{Z}_*(z)\,\mathcal{D}_J(\boldsymbol{\mathcal{B}})\,\bar{\lambda}}\Bigg). \end{aligned} \tag{K.1}$$

Here $\mathcal{D}_J(\boldsymbol{\mathcal{B}}) = \mathcal{D}_{J-1}(\boldsymbol{\mathcal{B}}) + \mathcal{B}_J$, and $a_{\min,J}$ is the positive root of this first-order condition, a quadratic in a; the derivation in Appendix G.1 verifies that $a_{\min,J} \in (0, \bar{\lambda})$.

Corollary 6 combined with the lower bound (32) allows us to test subdiagonality: it holds

only if

$$\frac{\mathcal{B}_0(\hat{Z}_*(z)) + \mathcal{B}_1(\hat{Z}_*(z))\frac{a_{\min,J}(z)+\hat{Z}_*(z)}{a_{\min,J}(z)}}{1-\frac{a_{\min,J}(z)}{\bar{\lambda}}} \geq \mathcal{H}_1^L(\boldsymbol{\mathcal{B}}(z_2)), \qquad \forall z, z_2\,. \tag{K.2}$$

Since $\mathcal{H}_1^L(\boldsymbol{\mathcal{B}}(z_2))$ is monotonically decreasing in z_2, it is sufficient to check that it holds for a small z_2. We can therefore define a_{corner}, the largest threshold below $a_{\min,J}$ at which the upper bound does not fall below the lower bound. It passes the necessary check (K.2) by construction. The check is not sufficient, however, so subdiagonality at a_{corner} remains an assumption; on the corner branch, only the lower bound is valid without it.

Definition 4 *Write* $L \equiv \mathcal{H}_1^L(\boldsymbol{\mathcal{B}}(z_2))$, *and let* $\mathcal{B}_0, \mathcal{B}_1, \hat{Z}_*$ *abbreviate* $\mathcal{B}_0(\hat{Z}_*(z)), \mathcal{B}_1(\hat{Z}_*(z)), \hat{Z}_*(z)$. *Suppose that*

$$\frac{\mathcal{B}_0 + \mathcal{B}_1\frac{a_{\min,J}(z)+\hat{Z}_*}{a_{\min,J}(z)}}{1-\frac{a_{\min,J}(z)}{\bar{\lambda}}} < L\,.$$

We define

$$a_{\text{corner}}(z, z_2) = \frac{\bar{\lambda}}{2L}\left[(L-\mathcal{B}_0-\mathcal{B}_1) - \sqrt{(\mathcal{B}_0+\mathcal{B}_1-L)^2 - \frac{4L\,\mathcal{B}_1\,\hat{Z}_*}{\bar{\lambda}}}\right] \tag{K.3}$$

to be the unique solution $a < a_{\min,J}(z)$ *of*

$$\frac{\mathcal{B}_0 + \mathcal{B}_1\frac{a+\hat{Z}_*}{a}}{1-\frac{a}{\bar{\lambda}}} = L\,. \tag{K.4}$$

Under the displayed condition the discriminant in a_{corner} is non-negative, so the root is real and $a_{\text{corner}} < a_{\min,J}$. We can then compute the upper bound that does not violate the subdiagonality condition (K.2) by following Algorithm 2.

Algorithm 2 Subdiagonal-validated bounds computation

Set $J_1 = J_2 = 1$, $\mathcal{Z} = \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$, and $z_2 = 10^{-4}$.

Compute $a_{\min}(z_1)$ for each $z_1 \in \mathcal{Z}$.

Compute

$$U_{\min}(z_1) = \frac{\mathcal{B}_0(\hat{Z}_*(z_1)) + \mathcal{B}_1(\hat{Z}_*(z_1)) \frac{a_{\min}(z_1) + \hat{Z}_*(z_1)}{a_{\min}(z_1)}}{1 - \frac{a_{\min}(z_1)}{\bar{\lambda}}}.$$

Compute the benchmark lower bound

$$L_2 = \mathcal{H}_1^L(\boldsymbol{\mathcal{B}}(z_2)).$$

if $U_{\min}(z_1) > L_2$ for every $z_1 \in \mathcal{Z}$ **then**

    Use the $a_{\min,J}$ branch for every grid point:

$$a^*(z_1) = a_{\min}(z_1), \qquad z_1 \in \mathcal{Z}.$$

    Compute the joint upper bound over $\mathcal{Z}$ using the $a_{\min,J}$-branch covariance formula.

else

    Use the corner branch (K.3) for every grid point:

$$a^*(z_1) = a_{\text{corner}}(z_1, z_2), \qquad z_1 \in \mathcal{Z}.$$

    Compute the joint upper bound over $\mathcal{Z}$ using the corner-branch covariance formula.

end if

The following lemmas furnish the covariance of the upper-bound estimator that Corollary L.3 combines across the grid, on the corner branch (Lemmas K.1–K.2), on the $a_{\min}$ branch (Lemmas K.3–K.4), and on a mixed grid (Lemma K.5).

Lemma K.1 (Delta-method gradient on the corner branch) *Define the corner upper bound by*

$$\mathcal{H}^U_{\text{corner}}(z_1, z_2) = \frac{\mathcal{B}_0(\hat{Z}_*(z)) + \ \mathcal{B}_1(\hat{Z}_*(z))\frac{a_{\text{corner}}(z_1,z_2)+\hat{Z}_*(z)}{a_{\text{corner}}(z_1,z_2)}}{1 - \frac{a_{\text{corner}}(z_1,z_2)}{\lambda}}$$

Then, as a function of the joint vector

$$\Big(\mathcal{B}_0(\hat{Z}_*(z_1)), \mathcal{B}_1(\hat{Z}_*(z_1)), \mathcal{B}_0(\hat{Z}_*(z_2)), \mathcal{B}_1(\hat{Z}_*(z_2))\Big),$$

the gradient of $\mathcal{H}^U_{\text{corner}}$ *is*

$$\nabla\mathcal{H}^U_{\text{corner}} = \big(0,\, 0,\, \partial_{\mathcal{B}_0(z_2)}\mathcal{H}^L_1(\boldsymbol{\mathcal{B}}(z_2)),\, \partial_{\mathcal{B}_1(z_2)}\mathcal{H}^L_1(\boldsymbol{\mathcal{B}}(z_2))\big).$$

In particular, for $J = 1$,

$$\mathcal{H}^L_1(\boldsymbol{\mathcal{B}}(z_2)) = \mathcal{B}_0(\hat{Z}_*(z_2)) + \frac{\mathcal{B}_0(\hat{Z}_*(z_2))\mathcal{B}_1(\hat{Z}_*(z_2))}{\mathcal{B}_0(\hat{Z}_*(z_2)) - \mathcal{B}_1(\hat{Z}_*(z_2))},$$

and hence

$$\partial_{\mathcal{B}_0(z_2)}\mathcal{H}^L_1 = 1 - \frac{\mathcal{B}_1(\hat{Z}_*(z_2))^2}{\Big(\mathcal{B}_0(\hat{Z}_*(z_2)) - \mathcal{B}_1(\hat{Z}_*(z_2))\Big)^2},$$

while

$$\partial_{\mathcal{B}_1(z_2)}\mathcal{H}^L_1 = \frac{\mathcal{B}_0(\hat{Z}_*(z_2))^2}{\Big(\mathcal{B}_0(\hat{Z}_*(z_2)) - \mathcal{B}_1(\hat{Z}_*(z_2))\Big)^2}.$$

Proof. By definition of $a_{\text{corner}}(z_1, z_2)$, we have

$$\frac{\mathcal{B}_0(\hat{Z}_*(z_1)) + \mathcal{B}_1(\hat{Z}_*(z_1))\frac{a_{\text{corner}}(z_1,z_2)+\hat{Z}_*(z_1)}{a_{\text{corner}}(z_1,z_2)}}{1 - \frac{a_{\text{corner}}(z_1,z_2)}{\lambda}} = \mathcal{H}_1^L(\boldsymbol{\mathcal{B}}(z_2)).$$

Therefore,

$$\mathcal{H}^U_{\text{corner}}(z_1, z_2) = \mathcal{H}_1^L(\boldsymbol{\mathcal{B}}(z_2)).$$

We now differentiate this identity with respect to the joint vector

$$\Big(\mathcal{B}_0(\hat{Z}_*(z_1)), \mathcal{B}_1(\hat{Z}_*(z_1)), \mathcal{B}_0(\hat{Z}_*(z_2)), \mathcal{B}_1(\hat{Z}_*(z_2))\Big).$$

Since the right-hand side depends only on

$$\mathcal{B}_0(\hat{Z}_*(z_2)) \quad \text{and} \quad \mathcal{B}_1(\hat{Z}_*(z_2)),$$

we immediately obtain

$$\partial_{\mathcal{B}_0(z_1)}\mathcal{H}^U_{\text{corner}}(z_1, z_2) = 0, \qquad \partial_{\mathcal{B}_1(z_1)}\mathcal{H}^U_{\text{corner}}(z_1, z_2) = 0.$$

Moreover,

$$\partial_{\mathcal{B}_0(z_2)}\mathcal{H}^U_{\text{corner}}(z_1, z_2) = \partial_{\mathcal{B}_0(z_2)}\mathcal{H}_1^L(\boldsymbol{\mathcal{B}}(z_2)),$$

and

$$\partial_{\mathcal{B}_1(z_2)}\mathcal{H}^U_{\text{corner}}(z_1, z_2) = \partial_{\mathcal{B}_1(z_2)}\mathcal{H}_1^L(\boldsymbol{\mathcal{B}}(z_2)).$$

For $J = 1$,

$$\mathcal{H}_1^L(\boldsymbol{\mathcal{B}}(z_2)) = \mathcal{B}_0(\hat{Z}_*(z_2)) + \frac{\mathcal{B}_0(\hat{Z}_*(z_2))\mathcal{B}_1(\hat{Z}_*(z_2))}{\mathcal{B}_0(\hat{Z}_*(z_2)) - \mathcal{B}_1(\hat{Z}_*(z_2))}.$$

Thus,

$$\partial_{\mathcal{B}_0(z_2)}\mathcal{H}_1^L(\boldsymbol{\mathcal{B}}(z_2)) = 1 + \frac{\mathcal{B}_1(\hat{Z}_*(z_2))\left(\mathcal{B}_0(\hat{Z}_*(z_2)) - \mathcal{B}_1(\hat{Z}_*(z_2))\right) - \mathcal{B}_0(\hat{Z}_*(z_2))\mathcal{B}_1(\hat{Z}_*(z_2))}{\left(\mathcal{B}_0(\hat{Z}_*(z_2)) - \mathcal{B}_1(\hat{Z}_*(z_2))\right)^2}.$$

Simplifying gives

$$\partial_{\mathcal{B}_0(z_2)}\mathcal{H}_1^L(\boldsymbol{\mathcal{B}}(z_2)) = 1 - \frac{\mathcal{B}_1(\hat{Z}_*(z_2))^2}{\left(\mathcal{B}_0(\hat{Z}_*(z_2)) - \mathcal{B}_1(\hat{Z}_*(z_2))\right)^2}.$$

Similarly,

$$\partial_{\mathcal{B}_1(z_2)}\mathcal{H}_1^L(\boldsymbol{\mathcal{B}}(z_2)) = \frac{\mathcal{B}_0(\hat{Z}_*(z_2))\left(\mathcal{B}_0(\hat{Z}_*(z_2)) - \mathcal{B}_1(\hat{Z}_*(z_2))\right) + \mathcal{B}_0(\hat{Z}_*(z_2))\mathcal{B}_1(\hat{Z}_*(z_2))}{\left(\mathcal{B}_0(\hat{Z}_*(z_2)) - \mathcal{B}_1(\hat{Z}_*(z_2))\right)^2},$$

which simplifies to

$$\partial_{\mathcal{B}_1(z_2)}\mathcal{H}_1^L(\boldsymbol{\mathcal{B}}(z_2)) = \frac{\mathcal{B}_0(\hat{Z}_*(z_2))^2}{\left(\mathcal{B}_0(\hat{Z}_*(z_2)) - \mathcal{B}_1(\hat{Z}_*(z_2))\right)^2}.$$

This proves the result. □

Lemma K.2 (Joint covariance on the corner branch) *For each $z_i \in \mathcal{Z} = \{z_1, ..., z_K\}$, the subdiagonality corrected upper bound on the corner branch satisfies*

$$\mathcal{H}_{\text{corner}}^U(z_i, z_2) = \mathcal{H}_1^L(\widehat{\boldsymbol{\Theta}}(z_2)), \qquad z_i \in \mathcal{Z}$$

Consequently, the joint covariance matrix of

$$\big(\mathcal{H}^U_{\text{corner}}(z_1, z_2), \ldots, \mathcal{H}^U_{\text{corner}}(z_K, z_2)\big)$$

is

$$\Sigma_{\mathcal{H}^U,\mathcal{Z}} = \sigma_L^2(z_2)\mathbf{1}_K\mathbf{1}_K^\top,$$

where

$$\sigma_L^2(z_2) = \nabla\mathcal{H}_1^L(\widehat{\boldsymbol{\Theta}}(z_2))^\top\Sigma_\Theta(z_2, z_2)\nabla\mathcal{H}_1^L(\widehat{\boldsymbol{\Theta}}(z_2)),$$

Proof. The proof follows directly from Lemma K.1. □

The next two lemmas furnish the analogous covariance on the $a_{\min}$ (smooth) branch, where the threshold is the unconstrained minimizer $a_{\min,J}$ rather than the subdiagonality-validity boundary a_{corner}. The simplification that makes the corner branch rank-one is absent here—each grid point now carries its own threshold and its own estimator vector—but a different simplification takes its place: because $a_{\min,J}$ is an interior minimizer, the envelope theorem removes the need to differentiate through the data-dependence of $a_{\min,J}(\boldsymbol{\mathcal{B}})$.

Lemma K.3 (Delta-method gradient on the $a_{\min}$ branch) *Fix a ridge penalty z and let $a_{\min,J} = a_{\min,J}(\boldsymbol{\mathcal{B}}(z)) \in (0, \bar{\lambda})$ be the interior minimizer in (K.1). Define the $a_{\min}$-branch (smooth-branch) upper bound by*

$$\mathcal{H}^U_{\min,J}(z) = \frac{\mathcal{D}_{J-1}(\boldsymbol{\mathcal{B}}(z)) + \mathcal{B}_J(\hat{Z}_*(z))\frac{a_{\min,J}+\hat{Z}_*(z)}{a_{\min,J}}}{1-\frac{a_{\min,J}}{\bar{\lambda}}} = \min_{a\in(0,\bar{\lambda})} \frac{\mathcal{D}_{J-1}(\boldsymbol{\mathcal{B}}(z)) + \mathcal{B}_J(\hat{Z}_*(z))\frac{a+\hat{Z}_*(z)}{a}}{1-\frac{a}{\bar{\lambda}}}.$$

Treating $\hat{Z}_*(z)$ and $\bar{\lambda}$ as deterministic (the deterministic-equivalent limit of Lemma 1 and the super-consistent trace of Lemma A.5, respectively), the gradient of $\mathcal{H}^U_{\min,J}$ in its arguments

$\boldsymbol{\mathcal{B}}(z) = \big(\mathcal{B}_0(\hat{Z}_*(z)), \ldots, \mathcal{B}_J(\hat{Z}_*(z))\big)$ *is*

$$\nabla \mathcal{H}^U_{\min,J}(\boldsymbol{\mathcal{B}}(z)) = \frac{1}{1 - \frac{a_{\min,J}}{\bar{\lambda}}} \Big(\underbrace{1, \ \ldots, \ 1}_{J \ \text{entries}}, \ \ \frac{a_{\min,J} + \hat{Z}_*(z)}{a_{\min,J}} \Big)^\top.$$

In particular, for $J = 1$, *since* $\mathcal{D}_0(\boldsymbol{\mathcal{B}}) = \mathcal{B}_0$,

$$\nabla \mathcal{H}^U_{\min,1}(\boldsymbol{\mathcal{B}}(z)) = \frac{1}{1 - \frac{a_{\min,1}}{\bar{\lambda}}} \Big(1, \ \ \frac{a_{\min,1} + \hat{Z}_*(z)}{a_{\min,1}} \Big)^\top.$$

Proof. Write the bound at a free threshold a as $\mathcal{H}^U_J(\boldsymbol{\mathcal{B}}(z); a) = N(a)/D(a)$ with

$$N(a) = \mathcal{D}_{J-1}(\boldsymbol{\mathcal{B}}(z)) + \mathcal{B}_J(\hat{Z}_*(z)) \frac{a + \hat{Z}_*(z)}{a}, \qquad D(a) = 1 - \frac{a}{\bar{\lambda}},$$

so that $\mathcal{H}^U_{\min,J}(z) = \mathcal{H}^U_J\big(\boldsymbol{\mathcal{B}}(z); a_{\min,J}(\boldsymbol{\mathcal{B}}(z))\big)$ is the value of the minimization over $a \in (0, \bar{\lambda})$. Since $a_{\min,J}$ is the interior root of the first-order condition $\partial_a \mathcal{H}^U_J = 0$ that defines (K.1), the envelope theorem gives, for each component $\mathcal{B}_k \equiv \mathcal{B}_k(\hat{Z}_*(z))$,

$$\frac{d}{d\mathcal{B}_k} \mathcal{H}^U_{\min,J} = \partial_{\mathcal{B}_k} \mathcal{H}^U_J \big|_{a = a_{\min,J}} + \underbrace{\partial_a \mathcal{H}^U_J \big|_{a = a_{\min,J}}}_{=0} \frac{\partial a_{\min,J}}{\partial \mathcal{B}_k} = \partial_{\mathcal{B}_k} \mathcal{H}^U_J \big|_{a = a_{\min,J}}.$$

The total gradient therefore equals the partial gradient at fixed $a = a_{\min,J}$, and no derivative of $a_{\min,J}(\boldsymbol{\mathcal{B}})$ is required—this is the analogue, on the $a_{\min}$ branch, of the identity $\mathcal{H}^U_{\text{corner}} = \mathcal{H}^L_1(\boldsymbol{\mathcal{B}}(z_2))$ used in Lemma K.1. Because $\mathcal{D}_{J-1}(\boldsymbol{\mathcal{B}}) = \sum_{j=0}^{J-1} \mathcal{B}_j$ enters N linearly with unit coefficients while $\mathcal{B}_J$ enters with coefficient $\frac{a + \hat{Z}_*(z)}{a}$, and D does not depend on $\boldsymbol{\mathcal{B}}$,

$$\partial_{\mathcal{B}_k} \mathcal{H}^U_J \big|_{a = a_{\min,J}} = \frac{1}{1 - a_{\min,J}/\bar{\lambda}}, \quad k = 0, \ldots, J-1, \qquad \partial_{\mathcal{B}_J} \mathcal{H}^U_J \big|_{a = a_{\min,J}} = \frac{1}{1 - a_{\min,J}/\bar{\lambda}} \frac{a_{\min,J} + \hat{Z}_*(z)}{a_{\min,J}}.$$

Collecting these coordinates yields the stated gradient; the $J = 1$ case is the specialization $\mathcal{D}_0 = \mathcal{B}_0$. □

Lemma K.4 (Joint covariance on the $a_{\min}$ branch) *On the $a_{\min}$ branch each grid point uses its own threshold, $a^*(z_i) = a_{\min,J}(\boldsymbol{\mathcal{B}}(z_i))$, so $\mathcal{H}^U_{\min,J}(z_i)$ is a function of $\boldsymbol{\mathcal{B}}(z_i)$ alone. Consequently the joint asymptotic covariance matrix of*

$$T^{1/2}\big(\mathcal{H}^U_{\min,J}(\widehat{\boldsymbol{\Theta}}(z_i)) - \mathcal{H}^U_{\min,J}(\boldsymbol{\mathcal{B}}(z_i))\big)_{i=1}^K$$

is the delta-method sandwich

$$\big(\Sigma^U_{\mathcal{H},\min}\big)_{ij} = \nabla\mathcal{H}^U_{\min,J}(\boldsymbol{\mathcal{B}}(z_i))^\top\, \Sigma_\Theta(z_i,z_j)\, \nabla\mathcal{H}^U_{\min,J}(\boldsymbol{\mathcal{B}}(z_j)), \qquad i,j=1,\dots,K,$$

with $\nabla\mathcal{H}^U_{\min,J}$ from Lemma K.3 and $\Sigma_\Theta(z_i,z_j)$ the joint covariance of $T^{1/2}(\widehat{\boldsymbol{\Theta}}(z_i)-\boldsymbol{\mathcal{B}}(z_i))$ and $T^{1/2}(\widehat{\boldsymbol{\Theta}}(z_j)-\boldsymbol{\mathcal{B}}(z_j))$ from Theorem H.1. Unlike the rank-one corner-branch covariance of Lemma K.2, $\Sigma^U_{\mathcal{H},\min}$ is generically full rank. Writing $G = \operatorname{blockdiag}\big(\nabla\mathcal{H}^U_{\min,J}(\boldsymbol{\mathcal{B}}(z_i))^\top\big)_{i=1}^K \in \mathbb{R}^{K\times K(J+1)}$ and $\widetilde{\Sigma} = \big(\Sigma_\Theta(z_i,z_j)\big)_{i,j=1}^K$ for the stacked estimator $(\widehat{\boldsymbol{\Theta}}(z_i))_{i=1}^K$, we have $\Sigma^U_{\mathcal{H},\min} = G\,\widetilde{\Sigma}\,G^\top$; since each gradient block is nonzero, G has full row rank K, so $\Sigma^U_{\mathcal{H},\min}$ is positive definite whenever $\widetilde{\Sigma}$ is. By contrast, the corner-branch map has rank one because all K bounds collapse onto the single vector $\widehat{\boldsymbol{\Theta}}(z_2)$.

Proof. Each $\mathcal{H}^U_{\min,J}(z_i)$ depends on the data only through $\widehat{\boldsymbol{\Theta}}(z_i)$, with gradient given by Lemma K.3. The joint CLT of Theorem H.1 for $(\widehat{\boldsymbol{\Theta}}(z_i))_{i=1}^K$ together with the delta method yields the stated sandwich, exactly as in Corollary L.2 with $\nabla\mathcal{H}^L_J$ replaced by $\nabla\mathcal{H}^U_{\min,J}$; the rank claim is the factorization $\Sigma^U_{\mathcal{H},\min} = G\widetilde{\Sigma}G^\top$ with G of full row rank K. □

The final lemma covers a grid on which some ridge penalties are validated on the corner branch and the rest on the $a_{\min}$ branch. It is the common generalization of Lemmas K.2 and K.4: each bound is, to first order, a linear functional of a single estimator vector—$\widehat{\boldsymbol{\Theta}}(z_i)$ on the $a_{\min}$ branch and the common benchmark $\widehat{\boldsymbol{\Theta}}(z_2)$ on the corner branch—and the cross covariances follow from the joint CLT.

Lemma K.5 (Joint covariance on a mixed grid) *Partition the grid* $\mathcal{Z} = \{z_1, \ldots, z_K\}$ *into* $\mathcal{Z}_{\min}$*, the points evaluated on the* $a_{\min}$ *branch, and* $\mathcal{Z}_{\text{corner}}$*, those evaluated on the corner branch against the common benchmark* z_2*. For each* z_i *define a representative ridge* ζ_i *and a gradient* g_i *by*

$$(\zeta_i,\, g_i) = \begin{cases} \big(z_i, \quad \nabla\mathcal{H}^U_{\min,J}(\boldsymbol{\mathcal{B}}(z_i))\big), & z_i \in \mathcal{Z}_{\min}, \\ \big(z_2, \quad \nabla\mathcal{H}^L_1(\boldsymbol{\mathcal{B}}(z_2))\big), & z_i \in \mathcal{Z}_{\text{corner}}, \end{cases}$$

with $\nabla\mathcal{H}^U_{\min,J}$ *from Lemma K.3 and* $\nabla\mathcal{H}^L_1$ *the lower-bound gradient of Corollary L.2. Then the joint asymptotic covariance matrix of the upper-bound vector* $T^{1/2}\big(\mathcal{H}^U_J(\widehat{\boldsymbol{\Theta}}(z_i)) - \mathcal{H}^U_J(\boldsymbol{\mathcal{B}}(z_i))\big)_{i=1}^K$ *is*

$$\big(\Sigma^U_{\mathcal{H}}\big)_{ij} = g_i^\top\, \Sigma_\Theta(\zeta_i, \zeta_j)\, g_j, \qquad i, j = 1, \ldots, K,$$

where $\Sigma_\Theta(\zeta_i, \zeta_j)$ *denotes the joint-CLT covariance of the components of* $\widehat{\boldsymbol{\Theta}}(\zeta_i)$ *and* $\widehat{\boldsymbol{\Theta}}(\zeta_j)$ *on which* g_i *and* g_j *act—the* $J{+}1$ *components* $(\mathcal{B}_0, \ldots, \mathcal{B}_J)$ *on the* $a_{\min}$ *branch and the two components* $(\mathcal{B}_0, \mathcal{B}_1)$ *of the* $J{=}1$ *benchmark on the corner branch. The relevant block is thus* $(J{+}1) \times (J{+}1)$*,* 2×2*, or, when an* $a_{\min}$ *point with* $J > 1$ *meets a corner point, the rectangular* $(J{+}1) \times 2$ *cross-covariance; in the implementation of Algorithm 2, where* $J{=}1$ *on both branches, every block is* 2×2*.*

Proof. By construction each upper bound depends on the data through a single estimator vector. For $z_i \in \mathcal{Z}_{\min}$ this is $\widehat{\boldsymbol{\Theta}}(z_i)$, with first-order coefficient $g_i = \nabla\mathcal{H}^U_{\min,J}(\boldsymbol{\mathcal{B}}(z_i))$ from Lemma K.3. For $z_i \in \mathcal{Z}_{\text{corner}}$, Lemma K.2 gives $\mathcal{H}^U_{\text{corner}}(z_i, z_2) = \mathcal{H}^L_1(\widehat{\boldsymbol{\Theta}}(z_2))$, a function of $\widehat{\boldsymbol{\Theta}}(z_2)$ alone, with first-order coefficient $g_i = \nabla\mathcal{H}^L_1(\boldsymbol{\mathcal{B}}(z_2))$. Writing each bound to first order as $g_i^\top\big(\widehat{\boldsymbol{\Theta}}(\zeta_i) - \boldsymbol{\mathcal{B}}(\zeta_i)\big)$ and applying the joint CLT (Theorem H.1) and the delta method gives the cross term $g_i^\top\Sigma_\Theta(\zeta_i, \zeta_j)\, g_j$. □

Remark K.1 Lemma K.5 nests the two homogeneous cases. If $\mathcal{Z}_{\text{corner}} = \mathcal{Z}$, then $\zeta_i \equiv z_2$ and $g_i \equiv \nabla\mathcal{H}_1^L(\boldsymbol{\mathcal{B}}(z_2))$ for all i, so $(\Sigma_{\mathcal{H}}^U)_{ij} = \nabla\mathcal{H}_1^L(\boldsymbol{\mathcal{B}}(z_2))^\top \Sigma_\Theta(z_2, z_2)\nabla\mathcal{H}_1^L(\boldsymbol{\mathcal{B}}(z_2)) = \sigma_L^2(z_2)$ for every i, j (the gradient evaluated at $\boldsymbol{\mathcal{B}}(z_2)$, equivalently at the consistent plug-in $\widehat{\boldsymbol{\Theta}}(z_2)$ of Lemma K.2), and $\Sigma_{\mathcal{H}}^U = \sigma_L^2(z_2)\,\mathbf{1}_K\mathbf{1}_K^\top$, recovering Lemma K.2. If $\mathcal{Z}_{\min} = \mathcal{Z}$, it reduces to Lemma K.4. Algorithm 2 selects a single branch for the entire grid, so one of these two homogeneous cases applies in the current implementation; the mixed formula is what a per-z branch choice would require.

L Using Multiple z Values

We evaluate the bounds on a grid of ridge penalties $z_1, \ldots, z_K$ and combine them, which requires the confidence limits to correct for the search across the grid. At each z the point estimates bracket the target, $\mathcal{H}_J^L(\boldsymbol{\mathcal{B}}(z)) \le \mathcal{R}_*^2 \le \mathcal{H}_J^U(\boldsymbol{\mathcal{B}}(z))$, the upper bound using the threshold a^* of Appendix K, so the largest lower bound and the smallest upper bound across the grid stay valid for $\mathcal{R}_*^2$. The following exact result turns either extremum into a one-sided confidence bound that adjusts for the multiplicity: applied with $X_i = \mathcal{H}_J^L(\hat{\boldsymbol{\Theta}}(\boldsymbol{z_i}))$ and $Y = \mathcal{R}_*^2$ it gives the lower confidence bound of Corollary L.2, and applied to the negated upper bounds $X_i = -\mathcal{H}_J^U(\hat{\boldsymbol{\Theta}}(\boldsymbol{z_i}))$ with $Y = -\mathcal{R}_*^2$ the upper confidence bound of Corollary L.3.

Theorem L.1 (Exact lower confidence bound in the multivariate Gaussian case) *Let*

$$X = (X_1, \ldots, X_d)^\top \sim N(\mu, \Sigma),$$

where $\mu = (\mu_1, \ldots, \mu_d)^\top \in \mathbb{R}^d$ *is unknown and* $\Sigma \in \mathbb{R}^{d\times d}$ *is known and positive semidefinite, with diagonal entries* $\Sigma_{ii} > 0$. *Define*

$$\theta := \max_{1\le i\le d} \mu_i,$$

and suppose that $Y \in \mathbb{R}$ *satisfies*

$$Y > \theta.$$

Let $D := \mathrm{diag}(\Sigma_{11}^{1/2}, \ldots, \Sigma_{dd}^{1/2})$*, and let*

$$R := D^{-1}\Sigma D^{-1}$$

be the corresponding correlation matrix. For $\alpha \in (0,1)$*, let* $c_{1-\alpha}(R)$ *be the unique real number satisfying*

$$\mathbb{P}_R(Z_1 \le c_{1-\alpha}(R), \ldots, Z_d \le c_{1-\alpha}(R)) = 1 - \alpha,$$

where $Z = (Z_1, \ldots, Z_d)^\top \sim N(0, R)$*. Define*

$$L_\alpha(X) := \max_{1 \le i \le d} \Big(X_i - c_{1-\alpha}(R)\sqrt{\Sigma_{ii}} \Big).$$

Then

$$\inf_{\mu:\, \max_i \mu_i < Y} \mathbb{P}_\mu(L_\alpha(X) \le Y) \ge 1 - \alpha.$$

Equivalently, $[L_\alpha(X), \infty)$ *is an exact one-sided* $(1-\alpha)$*-confidence interval for* Y*, uniformly over all* (μ, Y) *such that* $Y > \max_i \mu_i$*.*

Moreover,

$$\inf_{\mu:\, \max_i \mu_i = Y} \mathbb{P}_\mu(L_\alpha(X) \le Y) = 1 - \alpha,$$

so the bound is sharp.

Proof. Fix (μ, Y) such that $Y > \max_i \mu_i$. Since $\mu_i \le \theta < Y$ for every i,

$$\left\{ \frac{X_i - \mu_i}{\sqrt{\Sigma_{ii}}} \le c_{1-\alpha}(R) \ \forall i \right\} \subseteq \left\{ X_i - c_{1-\alpha}(R)\sqrt{\Sigma_{ii}} \le Y \ \forall i \right\}.$$

Therefore,

$$\begin{aligned} \mathbb{P}_\mu(L_\alpha(X) \le Y) &= \mathbb{P}_\mu\Big(X_i - c_{1-\alpha}(R)\sqrt{\Sigma_{ii}} \le Y \ \forall i \Big) \\ &\ge \mathbb{P}_\mu\left(\frac{X_i - \mu_i}{\sqrt{\Sigma_{ii}}} \le c_{1-\alpha}(R) \ \forall i \right). \end{aligned}$$

Now

$$\left(\frac{X_1 - \mu_1}{\sqrt{\Sigma_{11}}}, \ldots, \frac{X_d - \mu_d}{\sqrt{\Sigma_{dd}}} \right)^\top \sim N(0, R),$$

hence by the definition of $c_{1-\alpha}(R)$,

$$\mathbb{P}_\mu\left(\frac{X_i - \mu_i}{\sqrt{\Sigma_{ii}}} \le c_{1-\alpha}(R) \ \forall i \right) = 1 - \alpha.$$

This proves

$$\mathbb{P}_\mu(L_\alpha(X) \le Y) \ge 1 - \alpha.$$

Taking the infimum over all μ such that $\max_i \mu_i < Y$ yields

$$\inf_{\mu:\, \max_i \mu_i < Y} \mathbb{P}_\mu(L_\alpha(X) \le Y) \ge 1 - \alpha.$$

To prove sharpness, take $\mu_i = Y$ for every i, so that $\max_i \mu_i = Y$. Then

$$\mathbb{P}_\mu(L_\alpha(X) \le Y) = \mathbb{P}_\mu\Big(X_i - c_{1-\alpha}(R)\sqrt{\Sigma_{ii}} \le Y \ \forall i \Big) = \mathbb{P}_\mu\left(\frac{X_i - \mu_i}{\sqrt{\Sigma_{ii}}} \le c_{1-\alpha}(R) \ \forall i \right) = 1 - \alpha,$$

by the definition of $c_{1-\alpha}(R)$. Hence $\inf_{\mu:\,\max_i \mu_i = Y} \mathbb{P}_\mu(L_\alpha(X) \le Y) \le 1-\alpha$, and combined with the inequality established above, this infimum equals $1-\alpha$. This establishes sharpness. □

Corollary L.2 (Lower confidence bound) *Pick a grid $\{z_1, \cdots, z_K\}$ and let $\Sigma_{\mathcal{H}}$ be the asymptotic covariance matrix of the lower-bound vector $T^{1/2}(\mathcal{H}_J^L(\hat{\boldsymbol{\Theta}}(\boldsymbol{z_i})) - \mathcal{H}_J^L(\boldsymbol{\mathcal{B}}(\boldsymbol{z_i})))_{i=1}^K$. By the joint CLT (Theorem H.1) and the delta method,*

$$(\Sigma_{\mathcal{H}})_{ij} \;=\; \nabla\mathcal{H}_J^L\big(\boldsymbol{\mathcal{B}}(z_i)\big)^\top \Sigma_\Theta(z_i, z_j)\, \nabla\mathcal{H}_J^L\big(\boldsymbol{\mathcal{B}}(z_j)\big), \qquad i,j = 1,\ldots,K, \tag{L.1}$$

where $\Sigma_\Theta(z_i, z_j)$ is the joint asymptotic covariance of $T^{1/2}(\hat{\boldsymbol{\Theta}}(z_i) - \boldsymbol{\mathcal{B}}(z_i))$ and $T^{1/2}(\hat{\boldsymbol{\Theta}}(z_j) - \boldsymbol{\mathcal{B}}(z_j))$, and $\nabla\mathcal{H}_J^L$ is the gradient of the lower bound in its arguments $\boldsymbol{\Theta} = (\Theta_0, \ldots, \Theta_J)$; for $J = 1$,

$$\nabla\mathcal{H}_1^L(\boldsymbol{\mathcal{B}}) = \Big(1 - \tfrac{\mathcal{B}_1^2}{(\mathcal{B}_0 - \mathcal{B}_1)^2},\quad \tfrac{\mathcal{B}_0^2}{(\mathcal{B}_0 - \mathcal{B}_1)^2}\Big)^\top.$$

Let

$$\mathcal{R}_{\mathcal{H}} \;=\; \operatorname{diag}(\Sigma_{\mathcal{H}})^{-1/2} \Sigma_{\mathcal{H}} \operatorname{diag}(\Sigma_{\mathcal{H}})^{-1/2}. \tag{L.2}$$

Let $Z = (Z_1, \cdots, Z_K) \sim N(0, \mathcal{R}_{\mathcal{H}})$ and let $c_{1-\alpha}$ be the unique real number satisfying

$$\mathbb{P}_{\mathcal{R}_{\mathcal{H}}}(Z_1 \le c_{1-\alpha}, \ldots, Z_K \le c_{1-\alpha}) = 1 - \alpha\,.$$

Let

$$L_\alpha := \max_{1 \le i \le K} \Big(\mathcal{H}_J^L(\hat{\boldsymbol{\Theta}}(\boldsymbol{z_i})) - c_{1-\alpha}(\mathcal{R}_{\mathcal{H}}) \sqrt{\Sigma_{\mathcal{H},ii}/T} \Big).$$

Then, with $\Sigma_{\mathcal{H}}$ replaced by a consistent estimator, L_α is an asymptotic one-sided $(1-\alpha)$

lower confidence bound for $\mathcal{R}_*^2$*:*

$$\liminf_{T\to\infty} \mathbb{P}\big(L_\alpha \le \mathcal{R}_*^2\big) \ge 1-\alpha\,.$$

Corollary L.3 (Upper confidence bound) *Let* $\Sigma_{\mathcal{H}}^U$ *be the asymptotic covariance matrix of the upper-bound vector* $T^{1/2}(\mathcal{H}_J^U(\hat{\boldsymbol{\Theta}}(\boldsymbol{z_i})) - \mathcal{H}_J^U(\boldsymbol{\mathcal{B}}(\boldsymbol{z_i})))_{i=1}^K$*, with correlation* $\mathcal{R}_{\mathcal{H}}^U$ *and equicoordinate quantile* $c_{1-\alpha}(\mathcal{R}_{\mathcal{H}}^U)$*. Applying the Theorem to the negated upper bounds, the statistic*

$$U_\alpha := \min_{1\le i\le K}\Big(\mathcal{H}_J^U(\hat{\boldsymbol{\Theta}}(\boldsymbol{z_i})) + c_{1-\alpha}(\mathcal{R}_{\mathcal{H}}^U)\sqrt{\Sigma_{\mathcal{H},ii}^U/T}\Big)$$

is, with $\Sigma_{\mathcal{H}}^U$ *replaced by a consistent estimator, an asymptotic one-sided* $(1-\alpha)$ *upper confidence bound for* $\mathcal{R}_*^2$*:*

$$\liminf_{T\to\infty} \mathbb{P}\big(U_\alpha \ge \mathcal{R}_*^2\big) \ge 1-\alpha\,.$$

The covariance $\Sigma_{\mathcal{H}}^U$ *is the sandwich of Corollary L.2 with the gradient of the upper bound in place of* $\nabla\mathcal{H}_J^L$*. On the corner branch of Appendix K, where every upper bound equals* $\mathcal{H}_1^L(\boldsymbol{\mathcal{B}}(z_2))$*, it collapses to the rank-one* $\sigma_L^2(z_2)\,\mathbf{1}\mathbf{1}^\top$ *of Lemma K.2 and* $c_{1-\alpha}(\mathcal{R}_{\mathcal{H}}^U)$ *reduces to the univariate quantile. On the* $a_{\min}$ *branch the gradient is the envelope-theorem expression of Lemma K.3—no derivative of* $a_J^*(\boldsymbol{\Theta})$ *is needed—and* $\Sigma_{\mathcal{H}}^U$ *is the generically full-rank sandwich of Lemma K.4; a grid mixing the two branches is covered by Lemma K.5.*

M Simulations

We evaluate the finite-sample performance of our bounds using a Monte Carlo experiment. The main question is whether the upper and lower bounds can recover the true Sharpe ratio at the sample sizes and complexity ratios $c = P/T$ relevant for the empirical application. We

find that both bounds concentrate near the true value, with the gap between them narrowing as T grows. All estimates in this section are produced using Algorithm 1 in Appendix J.

One feature of the design deserves comment upfront. In both the simulations that follow and the empirical application of Section 5, the base assets fed into our estimator are managed portfolios. This is important for Assumption 1: individual stock returns have conditional means that move with observable characteristics, while managed portfolios formed from a rich cross-section of stocks do not. Indeed, Didisheim et al. (2024) establish that, in the limit as the number of underlying stocks $N \to \infty$, managed portfolio returns satisfy Assumption 1. The DGP below mirrors this construction.

M.1 Data-generating process

The DGP generates stock characteristics from a latent factor model, specifies conditional expected returns as a nonlinear function of these characteristics, and forms managed portfolios by interacting characteristics with realized returns exactly as in Section 5. To calibrate the experiment, we draw on monthly data on 132 stock characteristics for U.S. stocks over the period 1963 to 2025 (from Jensen et al., 2023, JKP henceforth).[16] Empirically, we find that stock characteristics are not independent from each other: 25% of the principal components of JKP characteristics explain approximately 80% of the total variance. We refer to this low-dimensional structure as the effective characteristic dimension and replicate it in our simulations.

At each period $t = 1, \ldots, T$, we generate d characteristics $X_{i,t} \in \mathbb{R}^d$ for each of $i = 1, \ldots, N$ stocks from a latent factor structure with $d_{\text{eff}} < d$ effective factors, matching the low-dimensional structure documented above. Let $\zeta_{i,t} \sim \mathcal{N}(0, I_{d_{\text{eff}}})$ and define

$$X_{i,t} \;=\; \Lambda^\top \text{diag}(\sigma_1, \ldots, \sigma_{d_{\text{eff}}})\zeta_{i,t} \;+\; \eta_{i,t},$$

[16] JKP characteristics for individual stocks can be downloaded at https://wrds-www.wharton.upenn.edu/pages/get-data/contributed-data-forms/global-factor-data/. Detailed documentation and further factor portfolio data are available at jkpfactors.com.

where $\Lambda \in \mathbb{R}^{d_{\text{eff}} \times d}$ has orthonormal rows, and $\eta_{i,t} \sim \mathcal{N}(0, \sigma_\eta^2 I_d)$. The factor variances follow a power-law decay and are normalized to sum to one:

$$\sigma_k^2 \;=\; \frac{k^{-3/2}}{\sum_{j=1}^{d_{\text{eff}}} j^{-3/2}}, \qquad k = 1, \ldots, d_{\text{eff}}.$$

The noise variance σ_η^2 is chosen so that the first d_{eff} population principal components explain a fraction $\phi \in (0, 1)$ of total variance. Since the population eigenvalues are $\{\sigma_k^2 + \sigma_\eta^2\}_{k=1}^{d_{\text{eff}}}$ and σ_η^2 (repeated $d - d_{\text{eff}}$ times), this implies

$$\frac{\sum_{k=1}^{d_{\text{eff}}} (\sigma_k^2 + \sigma_\eta^2)}{\sum_{k=1}^{d_{\text{eff}}} (\sigma_k^2 + \sigma_\eta^2) + (d - d_{\text{eff}})\sigma_\eta^2} \;=\; \phi,$$

which yields

$$\sigma_\eta^2 \;=\; \frac{1 - \phi}{\phi\, d - d_{\text{eff}}}, \qquad \text{provided } d_{\text{eff}} < \phi d.$$

This construction ensures that a small number of principal components explain most of the variation in characteristics, consistent with empirical evidence. Conditional expected stock returns are generated from the characteristics via

$$\alpha_{i,t} \;=\; \alpha \, \sin\!\big(\gamma \, w^\top X_{i,t}\big) \, T^{-1/4},$$

where $w \in \mathbb{R}^d$ is a unit-norm vector drawn once per simulation and $\gamma > 0$ controls the signal nonlinearity. The sinusoidal mapping is bounded, preventing outlier alphas from dominating the cross-section, and non-monotone, so that the relationship between characteristics and expected returns is richer than what a monotone function can capture. Because the managed portfolios constructed below use ReLU activations, a different nonlinearity in the DGP ensures that the portfolios cannot replicate the true conditional mean exactly.

Stock returns follow an n_F-factor model,

$$R_{i,t+1} \;=\; \alpha_{i,t} \;+\; \beta_i^\top F_{t+1} \;+\; \varepsilon_{i,t+1},$$

where

$$F_{t+1} \sim \mathcal{N}(\mu_F \mathbf{1}_{n_F}, \sigma_F^2 I_{n_F}), \qquad \beta_i \sim \text{Uniform}([-1,1]^{n_F}), \qquad \varepsilon_{i,t+1} \sim \mathcal{N}(0, \sigma_\varepsilon^2).$$

The conditional mean and covariance of stock returns are

$$\mu_{i,t} \;=\; \alpha_{i,t} + \beta_i^\top \mathbb{E}[F_{t+1}] \;=\; \alpha_{i,t} + \beta_i^\top (\mu_F \mathbf{1}_{n_F}),$$

collected as $\mu_t = (\mu_{1,t}, \ldots, \mu_{N,t})^\top$, and

$$\Sigma = \sigma_F^2 \beta\beta^\top + \sigma_\varepsilon^2 I_N,$$

an $N \times N$ matrix distinct from the $P \times P$ asset-space covariance of Assumption 1. Stock-level μ_t is time-varying; the managed-portfolio returns in (M.1) inherit the i.i.d. structure of Assumption 1 in the large-N limit.

The $T^{-1/4}$ attenuation in the alpha specification is a local-alternatives device. At the calibration, $\alpha = 15$ dominates $\beta_i^\top(\mu_F \mathbf{1}_{n_F})$, so $\mu_{i,t} \approx \alpha_{i,t}$ and $\|\mu_t\|^2$ is of order $N\alpha^2 T^{-1/2}$. The rank-n_F correction in Σ is small relative to $\sigma_\varepsilon^2 I_N$, so $\mathcal{R}_*^2 \approx \|\mu_t\|^2/\sigma_\varepsilon^2 \propto N\alpha^2 T^{-1/2}$. The population Sharpe ratio shrinks at the same rate as the estimation noise, keeping the relative difficulty of the inference problem comparable across sample sizes.

We construct managed portfolios from returns and characteristics as follows. Let $X_t \in \mathbb{R}^{N\times d}$ denote the cross-section of characteristics at time t and $R_{t+1} \in \mathbb{R}^N$ the cross-section of returns at time $t+1$. Draw a random matrix $\Omega \in \mathbb{R}^{P\times d}$ once per simulation with rows

$\omega_p^\top$ satisfying

$$\omega_{pj} \sim \mathcal{N}(0,1).$$

The p-th managed portfolio return is

$$R^F_{t+1,p} \;=\; \frac{1}{N}\sum_{i=1}^{N} R_{i,t+1}\,\mathrm{ReLU}\left(\frac{1}{2}\sqrt{\frac{2}{d}}\,X_{i,t}^\top\omega_p\right), \qquad p=1,\ldots,P, \tag{M.1}$$

where $\mathrm{ReLU}(x) = \max(x,0)$ is applied element-wise.[17] In the empirical work we will be forming portfolios in exactly the same way.

M.2 Bounds

For each simulation draw, we compute the GRS lower and upper bounds with $J = 1$ and a grid of ridge penalties $(z_1,\cdots,z_4) = (10^{-4},10^{-3},10^{-2},10^{-1})$, and the data-driven thresholds $a_1^*(\hat{\mathbf{\Theta}}(z_k))$ from Appendix K, for each $z_k, k = 1,\cdots,4$. We then combine the four bounds following Appendix L: the lower confidence limit applies the equicoordinate quantile $c_{1-\alpha}(\mathcal{R}_{\mathcal{H}})$ to the maximum of the lower bounds, and the upper confidence limit applies $c_{1-\alpha}(\mathcal{R}^U_{\mathcal{H}})$ to the minimum of the upper bounds, each at the one-sided 2.5% level, correcting for the search across the grid.

We report the relative error

$$\frac{\widehat{SR} - SR^*}{SR^*}, \tag{M.2}$$

[17] The Rectified Linear Unit (ReLU) is a piecewise-linear activation function widely used in neural networks. Applied element-wise to $X_t^\top\omega_p$, it introduces nonlinearity into the portfolio weights while remaining computationally tractable. The pre-factor $1/2\cdot\sqrt{2/d}$ can be written as $\varsigma\sqrt{2/d}$ with $\varsigma = 0.5$. The $\sqrt{2/d}$ component is the He initialization scale (He, Zhang, Ren and Sun, 2015), standard in neural tangent kernel analyses: it sets the weight variance to $2/d$, which preserves signal magnitude through ReLU layers and prevents the random features from degenerating as d grows. The additional scale $\varsigma \in (0,1]$ controls the operating region of the activation: smaller ς concentrates the ReLU arguments closer to zero, where the activation switches between its active and inactive regimes, effectively tuning the degree of nonlinearity in the portfolio weights.

where $\widehat{SR}$ denotes the bound and SR^* is the oracle Sharpe ratio.

The oracle return is computed from the conditionally efficient portfolio,

$$R^*_{t+1} \;=\; \frac{\mu_t^\top \Sigma^{-1} R_{t+1}}{1 + \mu_t^\top \Sigma^{-1} \mu_t},$$

using the μ_t and Σ defined in the DGP above. The oracle annualized Sharpe ratio is

$$SR^* \;=\; \sqrt{12}\,\frac{\bar{R}^*}{\hat{\sigma}(R^*)}.$$

This benchmark differs from the estimand. Our bounds target the *unconditional* maximal Sharpe ratio spanned by the P managed portfolios of (M.1), whereas SR^* is the realized Sharpe ratio of the *conditionally* efficient portfolio formed from the underlying stocks. The conditional stock-level frontier weakly dominates the unconditional managed-portfolio frontier, so SR^* is an upper benchmark that the estimand need not attain. The relative errors in (M.2) therefore mix two things: the tightness of the bounds around their own estimand, and the gap between that estimand and the stock-level conditional oracle. This matters for the level of the reported errors, though not for their comparative statics in T and P, which is what the design is used for below.

M.3 Simulation results

We simulate returns on 500 stocks and generate $P = 1{,}000$ or $5{,}000$ managed portfolios from them; we evaluate samples of size $T = 120$, 240, or 360 months. The remaining simulation hyperparameters are summarized in Table 1.

The grid $z \in [10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}]$ in Table 1 spans the usable range of penalties. The smallest value, $z = 10^{-4}$, is as small as the numerical stability of $(zI + \hat{\Sigma})^{-1}$ permits; larger penalties widen the bounds without tightening coverage, and beyond $z = 10^{-1}$ they turn vacuous as shrinkage erases the eigenvalues' informational content. We do not single out one

Symbol	Description	Value
N	Number of stocks	500
d	Characteristic dimension	100
d_{eff}	Effective characteristic dimension (latent factors)	25
ϕ	Fraction of variance explained by factors	0.8
T	Time-series length	120, 240, 360
P	Number of managed portfolios	1000, 5000
J	Bound order	1
z	Ridge penalties grid	$[10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}]$
α	Alpha strength	15.0
γ	Nonlinearity scale	0.5
ς	Random feature scale (He/NTK)	0.5
n_F	Number of risk factors	1
μ_F	Annual factor mean	7%
σ_F	Annual factor volatility	14%
σ_ε	Idiosyncratic volatility	1
S	Number of Monte Carlo draws	2,000

Table 1: Simulation hyperparameters.

penalty: we evaluate the bounds at all four grid values and combine them as described above, taking the largest lower bound and the smallest upper bound, a combination at least as tight as any single z. The combined bounds concentrate near the oracle Sharpe ratio (Figure 6), and the empirical application in Section 5 carries the same grid and the same combination.

Figure 6 compares the distributions of the relative error (M.2) for the lower and upper bounds across combinations of T (rows) and P (columns). Each panel contains two

distributions, corresponding to the point-estimate and CLT-based bounds, with the CLT-based distribution displayed in orange.

Several patterns emerge. The lower bound is centered to the left of zero, as it should be. Increasing T tightens it visibly: the distribution narrows and shifts toward the true Sharpe ratio as the sample grows. The effect of increasing P on the lower bound is more modest. The upper bound, by contrast, is centered to the right of zero and responds differently to T and P. Increasing T tightens it, but increasing P pushes it further from the true value. Thus, P and T pull in opposite directions for the upper bound: more test assets enrich the frontier but also worsen the bound.

Figure 7 displays the corresponding distribution of the optimal $a_1^*/\bar{\lambda}$ from Appendix K, computed at the penalty z_k selected as in Appendix L.

First, $a^*/\bar{\lambda}$ decreases with T for fixed P. As T grows, the complexity ratio $c = P/T$ falls and the effective regularization $\hat{Z}_*(z)$ shrinks: with more data, the ridge estimator captures more of the eigenvalue spectrum, so the optimal threshold is correspondingly lower. Second, $a^*/\bar{\lambda}$ increases with P for fixed T. Here the argument is less direct because $\bar{\lambda}$ itself changes with P. However, a larger P raises both c and $\hat{Z}_*(z)$; in addition, more of the spectrum falls into the small-eigenvalue range, increasing $\mathcal{B}_J$ relative to $\mathcal{D}_J$. The simulations confirm that these effects dominate, pushing $a^*/\bar{\lambda}$ up. The same mechanism produces the widening of the upper bound with P visible in Figure 6: a higher complexity ratio $c = P/T$ inflates a^*, which in turn inflates the upper bound.

Medians in Figure 7 range from 0.1 at $(P, T) = (1{,}000, 360)$ to 0.36 at $(P, T) = (5{,}000, 120)$, so chosen subdiagonality accounts for between 10% and 36% of $\mathcal{H}_J^U$ across the simulation grid. The data pin down the remaining 64% to 94%, so the upper bound rests primarily on estimable quantities.

The analysis above and the empirical application in Section 5 both use $J = 1$. We now evaluate the approximation error associated with this choice. In principle, higher values of

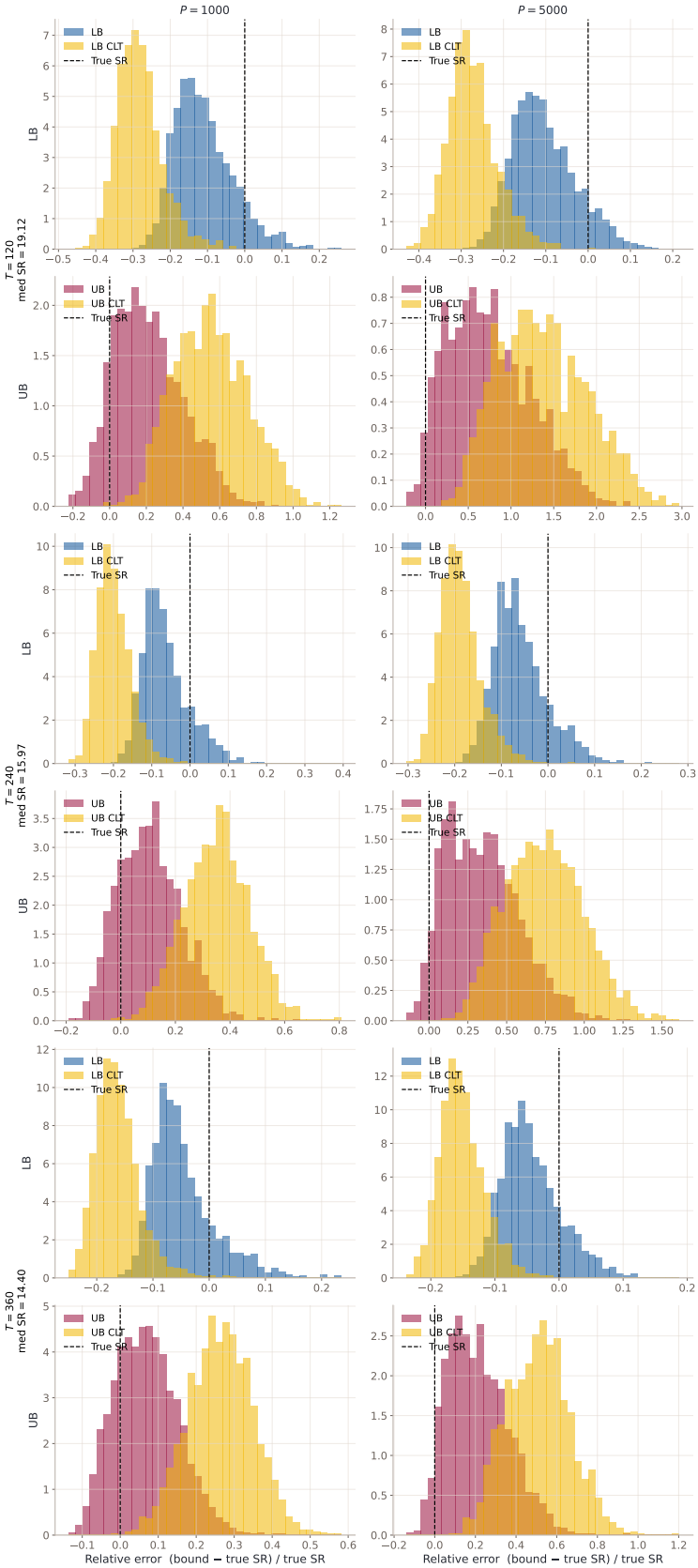


Figure 6: Lower and upper bound relative error distributions for $T \in \{120, 240, 360\}$ and $P \in \{1{,}000, 5{,}000\}$ nonlinearly managed random feature portfolios. Both bounds tighten with T; the upper bound widens with P, reflecting the cost of higher complexity for the bounding scheme.

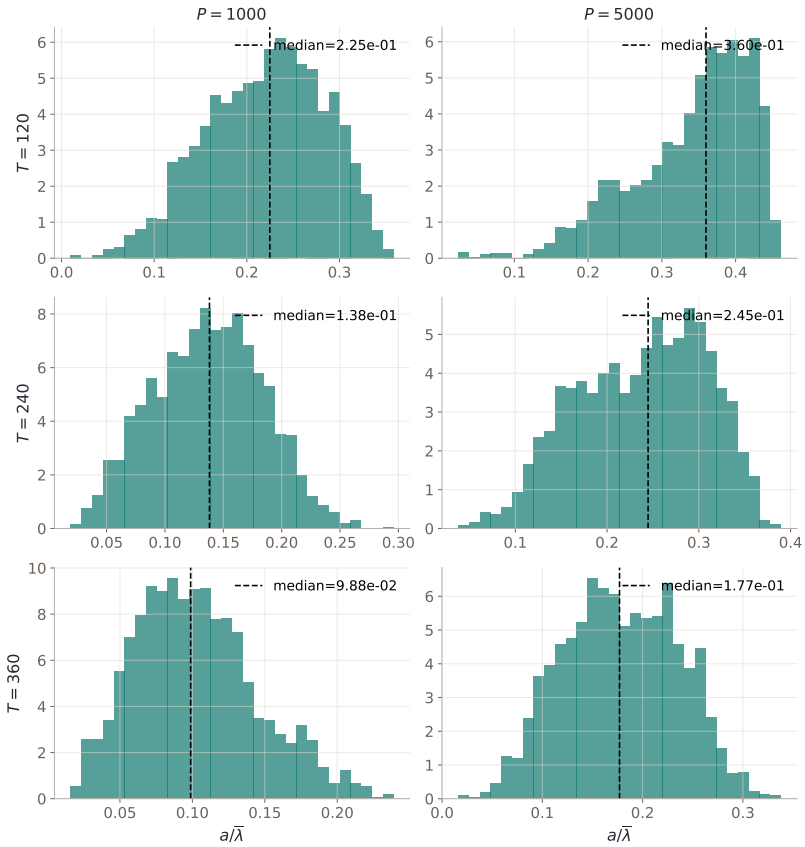


Figure 7: Distribution of $a^*/\bar{\lambda}$ for $T \in \{120, 240, 360\}$ and $P \in \{1{,}000, 5{,}000\}$, computed for nonlinearly managed random feature portfolios under the power-law eigenvalue decay of Appendix M. Dashed lines indicate medians.

J recover progressively finer detail from the eigenvalue–Sharpe ratio relationship, tightening both bounds. In practice, the CLT-based confidence bands for the higher-order statistics $\hat{\Theta}_j^d(z)$ widen with j: each additional order amplifies estimation noise. For T in the range 120–360, the confidence bands at $J = 2$ are too wide to be informative.

To assess the potential loss from the first-order approximation, we run a separate experiment with $T = 1{,}080$ (90 years of monthly data), holding all other parameters at the values in Table 1. At this sample size, the CLT approximation for $J = 2$ produces reasonable confidence bands, permitting a direct comparison with $J = 1$.

Figure 8 reports the results for $P \in \{1{,}000, 5{,}000\}$. Moving from $J = 1$ to $J = 2$ shifts

both bound distributions toward zero relative error: the lower bound moves closer to the true value from below, and the upper bound from above. The magnitude of the improvement, however, is small. For both values of P, the $J = 1$ bounds already concentrate near the true Sharpe ratio, and the second-order term adds only a marginal correction. This is consistent with the monotone decay of $\mathcal{B}_j$: $\mathcal{B}_1$ captures the dominant correction beyond $\mathcal{B}_0$, and higher-order terms contribute progressively less. We use $J = 1$ in the empirical analysis: the simulations show the tradeoff between statistical precision and approximation accuracy is favorable.

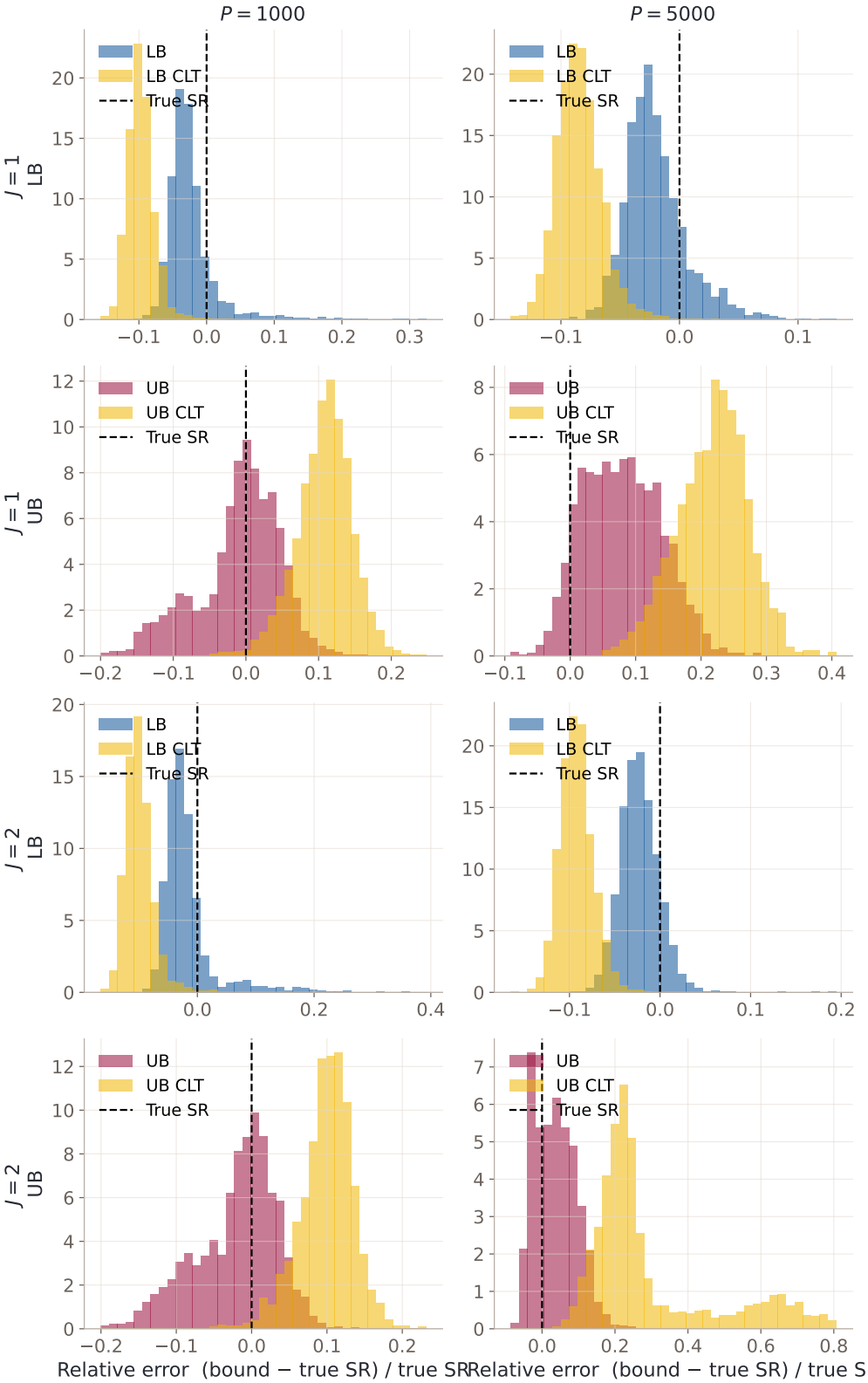


Figure 8: Lower and upper bound relative error distributions for $J \in \{1, 2\}$, $T = 1{,}080$, and $P \in \{1{,}000, 5{,}000\}$ nonlinearly managed random feature portfolios. The second-order correction ($J = 2$) yields only a marginal improvement over $J = 1$, suggesting that omitting the second-order term is not detrimental.